



Fluid Mechanics: an Introduction and some Relevant Applications

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Preface

The following notes are intended to provide the foundations for the study of Fluid Mechanics as a branch of applied physics and its applications in engineering. The approach taken is that set forth by Amable Liñán in his “Apuntes de Mecánica de Fluidos” (Publicaciones de la E.T.S.I. Aeronáuticos, Madrid 1964). The examples are mostly taken from the exams given at Universidad Carlos III over the past 15 years.

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Chapter 1

Introductory remarks and starting assumptions

Solids, liquids and gases

At macroscopic level, the most noticeable difference between solids and fluids lies in their different deformation capability. Under the action of a small external force, solids respond with a small deformation. Such behavior arises because in solids the internal force counteracting the external force is proportional to the deformation. Fluids, on the other hand, are easily deformed under the action of external forces. The internal resistive force turns out to be proportional to the rate of deformation, rather than to the deformation itself. As a direct consequence, fluids adapt their shape to that of the containers limiting them.

The differences between liquids and gases are much less fundamental. A first difference concerns their density: the density of liquids is much larger than that of gases (for given conditions of temperature and pressure), which in turn causes the force required to accelerate liquids to be much larger than that involved in the motion of gases. Concerning the mechanical properties of liquids and gases, perhaps the most fundamental difference lies in their compressibility. The increment of density associated with the application of a given overpressure is much less pronounced in liquids than it is in gases, which is expressed by the inequality

$$\left(\frac{\partial \rho}{\partial p}\right)_{T,l} \ll \left(\frac{\partial \rho}{\partial p}\right)_{T,g}, \quad (1.1)$$

where ρ , p and T represent the density, pressure and temperature, respectively. As an illustrative example, consider a balloon full of air. We all know that it is relatively simple to reduce the volume of the balloon by appropriately pressing with our hands, that is, we can easily increase the gas density by applying an overpressure. Things are quite different when the balloon is full of water instead: the balloon deforms under the action of an overpressure, but its volume- and therefore the liquid density- remains virtually unaltered¹. Similarly, by varying the temperature while maintaining the pressure constant one may change the density of a gas appreciably, whereas the same temperature variation would only slightly modify the density of a given liquid. Because of their low compressibility, under most conditions one may study the motion of liquids as if they had constant density. When we introduce that assumption, we shall say that the liquid behaves as a “perfect liquid”.

¹In fact, the overpressure needed to halve the specific volume of water is on the order of 10^6 atmospheres.

All of the macroscopic properties mentioned above are the result of the different microscopic structure exhibited by solids, liquids and gases. The force exerted between two molecules is known to be a function of the distance between their centers d , with a variation that is schematically represented in Fig. 1.1.

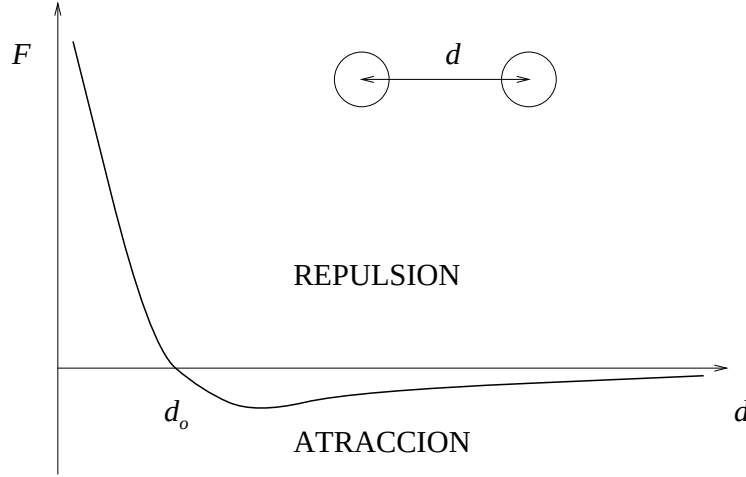


Figure 1.1: Schematic representation of the force between two molecules as a function of the distance between their centers.

When the distance becomes very small, the two molecules tend to repel each other, whereas an attraction force appears for values of d larger than a critical value d_o , on the order of the characteristic molecular size $d_o \simeq 3 \times 10^{-10}$ m, for which the system of two molecules would remain in stable equilibrium.

For known values of the density ρ and molecular mass W , it is straightforward to compute the average distance between two molecules according to $d = (\rho N_A / W)^{-1/3}$, where $N_A = 6.023 \times 10^{23}$ molecules/mol is the so-called Avogadro's number. The computation reveals that for gases at normal ambient temperature and pressure $d \simeq 3 \times 10^{-9}$ m $\simeq 10d_o$, while the molecules of liquids and solids are much more closely packed, with characteristic inter-molecular distances $d \simeq d_o$. Gas molecules can be expected to experience very weak attraction forces and, in the first approximation, one may assume that they move freely, interacting with the surrounding molecules only when collisions take place. This microscopic view explains the high compressibility of gases, as well as their deformation capability and their ability to fill up all of the available space in the container. In solids and liquids the intermolecular forces can be expected to be much more important. The repulsion force keeps molecules from getting closer, which explains the low compressibility of liquids and solids. Their different deformation capability emerges because, although the intermolecular distance is similar, the liquid molecules are in continuous relative motion with respect to their neighboring counterparts, whereas the molecules in a solid occupy fixed relative positions. It is worth mentioning that, although water is clearly a liquid, other substances exhibit a dual behavior and are not easily classified as solid or liquid. Paint is one such example: if we let it sit for long enough it will end up behaving as a highly elastic solid, readily recovering the liquid state when stirred at a sufficiently fast rate.

The fluid as a continuum: fluid particles.

The matter in a given fluid is discretely distributed: molecules in gases are separated by large distances, and, even for liquids, whose molecules are closely packed together, the mass is concentrated in the nuclei. The large number of molecules involved appears as a key complicating feature of the fluid motion (at normal ambient conditions, there exist of the order of 3×10^{16} molecules in a cubic millimeter of air and about one thousand times more in a cubic millimeter of water). Clearly, it is hopeless to try to describe the overall dynamics by accounting for the fate of each individual molecule, a computation that would require in principle integration of Newton's second law for each single molecule to determine its location and velocity, with account taken of intermolecular forces and collisions. This is a phenomenal computation that is out of reach of present and foreseeable future computational capabilities. A different perspective, based on the so-called continuum hypotheses, is to be adopted instead.

In previous years we have studied systems with uniform properties that had only a few degrees of freedom. For instance, in classical equilibrium thermodynamics the state of a gas within a given container was described by giving its density and temperature, both assumed to be uniform, while in mechanics the motion of a rigid body was determined in terms of its velocity and angular velocity, that is, a total of six degrees of freedom. In fluids, our experience tells us that things are not quite so simple. We have all seen the motion of suspended dust particles as they are made visible by a beam of sun light crossing a window, which allows us to visualize the complex air motion driven by buoyancy in a room. Clearly, a single velocity vector is not sufficient to describe this system, characterized by large spatial and temporal variations.

In the following discussion, we will use the concept of macroscopic distance, L , defined for a given flow field as the characteristic distance that needs to be traveled to observe *significant* variations of the different flow properties. For instance, for the buoyancy dominated air flow in your bedroom one could tentatively say that significant velocity variations occur over distances on the order of $L = 10$ cm. This macroscopic distance is of interest when describing the flow field because it tells you how detailed your description must be to give a realistic picture of the flow. For the bedroom example, with velocity variations occurring over distances of the order of $L = 10$ cm, to describe the velocity field at a given instant of time it would be enough to provide the velocity vector at points separated 1 cm apart (or perhaps 1 mm apart, if one wants the description to be very accurate). Based on this observation, it is natural to approach the study of fluid motion by dividing the fluid in small parcels called *fluid particles* with respect to which we shall define the concepts of velocity, density, etc. Each particle is centered at a location $\bar{x}$ at a time t . Their size must be much smaller than the characteristic macroscopic distance L of the fluid system under study, so that the velocity, density, etc of the system can be described accurately as continuous functions of $\bar{x}$ and t in terms of the properties of each fluid particle. The assumption that the different fluid properties can be described as continuous functions (although they are not, as mentioned above in connection with the mass distribution) is the so-called *continuum hypotheses*, also used in other branches of mechanics, such as elasticity.

To investigate the conditions of validity of the continuum hypotheses it is of interest to consider as an example the case of a gas and the concept of density. For a fluid particle of volume δV centered at $\bar{x}$ at time t the density is defined according to $\rho = \sum m_i / \delta V$, where $\sum m_i$ is the sum of the masses of all molecules contained in the fluid particle. For the proposed continuum description to make sense, the value of ρ should be independent of δV , so that the value of the density at time t and position $\bar{x}$ can be defined uniquely as $\rho(\bar{x}, t)$. To see whether that

independence is verified, we plot in Fig. 1.2 the variation of $\sum m_i/\delta V$ as a function of the fluid particle size $(\delta V)^{1/3}$.

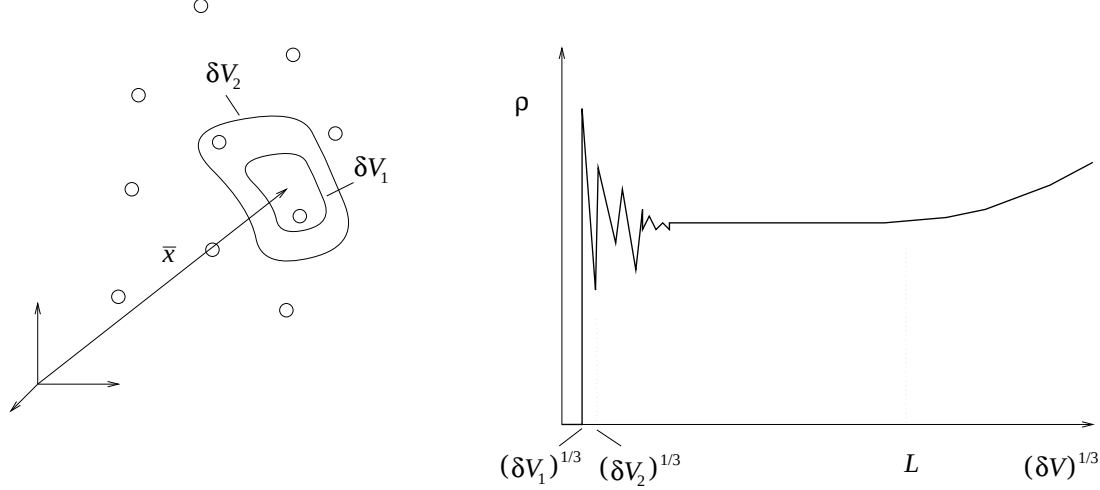


Figure 1.2: The concept of fluid particle.

When the size of the fluid particle is much smaller than the characteristic intermolecular distance d , it is very likely that the fluid particle be empty, and the associated density defined as $\sum m_i/\delta V$ be identically zero. As the size increases, a critical value is reached for which the first molecule enters the particle, leading to an abrupt change in the density. A decrease in density would follow for increasing values of $(\delta V)^{1/3}$, until the second molecule enters the fluid particle, leading to a new jump in the density. Subsequent jumps would be progressively smaller for increasing values of δV , and would become almost inappreciable when the fluid-particle size $(\delta V)^{1/3}$ reaches values much larger than d , when the particle contains a large number of molecules $\delta V/d^3 \gg 1$. The plateau value, defining uniquely the density at time t and position $\bar{x}$, remains independent of δV for increasing values of the particle size, until the particle is so large that it begins to embody fluid parcels with truly different properties, i.e., for $(\delta V)^{1/3} \sim L$. According to the figure, uniquely defined flow properties based on the fluid particle concept can be defined if and only if the fluid-particle size falls in the intermediate range

$$d \ll (\delta V)^{1/3} \ll L. \quad (1.2)$$

The development therefore depends critically on the condition $L \gg d$, which enables the introduction of fluid particles with well defined properties. Since for air at normal ambient density $d \simeq 3.4 \times 10^{-9}$ m, most flows of engineering interest easily satisfy the required condition and are therefore amenable to a continuum description. Note that, in liquid flow, with characteristic intermolecular distances $d \simeq d_o \simeq 3 \times 10^{-10}$ m, the condition $L \gg d$ can be satisfied even more easily than it is for gases. Extreme examples for which the continuum description is not applicable because $L \sim d$ include rarefied flows over reentering vehicles in the upper atmosphere, for which the average distance between neighboring molecules become very large, and the tiny flows appearing in micro and nanotechnology applications.

Density, velocity and internal energy

The concept of fluid particle allows us to define the density of the fluid according to

$$\rho(\bar{x}, t) = \lim_{\delta V \rightarrow 0} \frac{\sum m_i}{\delta V}, \quad (1.3)$$

subject to the additional condition $(\delta V)^{1/3} \gg d$ needed to avoid the discrete character of the fluid at microscopic level. Similarly, the fluid velocity can be defined as the mass averaged value of the velocity of the molecules contained within the fluid particle

$$\bar{v} = \lim_{\delta V \rightarrow 0} \frac{\sum m_i \bar{v}_i}{\sum m_i}, \quad (1.4)$$

which is the velocity of the center of mass for the given set of molecules. On the other hand, the energy per unit mass contained inside δV is given by $\sum E_i / \sum m_i$, where $E_i = m_i |\bar{v}_i|^2 / 2 + E_{v_i} + E_{r_i} + \dots$ represents the energy of each individual molecule, obtained as the sum of the kinetic energy associated with translation $m_i |\bar{v}_i|^2 / 2$, those corresponding to rotation E_{r_i} and vibration E_{v_i} (for non-monoatomic molecules), etc. It is customary to write the energy per unit mass by accounting separately for the kinetic energy associated with the flow velocity $|\bar{v}|^2 / 2$, thereby giving

$$\lim_{\delta V \rightarrow 0} \frac{\sum E_i}{\sum m_i} = e + |\bar{v}|^2 / 2, \quad (1.5)$$

where

$$e = \lim_{\delta V \rightarrow 0} \frac{\sum m_i |\bar{v}_i - \bar{v}|^2 / 2 + E_{v_i} + E_{r_i} + \dots}{\sum m_i} \quad (1.6)$$

is the so-called internal energy (the derivation of (1.6) is left as an exercise for the reader). The first term above, $|\bar{v}_i - \bar{v}|^2 / 2$, corresponds to the kinetic energy of the random molecular motion, a quantity directly associated with the definition of temperature.

Local thermodynamic equilibrium

Classical thermodynamics deals with systems that are in mechanical and thermal equilibrium, for which all thermodynamic properties are spatially uniform and constant in time. When the evolution of a given system is studied with the laws of classical thermodynamics, it is assumed that the process is so slow that the system at each instant can be considered to be in equilibrium. A very useful result for homogeneous systems in equilibrium is that their thermodynamic state is completely determined by giving the values of any two independent state variables, while the corresponding values of the other state variables can be computed through the equations of state. Fluid mechanics deals with systems that are not in thermodynamic equilibrium, with fluid properties changing both spatially and temporally. It is however found that the results of equilibrium thermodynamics are approximately valid for most of the non-equilibrium states found in fluid mechanics, in that an observer moving with the local fluid velocity can describe the fluid state in terms of thermodynamic variables that are inter-related by the same equations of state that apply for systems in strict equilibrium. This hypothesis of *local thermodynamic equilibrium* can be rigorously justified through the kinetic energy for gases, whereas for liquids it is substantiated by ample experimental evidence.

In the case of gases, molecules exchange momentum and energy through collisions, adjusting in that way their state to the local state of molecular motion. Molecular collisions constitute therefore the mechanism that enables thermodynamic equilibrium to be attained in a gas. The average distance travelled by a molecule between two subsequent collisions, called the mean free path λ , can be estimated by equating the volume swept by the molecule in its displacement between collisions, $d_o^2 \lambda$, to the volume per molecule that exists in the gas, d^3 , to give $\lambda/d \simeq (d/d_o)^2$, yielding for instance $\lambda \simeq 4 \times 10^{-7}$ m for air in normal ambient conditions. The corresponding time between collisions τ can be estimated by anticipating that molecules move with an average velocity a close to that of sound, giving for example $\tau = \lambda/a \simeq 10^{-9}$ s for normal ambient conditions.

For gas systems in which the mean free path λ is much smaller than the macroscopic length L each molecule will undergo a large number of collisions before reaching regions with different macroscopic properties. As a result, the fluid state remains at all times very close to the equilibrium conditions corresponding to the local values of density and internal energy. Hence, the criterion to be satisfied in a gas for the local thermodynamic equilibrium approximation to apply is

$$\frac{\lambda}{L} \ll 1 \quad (1.7)$$

where λ/L is the so-called Knudsen number. Likewise, for unsteady gas flow, if $\mathcal{T}$ denotes the characteristic time for variation of the macroscopic gas properties, the condition to be satisfied for local thermodynamic equilibrium is that $\mathcal{T} \gg \tau$. Note that the criterion given in (1.7) is more restrictive than that associated with the continuum hypothesis (Ec. 1.2).

Thermodynamic variables and useful relationships

From the continuum hypothesis, we have been able to define the flow velocity $\bar{v}(\bar{x}, t)$ together with the density $\rho(\bar{x}, t)$ and internal energy $e(\bar{x}, t)$. By incorporating now the assumption of local thermodynamic equilibrium, the remaining thermodynamic variables are determined through the equations of state. For instance, the local value of the entropy per unit mass $s(\bar{x}, t)$ can be obtained through an equation of state $s = s(e, \rho)$, which can be alternatively written in the form $e = e(s, \rho)$. Since

$$de = Tds - pd(1/\rho) \quad (1.8)$$

the corresponding temperature and pressure follow from

$$T = \left(\frac{\partial e}{\partial s} \right)_\rho \quad (1.9)$$

and

$$p = - \left(\frac{\partial e}{\partial \rho^{-1}} \right)_s. \quad (1.10)$$

Analogously, the enthalpy per unit mass can be defined as $h = e + p/\rho$.

Since a complete account of thermodynamic variables and equations of state can be found in any basic book of classical thermodynamics, we shall limit the presentation here to a brief summary of the equations of state that are most useful for the analysis of problems of fluid motion. The treatment is restricted to two idealized fluid states, the so-called perfect liquids and perfect gases.

Perfect liquids

A perfect liquid satisfies that its density ρ and specific heat c are constants, so that one can write

$$\rho = \rho_o \quad (1.11)$$

and

$$e = cT + e_o, \quad (1.12)$$

where e_o is the internal energy associated with the absolute zero temperature. From the definition of enthalpy, it follows that

$$h = cT + e_o + p/\rho_o, \quad (1.13)$$

whereas by integration of 1.8 one obtains

$$s = c \ln(T) + s_o, \quad (1.14)$$

where s_o is an integration constant defining an arbitrary reference state. Many liquids behave as perfect over relatively large intervals of pressure and temperature. For instance, in most applications water can be assumed to be a perfect liquid with density $\rho_o = 10^3 \text{ kg/m}^3$ and specific heat $c = 4180 \text{ J/(kg K)}$.

Perfect gases

The equation of state for a perfect gas is

$$p/\rho = R_g T, \quad (1.15)$$

where the gas constant $R_g = R^o/W$ is obtained as the ratio of the universal gas constant $R^o = 8.314 \text{ J/(mol K)}$ and the mean molecular mass of the gas W . The thermal equations of state for the internal energy, enthalpy, and entropy are given by

$$e = c_v T + e_o, \quad (1.16)$$

$$h = c_p T + e_o, \quad (1.17)$$

$$s = c_v \ln(p/\rho^\gamma) + s_o, \quad (1.18)$$

where c_v and $c_p = c_v + R_g$ are, respectively, the specific heats at constant volume and at constant pressure. The ratio of specific heats $\gamma = c_p/c_v$ equals $\gamma = 7/5$ for diatomic gases. Treating air as a perfect gas with $R_g = 287 \text{ J/(kg K)}$ and $c_v = 717 \text{ J/(kg K)}$ is an excellent approximation under most conditions of practical interest. At extremely high pressures or extremely low temperatures, however, the equation of state (1.15) must be replaced by a more complex expression (e.g., Van der Waals equation). Also, the assumption of constant specific heats must be relaxed in systems where the temperature changes appreciably, as occurs for instance in combustion processes.

Chapter 2

Flow kinematics

Vector and tensor formulae

This introductory section presents a brief account of different definitions of vector and tensor analysis that will be used in the following chapters. In the following, we shall consider an orthogonal coordinate system with unit vectors $(\bar{e}_1, \bar{e}_2, \bar{e}_3)$ at each point. In that coordinate system, any vector $\bar{a}$ admits a representation as a function of three scalars a_i such that $\bar{a} = \sum_i a_i \bar{e}_i$. Some common operations between two vectors $\bar{a}$ and $\bar{b}$ include the **scalar product** $\bar{a} \cdot \bar{b} = \sum_i a_i b_i$ and the **cross product**

$$\bar{a} \wedge \bar{b} = \begin{vmatrix} \bar{e}_1 & \bar{e}_2 & \bar{e}_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2 b_3 - a_3 b_2) \bar{e}_1 + (a_3 b_1 - a_1 b_3) \bar{e}_2 + (a_1 b_2 - a_2 b_1) \bar{e}_3. \quad (2.1)$$

A second-order tensor $\bar{\bar{A}}$ is to be interpreted in the framework of the present course as a linear operator that, when acting over a vector $\bar{a}$, gives as result another vector $\bar{b} = \bar{\bar{A}} \cdot \bar{a}$ with components $b_i = \sum_j A_{ij} a_j$. The transverse tensor $\bar{\bar{A}}^T$, with components $(\bar{\bar{A}}^T)_{ij} = (\bar{\bar{A}})_{ji}$, satisfies $\bar{a} \cdot \bar{\bar{A}} = \bar{\bar{A}}^T \cdot \bar{a}$. The tensor $\bar{\bar{I}}$ with components $I_{ii} = 1$ y $I_{ij} = 0$ if $i \neq j$ is the so-called identity tensor, which verifies $\bar{\bar{I}} \cdot \bar{a} = \bar{a}$ for any given vector $\bar{a}$. Of interest in the following development are the **dyadic product of two vectors**, $\bar{a}\bar{b}$, which produces a tensor of components $(\bar{a}\bar{b})_{ij} = a_i b_j$, and the **contraction of two tensors**, $\bar{\bar{A}} : \bar{\bar{B}}$ which gives as a result a scalar $\bar{\bar{A}} : \bar{\bar{B}} = \sum_i \sum_j A_{ij} B_{ij}$. Note that this last result is quite different from the product of the two tensors, $\bar{\bar{A}} \cdot \bar{\bar{B}}$, which is another tensor of components $(\bar{\bar{A}} \cdot \bar{\bar{B}})_{ij} = \sum_k A_{ik} B_{kj}$. Also of interest is the **cross product of a vector $\bar{a} = (a_1, a_2, a_3)$ and a tensor $\bar{\bar{A}}$** , which gives a tensor according to

$$\bar{a} \wedge \bar{\bar{A}} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \bar{\bar{A}}. \quad (2.2)$$

Besides cartesian (rectangular) coordinates $(x_1 = x, x_2 = y, x_3 = z)$, two other types of orthogonal curvilinear coordinates will be found to be useful: cylindrical coordinates $(x_1 = r, x_2 = \theta, x_3 = z)$ and spherical coordinates $(x_1 = r, x_2 = \theta, x_3 = \phi)$, which are schematically represented in Fig. 2.1, along with their associated unit vectors $(\bar{e}_1, \bar{e}_2, \bar{e}_3)$.

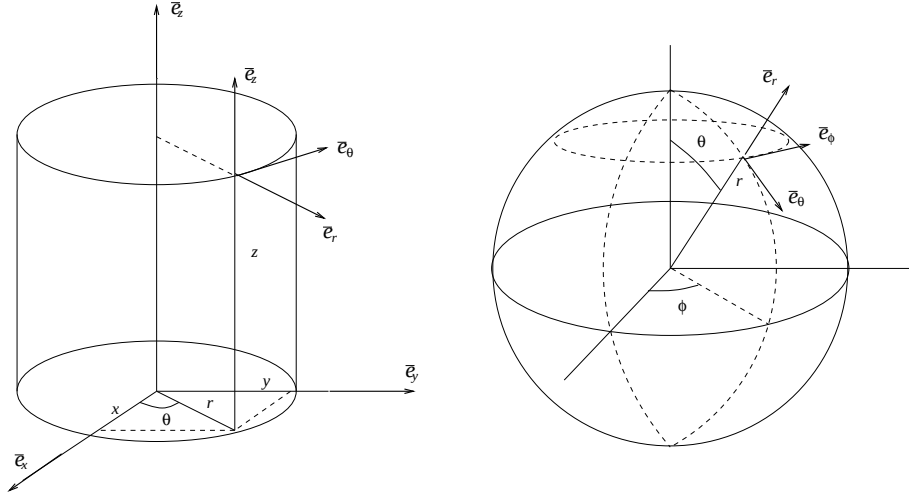


Figure 2.1: Schematic representation of relevant coordinate systems.

The differential line elements associated for each of the three coordinate systems considered are, respectively, $d\bar{x} = (dx, dy, dz)$, $d\bar{x} = (dr, r d\theta, dz)$ and $d\bar{x} = (dr, r d\theta, r \sin\theta d\phi)$. The coefficients that premultiply each of the differentials in the above expressions are the so-called scale factors (h_1, h_2, h_3) , which reduce to $(h_1 = 1, h_2 = 1, h_3 = 1)$ for cartesian coordinates, to $(h_1 = 1, h_2 = r, h_3 = 1)$ for cylindrical coordinates, and to $(h_1 = 1, h_2 = r, h_3 = r \sin\theta)$ for spherical coordinates. The introduction of these scale factors enables the differential operators to be written in a compact form independent of the coordinate system. Thus, the **gradient of a scalar function** Φ is in general given by

$$\nabla\Phi = \left(\frac{1}{h_1} \frac{\partial\Phi}{\partial x_1}, \frac{1}{h_2} \frac{\partial\Phi}{\partial x_2}, \frac{1}{h_3} \frac{\partial\Phi}{\partial x_3} \right), \quad (2.3)$$

while its **laplacian** can be expressed in the form

$$\nabla^2\Phi = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial x_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial\Phi}{\partial x_1} \right) + \frac{\partial}{\partial x_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial\Phi}{\partial x_2} \right) + \frac{\partial}{\partial x_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial\Phi}{\partial x_3} \right) \right]. \quad (2.4)$$

Similarly, the **divergence of a vector function** $\bar{a} = a_1 \bar{e}_1 + a_2 \bar{e}_2 + a_3 \bar{e}_3$ is

$$\nabla \cdot \bar{a} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial x_1} (h_2 h_3 a_1) + \frac{\partial}{\partial x_2} (h_1 h_3 a_2) + \frac{\partial}{\partial x_3} (h_1 h_2 a_3) \right], \quad (2.5)$$

while its **curl** is given by

$$\begin{aligned} \nabla \wedge \bar{a} &= \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \bar{e}_1 & h_2 \bar{e}_2 & h_3 \bar{e}_3 \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ h_1 a_1 & h_2 a_2 & h_3 a_3 \end{vmatrix} = \frac{1}{h_2 h_3} \left[\frac{\partial}{\partial x_2} (h_3 a_3) - \frac{\partial}{\partial x_3} (h_2 a_2) \right] \bar{e}_1 + \\ &\quad \frac{1}{h_1 h_3} \left[\frac{\partial}{\partial x_3} (h_1 a_1) - \frac{\partial}{\partial x_1} (h_3 a_3) \right] \bar{e}_2 + \frac{1}{h_1 h_2} \left[\frac{\partial}{\partial x_1} (h_2 a_2) - \frac{\partial}{\partial x_2} (h_1 a_1) \right] \bar{e}_3. \end{aligned} \quad (2.6)$$

Also, the **gradient of a vector function** produces a tensor according to

$$(\nabla \bar{a})_{ii} = \frac{1}{h_i} \frac{\partial a_i}{\partial x_i} + \sum_{k \neq i} \frac{a_k}{h_i h_k} \frac{\partial h_i}{\partial x_k} \quad \text{and} \quad (\nabla \bar{a})_{ij} = \frac{1}{h_i} \frac{\partial a_j}{\partial x_i} - \frac{a_i}{h_i h_j} \frac{\partial h_i}{\partial x_j} \quad \text{if } i \neq j, \quad (2.7)$$

while the **divergence of a tensor** $\nabla \cdot \bar{\bar{A}}$ gives a vector function of components

$$(\nabla \cdot \bar{\bar{A}})_i = \frac{h_i}{h} \sum_j \frac{\partial}{\partial x_j} \left(\frac{h A_{ij}}{h_i h_j} \right) + \sum_j \frac{A_{ij} + A_{ji}}{h_i h_j} \frac{\partial h_i}{\partial x_j} - \sum_j \frac{A_{jj}}{h_i h_j} \frac{\partial h_j}{\partial x_i}, \quad (2.8)$$

where $h = h_1 h_2 h_3$.

The so-called Gauss formula

$$\int_{\Sigma} (\bar{n} \circ \phi) d\sigma = \int_V (\nabla \circ \phi) dV, \quad (2.9)$$

where ϕ represents a scalar, vector or tensor function, V is a domain enclosed by the closed surface Σ , and $\circ$ denotes any of the products defined above, will be found to be useful in manipulating the conservation equations.

Some preliminary concepts

Two different approaches, called respectively **Eulerian and Lagrangian descriptions**, can be adopted when describing the velocity field. The Lagrangian approach takes a discrete point of view by treating the fluid as an ensemble of fluid particles. The flow field is described in terms of the fluid-particle trajectories

$$\bar{x} = \bar{x}_T(\bar{x}_o, t), \quad (2.10)$$

which are a function of their initial position $\bar{x}_o$ and time t , with the velocity and acceleration being computed by straightforward derivation to give $\bar{v} = \partial \bar{x}_T / \partial t$ and $\bar{a} = \partial^2 \bar{x}_T / \partial t^2$. This Lagrangian approximation, which might be of use in specific applications (e.g., for the analysis of the disperse phase in two-phase problems), leads in general to complex conservation laws. For that reason, in the following derivation we shall adopt instead the Eulerian description, which assumes the velocity to be a continuous vector field $\bar{v}(\bar{x}, t)$ as a function of the position $\bar{x}$ and time t .

A flow field is said to be uniform when the spatial differences of the different flow properties are strictly zero. Similarly, a fluid flow is said to be steady when it does not change in time. Thus, a **uniform velocity** field satisfies $\bar{v} = \bar{v}(t)$, whereas a **steady velocity** field is such that $\bar{v} = \bar{v}(\bar{x})$. At times, the steadiness of a given flow depends on the observer. For instance, for a flow over an aircraft moving with constant velocity, the flow is clearly unsteady for an observer standing on the ground, whereas for the pilot the flow is effectively steady. A moving frame of reference in that case would clearly facilitate the analysis of the flow.

A stagnation point is defined as a point where the velocity is zero (all three components). For a given velocity field $\bar{v} = \bar{v}(\bar{x}, t)$, the stagnation points are determined by the equation

$$\bar{v}(\bar{x}, t) = 0. \quad (2.11)$$

Clearly, the location of the stagnation points depends on the reference frame selected.

Trajectories and path lines

In its motion, a fluid particle initially located at $\bar{x} = \bar{x}_o$ changes its position in time as expressed in (2.10). Given the velocity field $\bar{v}(\bar{x}, t)$, the location of the fluid particle at each instant of time is determined by integration of

$$\frac{d\bar{x}}{dt} = \bar{v}(\bar{x}, t) \quad (2.12)$$

with initial condition $\bar{x} = \bar{x}_o$ at $t = t_o$. This vector equation can be alternatively expressed in its three components

$$dt = \frac{h_1 dx_1}{v_1(x_1, x_2, x_3, t)} = \frac{h_2 dx_2}{v_2(x_1, x_2, x_3, t)} = \frac{h_3 dx_3}{v_3(x_1, x_2, x_3, t)} \quad (2.13)$$

to be integrated subject to $x_1 = x_{1_o}$, $x_2 = x_{2_o}$ and $x_3 = x_{3_o}$. The solution can be expressed in the general form (2.10), which reduces to

$$\begin{aligned} x &= x_T(x_o, y_o, z_o, t) \\ y &= y_T(x_o, y_o, z_o, t) \\ z &= z_T(x_o, y_o, z_o, t) \end{aligned} \quad (2.14)$$

for cartesian coordinates.

The trajectory contains information about the **path** followed by the particle and also about the rate with which it travels along it. The equations that describe the path lines can be obtained by eliminating the time in (2.10). For instance, for cartesian coordinates, once the time has been eliminated in (2.14), the following two equations emerge

$$f(x_o, y_o, z_o, x, y, z) = g(x_o, y_o, z_o, x, y, z) = 0 \quad (2.15)$$

defining two surfaces whose intersection is the path line followed by the fluid particle initially located at (x_o, y_o, z_o) .

Fluid lines, fluid surfaces and fluid volumes

Fluid particles initially located along the curve

$$\bar{x} = \bar{x}_l(\lambda), \quad (2.16)$$

where λ denotes a parameter describing the curve¹, will continue to form a curve at any subsequent time. The equation describing the evolution of a **fluid line** is obtained by writing the equation for the trajectories, given in (2.10), for the fluid particles located at the initial time $t = t_o$ along the curve defined in (2.16) to yield

$$\bar{x} = \bar{x}_T(\bar{x}_l(\lambda), t). \quad (2.17)$$

Similarly, fluid particles forming initially a surface

$$\bar{x} = \bar{x}_s(\alpha, \beta), \quad (2.18)$$

where α y β are parameters² will continue to form a surface (**fluid surface**). The corresponding fluid-surface equation

$$\bar{x} = \bar{x}_T(\bar{x}_s(\alpha, \beta), t), \quad (2.19)$$

¹Recall that a curve can be in general expressed in the parametric form $\bar{x} = \bar{x}_l(\lambda)$ with $\lambda_1 < \lambda < \lambda_2$. For instance, the vertical axis is given by the expression $x = 0, y = 0, z = \lambda$ ($-\infty < \lambda < \infty$), while the parametric expression for a circle of radius R contained on the horizontal plane and centered at the origin is $x = R \cos \lambda, y = R \sin \lambda, z = 0$ ($0 < \lambda < 2\pi$).

²In general, a surface can be defined in terms of a pair of parameters. For instance, a sphere of radius R centered at the origin admits the parametric representation $x = R \sin \alpha \cos \beta, y = R \sin \alpha \sin \beta, z = R \cos \alpha$ ($0 < \alpha < \pi, 0 < \beta < 2\pi$).

can be simplified by eliminating the parameters α and β , yielding in general an implicit equation of the form

$$f(\bar{x}, t) = 0. \quad (2.20)$$

If the surface is initially closed, it can be expected to remain closed in its evolution. The finite volume of fluid bounded by a closed fluid surface is called a **fluid volume**, a useful concept to be used later when deriving the conservation laws governing the fluid motion. In particular, in view of the definition of the fluid velocity as the center of mass of the fluid particle (see (1.4)), it is clear that mass cannot cross a fluid surface, which implies that **the mass of a fluid volume remains constant**, thereby providing the first conservation law of fluid mechanics.

Stream lines, stream surfaces and stream tubes

The **stream lines** of a flow field at a given instant of time are lines that are tangent to the local velocity vector at each point. This tangency condition can be expressed in the form

$$\frac{h_1 dx_1}{v_1(x_1, x_2, x_3, t)} = \frac{h_2 dx_2}{v_2(x_1, x_2, x_3, t)} = \frac{h_3 dx_3}{v_3(x_1, x_2, x_3, t)} \quad (2.21)$$

The two integration constants for the above pair of differential equations serve to identify a specific stream line. For instance, one could identify a stream line by selecting the location (x_o, y_o) at which it crosses the horizontal plane $z = 0$. If two stream lines intersect, the crossing point is necessarily a stagnation point (why?). There are two additional concepts related to the stream lines that are worth mentioning. The surface formed by all streamlines intersecting a given line is called a **stream surface**. If the line selected is closed, the resulting stream surface would form a **stream tube**.

Although Ecs. (2.13) and (2.21) look similar, they represent two very different physical and mathematical concepts. The time t is an integration variable in the initial value problem defined by (2.13), whereas it merely plays the role of a parameter in the integration of (2.21). To find a trajectory, we follow the temporal evolution of the fluid particle. To find a stream line, we *freeze* the time by looking at the velocity field existing at a specific instant.

Path lines and stream lines differ in a general unsteady flow. However, for a steady velocity field $\bar{v} = \bar{v}(\bar{x})$ or, more generally, for a velocity field of the form $\bar{v} = f(t)\bar{V}(\bar{x})$ with $f(t)$ representing an arbitrary function of time, stream lines and path lines can be shown to coincide (show it!).

Material derivative

To determine the variation of any intensive scalar property ϕ following the fluid flow, such as its density, pressure or temperature, one must account for the fluid motion by considering the infinitesimal displacement of a fluid particle, as illustrated in Fig. 2.2. At time t the fluid particle is located at $\bar{x}$, so that the initial value of the intensive property is given by $\phi(\bar{x}, t)$. A time increment dt later, the particle has moved to occupy a new position given by $\bar{x} + d\bar{x}$ ($d\bar{x} = \bar{v}dt$), and the new value of ϕ would be given by $\phi(\bar{x} + d\bar{x}, t + dt)$. Neglecting small terms in a Taylor expansion of ϕ about $(\bar{x}, t)$ enables the variation of ϕ for the fluid particle to be written in the approximate form

$$d\phi = \phi(\bar{x} + d\bar{x}, t + dt) - \phi(\bar{x}, t) = dt \frac{\partial \phi}{\partial t} + d\bar{x} \cdot \nabla \phi. \quad (2.22)$$

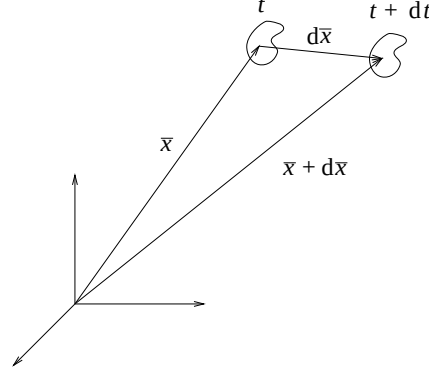


Figure 2.2: Infinitesimal displacement of a fluid particle.

Dividing the above expression by dt and identifying $\bar{v} = d\bar{x}/dt$ leads to

$$\frac{D\phi}{Dt} = \frac{\partial\phi}{\partial t} + \bar{v} \cdot \nabla\phi, \quad (2.23)$$

where the material derivative operator

$$\frac{D()}{Dt} = \frac{\partial()}{\partial t} + \bar{v} \cdot \nabla() \quad (2.24)$$

expresses the variation in time of any intensive scalar property following the fluid particle. The first term on the right-hand side represents the local temporal variation, while the second term is the so-called convective derivative, related to the motion of the fluid particle. Changes in the properties of a fluid particle may arise because the flow is unsteady and/or because the fluid particle moves through a nonuniform field³.

Acceleration

The concept of material derivative can be extended to determine the time evolution of a vector property. In particular, we shall define the acceleration vector as

$$\bar{a} = \frac{D\bar{v}}{Dt} = \frac{\partial\bar{v}}{\partial t} + \bar{v} \cdot (\nabla\bar{v}), \quad (2.25)$$

where the gradient of velocity $\nabla\bar{v}$ is a tensor. A convenient way to express (2.25) is

$$\bar{a} = \frac{\partial\bar{v}}{\partial t} + \nabla(|\bar{v}|^2/2) - \bar{v} \wedge (\nabla \wedge \bar{v}), \quad (2.26)$$

³Note that the concept of material derivative serves to derive the differential equation to be satisfied by the function $f(\bar{x}, t)$ corresponding to the fluid surface $f(\bar{x}, t) = 0$ (see Ec. (2.20)). The fluid surface therefore divides the flow field in two regions where the sign of $f(\bar{x}, t)$ is different. Since the fluid particles defining the fluid surface maintain a constant value $f(\bar{x}, t) = 0$ in their evolution, the function $f(\bar{x}, t)$ must satisfy the equation

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \bar{v} \cdot \nabla f = 0.$$

which is valid regardless of the coordinate system selected.

In cartesian coordinates, the components of the gradient of velocity are given by $(\nabla \bar{v})_{ij} = \partial v_j / \partial x_i$, so that each component of the acceleration a_i reduces to the material derivative of the corresponding velocity component according to

$$a_i = \frac{Dv_i}{Dt} = \frac{\partial v_i}{\partial t} + v_j \frac{\partial v_i}{\partial x_j}. \quad (2.27)$$

Note that this last result does not hold when cylindrical or spherical coordinates are used.

For the description of certain flow fields, it is at times convenient to employ a non-inertial reference frame, that is, a reference frame that is accelerating and/or rotating with respect to the *laboratory* frame of reference. In that case, when writing the fluid acceleration one must account for the additional inertial contribution

$$\bar{a}_s = \bar{a}_o + \frac{d\bar{\Omega}}{dt} \wedge \bar{x} + \bar{\Omega} \wedge (\bar{\Omega} \wedge \bar{x}) + 2\bar{\Omega} \wedge \bar{v}, \quad (2.28)$$

where $\bar{a}_o$ and $\bar{\Omega}$ are the acceleration and angular velocity of the non-inertial reference frame.

Circulation and vorticity

The **circulation** Γ along a curve L is defined as the integral

$$\Gamma = \int_L \bar{v} \cdot d\bar{l}, \quad (2.29)$$

where the vector $d\bar{l}$ represents the differential line element. Physically, the circulation gives as a measure of the fluid motion along the direction of the curve. If $\bar{x} = \bar{x}_l(\lambda)$ ($\lambda_1 < \lambda < \lambda_2$) is the parametric representation of the curve, then

$$d\bar{l} = \frac{d\bar{x}_l}{d\lambda} d\lambda, \quad (2.30)$$

so that (2.29) reduces to

$$\Gamma = \int_L \bar{v}(\bar{x}, t) \cdot d\bar{l} = \int_{\lambda_1}^{\lambda_2} \bar{v}(\bar{x}_l(\lambda), t) \cdot \frac{d\bar{x}_l}{d\lambda} d\lambda. \quad (2.31)$$

If one considers a closed curve, then according to Stokes theorem

$$\Gamma = \oint_L \bar{v} \cdot d\bar{l} = \int_{\Sigma} (\nabla \wedge \bar{v}) \cdot \bar{n} d\sigma, \quad (2.32)$$

where Σ represents any surface bounded by L and $d\sigma$ is a surface differential element with normal unit vector $\bar{n}$, defined for the surface to have positive orientation according to the so-called right-hand rule; see Fig. 2.3. For the last equation to hold, the velocity field $\bar{v}(\bar{x}, t)$ must be continuous and the surface Σ must be completely contained within the fluid domain. The vector $\bar{\omega} = \nabla \wedge \bar{v}$ is called **vorticity**.

The circulation along a closed curve limiting an infinitesimally small surface $d\sigma$ contained in a plane normal to $\bar{n}$ reduces to

$$\oint_L \bar{v} \cdot d\bar{l} = (\nabla \wedge \bar{v}) \cdot \bar{n} d\sigma. \quad (2.33)$$

Hence, $(\nabla \wedge \bar{v})_n = (\nabla \wedge \bar{v}) \cdot \bar{n}$ turns out to be the circulation per unit surface around a curve normal to $\bar{n}$. At a given point, the circulation will be maximum when the curve is contained in a plane normal to $\nabla \wedge \bar{v}$.

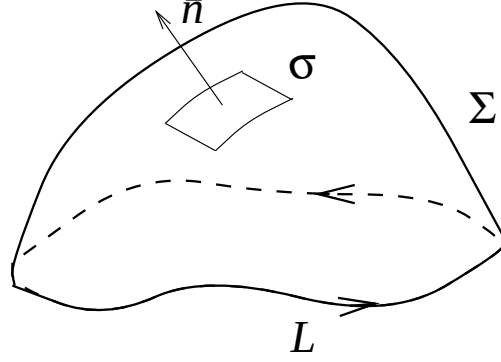


Figure 2.3: Application of Stokes theorem for the computation of the circulation.

Irrotational flow and velocity potential

The fluid motion is said to be irrotational when the vorticity is identically zero everywhere, i.e., $\bar{\omega} = \nabla \wedge \bar{v} = 0$. In vector calculus we have seen that for any irrotational vector field one can define a scalar function $\varphi(\bar{x})$ ($d\varphi = \bar{v} \cdot d\bar{x}$), called potential function, such that

$$\bar{v} = \nabla \varphi. \quad (2.34)$$

The potential is defined by (2.34), except for an integration constant, whose value, a function of time, can be selected arbitrarily. Clearly, the introduction of the velocity potential will simplify notably the description of irrotational motions, because we are effectively replacing the computation of a three-component vector field (the velocity $\bar{v}$) by that of a scalar field (the potential φ).

If the circulation along any closed curve is zero, the flow is necessarily irrotational. The demonstration begins by considering an infinitesimally small closed curve around a given point. According to (2.33), the circulation around the curve will be zero, regardless of the orientation $\bar{n}$, if $\nabla \wedge \bar{v} = 0$. The opposite is not necessarily true, that is, even though $\nabla \wedge \bar{v} = 0$ everywhere in the flow field, the circulation around a closed curve might not be zero. This is so because the validity of (2.32) is restricted to surfaces Σ that are completely contained in the fluid domain. Therefore, for an irrotational field, one can claim that the circulation around a given closed curve is zero if and only if we can find a surface bounded by the curve that is completely contained in the flow field, that is, when the curve is continuously reducible to a point (the fluid domain is simply connected). This feature of irrotational motion arises in the study of flows over airfoils, relevant to the design of wings, compressor and turbine blades, etc. In the two-dimensional flow field that appears, the circulation around a closed curve circling the airfoil - which is non-reducible to a point - is non-zero, and turns out to be linearly proportional to the lift force provided by the airfoil.

Convective flux

Mass, momentum and energy are transported by the moving fluid. To quantify in general this transport rate, we introduce $\phi(\bar{x}, t)$ to represent a given fluid magnitude expressed per unit volume, with particular cases of interest being the mass, momentum and energy per unit volume

($\phi = \rho$, $\phi = \rho\bar{v}$ and $\phi = \rho(e + |\bar{v}|^2/2)$). With this definition, the total amount of a given quantity existing within a given volume V is given by $\int_V \phi dV$ (e.g., the total amount of mass contained in a volume is $\int_V \rho dV$).

Let us now consider the fixed surface Σ_o shown in the figure. The fluid will cross the surface in its motion, transporting across mass, momentum and energy. The volume of fluid crossing in a

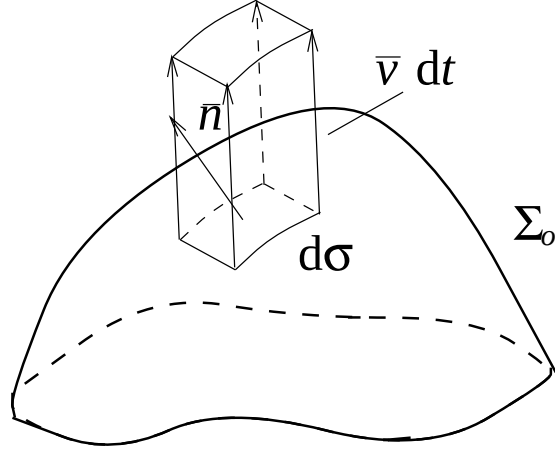


Figure 2.4: Convective flux.

time dt through the differential surface element $d\sigma$ oriented perpendicular to $\bar{n}$ is that contained in the parallelepiped of base $d\sigma$ and side $\bar{v}dt$ shown in the figure, which can be computed as $\bar{v} \cdot \bar{n} d\sigma dt$. The amount of mass, momentum and energy crossing with the fluid is therefore given by $\phi \bar{v} \cdot \bar{n} d\sigma dt$, and the total convective flux across the surface (mass, momentum or energy that crosses the surface per unit time) is obtained by adding the contribution of the different surface elements and dividing the result by the unit time dt to give

$$\int_{\Sigma_o} \phi \bar{v} \cdot \bar{n} d\sigma. \quad (2.35)$$

Note that, with $\phi = 1$, the above integral serves to compute the volume of fluid that crosses Σ_o per unit time, the **volume flux**

$$Q = \int_{\Sigma_o} \bar{v} \cdot \bar{n} d\sigma. \quad (2.36)$$

When $\phi = \bar{v}$, we find inside the integral (2.35) the so-called momentum flux tensor $\rho \bar{v} \bar{v}$.

If Σ_o is a closed surface (and $\phi \bar{v}$ is a continuous function), Gauss formula (2.9) enables the convective flux to be written in the form

$$\int_{\Sigma_o} \phi \bar{v} \cdot \bar{n} d\sigma = \int_{V_o} \nabla \cdot (\phi \bar{v}) dV, \quad (2.37)$$

where V_o is the volume bounded by Σ_o . If we now consider an infinitesimally small volume, it then follows that $\nabla \cdot (\phi \bar{v})$ is the loss rate of mass, momentum or energy per unit time due to the fluid outflow. Similarly, $\nabla \cdot \bar{v}$ is the amount of volume that abandons the unit volume per unit time (the expansion rate). Note that, for a perfect liquid, whose density is strictly constant, the condition that the specific volume cannot change requires that the condition

$$\nabla \cdot \bar{v} = 0 \quad (2.38)$$

be satisfied at each point.

The concept of convective flux can be extended to a moving surface $\Sigma_c(t)$ whose points are moving with velocity $\bar{v}_c(\bar{x}_c, t)$ by replacing the fluid velocity $\bar{v}$ with the relative velocity $\bar{v} - \bar{v}_c$ when computing the flux to give

$$\int_{\Sigma_c(t)} \phi(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma. \quad (2.39)$$

The stream function

The description of flows that are either planar or axisymmetric can be simplified when the velocity field is *solenoidal* (i.e., satisfies $\nabla \cdot \bar{v} = 0$), as occurs in constant-density fluids (see the discussion leading to (2.63)). To present the development, we restrict our attention to the case of a planar flow described in a cartesian $x - y$ coordinate system. Extensions to steady gas flow, for which the continuity equation becomes $\nabla \cdot (\rho \bar{v}) = 0$, and also to polar coordinates and axisymmetric flows can be performed, but will not be considered further in this introductory section.

For a planar flow $\nabla \cdot \bar{v} = 0$ reduces to

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0. \quad (2.40)$$

To solve the problem, it is convenient to introduce a scalar function ψ , called **stream function** defined from the two equations

$$v_x = \frac{\partial \psi}{\partial y} \quad \text{and} \quad v_y = -\frac{\partial \psi}{\partial x}, \quad (2.41)$$

which determine ψ upon integration, except for an arbitrary additive constant. Straightforward substitution of (2.41) into (2.40) reveals that the velocity field deriving from a stream function satisfies automatically the condition of zero expansion rate. An important property of the stream function is that the isolines $\psi = \text{constant}$ are stream lines, as can be seen by noticing that along an isoline

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = -v_y dx + v_x dy = 0. \quad (2.42)$$

The last of these equations can be written as

$$\frac{dx}{v_x} = \frac{dy}{v_y}, \quad (2.43)$$

thereby demonstrating that isolines and stream lines coincide.

Another useful property of ψ is that the difference $\psi_2 - \psi_1$ between the value of the stream function corresponding to two different stream lines equals the volumetric flux (2.36) for the planar stream tube defined by the two streamlines. Given a curve with end points on each one of the stream lines, as shown in the figure, the value of $\psi_2 - \psi_1$ is given by the integral

$$\psi_2 - \psi_1 = \int_1^2 d\psi = -v_y dx + v_x dy, \quad (2.44)$$

as can be seen from (2.42). Bearing in mind that $(dy, -dx) = \bar{n} dl$, where $\bar{n}$ is the unit vector normal to the integration curve, the previous equation can finally be rewritten in the form

$$\psi_2 - \psi_1 = \int_1^2 \bar{v} \cdot \bar{n} dl, \quad (2.45)$$

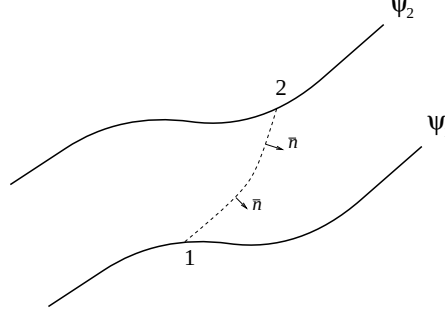


Figure 2.5: Volumetric flux between two stream lines.

where the integral on the right-hand side is the volumetric flux circulating between the two stream surfaces considered (per unit length perpendicular to the plane)

Relative motion near a point

As previously mentioned, the force exerted by a portion of fluid on an adjacent portion of fluid is proportional to the rate at which the fluid is being deformed. To determine that rate, let us consider the motion of a differential element of fluid line $d\bar{x}$, whose ends are initially located at $\bar{x}$ and $\bar{x} + d\bar{x}$. The velocities of these two ends differ by a small amount $d\bar{v}$, which can be expressed in the first approximation as

$$d\bar{v} = d\bar{x} \cdot \nabla \bar{v} \quad (2.46)$$

in terms of the gradient of velocity $\nabla \bar{v}$. After an infinitesimally small time dt , the two ends of the fluid line element have moved to occupy new positions given by $\bar{x} + \bar{v}dt$ and $\bar{x} + d\bar{x} + (\bar{v} + d\bar{v})dt$, respectively. Clearly, besides a translation $\bar{v}dt$, the fluid element has undergone a distortion, with $d\bar{x}$ becoming $d\bar{x} + d\bar{v}dt$, as indicated in Fig. 2.6.

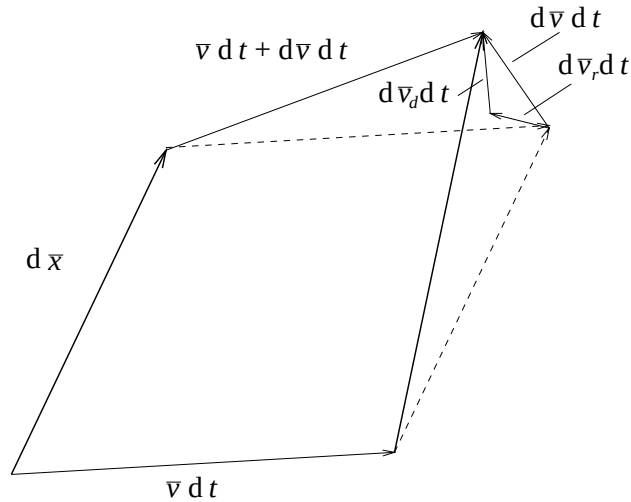


Figure 2.6: Motion of a differential element of fluid line.

The geometrical character of $d\bar{v}dt = d\bar{x} \cdot \nabla \bar{v}dt$ is best investigated by decomposing the velocity gradient into parts that are symmetrical and anti-symmetrical, according to

$$\nabla \bar{v} = \frac{1}{2}(\nabla \bar{v} + \nabla \bar{v}^T) + \frac{1}{2}(\nabla \bar{v} - \nabla \bar{v}^T) = \bar{\bar{T}}_d + \bar{\bar{T}}_r. \quad (2.47)$$

where the symmetrical tensor $\bar{\bar{T}}_d$ is called the rate-of-strain tensor. In cartesian coordinates, the two tensors $\bar{\bar{T}}_d$ and $\bar{\bar{T}}_r$ can be computed according to

$$\bar{\bar{T}}_d = \begin{bmatrix} \frac{\partial v_1}{\partial x_1} & \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right) & \frac{1}{2} \left(\frac{\partial v_3}{\partial x_1} + \frac{\partial v_1}{\partial x_3} \right) \\ \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right) & \frac{\partial v_2}{\partial x_2} & \frac{1}{2} \left(\frac{\partial v_3}{\partial x_2} + \frac{\partial v_2}{\partial x_3} \right) \\ \frac{1}{2} \left(\frac{\partial v_3}{\partial x_1} + \frac{\partial v_1}{\partial x_3} \right) & \frac{1}{2} \left(\frac{\partial v_3}{\partial x_2} + \frac{\partial v_2}{\partial x_3} \right) & \frac{\partial v_3}{\partial x_3} \end{bmatrix} \quad (2.48)$$

and

$$\bar{\bar{T}}_r = \begin{bmatrix} 0 & \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) & \frac{1}{2} \left(\frac{\partial v_3}{\partial x_1} - \frac{\partial v_1}{\partial x_3} \right) \\ -\frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) & 0 & \frac{1}{2} \left(\frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3} \right) \\ -\frac{1}{2} \left(\frac{\partial v_3}{\partial x_1} - \frac{\partial v_1}{\partial x_3} \right) & -\frac{1}{2} \left(\frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3} \right) & 0 \end{bmatrix} \quad (2.49)$$

This last expression can be rewritten in the form

$$\bar{\bar{T}}_r = \frac{1}{2} \begin{bmatrix} 0 & \omega_3 & -\omega_2 \\ -\omega_3 & 0 & \omega_1 \\ \omega_2 & -\omega_1 & 0 \end{bmatrix}, \quad (2.50)$$

in terms of the three components ω_1 , ω_2 and ω_3 of the vorticity vector $\bar{\omega} = \nabla \wedge \bar{v}$.

Introducing (2.47) into (2.46) yields

$$d\bar{v} = d\bar{x} \cdot \bar{\bar{T}}_d + d\bar{x} \cdot \bar{\bar{T}}_r = d\bar{v}_d + d\bar{v}_r. \quad (2.51)$$

The contribution of the anti-symmetrical tensor $\bar{\bar{T}}_r$ corresponds to a rotation of the fluid element $d\bar{x}$ with angular velocity $(\nabla \wedge \bar{v})/2$, as can be seen by writing $d\bar{v}_r$ in the form

$$d\bar{v}_r = d\bar{x} \cdot \bar{\bar{T}}_r = \frac{1}{2}(\nabla \wedge \bar{v}) \wedge d\bar{x} = \frac{1}{2} \bar{\omega} \wedge d\bar{x}. \quad (2.52)$$

Thus, if $\bar{\bar{T}}_d$ were identically zero, the motion of the fluid would be exactly that of a rigid body, that is, exclusively translation and rotation.

On the other hand, the symmetry of the rate-of-strain tensor $\bar{\bar{T}}_d$ enables the strain rate to be written in the form

$$d\bar{v}_d = \bar{\bar{T}}_d \cdot \bar{n} ds, \quad (2.53)$$

where the differential fluid element has been expressed as $d\bar{x} = \bar{n}ds$ in terms of its differential length ds and the unit vector $\bar{n}$. In general, the strain rate is not aligned with $\bar{n}$, implying that the element $d\bar{x}$ undergoes both extension and shear strain. The magnitude of the extension rate is given by the scalar product $\bar{n} \cdot \bar{\bar{T}}_d \cdot \bar{n} ds$, so that $\bar{n} \cdot \bar{\bar{T}}_d \cdot \bar{n}$ represents the extension rate per unit length along the direction defined by $\bar{n}$. On the other hand, the contribution of $d\bar{v}_d$ to the shear deformation is obtained by appropriately subtracting from the total strain rate the extension rate according to $[\bar{\bar{T}}_d \cdot \bar{n} - (\bar{n} \cdot \bar{\bar{T}}_d \cdot \bar{n}) \bar{n}]ds$.

There are three privileged directions of space, called principal directions of strain, along which the strain reduces to an extension, without any shear strain (the resulting vector $d\bar{v}_d$ is parallel to $\bar{n}$). These principal directions $\bar{n}_1$, $\bar{n}_2$ and $\bar{n}_3$, and their corresponding strain rates, λ_1 , λ_2 and λ_3 , are computed by solving the eigenvalue problem

$$\bar{\bar{T}}_d \cdot \bar{n} = \lambda \bar{n}. \quad (2.54)$$

In particular, the characteristic equation that determines λ_i ,

$$|\bar{\bar{T}}_d - \lambda \bar{\bar{I}}| = 0 \quad (2.55)$$

follows from imposing the existence of nontrivial solutions. The symmetry of the tensor $\bar{\bar{T}}_d$ guarantees the existence of three real roots for this equation, associated with three mutually perpendicular principal directions. Obviously, in the local reference frame defined by the three directions $\bar{n}_i$ the associated rate-of-strain tensor becomes diagonal, with diagonal components λ_1 , λ_2 and λ_3 .

Deformation of a square fluid element

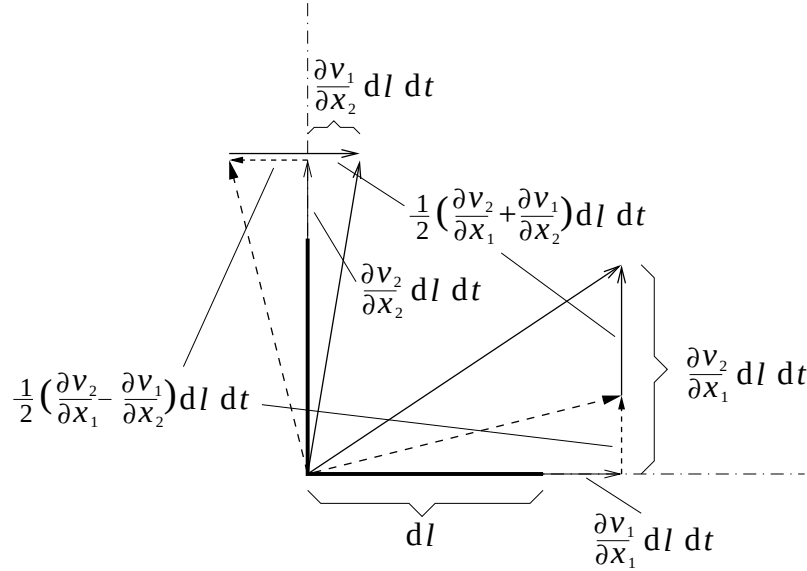


Figure 2.7: Deformation and rotation of an infinitesimally small fluid element of square shape.

Let us now consider the evolution of a square fluid element. The two sides, of initial length dl , evolve after a time dt from $d\bar{l}$ to $d\bar{l} + d\bar{l} \cdot \nabla \bar{v} dt$, so that

$$d\bar{l} \cdot \nabla \bar{v} = dl(1, 0) \begin{bmatrix} \frac{\partial v_1}{\partial x_1} & \frac{\partial v_2}{\partial x_1} \\ \frac{\partial v_1}{\partial x_2} & \frac{\partial v_2}{\partial x_2} \end{bmatrix} = dl \left(\frac{\partial v_1}{\partial x_1}, \frac{\partial v_2}{\partial x_1} \right) \quad (2.56)$$

and

$$d\bar{l} \cdot \nabla \bar{v} = dl(0, 1) \begin{bmatrix} \frac{\partial v_1}{\partial x_1} & \frac{\partial v_2}{\partial x_1} \\ \frac{\partial v_1}{\partial x_2} & \frac{\partial v_2}{\partial x_2} \end{bmatrix} = dl \left(\frac{\partial v_1}{\partial x_2}, \frac{\partial v_2}{\partial x_2} \right) \quad (2.57)$$

The observation of Fig. 2.7 helps to reveal the physical meaning of each of the terms appearing in the tensors

$$\bar{\bar{T}}_d = \begin{bmatrix} \frac{\partial v_1}{\partial x_1} & \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right) \\ \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right) & \frac{\partial v_2}{\partial x_2} \end{bmatrix} \quad (2.58)$$

and

$$\bar{\bar{T}}_r = \begin{bmatrix} 0 & \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) \\ -\frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) & 0 \end{bmatrix} \quad (2.59)$$

Recalling that the decomposition $d\bar{l} \cdot \nabla \bar{v} dt = d\bar{v}_d dt + d\bar{v}_r dt$ applies, with $d\bar{v}_d = d\bar{l} \cdot \bar{\bar{T}}_d$ and $d\bar{v}_r = d\bar{l} \cdot \bar{\bar{T}}_r$, one obtains for the horizontal side $d\bar{l}(1,0)$

$$d\bar{v}_d = d\bar{l} \left[\frac{\partial v_1}{\partial x_1}, \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right) \right] \quad \text{and} \quad d\bar{v}_r = d\bar{l} \left[0, \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) \right], \quad (2.60)$$

whereas for the vertical side

$$d\bar{v}_d = d\bar{l} \left[\frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right), \frac{\partial v_2}{\partial x_2} \right] \quad \text{and} \quad d\bar{v}_r = d\bar{l} \left[-\frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right), 0 \right]. \quad (2.61)$$

According to the figure, the elements along the diagonal of the rate-of-strain tensor represent the rate of extension per unit length in the directions of the three axes. Besides, the sides have undergone rotations of angles $dt(\partial v_2/\partial x_1)$ and $-dt(\partial v_1/\partial x_2)$, so that the average angular velocity for the square element in its plane is given by $\frac{1}{2}(\partial v_2/\partial x_1 - \partial v_1/\partial x_2)$. Therefore, the off-diagonal elements of the anti-symmetrical tensor $\bar{\bar{T}}_r$ represent the three components of the angular velocity of the fluid element. On the other hand, the angle formed by the fluid-element sides, initially perpendicular, have decreased an amount $(\partial v_2/\partial x_1 + \partial v_1/\partial x_2)dt$, indicating that $(\partial v_2/\partial x_1 + \partial v_1/\partial x_2)$ is the shear strain rate (the rate at which the angle formed by the directions 1 and 2 is decreasing). In other words, the off-diagonal components in $\bar{\bar{T}}_d$ correspond to half of the shear strain rate perpendicular to the three axis of the reference frame.

Deformation of a cubic fluid element

The above figure considers the evolution of an infinitesimally small fluid element of cubic shape, with the simplified notation $\partial_i v_j = \partial v_j / \partial x_i$ adopted for convenience. The discussion given above for the square shape also applies here. In this case, the distortion of the fluid element associated with the straining motion produces a variation of volume from its initial value $d\bar{l}^3$. After a differential time dt , the new volume is given by

$$d\bar{l}^3 \begin{vmatrix} 1 + \partial_1 v_1 dt & \partial_1 v_2 dt & \partial_1 v_3 dt \\ \partial_2 v_1 dt & 1 + \partial_2 v_2 dt & \partial_2 v_3 dt \\ \partial_3 v_1 dt & \partial_3 v_2 dt & 1 + \partial_3 v_3 dt \end{vmatrix} \simeq d\bar{l}^3 + d\bar{l}^3 dt (\partial_1 v_1 + \partial_2 v_2 + \partial_3 v_3), \quad (2.62)$$

after small terms of order dt^2 and dt^3 are neglected. The previous equation reveals that the velocity divergence $\nabla \cdot \bar{v} = \partial_1 v_1 + \partial_2 v_2 + \partial_3 v_3$ represents physically the rate at which the volume of the fluid element is changing per unit volume. This quantity $\nabla \cdot \bar{v}$, equal to the trace of the

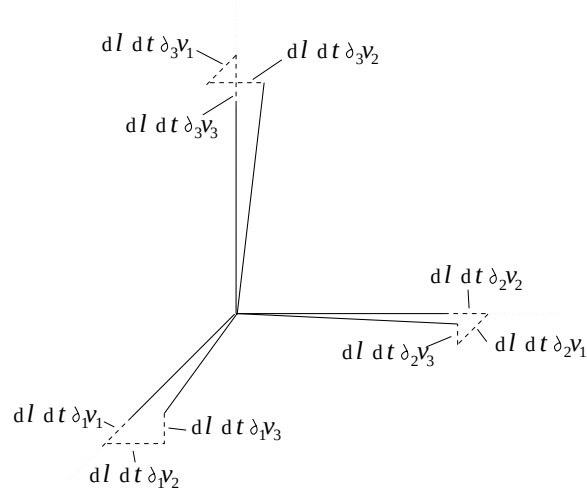


Figure 2.8: Deformation and rotation of a cubic fluid element of infinitesimally small size.

tensors $\nabla \bar{v}$ and $\bar{T}_d$, is called **rate of expansion** or **dilation rate**. For a perfect liquid, for which the volume of the fluid element does not change since the density is constant, the velocity field must satisfy

$$\nabla \cdot \bar{v} = 0, \quad (2.63)$$

a result that was anticipated earlier. Clearly, for the dilation velocity to be zero, the extension rates $\partial_1 v_1$, $\partial_2 v_2$ and $\partial_3 v_3$ cannot all three have the same sign. In other words, for the volume of an incompressible fluid particle to remain constant, a positive extension rate appears in one or two directions, with a compensating compression rate appearing in the other one (or two) directions to give a zero net expansion rate.

Chapter 3

Conservation laws

Reynolds transport theorem

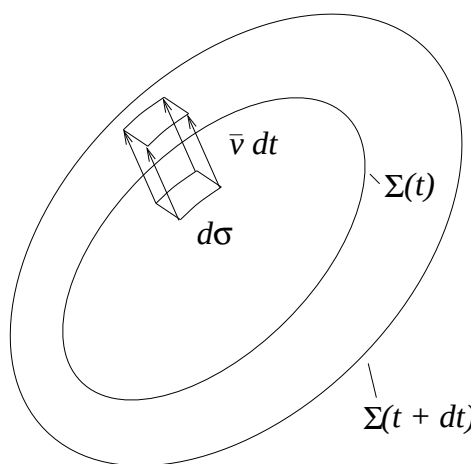


Figure 3.1: Evolution of a fluid volume.

To write the conservation equations, it is of interest to determine the variation with time of the amount of mass, momentum or energy contained in a fluid volume $V_f(t)$ limited externally by a closed fluid surface $\Sigma_f(t)$

$$\frac{d}{dt} \left[\int_{V_f(t)} \phi(\bar{x}, t) dV \right]; \quad \phi = \rho, \rho \bar{v}, \rho(e + |\bar{v}|^2/2). \quad (3.1)$$

Since both the integrand $\phi(\bar{x}, t)$ and the integration domain $V_f(t)$ vary with time, one can expect two different contributions associated, respectively, with the unsteadiness of the flow field and with the motion of the fluid, just as we saw earlier in connection with the material derivative. To quantify the two contributions, it is convenient to consider the evolution of the fluid volume between two successive instants of time t and $t + dt$, as sketched in Fig. 3.1.

Equation (3.1) can be written using the definition of limit

$$\frac{d}{dt} \left[\int_{V_f(t)} \phi(\bar{x}, t) dV \right] = \lim_{dt \rightarrow 0} \frac{\int_{V_f(t+dt)} \phi(\bar{x}, t+dt) dV - \int_{V_f(t)} \phi(\bar{x}, t) dV}{dt}, \quad (3.2)$$

which can also be split into two separate contributions according to

$$\lim_{dt \rightarrow 0} \frac{\int_{V_f(t)} [\phi(\bar{x}, t+dt) - \phi(\bar{x}, t)] dV}{dt} + \lim_{dt \rightarrow 0} \frac{\int_{V_f(t+dt)-V_f(t)} \phi(\bar{x}, t+dt) dV}{dt}. \quad (3.3)$$

In evaluating the first term we use the expansion $\phi(\bar{x}, t+dt) - \phi(\bar{x}, t) = (\partial\phi/\partial t)dt + O(dt^2)$ to give in the first approximation

$$\lim_{dt \rightarrow 0} \frac{\int_{V_f(t)} [\phi(\bar{x}, t+dt) - \phi(\bar{x}, t)] dV}{dt} = \int_{V_f(t)} \frac{\partial\phi(\bar{x}, t)}{\partial t} dV, \quad (3.4)$$

while the second integral can be evaluated as the convective flux crossing the surface $\Sigma_f(t)$ during the time interval dt to give

$$\lim_{dt \rightarrow 0} \frac{\int_{V_f(t+dt)-V_f(t)} \phi(\bar{x}, t+dt) dV}{dt} = \lim_{dt \rightarrow 0} \frac{\int_{\Sigma_f(t)} \phi \bar{v} \cdot \bar{n} d\sigma dt}{dt} = \int_{\Sigma_f(t)} \phi \bar{v} \cdot \bar{n} d\sigma, \quad (3.5)$$

finally yielding

$$\frac{d}{dt} \left[\int_{V_f(t)} \phi(\bar{x}, t) dV \right] = \int_{V_f(t)} \frac{\partial\phi(\bar{x}, t)}{\partial t} dV + \int_{\Sigma_f(t)} \phi(\bar{x}, t) \bar{v}(\bar{x}, t) \cdot \bar{n} d\sigma. \quad (3.6)$$

The two terms in (3.6) correspond to the previously anticipated contributions arising from flow unsteadiness and from fluid motion, respectively.

The solution of a given problem involves in general the variation with time of fluid properties in a finite region $V_c(t)$ that, depending on the problem, can be fixed or variable in time. That region is called a **control volume**, which is limited externally by a control surface $\Sigma_c(t)$, a closed surface whose points move with velocity $\bar{v}_c(\bar{x}_c, t)$. Since the conservation laws of fluid mechanics apply to fluid volumes, and not to control volumes, it is of interest to relate the time derivatives extended to $V_c(t)$ to those extended to a control volume $V_f(t)$ occupying the same space at the instant considered. For $V_c(t)$, the reasoning that led before to (3.6) allows us to write

$$\frac{d}{dt} \left[\int_{V_c(t)} \phi(\bar{x}, t) dV \right] = \int_{V_c(t)} \frac{\partial\phi(\bar{x}, t)}{\partial t} dV + \int_{\Sigma_c(t)} \phi(\bar{x}, t) \bar{v}_c(\bar{x}, t) \cdot \bar{n} d\sigma. \quad (3.7)$$

Bearing in mind that we chose the control volume $V_c(t)$ and its bounding surface $\Sigma_c(t)$ to coincide exactly with $V_f(t)$ and $\Sigma_f(t)$ at the instant considered, the integration domains that appear in (3.6) and (3.7) are identical, and straightforward subtraction of both expressions provides

$$\frac{d}{dt} \left[\int_{V_f(t)} \phi(\bar{x}, t) dV \right] = \frac{d}{dt} \left[\int_{V_c(t)} \phi(\bar{x}, t) dV \right] + \int_{\Sigma_c(t)} \phi(\bar{x}, t) (\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma. \quad (3.8)$$

This equation, to be used below when deriving the conservation equations for an arbitrary control volume, indicates that the time variation of the given flow property (mass, momentum or energy) in a fluid volume $V_f(t)$ equals the time variation in a control volume coinciding with $V_f(t)$ plus the convective flux across the bounding surface $\Sigma_c(t)$ (see (2.39)).

Mass conservation

The first conservation law of fluid mechanics states that the mass of a given fluid volume remains constant,

$$\frac{d}{dt} \left[\int_{V_f(t)} \rho dV \right] = 0, \quad (3.9)$$

which can be expressed with use made of (3.8) in the form

$$\frac{d}{dt} \left[\int_{V_c(t)} \rho dV \right] + \int_{\Sigma_c(t)} \rho(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = 0. \quad (3.10)$$

The physical meaning underlying the above equation becomes clear by writing the second integral on the right-hand side of the equation to give

$$\frac{d}{dt} \left[\int_{V_c(t)} \rho dV \right] = - \int_{\Sigma_c(t)} \rho(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = 0,$$

which states that the rate of increment of the mass contained in a control volume equals the mass flux entering across its bounding surface, $-\int_{\Sigma_c(t)} \rho(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma$. When the control volume is fixed in space, the above equation reduces to

$$\frac{d}{dt} \left[\int_{V_o} \rho dV \right] + \int_{\Sigma_o} \rho \bar{v} \cdot \bar{n} d\sigma = 0. \quad (3.11)$$

Momentum conservation

Volume and surface forces

The forces acting on a fluid (or on a solid) can be classified in two different categories: long-range forces, also called volume or mass forces, and short-range forces, also called surface forces. The former, including for instance gravity and inertial forces, decay slowly over distances much larger than the intermolecular distance d and therefore act over the whole fluid volume (for instance, gravity decays over distances that are of the order of the Earth radius). Short-range forces, on the other hand, have a direct molecular origin and decay rapidly with distance, their range of influence being of the order of the intermolecular distance d .

The mass force acting on a fluid particle of volume dV can be expressed in the form

$$\rho \bar{f}_m(\bar{x}, t) dV, \quad (3.12)$$

where $\bar{f}_m(\bar{x}, t)$ is the force acting per unit mass, which equals $\bar{f}_m(\bar{x}, t) = \bar{g}$ when gravity is the only mass force acting on the system. If a non-inertial reference frame is selected in the description, then inertial forces must be also considered to give

$$\bar{f}_m(\bar{x}, t) = \bar{g} - \bar{a}_o - \frac{d\bar{\Omega}}{dt} \wedge \bar{x} - \bar{\Omega} \wedge (\bar{\Omega} \wedge \bar{x}) - 2\bar{\Omega} \wedge \bar{v}, \quad (3.13)$$

with $\bar{a}_o$ and $\bar{\Omega}$ representing the acceleration and angular velocity of the non-inertial reference frame. To write (3.12) we have neglected the variation of $\bar{f}_m$ within the fluid particle, an accurate

approximation given the long-range nature of these forces. Some mass forces are conservative and derive from a potential according to $\bar{\mathbf{f}}_m = -\nabla U$, so that, for instance,

$$\bar{\mathbf{g}} - \bar{\mathbf{a}}_o - \bar{\boldsymbol{\Omega}} \wedge (\bar{\boldsymbol{\Omega}} \wedge \bar{\mathbf{x}}) = -\nabla[-\bar{\mathbf{g}} \cdot \bar{\mathbf{x}} + \bar{\mathbf{a}}_o \cdot \bar{\mathbf{x}} - (\bar{\boldsymbol{\Omega}} \wedge \bar{\mathbf{x}}) \cdot (\bar{\boldsymbol{\Omega}} \wedge \bar{\mathbf{x}})/2]. \quad (3.14)$$

On the other hand, short-range forces only reach a short distance of the order of d . For a gas, they are directly related with the flux of momentum emerging due to the random molecular motion. The molecules crossing a surface separating two adjacent portions of fluid transport their momentum and energy. This microscopic transport effect produces a surface force at the macroscopic level, and also gives rise to heat conduction. If the fluid is a liquid, then there exist additional contributions due to the forces between molecules located on both sides of the surface. Because of their short-range nature, the force acting on a fluid parcel is proportional to its surface, so that given a surface element $d\sigma$ with orientation $\bar{\mathbf{n}}$ separating two fluid elements the resulting force is given by

$$\bar{\mathbf{f}}_n(\bar{\mathbf{n}}, \bar{\mathbf{x}}, t) d\sigma, \quad (3.15)$$

where $\bar{\mathbf{f}}_n$ is the force per unit surface (**stress**), a function of $\bar{\mathbf{x}}, t$ and, also, of the orientation $\bar{\mathbf{n}}$. In the convention adopted, $\bar{\mathbf{f}}_n$ is the stress exerted by the fluid on the side of the surface element to which $\bar{\mathbf{n}}$ points on the fluid located on the opposite side.

Stress tensor

The dependence of the stress $\bar{\mathbf{f}}_n$ on $\bar{\mathbf{n}}$ can be shown to reduce to a linear relationship, given below in (3.19), involving a second-order symmetric tensor $\bar{\boldsymbol{\tau}}$ called the stress tensor. To derive this equation, one applies Newton's second law to the fluid element shown in Fig. 3.2, whose initial shape is that of a tetrahedron of sides dx_1, dx_2 and dx_3 and volume $dV = (1/6)dx_1dx_2dx_3$. Both

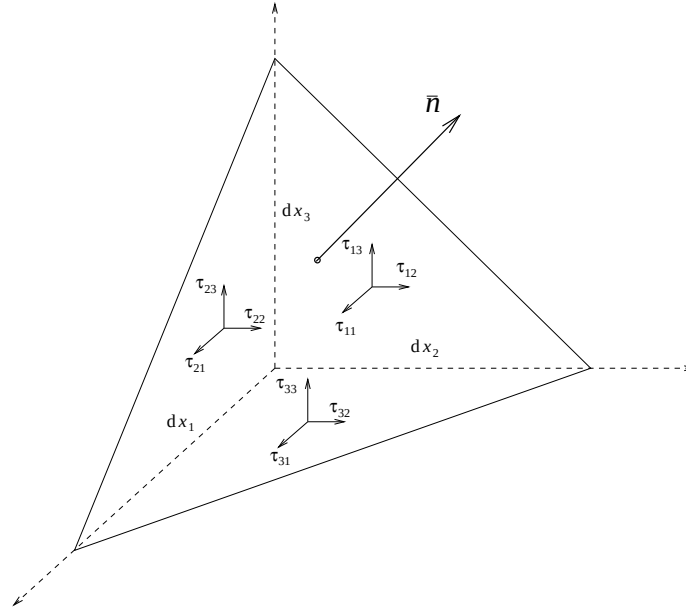


Figure 3.2: Stresses on a tetrahedral fluid element.

the inertial term (mass ρdV times acceleration $D\bar{\mathbf{v}}/Dt$) and the resultant mass force $\rho dV \bar{\mathbf{f}}_m$ are

proportional to the volume of the element, while the surface forces acting on the four sides of the tetrahedron are proportional to their surfaces. Hence, for vanishingly small values of $dx \ll 1$, the momentum equation reduces to a balance between the surface forces. In expressing the equilibrium balance, we shall use $\bar{f}_i = (\tau_{i1}, \tau_{i2}, \tau_{i3})$ ($i = 1, 2, 3$) to denote the stress (force per unit surface) acting on a surface normal to the unit vector $\bar{e}_i$ of the reference frame. Consequently, with the sign convention adopted, the force acting on the tetrahedronal fluid element through the face contained in a plane normal to $\bar{e}_i$ is given by $-dA_i \bar{f}_i$, whereas that applied through the oblique face is $dA \bar{f}_n$, with $\bar{n} = (n_1, n_2, n_3)$ denoting the unit vector normal to the oblique face. With this notation, the equilibrium equation reduces to

$$dA \bar{f}_n - dA_1 \bar{f}_1 - dA_2 \bar{f}_2 - dA_3 \bar{f}_3 = 0. \quad (3.16)$$

Using the geometrical identity $dA_i = n_i dA$, where dA denotes the surface area of the oblique face, allows us to rewrite the previous equation in the form

$$\bar{f}_n = n_1 \bar{f}_1 + n_2 \bar{f}_2 + n_3 \bar{f}_3 = \bar{n} \cdot \bar{\bar{\tau}}, \quad (3.17)$$

where

$$\bar{\bar{\tau}} = \begin{bmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} \end{bmatrix} \quad (3.18)$$

is the so-called stress tensor. Equation (3.17) indicates that the stress acting on a surface of orientation $\bar{n}$ can be expressed as a function of the nine different components of the tensor $\bar{\bar{\tau}}$, which are in principle a function of $\bar{x}$ and t .

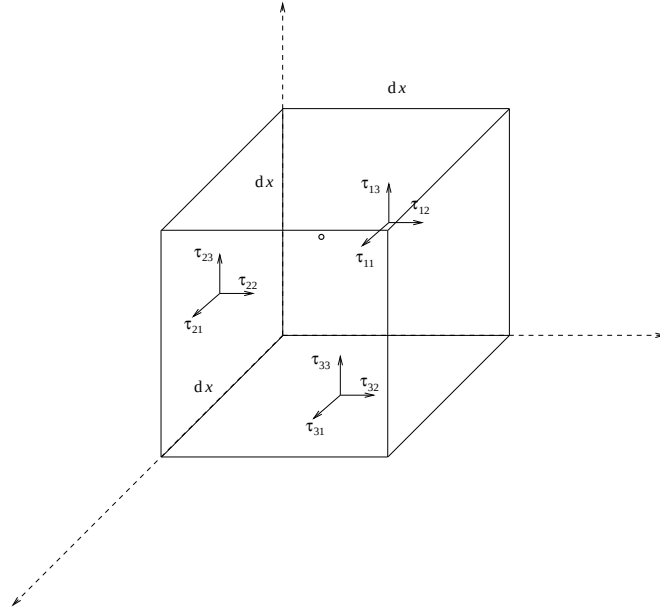


Figure 3.3: Stresses on a cubic element.

The result is even simpler, in that the stress tensor is actually symmetrical, and therefore contains only six different components. To show the symmetry, one applies the conservation of angular

momentum to the cubic fluid element of side dx shown in Fig. 3.3. In the limit $dx \rightarrow 0$, the surface forces and their associated momenta with respect to any point must be in equilibrium. For instance, the equation imposing a zero torque in the vertical direction reads $dx^3(\tau_{21} - \tau_{12}) = 0$, indicating that $\tau_{12} = \tau_{21}$, and similar arguments in the other two directions yield $\tau_{23} = \tau_{32}$ and $\tau_{13} = \tau_{31}$, thereby demonstrating the aforementioned symmetry. The stress $\bar{f}_n(\bar{n}, \bar{x}, t)$ can therefore be expressed in the form

$$\bar{f}_n = \bar{\tau} \cdot \bar{n} = \begin{bmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{12} & \tau_{22} & \tau_{23} \\ \tau_{13} & \tau_{23} & \tau_{33} \end{bmatrix} \cdot \bar{n}, \quad (3.19)$$

where the normal stresses τ_{11} , τ_{22} and τ_{33} as well as the shear (tangential) stresses τ_{12} , τ_{13} and τ_{23} are functions of the position and time. Their dependences on the strain rate and on the local thermodynamic state of the fluid are to be studied in the next section for the case of newtonian fluids.

The stress $\bar{\tau} \cdot \bar{n}$ can be split into its normal component $(\bar{n} \cdot \bar{\tau} \cdot \bar{n})\bar{n}$ and its tangential component $\bar{\tau} \cdot \bar{n} - (\bar{n} \cdot \bar{\tau} \cdot \bar{n})\bar{n}$, the latter called shear stress. There exist three directions, called the principal directions of stress, for which the resulting shear stress is identically zero, i.e., the only stress component is normal, a condition expressed mathematically by the eigenvalue problem

$$\bar{\tau} \cdot \bar{n} = \lambda \bar{n}. \quad (3.20)$$

The symmetry of the stress tensor $\bar{\tau}$ guarantees the existence of three real roots of the associated characteristic equation

$$|\bar{\tau} - \lambda \bar{I}| = 0. \quad (3.21)$$

These roots are the normal stresses in the principal directions, which are of course mutually orthogonal.

The surface force acting on a given surface can be obtained by extending (3.19) to give

$$\int_{\Sigma} \bar{f}_n d\sigma = \int_{\Sigma} \bar{\tau} \cdot \bar{n} d\sigma. \quad (3.22)$$

For a closed surface, Gauss formula (2.9) enables the previous equation to be written in the form

$$\int_{\Sigma} \bar{\tau} \cdot \bar{n} d\sigma = \int_V \nabla \cdot \bar{\tau} dV. \quad (3.23)$$

Evaluating now the previous equation for a differential fluid element we conclude that $\nabla \cdot \bar{\tau}$ is the resultant surface force per unit volume, so that $(\nabla \cdot \bar{\tau})dV$ is the surface force acting on a fluid particle of volume dV . Noting that the mass of the fluid particle is given by ρdV , straightforward application of Newton's second law provides

$$\rho dV \frac{D\bar{v}}{Dt} = \nabla \cdot \bar{\tau} dV + \rho \bar{f}_m dV, \quad (3.24)$$

which incorporates also the effect of the mass forces possibly acting on the fluid. By eliminating dV one arrives at the differential form of the momentum equation

$$\rho \frac{D\bar{v}}{Dt} = \nabla \cdot \bar{\tau} + \rho \bar{f}_m, \quad (3.25)$$

to be derived in the next chapter.

Navier-Poisson equation

If the fluid is at rest in a given reference frame, the surface forces act always normal to the surface and can be in general written as

$$\bar{f}_n = -p\bar{n}, \quad (3.26)$$

where p is the pressure, a thermodynamic variable introduced above when writing the equations of state. Correspondingly, the stress tensor reduces in this case to

$$\bar{\tau} = -p\bar{\mathbb{I}}, \quad (3.27)$$

where $\bar{\mathbb{I}}$ represents the identity tensor.

Note that the existence of strain rates is independent of the reference frame utilized in the description. Thus, if $\bar{\mathbb{T}}_d = \nabla \bar{v} - \bar{\mathbb{T}}_r$ is zero in some reference frame, then it is also zero when a different frame is used, so that the fluid motion reduces to a local translation and rotation as a rigid body. In an appropriately selected local reference frame the fluid would be at rest, so that one may conclude that the result $\bar{\tau} = -p\bar{\mathbb{I}}$ applies in general whenever $\bar{\mathbb{T}}_d = 0$.

When the rate-of-strain tensor $\bar{\mathbb{T}}_d$ is non-zero (which occurs independently of the reference frame employed), besides the pressure forces there exist additional surfaces forces, called viscous forces, that can be expressed in terms of the viscous stress tensor $\bar{\tau}'$, defined such that

$$\bar{\tau} = -p\bar{\mathbb{I}} + \bar{\tau}'. \quad (3.28)$$

It must be pointed out that, up to this point, no hypotheses have been made about the nature of the fluid, other than the existence of local thermodynamic equilibrium. However, in order to relate the viscous stress tensor to the kinematic state of the flow at a particular point, it is necessary to make assumptions about the functional form of this relationship. Although it could be quite complex, for the so-called **newtonian fluids**, which include most of the fluids of engineering interest (e.g., air and water), it is observed that there exists a linear proportionality between the viscous stresses τ'_{ij} and the strain rates,

$$\tau'_{ij} = \alpha_{ijkl}\gamma_{kl}, \quad (3.29)$$

with γ_{kl} representing each one of the components of the rate-of-strain tensor $\bar{\mathbb{T}}_d$. Notice that the stress tensor cannot depend on the fluid velocity, $\bar{v}$, nor on the rate-of-rotation tensor, $\bar{\mathbb{T}}_r$, as it is always possible to choose a reference frame in which both are zero. In principle, since both $\bar{\tau}'$ and $\bar{\mathbb{T}}_d$ are symmetrical, one should consider up to 36 proportionality constants α_{ijkl} , but for isotropic fluids the number reduces to only two, as we shall show below.

The principal directions of strain and the principal directions of stress coincide for an isotropic fluid, and could be used as a local reference frame to express $\bar{\tau}'$ and $\bar{\mathbb{T}}_d$ in diagonal form. In the following, τ'_i and γ_i represent the stresses and strain rates in the principal directions. Evaluating (3.29) for τ'_1 and bearing in mind the isotropy of the fluid one obtains $\tau'_1 = \alpha\gamma_1 + \lambda\gamma_2 + \lambda\gamma_3$, where the coefficients α and λ relate the strain rate in a given direction with the viscous stresses that appear in that direction and in the two perpendicular ones, respectively. Analogous expressions, involving also α and λ would be obtained for τ'_2 and τ'_3 . The result can be expressed in the compact form $\tau'_i = (\alpha - \lambda)\gamma_i + \lambda\nabla \cdot \bar{v}$ or, alternatively,

$$\bar{\tau}' = 2\mu\bar{\mathbb{T}}_d + \lambda(\nabla \cdot \bar{v})\bar{\mathbb{I}}, \quad (3.30)$$

where the constants $\mu = (\alpha - \lambda)/2$ and λ are the first and second viscosity coefficients. This last expression remains valid when the principal directions are replaced by any other orthogonal reference frame (remember that the expansion rate $\nabla \cdot \bar{v}$ is also invariant under the transformation). Hence, the viscous stress tensor can be in general written in the form

$$\bar{\tau}' = 2\mu\bar{T}_d + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v})\bar{I}, \quad (3.31)$$

which further simplifies for an incompressible fluid, for which the second term vanishes. Following the most commonly used notation, the above expression is written in terms of the bulk viscosity coefficient $\mu_B = \lambda + \frac{2}{3}\mu$. The expressions for $\nabla \bar{v}$ and $\nabla \cdot \bar{v}$ given previously can be used to express $\bar{\tau}'$ in terms of the derivatives of the different velocity components.

As can be seen, the viscous stresses depend on the strain rate, which appears explicitly in (3.31), and also on the local thermodynamic state, through the viscosity coefficients μ and μ_B . Besides these two coefficients, a related quantity, which becomes useful when intercomparing the molecular transport of mass, momentum and energy, is the so-called kinematic viscosity $\nu = \mu/\rho$. Characteristic values of ν for air and water at ambient pressure are, respectively, $\nu_a = 1.48 \times 10^{-5} \text{ m}^2/\text{s}$ and $\nu_w = 1.14 \times 10^{-6} \text{ m}^2/\text{s}$ at $T = 288 \text{ K}$, and $\nu_a = 2.24 \times 10^{-5} \text{ m}^2/\text{s}$ and $\nu_w = 0.31 \times 10^{-6} \text{ m}^2/\text{s}$ at $T = 368 \text{ K}$.

The existence of local thermodynamic equilibrium enables the dependence of μ and μ_B on the thermodynamic state of the fluid to be studied in terms of any two independent thermodynamic variables, for instance p and T . Experiments show that the dependence on the pressure is negligible for liquids and gases, a finding that agrees with the results of the kinetic theory of gases. As for the temperature dependence, the gas viscosity increases with temperature, because the faster random molecular motion enhances molecular momentum transport. For liquids, the intermolecular forces contribute fundamentally to the viscous stresses. With increasing temperatures the intermolecular distance also increases, causing the intermolecular force to drop and therefore producing a decrease in the viscosity (we have all seen how the oil viscosity drastically decreases as the cooking pan is heated on the stove). For both liquids and gases, so long as the temperature does not vary significantly, one may correctly assume that the viscosity coefficients remain constant, an assumption to be employed in the following development. As a final comment, it is worth mentioning that the bulk viscosity coefficient μ_B is identically zero for monoatomic gases, as indicated by the kinetic theory of gases.

Force and torque on a solid submerged in a fluid

The resultant force $\bar{F}$ of the pressure and viscosity forces acting on a surface Σ of normal vector $\bar{n}$ pointing towards the fluid can be computed from (3.22) and (3.28) to yield

$$\bar{F} = - \int_{\Sigma} p \bar{n} d\sigma + \int_{\Sigma} \bar{\tau}' \cdot \bar{n} d\sigma. \quad (3.32)$$

Similarly, the corresponding moment $\bar{M}$ of the pressure and viscosity forces with respect to a given point $\bar{x}_o$ is given by

$$\bar{M} = - \int_{\Sigma} (\bar{x} - \bar{x}_o) \wedge p \bar{n} d\sigma + \int_{\Sigma} (\bar{x} - \bar{x}_o) \wedge \bar{\tau}' \cdot \bar{n} d\sigma. \quad (3.33)$$

These equations can be used, in particular, to determine the force and torque that a fluid exerts on an solid body in contact with the fluid, with the surface Σ representing in that case the wetted surface of the body with the normal unit vector $\bar{n}$ pointing towards the fluid.

The momentum equation

Newton's second law applied to a fluid volume states that the rate of variation of the momentum equals the resultant force associated with the surface and volume forces acting on the fluid

$$\frac{d}{dt} \left[\int_{V_f(t)} \rho \bar{v} dV \right] = \int_{\Sigma_f(t)} \bar{\tau} \cdot \bar{n} d\sigma + \int_{V_f(t)} \rho \bar{f}_m dV. \quad (3.34)$$

Expressing the stress tensor $\bar{\tau}$ according to (3.28) and using Reynolds transport theorem (3.8) to extend the above equation for an arbitrary control volume $V_c(t)$ yields

$$\frac{d}{dt} \left[\int_{V_c(t)} \rho \bar{v} dV \right] + \int_{\Sigma_c(t)} \rho \bar{v} (\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = - \int_{\Sigma_c(t)} p \bar{n} d\sigma + \int_{\Sigma_c(t)} \bar{\tau}' \cdot \bar{n} d\sigma + \int_{V_c(t)} \rho \bar{f}_m dV. \quad (3.35)$$

The above equation states that the rate of increment of the momentum contained in a given control volume equals the sum of four contributions: the convective influx of momentum through its bounding surface $-\int_{\Sigma_c(t)} \rho \bar{v} (\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma$, the resultant pressure force $-\int_{\Sigma_c(t)} p \bar{n} d\sigma$, the resultant viscous force $\int_{\Sigma_c(t)} \bar{\tau}' \cdot \bar{n} d\sigma$ and the resultant mass force $\int_{V_c(t)} \rho \bar{f}_m dV$. Note that, for a fixed control volume, momentum conservation reduces to

$$\left[\int_{V_o} \frac{\partial \rho \bar{v}}{\partial t} dV \right] + \int_{\Sigma_o} \rho \bar{v} \bar{v} \cdot \bar{n} d\sigma = - \int_{\Sigma_o} p \bar{n} d\sigma + \int_{\Sigma_o} \bar{\tau}' \cdot \bar{n} d\sigma + \int_{V_o} \rho \bar{f}_m dV. \quad (3.36)$$

Angular momentum equation

Newton's second law can be alternatively expressed for the angular momentum in the following terms: the rate of variation of the angular momentum of a system equals the momentum of the external forces acting on it. For a fluid particle of mass ρdV located at $\bar{x}$ the angular momentum with respect to a given position $\bar{x}_o$ is given by $(\bar{x} - \bar{x}_o) \wedge \rho \bar{v} dV$. Therefore, when the overall fluid volume is considered, angular momentum conservation provides the integral equation

$$\frac{d}{dt} \left[\int_{V_f(t)} \rho (\bar{x} - \bar{x}_o) \wedge \bar{v} dV \right] = \int_{\Sigma_f(t)} (\bar{x} - \bar{x}_o) \wedge (\bar{\tau} \cdot \bar{n}) d\sigma + \int_{V_f(t)} \rho (\bar{x} - \bar{x}_o) \wedge \bar{f}_m dV, \quad (3.37)$$

which can be written for an arbitrary control volume $V_c(t)$ in the form

$$\begin{aligned} \frac{d}{dt} \left[\int_{V_c(t)} \rho (\bar{x} - \bar{x}_o) \wedge \bar{v} dV \right] &+ \int_{\Sigma_c(t)} \rho [(\bar{x} - \bar{x}_o) \wedge \bar{v}] [(\bar{v} - \bar{v}_c) \cdot \bar{n}] d\sigma = \\ &- \int_{\Sigma_c(t)} (\bar{x} - \bar{x}_o) \wedge (p \bar{n}) d\sigma + \int_{\Sigma_c(t)} (\bar{x} - \bar{x}_o) \wedge (\bar{\tau}' \cdot \bar{n}) d\sigma + \int_{V_c(t)} \rho (\bar{x} - \bar{x}_o) \wedge \bar{f}_m dV. \end{aligned} \quad (3.38)$$

Energy conservation

Heat transfer by conduction

We have seen above that the existence of strain rates in the flow field results in the appearance of viscous stresses between adjacent portions of fluid. Likewise, the existence of temperature

gradients gives rise to a heat flux (conduction) whose microscopic origin lies - as that of the viscous stresses- in the random molecular motion, with the work of intermolecular forces also contributing non-negligibly in the case of liquids. As expected, the amount of energy transported is proportional to the surface considered, so that the heat flux q_n (energy per unit time) through a surface element $d\sigma$ oriented perpendicular to the unit vector $\bar{n}$ is given by

$$q_n(\bar{n}, \bar{x}, t)d\sigma. \quad (3.39)$$

In the convention adopted, q_n is positive if the energy is transferred to the fluid element to which $\bar{n}$ points from the fluid located on the side which $\bar{n}$ points away from.

To investigate the dependence of q_n on the orientation let us consider once more the tetrahedral fluid element shown in Fig. 3.2. When its size becomes vanishingly small energy conservation can be shown to reduce *in the first approximation* to a balance between the heat fluxes through the four faces of the tetrahedron

$$q_n dA = q_1 dA_1 + q_2 dA_2 + q_3 dA_3. \quad (3.40)$$

In the notation adopted, $q_n dA$ is the flux through the oblique face with orientation $\bar{n}$ and q_i denotes the heat flux per unit surface through a plane perpendicular to $\bar{e}_i$, so that $-q_i dA_i$ is the energy added per unit time to the tetrahedron due to heat conduction through its face perpendicular to $\bar{e}_i$. Using the geometrical relationship $dA_i = n_i dA$ and dividing the resulting equation by dA yields finally

$$q_n = \bar{q} \cdot \bar{n}, \quad (3.41)$$

showing that the heat flux in direction $\bar{n} = (n_1, n_2, n_3)$ can be expressed in terms of a vector $\bar{q} = (q_1, q_2, q_3)$, termed the heat-flux vector, whose three components are in general functions of the position and time considered.

Extending (3.39) to the computation of the heat flux through a surface Σ yields

$$\int_{\Sigma} \bar{q} \cdot \bar{n} d\sigma. \quad (3.42)$$

For a closed surface whose interior is free from localized heat sources Gauss formula (2.9) provides

$$\int_{\Sigma} \bar{q} \cdot \bar{n} d\sigma = \int_V \nabla \cdot \bar{q} dV, \quad (3.43)$$

where V is the volume enclosed by Σ . Consideration of the limit of very small volumes reveals in this case that $\nabla \cdot \bar{q}$ is the rate of conductive heat loss per unit volume (energy that abandons the unit volume per unit time due to heat conduction).

Fourier law

The heat-flux vector can be computed according to Fourier law

$$\bar{q} = -k \nabla T, \quad (3.44)$$

stating a linear proportionality between the heat-flux vector and the local temperature gradient, with the minus sign indicating that the energy is transported from regions of higher temperature to regions of lower temperature. The proportionality constant k , called thermal conductivity, is

a thermodynamic property that depends on the temperature but is independent of the pressure. In gases, where the conductive heat transfer is directly linked with the random motion of the molecules, a rise in temperature produces an increase in the thermal conductivity, so that, for instance, for air $k_a = 0.025 \text{ W/(m K)}$ at $T = 288 \text{ K}$ and $k_a = 0.03 \text{ W/(m K)}$ at $T = 368 \text{ K}$. Liquids behave differently, because the work of intermolecular forces play a nonnegligible role, causing the resulting thermal conductivity to exhibit in most liquids a slight decrease with increasing temperature. For water, however, the opposite response is observed, so that, for instance, $k_w = 0.59 \text{ W/(m K)}$ at $T = 288 \text{ K}$ and $k_w = 0.68 \text{ W/(m K)}$ at $T = 368 \text{ K}$.

The thermal diffusivity $\alpha = k/(\rho c_p)$ - or $\alpha = k/(\rho c)$ for a liquid- constitutes a useful measure of heat transfer by conduction. Its dimensions (squared length per unit time) are those of the kinematic viscosity ν defined above, so that the ratio $Pr = \nu/\alpha$ is a nondimensional number, called the Prandtl number, that serves to compare the molecular transport rates of momentum and heat for a given fluid. In fluids with $Pr \gg 1$, such as lubricating oils, the transport of momentum is much more effective than that of heat. Conversely, fluids with $Pr \ll 1$ may transport heat at a rate that is much more rapid than that of momentum transport, that being for instance the case of liquid metals like mercury, for which the presence of free electrons provides a very effective means for energy transfer. For gases, the Prandtl number is of order unity (e.g., $Pr = 0.72$ for air in a wide range of temperatures), while for water the value, dependent on the temperature, tends to be somewhat larger than unity ($Pr = 8.14$ at $T = 288 \text{ K}$ and $Pr = 1.82$ at $T = 368 \text{ K}$).

The energy equation

As discussed previously, the energy of a fluid (per unit mass) can be expressed as the sum of the kinetic energy associated with the fluid velocity, $|\bar{v}|^2/2$, and the internal energy e . The total energy contained in a fluid volume V_f is therefore given by $\int_{V_f} \rho(e + |\bar{v}|^2/2) dV$. According to the first law of thermodynamics, this quantity may change due to heat addition -by heat conduction- through the bounding fluid surface $\Sigma_f(t)$ or by chemical reaction or radiation inside the fluid volume. Additionally, it might also change due to the mechanical work exerted by mass or surface forces. This rate balance can be expressed through the integral equation

$$\begin{aligned} \frac{d}{dt} \left[\int_{V_f(t)} \rho(e + |\bar{v}|^2/2) dV \right] = & - \int_{\Sigma_f(t)} p \bar{v} \cdot \bar{n} d\sigma + \int_{\Sigma_f(t)} \bar{v} \cdot \bar{\tau}' \cdot \bar{n} d\sigma \\ & + \int_{V_f(t)} \rho \bar{f}_m \cdot \bar{v} dV - \int_{\Sigma_f(t)} \bar{q} \cdot \bar{n} d\sigma + \int_{V_f(t)} (Q_c + Q_r) dV, \end{aligned} \quad (3.45)$$

where Q_c and Q_r represent, respectively, the rates of heat release (energy per unit volume per unit time) due to chemical reaction and radiation. Using Reynolds transport theorem enables the above equation to be written for an arbitrary control volume in the form

$$\begin{aligned} \frac{d}{dt} \left[\int_{V_c(t)} \rho(e + |\bar{v}|^2/2) dV \right] + \int_{\Sigma_c(t)} \rho(e + |\bar{v}|^2/2) (\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = & - \int_{\Sigma_c(t)} p \bar{v} \cdot \bar{n} d\sigma \\ & + \int_{\Sigma_c(t)} \bar{v} \cdot \bar{\tau}' \cdot \bar{n} d\sigma + \int_{V_c(t)} \rho \bar{f}_m \cdot \bar{v} dV - \int_{\Sigma_c(t)} \bar{q} \cdot \bar{n} d\sigma + \int_{V_c(t)} (Q_c + Q_r) dV. \end{aligned} \quad (3.46)$$

Note that, in computing Q_c , one needs to consider separate conservation equations to compute the fluid composition, with account taken of chemical reactions occurring between the different chemical species. On the other hand, the computation of $Q_r = -\nabla \cdot \bar{q}_r$ from the radiative heat flux $\bar{q}_r$ requires consideration of separate equations for radiation, accounting for local emission and absorption, with simplified limits arising in optically thin and optically thick environments. All these complexities are clearly beyond the scope of the present introductory course and are no longer considered here.

Chapter 4

The Navier-Stokes equations

In many engineering problems, approximate solutions concerning the overall properties of a fluid system can be obtained by application of the conservation equations of mass, momentum and energy written in integral form, given above in (3.10), (3.35) and (3.46), for a conveniently selected control volume. This approach necessitates in general introduction of simplifying assumptions, regarding in particular the spatial distributions of the different variables (e.g., uniform velocity at the open boundaries) and the neglect of terms that are anticipated to give a relatively small contribution to the overall balances. This integral approach is however not suitable when one is interested in computing local properties of the flow (e.g., distributions of velocity $\bar{v}(\bar{x}, t)$, density $\rho(\bar{x}, t)$, pressure $p(\bar{x}, t)$, etc). For that purpose, the conservation equations, that were written above for a control volume of finite size in the form

$$\frac{d}{dt} \left[\int_{V_f(t)} \rho dV \right] = 0, \quad (4.1)$$

$$\frac{d}{dt} \left[\int_{V_f(t)} \rho \bar{v} dV \right] = - \int_{\Sigma_f(t)} p \bar{n} d\sigma + \int_{\Sigma_f(t)} \bar{\tau}' \cdot \bar{n} d\sigma + \int_{V_f(t)} \rho \bar{f}_m dV, \quad (4.2)$$

$$\frac{d}{dt} \left[\int_{V_f(t)} \rho (e + |\bar{v}|^2/2) dV \right] = - \int_{\Sigma_f(t)} p \bar{v} \cdot \bar{n} d\sigma + \int_{\Sigma_f(t)} \bar{v} \cdot \bar{\tau}' \cdot \bar{n} d\sigma \quad (4.3)$$

$$+ \int_{V_f(t)} \rho \bar{f}_m \cdot \bar{v} dV - \int_{\Sigma_f(t)} \bar{q} \cdot \bar{n} d\sigma + \int_{V_f(t)} (Q_c + Q_r) dV,$$

need to be transformed with use made of the Gauss formula to express the rate balances occurring locally, providing a set of partial differential equations to be integrated with boundary and initial conditions.

The continuity equation

Using Gauss formula (2.9) it is straightforward to rewrite (4.1) in the form

$$\int_{V_f} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{v}) \right] dV = 0. \quad (4.4)$$

This equation is to be satisfied regardless of the choice of V_f if and only if the identity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{v}) = 0 \quad (4.5)$$

holds at each point of space. This is the continuity (or mass conservation) equation, stating that the sum of the rate of local density variation and the rate of mass loss by convective outflow equals zero. An alternative expression is

$$\frac{1}{\rho} \frac{D\rho}{Dt} = -\nabla \cdot \bar{v}, \quad (4.6)$$

indicating that the variation of density following the fluid particle is exclusively due to the rate of volume variation. Note that for an incompressible (constant-density) fluid, the continuity equation reduces to

$$\nabla \cdot \bar{v} = 0, \quad (4.7)$$

a result anticipated previously, whereas for steady gas flow one obtains

$$\nabla \cdot (\rho \bar{v}) = 0, \quad (4.8)$$

that is, the convective rate of mass loss per unit volume is zero.

The momentum equation

The derivation of the momentum equation in differential form follows the procedure used before to obtain (4.5), that is, Gauss formula (2.9) is used to rewrite all of the surface integrals appearing in (4.2) as volume integrals and the resulting integrand is set equal to zero to yield

$$\frac{\partial}{\partial t}(\rho \bar{v}) + \nabla \cdot (\rho \bar{v} \bar{v}) = -\nabla p + \nabla \cdot \bar{\tau}' + \rho \bar{f}_m. \quad (4.9)$$

Using now (4.5) and (2.26) provides

$$\rho \frac{D\bar{v}}{Dt} = \rho \left[\frac{\partial \bar{v}}{\partial t} + \nabla(|\bar{v}|^2/2) - \bar{v} \wedge (\nabla \wedge \bar{v}) \right] = -\nabla p + \nabla \cdot \bar{\tau}' + \rho \bar{f}_m, \quad (4.10)$$

which is simply Newton's second law expressed per unit volume of fluid, a result anticipated in (3.25).

For an incompressible fluid with constant viscosity $\nabla \cdot \bar{\tau}' = -\mu \nabla \wedge (\nabla \wedge \bar{v})$, so that (4.10) reduces to

$$\frac{D\bar{v}}{Dt} = -\nabla \left(\frac{p}{\rho} \right) - \nu \nabla \wedge (\nabla \wedge \bar{v}) + \bar{f}_m. \quad (4.11)$$

The viscous force per unit volume can be alternatively expressed as¹ $-\mu \nabla \wedge (\nabla \wedge \bar{v}) = \mu \nabla^2 \bar{v}$, yielding

$$\frac{D\bar{v}}{Dt} = -\nabla \left(\frac{p}{\rho} \right) + \nu \nabla^2 \bar{v} + \bar{f}_m, \quad (4.12)$$

¹It is to be noted that in cartesian coordinates each component of the vector $\nabla^2 \bar{v}$ is given simply by the laplacian of the corresponding velocity component, whereas additional terms appear in cylindrical and spherical coordinates, thereby complicating the computation.

Note that (4.7) and (4.11) suffice to determine the velocity and pressure fields for an incompressible flow with constant viscosity. For such flows, which include those involving water, these two equations are therefore decoupled from the energy equation, which could be used *a posteriori* to determine the temperature field. Also of interest is that, when the mass force is conservative, introduction of $\bar{f}_m = -\nabla U$ enables (4.11) to be written in the form

$$\frac{D\bar{v}}{Dt} = -\nabla \left(\frac{p + \rho U}{\rho} \right) - \nu \nabla \wedge (\nabla \wedge \bar{v}). \quad (4.13)$$

This last expression suggests that, for the motion of perfect liquids, introduction of the reduced pressure $P = p + \rho U$ (e.g., $P = p + \rho g z$ when $\bar{f}_m = \bar{g}$) may simplify the description significantly. For instance, when the pressure does not enter in the boundary conditions, the solution described with use made of P becomes entirely independent of the mass forces.

A well-known result arises for steady incompressible flows with conservative mass forces when the flow conditions are such that the effect of viscosity is negligible, so that the momentum equation reduces to

$$\nabla \left(\frac{p}{\rho} + U + \frac{|\bar{v}|^2}{2} \right) - \bar{v} \wedge (\nabla \wedge \bar{v}) = 0 \quad (4.14)$$

as can be seen by using (4.13) with $\nu = 0$ together with

$$\frac{D\bar{v}}{Dt} = \nabla(|\bar{v}|^2/2) - \bar{v} \wedge (\nabla \wedge \bar{v}), \quad (4.15)$$

the corresponding steady form of (2.26). The vector equation (4.14) can be projected along streamlines by multiplying at each point in space by the unit vector $\bar{v}/|\bar{v}|$. Since the contribution of the last term is identically zero, the projection reduces to

$$\frac{\partial}{\partial l} \left(\frac{p}{\rho} + U + \frac{|\bar{v}|^2}{2} \right) = 0, \quad (4.16)$$

where $\partial/\partial l = |\bar{v}|^{-1}(\bar{v} \cdot \nabla)$ denotes the derivative along a stream line. It then follows that **for frictionless steady flow of an incompressible fluid** the quantity

$$p + \rho U + \rho |\bar{v}|^2/2 = C_l \quad (4.17)$$

remains constant along any given stream line, with a value C_l that in general is different for different stream lines. Equation (4.17) is the so-called **Bernoulli's theorem**, that provides a useful relationship between the variations of kinetic energy, potential energy and pressure along streamlines. Note that, if the flow is irrotational, the constant C_l is the same for all streamlines, because $\nabla(p/\rho + U + |\bar{v}|^2/2) = 0$, as follows from (4.14).

The energy equation

Internal energy and kinetic energy conservation equations

The same transformation procedure used above in deriving (4.5) and (4.10) can be employed to derive from (4.3)

$$\frac{\partial}{\partial t} [\rho(e + |\bar{v}|^2/2)] + \nabla \cdot [\rho(e + |\bar{v}|^2/2)\bar{v}] = -\nabla \cdot (p\bar{v}) + \nabla \cdot (\bar{v} \cdot \bar{\tau}') + \rho \bar{f}_m \cdot \bar{v} - \nabla \cdot \bar{q} + Q_c + Q_r, \quad (4.18)$$

which can be rewritten by virtue of (4.5) in the simplified form

$$\rho \frac{D}{Dt} (e + |\bar{v}|^2/2) = -\nabla \cdot (p\bar{v}) + \nabla \cdot (\bar{v} \cdot \bar{\tau}') + \rho \bar{f}_m \cdot \bar{v} - \nabla \cdot \bar{q} + Q_c + Q_r. \quad (4.19)$$

If the mass force derives from a steady potential such that $\bar{f}_m = -\nabla U$ with $\partial U/\partial t = 0$, then the above equation admits the alternative writing

$$\rho \frac{D}{Dt} (e + |\bar{v}|^2/2 + U) = -\nabla \cdot (p\bar{v}) + \nabla \cdot (\bar{v} \cdot \bar{\tau}') - \nabla \cdot \bar{q} + Q_c + Q_r. \quad (4.20)$$

A separate equation for the mechanical (kinetic) energy $|\bar{v}|^2/2$ can be derived by multiplying (4.10) by $\bar{v}$. Since $\bar{v} \cdot [\bar{v} \wedge (\nabla \wedge \bar{v})] = 0$, the product provides the scalar equation

$$\rho \frac{D}{Dt} (|\bar{v}|^2/2) = -\bar{v} \cdot \nabla p + \bar{v} \cdot (\nabla \cdot \bar{\tau}') + \rho \bar{v} \cdot \bar{f}_m, \quad (4.21)$$

which is a purely mechanical energy law. As can be seen, the rate of variation of the kinetic energy following the fluid particle equals the work rate of the mass force $\rho \bar{v} \cdot \bar{f}_m$ plus the work rate of the surface forces associated with the translational motion of the fluid particle $\bar{v} \cdot (\nabla \cdot \bar{\tau}')$. Subtracting now (4.21) from (4.19) yields

$$\rho \frac{De}{Dt} = -p \nabla \cdot \bar{v} + \bar{\tau}' : \nabla \bar{v} - \nabla \cdot \bar{q} + Q_c + Q_r, \quad (4.22)$$

clearly indicating that the rate of variation of the internal energy is associated with heat addition and with the deformation work rate of the surface forces. The term $-p \nabla \cdot \bar{v}$ corresponds to the compression work, which is a reversible contribution, in that when the fluid volume is reduced ($\nabla \cdot \bar{v} < 0$) the compression work produces an increase in the internal energy; conversely, the internal energy decreases for positive expansion rates $\nabla \cdot \bar{v} > 0$. The term $\phi_v = \bar{\tau}' : \nabla \bar{v}$ is the deformation work due to viscous forces. It is non-negative ($\phi_v \geq 0$)², and corresponds to the rate of mechanical energy dissipation per unit volume and unit time. Note that both $-p \nabla \cdot \bar{v}$ and $\bar{\tau}' : \nabla \bar{v}$ are independent of the orientation, position and motion of the reference frame.

Integral balance equations for mechanical and internal energy

The integral conservation equation for the energy (4.3) is written for the combined contribution $(e + |\bar{v}|^2/2)$. It is interesting to write separate integral equations for the mechanical and internal energy. To that end, one may integrate (4.21) in the fluid volume $V_f(t)$, bearing in mind the equation

$$\rho \frac{D}{Dt} (|\bar{v}|^2/2) = \frac{\partial}{\partial t} (\rho |\bar{v}|^2/2) + \nabla \cdot (\rho \bar{v} |\bar{v}|^2/2) \quad (4.23)$$

²From the definition of contraction of two tensors $\bar{\bar{A}} : \bar{\bar{B}} = \sum_i \sum_j A_{ij} B_{ij}$ it can be shown that

$$\phi_v = 2\mu \bar{\bar{T}}_d : \bar{\bar{T}}_d + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v})^2 = \frac{2\mu}{3}[(\gamma_{11} - \gamma_{22})^2 + (\gamma_{11} - \gamma_{33})^2 + (\gamma_{22} - \gamma_{33})^2 + 6(\gamma_{12}^2 + \gamma_{13}^2 + \gamma_{23}^2)] + \mu_B(\nabla \cdot \bar{v})^2,$$

thereby demonstrating that $\phi_v \geq 0$ provided $\mu > 0$ and $\mu_B > 0$.

and also (3.6), to give

$$\begin{aligned} \frac{d}{dt} \left[\int_{V_f(t)} \rho |\bar{v}|^2 / 2 dV \right] = & - \int_{\Sigma_f(t)} p \bar{v} \cdot \bar{n} d\sigma + \int_{\Sigma_f(t)} \bar{v} \cdot \bar{\tau}' \cdot \bar{n} d\sigma + \int_{V_f(t)} \rho \bar{f}_m \cdot \bar{v} dV \\ & - \left[\int_{V_f(t)} -p \nabla \cdot \bar{v} dV + \int_{V_f(t)} \phi_v dV \right]. \end{aligned} \quad (4.24)$$

Similarly, one may use (4.22) to derive

$$\begin{aligned} \frac{d}{dt} \left[\int_{V_f(t)} \rho e dV \right] = & - \int_{\Sigma_f(t)} \bar{q} \cdot \bar{n} d\sigma + \int_{V_f(t)} (Q_c + Q_r) dV \\ & + \left[- \int_{V_f(t)} p \nabla \cdot \bar{v} dV + \int_{V_f(t)} \phi_v dV \right]. \end{aligned} \quad (4.25)$$

The way we have written (4.24) and (4.25) is intended to give a clear picture of the balances of work and energy in the flow field. Thus, the work done by mass forces in the interior of the fluid volume as well as that done by surface forces on its bounding surface (the right-hand-side terms in the first line of (4.24)) are devoted to increase the net amount of mechanical energy contained in the fluid volume. On the other hand, the heat added by conduction, radiation and chemical reaction (the right-hand-side terms in the first line of (4.25)) is employed directly to increase the total internal energy. Viscous dissipation as well as compression work, written for clarity in separate lines in (4.24) and (4.25), are the mechanisms transforming mechanical energy into internal energy, with only the latter being reversible.

Enthalpy conservation equation

Equations (4.5) and (4.10) together with (4.19) are the so-called Navier-Stokes equations governing fluid motion. The energy equation admits alternative forms, that may be more convenient than (4.19) for the analysis of specific problems. In particular, often in the analysis of gas flows it is of interest to use the enthalpy $h = e + p/\rho$ as a replacement for the internal energy e . In the derivation, one must first use the continuity equation (4.5) to give

$$\nabla \cdot (p \bar{v}) = \rho \frac{D}{Dt} (p/\rho) - \frac{\partial p}{\partial t}. \quad (4.26)$$

Combining now this result with (4.19) yields

$$\rho \frac{D}{Dt} (h + |\bar{v}|^2/2) = \frac{\partial p}{\partial t} + \nabla \cdot (\bar{v} \cdot \bar{\tau}') + \rho \bar{f}_m \cdot \bar{v} - \nabla \cdot \bar{q} + Q_c + Q_r, \quad (4.27)$$

which can also be written with use made of (4.21) as a separate equation for the enthalpy

$$\rho \frac{Dh}{Dt} = \frac{Dp}{Dt} + \phi_v - \nabla \cdot \bar{q} + Q_c + Q_r. \quad (4.28)$$

Note that, when mass forces are conservative and steady, Eq. (4.27) reduces to

$$\rho \frac{D}{Dt} (h + |\bar{v}|^2/2 + U) = \frac{\partial p}{\partial t} + \nabla \cdot (\bar{v} \cdot \bar{\tau}') - \nabla \cdot \bar{q} + Q_c + Q_r, \quad (4.29)$$

indicating that, when heat addition, temporal pressure variations and viscous dissipation are all simultaneously negligible, the combination $h + |\bar{v}|^2/2 + U$ remains constant in the evolution of each fluid particle.

Entropy conservation equation

It is also of interest to express the energy equation in terms of the entropy s . Since $Tds = dh - dp/\rho$, it is straightforward to use (4.28) to write

$$\rho T \frac{Ds}{Dt} = \phi_v - \nabla \cdot \bar{q} + Q_c + Q_r, \quad (4.30)$$

indicating that the entropy of a fluid particle increases due to viscous dissipation and heat addition, while the rate of compression work $-p\nabla \cdot \bar{v}$ is reversible and therefore does not change the entropy of the fluid particle. Equation (4.30) can be seen as an application of the second principle of thermodynamics, enabling changes of entropy to be quantified for fluids out of equilibrium. One may divide (4.30) by T and integrate the resulting equation in $V_f(t)$ to give

$$\begin{aligned} \frac{d}{dt} \left[\int_{V_f(t)} \rho s dV \right] &= \int_{V_f(t)} (\phi_v/T) dV - \int_{V_f(t)} [(\bar{q} \cdot \nabla T)/T^2] dV \\ &\quad - \int_{\Sigma_f(t)} (\bar{q}/T) \cdot \bar{n} d\sigma + \int_{V_f(t)} (Q_c + Q_r)/T dV, \end{aligned} \quad (4.31)$$

for the variation of the total entropy contained in a fluid volume. The term involving $(Q_c + Q_r)/T$ represents a volumetric source of entropy associated with chemical reaction and radiation. On the other hand, $\bar{q} \cdot \bar{n}/T$ is the entropy flux across the surface due to heat conduction, so that $-\int_{\Sigma_f(t)} (\bar{q}/T) \cdot \bar{n} d\sigma$ is the corresponding rate of entropy gain. Viscous dissipation gives a nonnegative contribution $\int_{V_f(t)} (\phi_v/T) dV$. Finally, the term $-\int_{V_f(t)} [(\bar{q} \cdot \nabla T)/T^2] dV$ is an additional source of entropy that appears due to conduction, which can be shown to be irreversible with use made of (3.44).

Mathematical description of fluid flows

It is a good time to summarize now what we have done so far to derive a rigorous mathematical framework for the description of fluid-flow problems. The derivation given above began by introducing the continuum hypothesis, which was instrumental in defining the concepts of density ρ , velocity $\bar{v}$, and internal energy e as continuous functions of the position $\bar{x}$ and time t . Next, by introducing the hypothesis of local thermodynamic equilibrium, the local values of all of the remaining thermodynamic variables (temperature, pressure, enthalpy, etc) were automatically defined in terms of ρ and e for the non-equilibrium states found in fluid mechanics problems. This hypothesis enables, in particular, ρ and e to be replaced by p and T as fundamental thermodynamic variables for the flow-field description. We have also assumed that the viscous stress tensor $\bar{\tau}'$ and the heat-flux vector $\bar{q}$ are linear isotropic functions of the rate of strain and the temperature gradient, respectively. In the constitutive equations arising from this assumption (Navier-Poisson and Fourier laws) there appear three transport coefficients μ , μ_B and k that are assumed to be functions of the local thermodynamic state of the fluid. Although some

of the hypotheses introduced can be expected to fail under extreme conditions (rarefied flows, nanofluidics applications, etc), the anticipated range of validity of the mathematical formulation derived covers most applications of interest in engineering.

Summary of conservation equations, equations of state and constitutive equations

To solve a given flow problem, the conservation equations of continuity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{v}) = 0, \quad (4.32)$$

momentum

$$\rho \frac{D\bar{v}}{Dt} = -\nabla p + \nabla \cdot \bar{\tau}' + \rho \bar{f}_m, \quad (4.33)$$

and energy

$$\rho \frac{De}{Dt} = -p \nabla \cdot \bar{v} + \bar{\tau}' : \nabla \bar{v} - \nabla \cdot \bar{q} + Q_c + Q_r, \quad (4.34)$$

must be supplemented with the equations of state

$$\rho = \rho(p, T) \quad \text{and} \quad e = e(p, T), \quad (4.35)$$

together with the constitutive equations

$$\bar{\tau}' = \mu(\nabla \bar{v} + \nabla \bar{v}^T) + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v})\bar{\mathbf{I}} \quad \text{and} \quad \bar{q} = -k \nabla T \quad (4.36)$$

and associated state functions

$$\mu = \mu(T), \quad \mu_B = \mu_B(T) \quad \text{and} \quad k = k(T). \quad (4.37)$$

In particular, the state equations (4.35) reduce to

$$\rho = \rho_o \quad \text{and} \quad e = e_o + cT \quad (4.38)$$

for a perfect liquid and to

$$\rho = \frac{p}{R_g T} \quad \text{and} \quad e = e_o + c_v T \quad (4.39)$$

for a perfect gas.

The vector expressions presented earlier in Chapter 2 enable (4.32), (4.33) and (4.34) to be expressed in any system of orthogonal curvilinear coordinates. For a perfect liquid with constant viscosity, Eqs. (4.32), (4.33) and (4.34) admit the simplified form

$$\nabla \cdot \bar{v} = 0 \quad (4.40)$$

$$\rho \frac{D\bar{v}}{Dt} = -\nabla p - \mu \nabla \wedge (\nabla \wedge \bar{v}) + \rho \bar{f}_m \quad (4.41)$$

$$\rho c \frac{DT}{Dt} = 2\mu \bar{\mathbf{T}}_d : \bar{\mathbf{T}}_d + k \nabla^2 T + Q_c + Q_r, \quad (4.42)$$

which are written out below for cartesian, cylindrical and spherical coordinates.

Initial and boundary conditions

Equations (4.32), (4.33) and (4.34) describe all fluid motions (within the range of validity of the various assumptions introduced in their derivation). The difference between two flow problems lies in their initial and boundary conditions, which need to be specified to enable the integration of the Navier-Stokes equations to be performed.

For unsteady fluid motion, at the initial instant we must define the velocity field $\bar{v} = \bar{v}_o(\bar{x})$ together with the thermodynamic state, given for instance in terms of $p = p_o(\bar{x})$ and $T = T_o(\bar{x})$ (or any other pair of independent thermodynamic variables). For an incompressible fluid, the initial velocity field must necessarily satisfy $\nabla \cdot \bar{v}_o = 0$. Because of the presence of the term $\partial\rho/\partial t$ in (4.32), in the case of gas flow there is no such restriction. Clearly, no initial conditions are required for the study of steady or periodic fluid flows.

The Navier-Stokes equations determine the flow evolution *within* the flow field, and must be therefore complemented with appropriate boundary conditions on the boundaries of the flow field. In typical configurations, the fluid may be confined by solid walls, as occurs in ducted flows. The fluid velocity in contact with the wall will be assumed to be equal to the wall velocity, i.e., at $\bar{x} = \bar{x}_p$ we impose $\bar{v} = \bar{v}_p$, which reduces to $\bar{v} = 0$ when the wall is at rest relative to the reference frame used in the description. This so-called **non-slip condition** comes from assuming that the interaction of the molecules near the wall with the wall is equivalent to that occurring between neighboring particles. For the velocity field to be continuous, no velocity discontinuity is permitted between the wall and the adjacent fluid. There exists ample experimental evidence confirming the validity of the non-slip condition, at least for flows in which the mean free path λ is much shorter than the macroscopic length of the flow field L . Note that, when this is not the case, the non-slip condition must be relaxed, along with the condition of local thermodynamic equilibrium, thereby complicating the associated analysis.

We shall also assume that there exists local thermodynamic equilibrium at the wall, so that the fluid temperature in contact with the wall is that of the wall ($T = T_p$ at $\bar{x} = \bar{x}_p$). There must also exist an equilibrium of the heat fluxes at the wall surface, providing an additional equation that reduces to $\bar{q} \cdot \bar{n} = \partial T / \partial n = 0$ at $\bar{x} = \bar{x}_p$ when the wall is thermally insulated (adiabatic wall). In principle, in integrating the conservation equations for the fluid, one could use alternatively the temperature distribution T_p or the heat flux $\partial T / \partial n$ as boundary condition for the temperature at the wall $\bar{x} = \bar{x}_p$. In many problems, however, neither quantity is known a priori, and they have to be determined as part of the solution, which involves the integration of the Navier-Stokes equations for the fluid together with the heat conduction equation in the solid, coupled with the conditions of thermodynamic equilibrium and heat flux balance at the wall surface $\bar{x} = \bar{x}_p$.

The boundary conditions needed when the fluid extends to infinity (i.e., to distances much larger than the characteristic macroscopic size of the flow field L), as occurs in external aerodynamic problems, include specification of the boundary velocity field and its accompanying thermodynamic state. These external boundary conditions can be written in the form $\bar{v} = \bar{v}_\infty(\bar{x}, t)$, $p = p_\infty(\bar{x}, t)$, and $T = T_\infty(\bar{x}, t)$, for $|\bar{x}| \rightarrow \infty$. For instance, to study the motion of a uniform stream of velocity U_∞ over a solid body of characteristic size L , we shall impose $\bar{v} = U_\infty \bar{e}_x$, $p = p_\infty$, and $T = T_\infty$ at $|\bar{x}|/L \gg 1$.

As an example, consider the motion a liquid stream that moves with velocity $U_\infty(t)\bar{e}_x$ over a spherical body of radius R . If the density and transport properties of the fluid are constant and the effect of radiation and chemical reaction are negligible, the problem reduces to that of

integrating

$$\nabla \cdot \bar{v} = 0 \quad (4.43)$$

$$\rho \frac{\partial \bar{v}}{\partial t} + \rho \bar{v} \cdot \nabla \bar{v} = -\nabla p + \mu \nabla^2 \bar{v} + \rho g \bar{e}_z \quad (4.44)$$

$$\rho c \frac{\partial T}{\partial t} + \rho c \bar{v} \cdot \nabla T = 2\mu \bar{\mathbb{T}}_d : \bar{\mathbb{T}}_d + k \nabla^2 T. \quad (4.45)$$

Boundary conditions must be imposed on the surface of the body

$$|\bar{x}| = R : \bar{v} = T - T_w = 0, \quad (4.46)$$

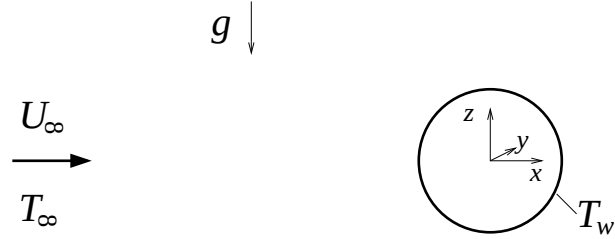
where T_w is the temperature on the surface of the body, different from the value T_∞ of the free stream. Far from the body the velocity should approach that of the free stream whereas the pressure equilibrates the gravity and inertial forces according to

$$|\bar{x}| \gg R : \bar{v} = U_\infty \bar{e}_x, T = T_\infty, p + \rho g z + \rho \frac{dU}{dt} x = \text{constant}. \quad (4.47)$$

If the fluid is initially at rest (i.e., $U = 0$ for $t \leq 0$), appropriate initial conditions are

$$t = 0 : \bar{v} = 0, T = T_\infty + \frac{T_w - T_\infty}{|\bar{x}|/R}, \quad (4.48)$$

where the temperature field is that obtained for a fluid at rest through solution of the corresponding reduced energy equation $\nabla^2 T = 0$ with account taken of the spherical symmetry.



Stress-tensor components for a newtonian fluid

The components of the viscous stress tensor can be determined from (3.31) to give

$$\tau'_{ij} = 2\mu\gamma_{ij} + (\mu_B - \frac{2}{3}\mu)\nabla \cdot \bar{v}\delta_{ij},$$

where $\gamma_{ij} = [(\nabla\bar{v})_{ij} + (\nabla\bar{v}^T)_{ij}]/2$ is the ij component of the rate-of-strain tensor $\bar{\bar{T}}_d$ and δ_{ij} represents the Kronecker delta ($\delta_{ii} = 1$, $\delta_{ij} = 0$ if $i \neq j$). From the expressions derived above in Chapter 2, it can be shown that

$$\gamma_{ii} = \frac{1}{h_i} \frac{\partial v_i}{\partial x_i} + \sum_{k \neq i} \frac{v_k}{h_i h_k} \frac{\partial h_i}{\partial x_k} \quad \text{and} \quad \gamma_{ij} = \frac{h_j}{2h_i} \frac{\partial}{\partial x_i} \left(\frac{v_j}{h_j} \right) + \frac{h_i}{2h_j} \frac{\partial}{\partial x_j} \left(\frac{v_i}{h_i} \right) \quad (\text{if } i \neq j).$$

Using this last equation together with

$$\nabla \cdot \bar{v} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial x_1} (h_2 h_3 v_1) + \frac{\partial}{\partial x_2} (h_1 h_3 v_2) + \frac{\partial}{\partial x_3} (h_1 h_2 v_3) \right],$$

one may write expressions for $\tau'_{ij} = \tau'_{ji}$ in different orthogonal coordinate systems.

Cartesian coordinates ($\nabla \cdot \bar{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$)

$$\begin{aligned} \tau'_{xx} &= 2\mu \frac{\partial v_x}{\partial x} + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v}) & \tau'_{xy} &= \mu \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) \\ \tau'_{yy} &= 2\mu \frac{\partial v_y}{\partial y} + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v}) & \tau'_{yz} &= \mu \left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \right) \\ \tau'_{zz} &= 2\mu \frac{\partial v_z}{\partial z} + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v}) & \tau'_{zx} &= \mu \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) \end{aligned}$$

Cylindrical coordinates ($\nabla \cdot \bar{v} = \frac{1}{r} \frac{\partial r v_r}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z}$)

$$\begin{aligned} \tau'_{rr} &= 2\mu \frac{\partial v_r}{\partial r} + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v}) & \tau'_{r\theta} &= \mu \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right] \\ \tau'_{\theta\theta} &= 2\mu \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v}) & \tau'_{\theta z} &= \mu \left(\frac{\partial v_\theta}{\partial z} + \frac{1}{r} \frac{\partial v_z}{\partial \theta} \right) \\ \tau'_{zz} &= 2\mu \frac{\partial v_z}{\partial z} + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v}) & \tau'_{zr} &= \mu \left(\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right) \end{aligned}$$

Spherical coordinates ($\nabla \cdot \bar{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi}$)

$$\begin{aligned} \tau'_{rr} &= 2\mu \frac{\partial v_r}{\partial r} + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v}) & \tau'_{r\theta} &= \mu \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right] \\ \tau'_{\theta\theta} &= 2\mu \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v}) & \tau'_{\theta\phi} &= \mu \left[\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\frac{v_\phi}{\sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} \right] \\ \tau'_{\phi\phi} &= 2\mu \left(\frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_r}{r} + \frac{v_\theta \cot \theta}{r} \right) + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v}) & \tau'_{\phi r} &= \mu \left[\frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} + r \frac{\partial}{\partial r} \left(\frac{v_\phi}{r} \right) \right] \end{aligned}$$

Navier-Stokes equations for a perfect liquid (μ and k constants)

CARTESIAN COORDINATES (x, y, z)

Continuity equation

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

Momentum equation

$$\begin{aligned} \rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) &= -\frac{\partial p}{\partial x} + \mu \left[\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right] + \rho f_{m_x} \\ \rho \left(\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right) &= -\frac{\partial p}{\partial y} + \mu \left[\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2} \right] + \rho f_{m_y} \\ \rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) &= -\frac{\partial p}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho f_{m_z} \end{aligned}$$

Energy equation

$$\rho c \left(\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = \phi_v + k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + Q_c + Q_r$$

where

$$\phi_v = \mu \left[2 \left(\frac{\partial v_x}{\partial x} \right)^2 + 2 \left(\frac{\partial v_y}{\partial y} \right)^2 + 2 \left(\frac{\partial v_z}{\partial z} \right)^2 + \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right)^2 + \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right)^2 + \left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \right)^2 \right]$$

Navier-Stokes equations for a perfect liquid (μ and k constants)

CYLINDRICAL COORDINATES (r, θ, z)

Continuity equation

$$\frac{1}{r} \frac{\partial r v_r}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} = 0$$

Momentum equation

$$\begin{aligned} \rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z \frac{\partial v_r}{\partial z} \right) = \\ - \frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho f_{m_r} \end{aligned}$$

$$\begin{aligned} \rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r v_\theta}{r} + v_z \frac{\partial v_\theta}{\partial z} \right) = \\ - \frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho f_{m_\theta} \end{aligned}$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) = - \frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho f_{m_z}$$

Energy equation

$$\rho c \left(\frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + v_z \frac{\partial T}{\partial z} \right) = \phi_v + k \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right] + Q_c + Q_r,$$

where

$$\begin{aligned} \phi_v = \mu \left\{ 2 \left[\left(\frac{\partial v_r}{\partial r} \right)^2 + \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right)^2 + \left(\frac{\partial v_z}{\partial z} \right)^2 \right] + \right. \\ \left. \left(\frac{1}{r} \frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \right)^2 + \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]^2 + \left(\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right)^2 \right\} \end{aligned}$$

Navier-Stokes equations for a perfect liquid (μ and k constants)

SPHERICAL COORDINATES (r, θ, ϕ)

Continuity equation

$$\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} = 0$$

Momentum equation

$$\begin{aligned} \rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{v_\theta^2 + v_\phi^2}{r} \right) = & -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) \right) + \right. \\ & \left. \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v_r}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_r}{\partial \phi^2} - \frac{2}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) - \frac{2}{r^2 \sin \theta} \frac{\partial v_\phi}{\partial \phi} \right] + \rho f_{m_r} \end{aligned}$$

$$\begin{aligned} \rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} + \frac{v_r v_\theta}{r} - \frac{v_\phi^2 \cot \theta}{r} \right) = & -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) + \right. \\ & \left. \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_\theta}{\partial \phi^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} - \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial v_\phi}{\partial \phi} \right] + \rho f_{m_\theta} \end{aligned}$$

$$\begin{aligned} \rho \left(\frac{\partial v_\phi}{\partial t} + v_r \frac{\partial v_\phi}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\phi}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_r v_\phi}{r} + \frac{v_\theta v_\phi \cot \theta}{r} \right) = & -\frac{1}{r \sin \theta} \frac{\partial p}{\partial \phi} + \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\phi}{\partial r} \right) + \right. \\ & \left. \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\phi \sin \theta) \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_\phi}{\partial \phi^2} + \frac{2}{r^2 \sin \theta} \frac{\partial v_r}{\partial \phi} + \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial v_\theta}{\partial \phi} \right] + \rho f_{m_\phi} \end{aligned}$$

Energy equation

$$\begin{aligned} \rho c \left(\frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial T}{\partial \phi} \right) = \\ \phi_v + k \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} \right] + Q_c + Q_r, \end{aligned}$$

where

$$\begin{aligned} \phi_v = \mu \left\{ 2 \left[\left(\frac{\partial v_r}{\partial r} \right)^2 + \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right)^2 + \left(\frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_r}{r} + \frac{v_\theta \cot \theta}{r} \right)^2 \right] + \right. \\ \left. \left[\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\frac{v_\phi}{\sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} \right]^2 + \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]^2 + \left[\frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} + r \frac{\partial}{\partial r} \left(\frac{v_\phi}{r} \right) \right]^2 \right\} \end{aligned}$$

Chapter 5

Hydrostatics

In this short chapter we shall address the study of fluids that are in mechanical equilibrium, leaving aside effects of surface tension. After investigating the conditions necessary for the existence of mechanical equilibrium and writing the equations that apply to the description of the corresponding pressure and temperature fields, we will consider the case of a liquid, deriving in particular Arquimedes law for the force and momentum acting on a submerged body surrounded by a quiescent liquid. The case of gases at rest will be considered next by analyzing as a relevant aeronautical application the so-called Standard Atmosphere.

Equilibrium equations

For a fluid at rest with respect to a given reference frame (i.e., such that $\bar{v} = 0$ at every point), the continuity, momentum and energy equations reduce to

$$\frac{\partial \rho}{\partial t} = 0 \quad (5.1)$$

$$-\nabla p + \rho \bar{f}_m = 0 \quad (5.2)$$

$$\rho \frac{\partial e}{\partial t} = \nabla \cdot (k \nabla T) + Q_r + Q_c. \quad (5.3)$$

The first equation indicates that in a quiescent fluid the density cannot vary with time (spatial variations are in principle allowed). As can be seen, in the absence of fluid motion, momentum conservation requires that pressure forces be in equilibrium with mass forces. If a noninertial reference frame moving with acceleration $\bar{a}_o$ and angular velocity $\bar{\Omega}$ is used in the description of the problem, besides gravity one needs to account in principle for inertial forces in computing the pressure field, giving

$$\bar{f}_m(\bar{x}, t) = \bar{g} - \bar{a}_o - \bar{\Omega} \wedge (\bar{\Omega} \wedge \bar{x}) - \frac{d\bar{\Omega}}{dt} \wedge \bar{x}, \quad (5.4)$$

free from the Coriolis term $2\bar{\Omega} \wedge \bar{v}$, as corresponds to $\bar{v} = 0$. On the other hand, the energy equation (5.3) states that the increase in internal energy is due to energy addition due to heat conduction, radiation, or chemical reactions.

In general, the above equations, supplemented with state equations $\rho = \rho(p, T)$ and $e = e(p, T)$ and a state function for the variation of the thermal conductivity $k = k(T)$, must be integrated

with appropriate initial and boundary conditions. The only initial condition for $\bar{v}$ that is compatible with equilibrium is $\bar{v} = \bar{v}_i(\bar{x}) = 0$, whereas the nonslip condition requires that all solid boundaries be at rest. Integration of (5.1) with the initial density distribution $\rho_i(\bar{x})$ yields $\rho(\bar{x}, t) = \rho_i(\bar{x})$. The initial temperature distribution $T = T_i(\bar{x})$ should be used in integrating (5.3), along with appropriate boundary values $T = T_w$ on the confining walls. With surface tension neglected in this introductory study, the condition of mechanical equilibrium at the interphase between two immiscible fluids require that the pressure be equal on both sides of the separating interphase, where the temperature must also be equal, as implied by the condition of thermodynamic equilibrium.

Hydrostatics

For perfect liquids, (5.1) is satisfied identically, while (5.2) takes the alternative form

$$-\nabla(p/\rho) + \bar{f}_m = 0, \quad (5.5)$$

indicating that the only inertial forces compatible with equilibrium are conservative forces deriving from a potential. That is the case of the first three terms in (5.4), which can be expressed in the form

$$\bar{f}_m = -\nabla U \quad (5.6)$$

in terms of the potential

$$U = -\bar{g} \cdot \bar{x} + \bar{a}_o \cdot \bar{x} - (\bar{\Omega} \wedge \bar{x}) \cdot (\bar{\Omega} \wedge \bar{x})/2. \quad (5.7)$$

However, the inertial force $d\bar{\Omega}/dt \wedge \bar{x}$ is nonconservative. Therefore, for the liquid to be in equilibrium, the angular velocity Ω must remain constant.

In equilibrium, integration (5.5) with use made of (5.6) yields

$$p + \rho U = C(t), \quad (5.8)$$

where $C(t)$ is the value of $p + \rho U$ at any spatial location. To complete the solution, one should integrate (5.3) for the temperature field. For instance, in the absence of radiation or chemical reaction and with constant thermal conductivity, the equation reduces to the familiar heat equation

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T, \quad (5.9)$$

where $\alpha = k/(\rho c_p)$ is the thermal diffusivity.

Forces on a surface. Archimedes principle

Consider now the case of a liquid at rest under the sole action of gravity, whose potential is given by $U = \rho g z$, with z representing the vertical coordinate, defined such that $\bar{g} = -g\bar{e}_z$. If the liquid is limited by a gas at pressure p_a , then the pressure given by (5.8) reduces to

$$p + \rho g z = p_a, \quad (5.10)$$

where z is measured from the free surface separating the liquid and the air. The force and the moment with respect to a given point $\bar{x}_o$ exerted by the pressure on a given surface Σ with unit normal vector $\bar{n}$ oriented towards the fluid are given by

$$\bar{F} = - \int_{\Sigma} p \bar{n} d\sigma \quad \text{and} \quad \bar{M} = - \int_{\Sigma} (\bar{x} - \bar{x}_o) \wedge p \bar{n} d\sigma. \quad (5.11)$$

These expressions can be rewritten with use made of (5.10) in the form

$$\bar{F} = - \int_{\Sigma} p_a \bar{n} d\sigma + \rho g \int_{\Sigma} z \bar{n} d\sigma \quad (5.12)$$

$$\bar{M} = - \int_{\Sigma} (\bar{x} - \bar{x}_o) \wedge p_a \bar{n} d\sigma + \rho g \int_{\Sigma} (\bar{x} - \bar{x}_o) \wedge z \bar{n} d\sigma. \quad (5.13)$$

The first term in (5.12) is the contribution of the gas pressure, while the second corresponds to the weight of the liquid column of height $-z$ standing above the surface element. To obtain the projection of the force in a given direction, defined by the unit vector $\bar{m}$, we perform the scalar product

$$F_m = \bar{F} \cdot \bar{m} = - \int_{\Sigma} (p_a - \rho g z) \bar{n} \cdot \bar{m} d\sigma = - \int_{\Sigma} (p_a - \rho g z) d\sigma_m, \quad (5.14)$$

where $d\sigma_m$ is the projection of $d\sigma$ on a plane normal to $\bar{m}$.

If the surface Σ is enclosed, as occurs for instance when computing the force over a submerged solid body, we may imagine that the fluid extends to the interior of the body and apply Gauss formula to obtain

$$\bar{F} = - \int_V \nabla p_a dV + \rho g \int_V \nabla z dV = \rho g V \bar{e}_z. \quad (5.15)$$

Similarly, the moment of the pressure forces with respect to a point $\bar{x}_o$ can be written from (5.13) as

$$\begin{aligned} \bar{M} &= \int_{\Sigma} \bar{n} \wedge [(p_a - \rho g z)(\bar{x} - \bar{x}_o)] d\sigma = \int_V \nabla \wedge [(p_a - \rho g z)(\bar{x} - \bar{x}_o)] dV = \\ &= -\rho g \bar{e}_z \wedge \int_V (\bar{x} - \bar{x}_o) dV = (\bar{x}_g - \bar{x}_o) \wedge \rho g V \bar{e}_z, \end{aligned} \quad (5.16)$$

where $\bar{x}_g = V^{-1} \int_V \bar{x} dV$ is the center of gravity of the fluid being displaced by the body. Equations (5.15) and (5.16) constitute the mathematical statement of Archimedes principle: the buoyancy force acting on a submerged object is equal to the weight of the fluid displaced by the object. Its point of application is the center of gravity of the displaced liquid, also called centroid (not necessarily coincident with the center of gravity of the object).

The relative position of the centroid and of the center of gravity of the object is fundamental in connection with the stability of submerged and floating bodies. Thus, for a body in equilibrium, we talk about *stable* equilibrium if under small perturbations the body reacts to recover the original equilibrium position. Consider for instance the stability of the submerged body of volume V sketched in Fig. 5.1. In equilibrium, the buoyancy force $\rho g V \bar{e}_z$, applied in the centroid A , must equate the weight $-W \bar{e}_z$, applied in the center of gravity of the body B which, in equilibrium, must be aligned vertically with the centroid. For stable equilibrium, A must lie above B , so that any rotating motion gives rise to a restoring torque that brings the body back to equilibrium. Clearly, for increased stability, the center of gravity must be placed as low as possible.

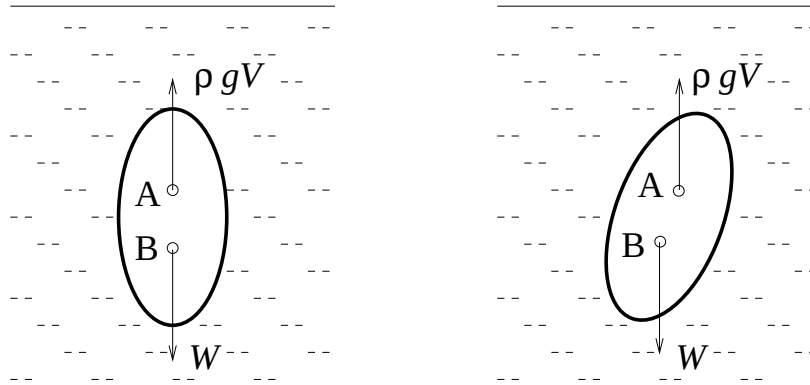


Figure 5.1: Stability of a submerged body.

The standard atmosphere

For a gas in equilibrium under the action of gravity, (5.2) gives $\partial p / \partial x = \partial p / \partial y = 0$ and therefore

$$\frac{dp}{dz} = -\rho g. \quad (5.17)$$

Owing to its low density, the variation of the air pressure with height is very small, so that for many applications in which the height difference is on the order of meters, it is a good approximation to neglect pressure differences altogether and use a constant uniform value for the pressure of a gas at rest. To observe appreciable variations of p one needs to consider distant points separated by large vertical distances of the order of several kilometers, as occurs when we move across the atmosphere.

The atmosphere is the layer of air that surrounds the Earth, which is relatively thin (its thickness is only a few tenths of kilometers, to be compared with the average Earth radius, 6,371 Km). The atmospheric turbulence ensures sufficient mixing for the air composition, given approximately by 78% of N_2 , 21 % of O_2 , and 1% de Ar, to be roughly constant up to very large altitudes. The corresponding mean molecular weight $W_{\text{air}} = 0.029 \text{ Kg/mol}$ yields $R_g = R^\circ / W = 287 \text{ J/(kg}\cdot\text{K)}$ for the gas constant, to be used in writing the equation of state

$$\frac{p}{\rho} = R_g T. \quad (5.18)$$

Solving (5.18) for the density and substituting the result into (5.17) yields

$$\frac{dp}{dz} = -\frac{p}{R_g T} g \quad \rightarrow \quad \frac{dp}{p} + \frac{g}{R_g T} dz = 0 \quad (5.19)$$

as an equation to describe the pressure variation with the height z . Note that specification of the function $T = T(z)$ is required prior to integration.

The approximate temperature variation in the atmosphere includes an initial linear decrease in the troposphere, followed by a layer of constant temperature in the stratosphere, which ends at $z = 20 \text{ Km}$ with a upper region where the temperature increases slightly. This can be expressed in the form

$$T = \begin{cases} T_0 - \alpha z & 0 < z < 11000 \text{ m} & \text{(troposphere)} \\ T_{11} & 11000 < z < 20000 \text{ m} & \text{(stratosphere)} \end{cases} \quad (5.20)$$

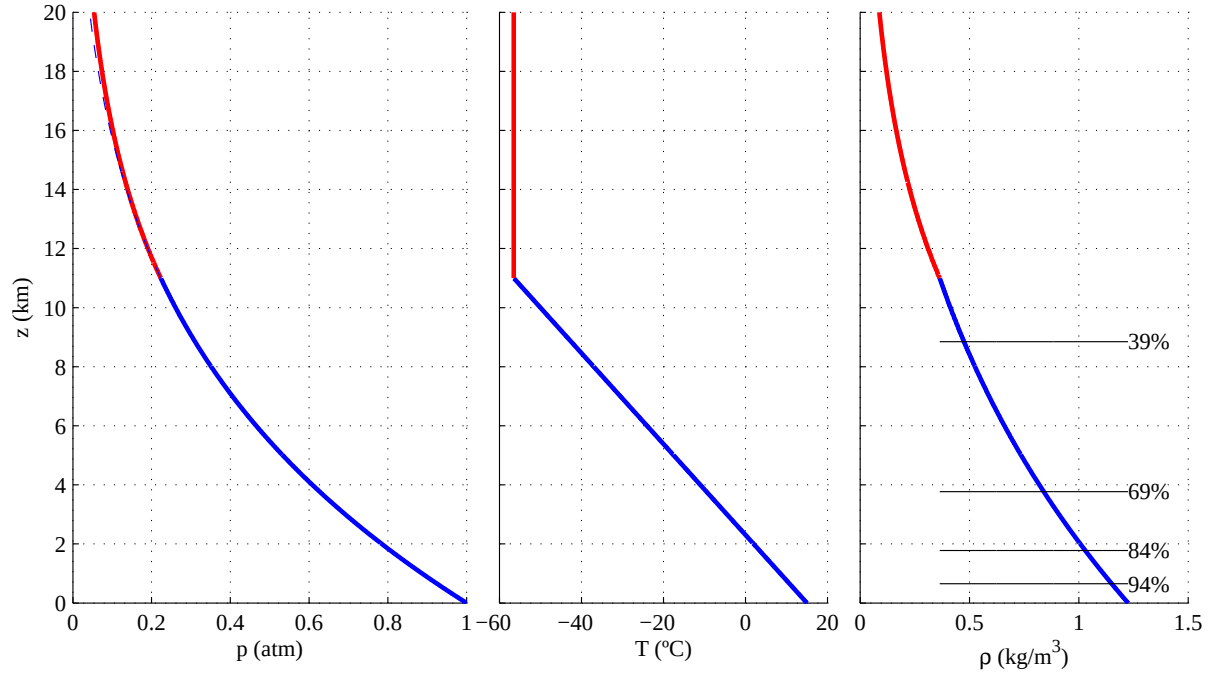


Figure 5.2: Temperature, pressure, and density distributions across the standard atmosphere. The red and blue lines are used to represent, respectively, the troposphere ($0 < z < 11$ km) and the stratosphere ($11 < z < 20$ km).

where T_0 is the temperature at sea level ($z = 0$), α is the temperature gradient in the troposphere, and $T_{11} = T_0 - \alpha 11000$ is the temperature in the stratosphere. Although the different values may change slightly from day to day, there exists a set of internationally accepted values for the computation of the so-called Standard Atmosphere, used by airplane and engine manufacturers for product specification. These values are $T_0 = 288.16$ K and $\alpha = 6.5 \cdot 10^{-3}$ K/m, leading to $T_{11} = 288.16 - 6.5 \cdot 11 = 216.66$ K.

Introducing in (5.19) the variation for the temperature in the troposphere yields

$$\int \frac{dp}{p} = -\frac{g}{R_g} \int \frac{dz}{T_0 - \alpha z} \quad (5.21)$$

which can be integrated to give

$$\ln p - \frac{g}{R_g \alpha} \ln(T_0 - \alpha z) = C \quad (5.22)$$

The value of the constant of integration C is determined by evaluating (5.22) at sea level, $z = 0$, where the pressure is $p = p_a = 1$ atm, yielding

$$C = \ln p_a - \frac{g}{R_g \alpha} \ln T_0. \quad (5.23)$$

Finally, substituting this value into (5.22) provides the expression

$$p = p_a \left(1 - \frac{\alpha z}{T_0}\right)^{\frac{g}{R_g \alpha}} \quad \text{for } 0 < z < 11000 \text{ m} \quad (5.24)$$

for the pressure evolution in the troposphere, where z represents the height above sea level. Note that at $z = 11$ Km, the above equation predicts $p = p_{11\text{ km}} = p_a \left(\frac{T_{11}}{T_0} \right)^{\frac{g}{R_g \alpha}}$. In the stratosphere, where $T = T_{11}$ remains constant, straightforward integration of (5.19) with the condition $p = p_{11\text{ km}}$ at $z = 11000$ yields

$$p = p_{11\text{ km}} \exp \left[-\frac{g}{R_g T_{11}} (z - 11000) \right] \quad \text{for } 11000 < z < 20000 \text{ m} \quad (5.25)$$

Figure 5.2 represent the decrease with height of the pressure and density for the standard atmosphere, the latter computed with use made of (5.20), (5.24), and (5.25) together with (5.18). For illustrative purposes, dashed lines are utilized in the plot to indicate conditions in Madrid (650 m), Navacerrada (1780 m), Teide (3718), and Everest (8848 m). Note that at the top of the Everest climbers find a density that is only 39% of that found at sea level, a low value that readily leads to death by oxygen deprivation after only a few hours of exposure.

Chapter 6

Dimensional Analysis

After introducing the subject of dimensional analysis through a simple illustrative example from classical mechanics, we shall present the so-called Π (or Vaschy-Buckingham) theorem, which is next applied to different problems of fluid mechanics. The nondimensionalization of the conservation equations is also introduced as an alternative method to expose the parametric dependence of a fluidmechanical problem. Finally, physical similarity is studied along with its application in scaling experiments.

Dimensions of a physical magnitude

We consider the motion of a point particle of mass m that falls from rest from an initial height h under the action of gravity. To compute the collision time t_c at which it impacts the ground as well as the collision velocity at that instant v_c one may integrate Newton's second law for the motion of the particle $m d^2z/dt^2 = -mg$ with initial conditions $z = h$ and $dz/dt = 0$ at $t = 0$ to give $dz/dt = -gt$ and $z = h - gt^2/2$ for the evolution with time of the particle velocity and of the instantaneous distance to the ground z . Equating the last expression to zero yields $t_c = \sqrt{2h/g}$, which can be used to give $v_c = gt_c = \sqrt{2hg}$. The objective of this introductory section is that of showing how the solution can be determined almost completely by a simple reasoning based on dimensional arguments.

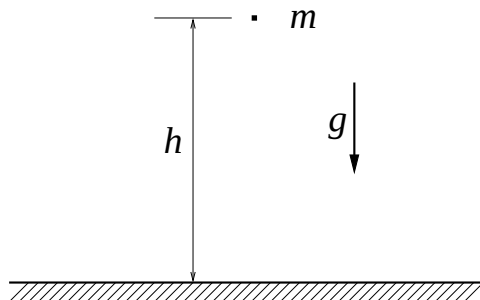


Figure 6.1: The fall of a point mass.

As a first step in solving any given problem in engineering or physics we need to identify the dependences of the unknown quantities, i.e., the governing parameters that influence the solution.

For the problem at hand, one would in principle guess that the values of the collision time and collision velocity are functions of the mass of the body m , the falling distance h , and the gravity g (because that is after all the cause of the motion), so that we write

$$t_c = t_c(m, h, g) \quad \text{and} \quad v_c = v_c(m, h, g). \quad (6.1)$$

The numbers t_c , v_c , h , m and g represent the magnitudes of physical quantities. To obtain these numbers we need to define a system of fundamental units of measurement. For instance, one may select the second, the minute or the hour as the unit of time. The value of t_c is obtained by comparing the collision time with the unit of time, giving a number that depends on the unit of time selected (for instance, $t_c = 180$ seconds would be $t_c = 3$ minutes or $t_c = 0.05$ hours). Similarly, to complete the system of units we define a unit of length (centimeter, or inch, or yard) and a unit of mass (kilogram, or pound, or ton). The comparison of the initial height with the unit length would provide the number h while the comparison of the mass of the body with the unit of mass would give the number m . From the fundamental units of length and time one may construct a unit of velocity, as the velocity of a moving body that travels the unit of length in the unit time. Comparing the impact velocity of the body with the unit of velocity would give the number v_c (e.g., $v_c = 360$ Km/h or $v_c = 100$ m/s). Correspondingly, we shall define the unit of acceleration as the that of a body that increments its velocity by the unit of velocity in the unit of time. The unit of acceleration is to be compared with gravity to give for instance on the Earth surface $g \simeq 9.8$ m/s² or $g = 35.28$ Km/min².

Changing the units of measurement changes the resulting numerical values of t_c , v_c , h , m and g . In particular, if the unit of length is decreased by a factor L , the unit of time is decreased by a factor T , and the unit of mass is decreased by a factor M , the numbers h , t_c and m will increase by a factor L , T , and M , respectively (e.g., changing the unit of time from the minute to the second, which corresponds to decreasing it by a factor of 60, would cause the numerical value t_c to increase from $t_c = 3$ minutes to $t_c = 180$ seconds). It is easy to verify that the change of units proposed would increase the numerical value v_c in a factor LT^{-1} and the numerical value g by a factor LT^{-2} . These abstract positive numbers are the dimensions of the different physical quantities. Thus, we shall say that the dimensions of h , t_c , and m are, respectively, L , T and M , and this is represented in the form $[h] = L$, $[t_c] = T$, and $[m] = M$. Similarly, v_c has dimensions $[v_c] = LT^{-1}$ and g has dimensions $[g] = LT^{-2}$. The dimensions are always power-law monomials of the form $L^\alpha T^\beta M^\gamma$, where the constants α , β and γ are in general rational numbers.

To advance in the solution, let us consider now the quantities $(h/g)^{1/2}$ and $(gh)^{1/2}$. It is easy to see that decreasing the unit length and the unit time in factors L and T causes $(h/g)^{1/2}$ and $(gh)^{1/2}$ to increase by a factor T and LT^{-1} , respectively, so that we may write $[(h/g)^{1/2}] = T$ and $[(gh)^{1/2}] = LT^{-1}$. Correspondingly, the ratios

$$\Pi_1 = \frac{t_c}{(h/g)^{1/2}} \quad (6.2)$$

and

$$\Pi_2 = \frac{v_c}{(gh)^{1/2}} \quad (6.3)$$

have dimensions $[\Pi_1] = 1$ and $[\Pi_2] = 1$. They are dimensionless quantities. Their fundamental property is that their numerical value is independent of the specific system of units employed (i.e., changing the units of length and time may change t_c and $(h/g)^{1/2}$, but it leaves the value

of $t_c/(h/g)^{1/2}$ unperturbed). Using the presumed equations (6.1) we can now write

$$\Pi_1 = \frac{t_c(m, h, g)}{(h/g)^{1/2}} = \Pi_1(m, h, g) \quad (6.4)$$

and

$$\Pi_2 = \frac{v_c(m, h, g)}{(gh)^{1/2}} = \Pi_2(m, h, g) \quad (6.5)$$

for the parametric dependences of Π_1 and Π_2 . Note that Π_1 and Π_2 are dimensionless quantities, independent therefore of the system of fundamental units employed, whereas m , h , and g are dimensional quantities, whose numerical value changes when we change the units of measurement according to $[m] = M$, $[h] = L$, and $[g] = LT^{-2}$. This can be used to simplify the functional dependences given above. In particular, by modifying only the unit of mass, while leaving the units of length and time unaltered, one may change the value of m arbitrarily while maintaining the numbers h and g constant. Since Π_1 and Π_2 are independent of the system of units - and therefore remain constant when the unit of mass changes- it is clear that the presumed functional dependences (6.4) and (6.5) are incorrect, in that Π_1 and Π_2 are not a function of m , so that we reduce the number of independent variables to give $\Pi_1 = \Pi_1(h, g)$ and $\Pi_2 = \Pi_2(h, g)$. We continue the reasoning by decreasing the unit of time by a factor T , which increases the numerical value of g in a factor T^{-2} , while h (and of course Π_1 and Π_2) remains unaltered. We can then conclude that Π_1 and Π_2 are not a function of g either. Finally, by modifying the unit of length, we can change arbitrarily h and, again, that does not result in changes of Π_1 and Π_2 , so that these two quantities are constant, independent of all the presumed parameters, as expressed in

$$\frac{t_c}{(h/g)^{1/2}} = C_1 \quad (6.6)$$

and

$$\frac{v_c}{(gh)^{1/2}} = C_2. \quad (6.7)$$

Dimensional analysis alone leads to (6.6) and (6.7), but does not provide the values of C_1 and C_2 . They could be obtained, for instance, from numerical integration of Newton's second law, as shown above, giving $C_1 = C_2 = \sqrt{2}$. They would also be determined by carrying out a single experiment, by dropping a given point object of mass m from a given height h in a given planet with gravity g . Measuring the resulting impact velocity and collision time and using (6.6) and (6.7) gives then the values of C_1 and C_2 . Note that, once they are determined, the laws given in (6.6) and (6.7) serve to describe the fall of point objects of any mass, from any height on any planet. This is an example of how arguments based on the concept of dimension, possibly together with a limited number of experiments (one in this case) may determine the solution of an engineering problem. As shown below, the methodology used can be generalized, providing a rigorous procedure for the analysis of any given physical problem.

Physical quantities with independent dimensions

The number of units in the fundamental system of units depends on the character of the problem. In a geometrical problem, it is enough to introduce a unit of length, whereas for a kinematic problem we need to add one unit to measure time. In mechanics, the system includes three units (for mass, length and time, for instance), and a larger number is required for describing

thermal problems, when an additional unit must be introduced to measure temperature, or electromagnetic problems, involving a unit of electric charge. The presentation below will focus on problems of mechanical character, for which the dimensions of all physical quantities admit a power-law monomial representation of the form $[a] = L^\alpha T^\beta M^\gamma$.

We say that the k physical quantities $a_1, \dots, a_k$ have independent dimensions when the dimensions of none of them can be expressed as a product of powers of the dimensions of the remaining $k - 1$ quantities. For instance, consider the set of three physical quantities formed by the density ($[\rho] = ML^{-3}$), velocity ($[U] = LT^{-1}$) and viscosity ($[\mu] = ML^{-1}T^{-1}$). Do they have independent dimensions? To find the answer, we begin by assuming the converse. If their dimensions were dependent we could find two numbers x and y such that $[\rho] = [U]^x[\mu]^y$. Substituting the dimensions of the different quantities yields

$$L^{-3}M^1T^0 = (L^1M^0T^{-1})^x(L^{-1}M^1T^{-1})^y, \quad (6.8)$$

which provides the three equations

$$-3 = x - y, \quad 1 = y \quad \text{and} \quad 0 = -x - y. \quad (6.9)$$

The reader can easily verify that it is not possible to find x and y satisfying simultaneously all three equations, so that we conclude that the quantities ρ , U and μ in fact have independent dimensions. As an example of three quantities that do not have independent dimensions, consider the pressure ($[p] = ML^{-1}T^{-2}$), density ($[\rho] = ML^{-3}$), and velocity ($[U] = LT^{-1}$). In this case, there exist values of $x = 1$ and $y = 2$ such that $[p] = [\rho][U]^2$.

Note that the equations (6.9) can also be expressed in the alternative form

$$(-3, 1, 0) = x(1, 0, -1) + y(-1, 1, -1), \quad (6.10)$$

revealing that ρ , U and μ have independent dimensions if the vectors $(-3, 1, 0)$, $(1, 0, -1)$ and $(-1, 1, -1)$ formed by the powers in the dimensions of each of the three quantities are linearly independent. Therefore, to check in general whether the physical quantities $a_1, \dots, a_k$ have independent dimensions it is sufficient to see whether the rank of the matrix formed by the powers in the dimensions equals k . For the case of ρ , U and μ the corresponding matrix is

$$\begin{bmatrix} -3 & 1 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & -1 \end{bmatrix} \quad (6.11)$$

whose rank in fact equals 3, so that these three quantities have independent dimensions. However, for ρ , U and p the matrix becomes

$$\begin{bmatrix} -3 & 1 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & -2 \end{bmatrix} \quad (6.12)$$

whose rank equals 2, indicating that the three quantities ρ , U and p do not have independent dimensions.

The Π theorem (the Vaschy-Buckingham theorem)

To solve a given physical problem we need to compute an unknown physical quantity

$$a_0 = f(a_1, \dots, a_k, a_{k+1}, \dots, a_n), \quad (6.13)$$

which is a function of a number of governing parameters $a_1, \dots, a_k, a_{k+1}, \dots, a_n$. Let us assume that $k \leq n$ is the maximum number of governing parameters that have independent dimensions, with $a_1, \dots, a_k$ denoting the subset of parameters with independent dimensions. Consequently, the dimensions of the remaining parameters $a_{k+1}, \dots, a_n$ can be expressed in the form

$$\begin{aligned} [a_{k+1}] &= [a_1]^{\alpha_{k+1}} \dots [a_k]^{\gamma_{k+1}}, \\ [a_{k+2}] &= [a_1]^{\alpha_{k+2}} \dots [a_k]^{\gamma_{k+2}}, \\ &\vdots \\ [a_n] &= [a_1]^{\alpha_n} \dots [a_k]^{\gamma_n}. \end{aligned} \quad (6.14)$$

In general, $0 < k < n$, but sometimes $k = n$, as occurs in the analysis of the falling body presented earlier, or $k = n$, as occurs if all parameters $a_1, \dots, a_n$ are dimensionless. It can be shown that the dimensions of the governed parameter a_0 can also be expressed in the form

$$[a_0] = [a_1]^{\alpha_0} \dots [a_k]^{\gamma_0}. \quad (6.15)$$

We can now use (6.14) and (6.15) to define the dimensionless parameters

$$\Pi_0 = \frac{a_0}{a_1^{\alpha_0} \dots a_k^{\gamma_0}}, \Pi_1 = \frac{a_{k+1}}{a_1^{\alpha_{k+1}} \dots a_k^{\gamma_{k+1}}}, \Pi_2 = \frac{a_{k+2}}{a_1^{\alpha_{k+2}} \dots a_k^{\gamma_{k+2}}}, \dots, \Pi_{n-k} = \frac{a_n}{a_1^{\alpha_n} \dots a_k^{\gamma_n}} \quad (6.16)$$

Dividing both sides of (6.13) by $a_1^{\alpha_0} \dots a_k^{\gamma_0}$ and expressing $a_0, a_{k+1}, \dots, a_n$ in terms of $\Pi_0, \Pi_1, \dots, \Pi_{n-k}$ provide

$$\Pi_0 = \frac{f(a_1, \dots, a_k, a_{k+1}, \dots, a_n)}{a_1^{\alpha_0} \dots a_k^{\gamma_0}} = \frac{f(a_1, \dots, a_k, \Pi_1 a_1^{\alpha_{k+1}} \dots a_k^{\gamma_{k+1}}, \dots, \Pi_{n-k} a_1^{\alpha_n} \dots a_k^{\gamma_n})}{a_1^{\alpha_0} \dots a_k^{\gamma_0}}, \quad (6.17)$$

which can alternatively be rewritten as

$$\Pi_0 = \Pi_0(a_1, \dots, a_k, \Pi_1, \dots, \Pi_{n-k}). \quad (6.18)$$

Let us illustrate the procedure with an example taken from fluid mechanics. Consider the motion of a liquid along a long pipe of circular section. The variable of interest is the pressure loss per unit length

$$P_l = -\partial p / \partial x = f(Q, D, \rho, \mu) \quad (6.19)$$

which would be a function of the volume flux to be transported Q , with $[Q] = L^3 T^{-1}$, the transverse size of the pipe (given for instance by its diameter D , with $[D] = L$) and the properties of the fluid, given by its density $[\rho] = M L^{-3}$ and viscosity $[\mu] = M L^{-1} T^{-1}$. In this case, therefore, $n = 4$, whereas the number of parameters that have independent dimensions is $k = 3$, as can be easily demonstrated. If we select Q, D, ρ as the subset of governing parameters with independent dimensions, then the dimensions of the remaining parameters can be expressed as

$$[\mu] = [Q]^1 [D]^{-1} [\rho]^1 \quad (6.20)$$

and

$$[P_l] = [Q]^2 [D]^{-5} [\rho]^1, \quad (6.21)$$

corresponding in this example to equations (6.14) and (6.15) in the general presentation. It is then possible to define the dimensionless quantities

$$\Pi_0 = \frac{P_l}{\rho Q^2/D^5} \quad \text{and} \quad \Pi_1 = \frac{\mu}{\rho Q/D} \quad (6.22)$$

which can be substituted into (6.19) to give

$$\Pi_0 = \frac{f(Q, D, \rho, \Pi_1 \rho Q/D)}{\rho Q^2/D^5} = F(Q, D, \rho, \Pi_1), \quad (6.23)$$

which corresponds to (6.18).

We now proceed analogously as we did in the previous analysis of the falling body, keeping in mind that the dimensionless quantities Π_0 and Π_1 are independent of the system of units utilized, while the numbers Q , D and ρ represent dimensional magnitudes whose values change when the units of length, mass and time change according to $[Q] = L^3 T^{-1}$, $[D] = L$ and $[\rho] = M L^{-3}$. Therefore, if one decreases the unit of mass by a factor M , the value of ρ would increase by that same factor, whereas the values of Π_0 , Π_1 , Q and D remain unperturbed. In that way, it is therefore possible to change arbitrarily the value of ρ , while keeping constant Π_1 , Q and D , and the change does not affect the value of Π_0 , so that Π_0 is independent of ρ and we can write $\Pi_0 = F(Q, D, \Pi_1)$ in place of (6.23). We can now continue the simplification by decreasing the unit of time by a factor T , increasing consequently the value of Q by a factor T^{-1} (recall that $[Q] = L^3 T^{-1}$), while leaving unperturbed the values of Π_0 , Π_1 and D , so that we conclude that $\Pi_0 = F(D, \Pi_1)$. Similarly, by modifying the unit of length we arrive at the final result $\Pi_0 = F(\Pi_1)$, which can be written in the form

$$\frac{P_l}{\rho Q^2/D^5} = F\left(\frac{\mu}{\rho Q/D}\right). \quad (6.24)$$

The same procedure can be used to simplify (6.18). Thus, it is always possible to find a change in the system of units such that only the value of a_1 is modified, while those of $a_2, \dots, a_k$ remain unaltered. Since $\Pi_0, \Pi_1, \dots, \Pi_{n-k}$ are dimensionless quantities independent of the system of fundamental units, we see that modifying the governing parameter a_1 with all remaining governing parameters kept constant leads to no changes in Π_0 , so that Π_0 is not a function of a_1 . Proceeding sequentially it is possible to prove that the dimensionless quantity Π_0 is independent $a_2, \dots, a_k$, thereby simplifying the parametric dependence to

$$\Pi_0 = F(\Pi_1, \dots, \Pi_{n-k}). \quad (6.25)$$

The simplification of (6.13) to give (6.25) is formally stated in the so-called Π theorem: **For a given physical problem that can be expressed in the form (6.13), that is, in which we seek to find an unknown quantity a_0 as a function of the known values of its n different governing parameters $a_1, \dots, a_n$, it is always possible to reduce to $n - k$ the number of governing parameters, with k representing the number of governing parameters with independent dimensions. To write the simplified parametric expression, we use the k magnitudes with independent dimensions to express the remaining $n - k$ governing parameters and the unknown quantity a_0 in dimensionless form, yielding as result (6.25).**

Note that the simplification can be quite remarkable. For instance, for the flow in a pipe, application of the Π theorem allows us to reduce the number of parameters from four, i.e., (6.19),

to only one, i.e., (6.24). This simplification reduces drastically the number of experiments to be performed to characterize a given physical phenomenon. Thus, to determine the pressure loss per unit length P_l one could carry out a single series of experiments varying the flux Q and measuring the accompanying pressure loss P_l obtained in a given experimental setup with a given fluid. Representing now the results in the dimensionless form (6.24) we would have obtained the general law for pressure loss of any fluid in any pipe. To compute the pressure loss corresponding to a volume flux Q^* of a different fluid of density ρ^* and viscosity μ^* circulating along a different pipe of diameter D^* one begins by computing the value of the dimensionless governing parameter $\Pi_1^* = \mu^*/(\rho^*Q^*/D^*)$. Looking up in the plot $\Pi_0 = F(\Pi_1)$ we could then obtain the corresponding value of $\Pi_0^* = P_l^*/(\rho^*(Q^*)^2/(D^*)^5)$, which could be readily solved to determine the pressure loss per unit length $P_l^* = \Pi_0^*\rho^*(Q^*)^2/(D^*)^5$ corresponding to the specific conditions of interest.

Taylor's analysis of a nuclear explosion

In an atomic explosion a large amount of energy E is released almost instantaneously at a localized position. In the initial stages of explosion development, a very strong spherical shock wave is formed whose radius r_o increases with time t . In the beginning, the pressure p_o behind the shock is several thousands times larger than the ambient value, whose influence is therefore negligible, reducing in this case the general parametric dependence of the solution (6.13) to

$$r_o = f_1(E, \rho_a, t) \quad \text{and} \quad p_o = f_2(E, \rho_a, t). \quad (6.26)$$

The three governing parameters E , ρ_a , and t have independent dimensions, as can be verified for instance by computing the rank of the matrix

$$\begin{array}{c|ccc} & L & M & T \\ \hline E & 2 & 1 & -2 \\ \rho_a & -3 & 1 & 0 \\ t & 0 & 0 & 1 \end{array} \quad (6.27)$$

formed by the powers appearing in the dimension functions $[E] = ML^2T^{-2}$, $[\rho_a] = ML^{-3}$ and $[t] = T$. According to the Π theorem, we can in this case reduce the number of governing parameters from $n = 3$ to $n - k = 3 - 3 = 0$. Since $[r_o] = L = [E]^{1/5}[\rho_a]^{-1/5}[t]^{2/5}$ and $[p_o] = ML^{-1}T^{-2} = [E]^{2/5}[\rho_a]^{3/5}[t]^{-6/5}$, the solution simplifies to

$$\frac{r_o}{E^{1/5}\rho_a^{-1/5}t^{2/5}} = C_1 \quad \text{and} \quad \frac{p_o}{E^{2/5}\rho_a^{3/5}t^{-6/5}} = C_2, \quad (6.28)$$

which can also be written as

$$r_o = C_1 E^{1/5} \rho_a^{-1/5} t^{2/5} \quad \text{and} \quad p_o = C_2 E^{2/5} \rho_a^{3/5} t^{-6/5}. \quad (6.29)$$

The constants C_1 and C_2 must be determined by a separate analysis of compressible flow, involving integration of the corresponding conservation equations, or by experimental measurement. The shock wave progressively weakens as time goes by, so that its propagation velocity $dr_o/dt = (2/5)C_1 E^{1/5} \rho_a^{-1/5} t^{-3/5}$ and the overpressure p_o given in (6.29) continuously decrease. The solution is no longer valid as the overpressure decays to values of the order of the atmospheric

pressure p_a , which occurs for $t \sim E^{1/3} \rho_a^{1/2} / p_a^{5/6}$. The reader can use these ideas to analyze the shock wave induced by a lightning, which can be assumed to be equivalent to the instantaneous deposition along a line of a large amount of energy, with E representing the amount of energy released per unit length ($[E] = MLT^{-2}$).

Parametric dependence of the solution

Perhaps one of the most complicated steps for the successful application of the Π theorem is the correct identification of the governing parameters that influence the solution, that is, the writing of the starting equation (6.13). Considering too many parameters or missing an important one leads to incorrect results. While an educated guess is the only way to find the parametric dependence in some physical or engineering problems, in fluid mechanics there is always a simpler approach, consisting in writing the conservation equations with initial and boundary conditions that determine the solution. By inspection of the resulting mathematical problem one can then easily identify the governing parameters. Let us illustrate the method with an example.

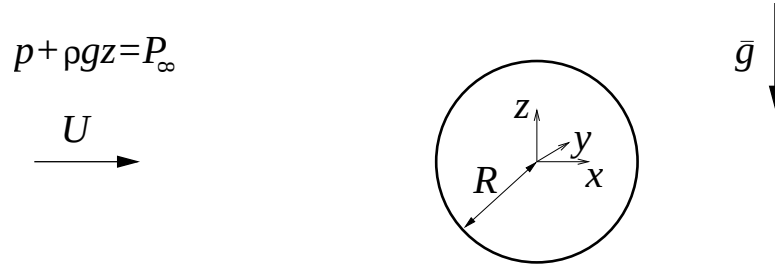


Figure 6.2: Fluid motion around a sphere.

Let us analyze the drag force F_x acting on a solid sphere moving with constant velocity U in an otherwise stagnant liquid of density ρ and viscosity μ whose pressure far from the sphere is given by the hydrostatic distribution $p + \rho gz = P_\infty$ in terms of the ambient reduced pressure P_∞ . In using the Π theorem, it might at first seem reasonable to assume the following parametric dependence

$$F_x = F_x(R, U, \rho, \mu, P_\infty, g). \quad (6.30)$$

To identify the maximum number of governing parameters with independent dimension we write the matrix

$$\begin{array}{c} R \\ U \\ \rho \\ \mu \\ p_\infty \\ g \end{array} \begin{array}{c} L \quad M \quad T \\ \left[\begin{array}{ccc} 1 & 0 & 0 \\ 1 & 0 & -1 \\ -3 & 1 & 0 \\ -1 & 1 & -1 \\ -1 & 1 & -2 \\ 1 & 0 & -2 \end{array} \right] \end{array} \quad (6.31)$$

for the powers of the dimensions $[R] = L$, $[U] = LT^{-1}$, $[\rho] = ML^{-3}$, $[\mu] = ML^{-1}T^{-1}$, $[P_\infty] = ML^{-1}T^{-2}$, and $[g] = LT^{-2}$. It is easy to see that in this case $k = 3$, by verifying for instance that the determinant of the matrix formed by the first three rows is nonzero (note that, the

rank of this 6×3 matrix can be at the most $k = 3$, equal to the number of columns). We can then select (R, U, ρ) as the subset $(a_1, \dots, a_k)$ of parameters with independent dimensions to be used for application of the Π theorem. According to what we have seen earlier, to derive the simplified expression (6.25), we have to use $(a_1, \dots, a_k)$ to define the dimensionless quantities Π_0 and $\Pi_1, \dots, \Pi_{n-k}$. For instance, in the definition of a dimensionless force we use $[F_x] = [\rho][U]^2[R]^2$ to give

$$\Pi_0 = \frac{F_x}{\rho U^2 R^2}. \quad (6.32)$$

Similarly, since $[\mu] = [\rho][U][R]$, $[P_\infty] = [\rho][U]^2$, and $[g] = [U]^2[R]^{-1}$ the dimensionless parameters associated with the viscosity, ambient reduced pressure, and gravity are

$$\Pi_1 = \frac{\mu}{\rho U R}, \quad \Pi_2 = \frac{P_\infty}{\rho U^2}, \quad \text{and} \quad \Pi_3 = \frac{g}{U^2/R} \quad (6.33)$$

so that, finally, the application of the Π theorem leads to

$$\frac{F_x}{\rho U^2 R^2} = f\left(\frac{\mu}{\rho U R}, \frac{P_\infty}{\rho U^2}, \frac{g}{U^2/R}\right). \quad (6.34)$$

As can be seen, we have reduced the number of parameters from six to only three. The result is however incorrect, in that the initial guess (6.30) includes two spurious parameters P_∞ and g that do not influence the solution and should not have been included in the first place. The application of the Π theorem to the correct initial functional dependence $F_x = F_x(R, U, \rho, \mu)$ yields

$$\frac{F_x}{\rho U^2 R^2} = f\left(\frac{\mu}{\rho U R}\right), \quad (6.35)$$

instead of (6.34). It is however not that easy to anticipate that the solution does not depend on the ambient pressure and gravity, unless we proceed as suggested above by writing the conservation equations with boundary conditions that determine the drag force prior to writing (6.30). In a reference frame moving with the sphere the computation of the pressure and velocity in the fluid requires integration of

$$\begin{aligned} \nabla \cdot \bar{v} &= 0 \\ \rho \bar{v} \cdot \nabla \bar{v} &= -\nabla p + \mu \nabla^2 \bar{v} - \rho g \bar{e}_z \end{aligned} \quad \begin{cases} |\bar{x}| \rightarrow \infty : p = P_\infty - \rho g z, \bar{v} = U \bar{e}_x \\ |\bar{x}| = R : \bar{v} = 0 \end{cases} \quad (6.36)$$

Once p and $\bar{v}$ are determined, the total force on the sphere would be computed from

$$\bar{F} = - \int_{|\bar{x}|=R} p \bar{n} d\sigma + \int_{|\bar{x}|=R} \mu (\nabla \bar{v} + \nabla \bar{v}^T) \cdot \bar{n} d\sigma, \quad (6.37)$$

where $\bar{\tau}' = \mu(\nabla \bar{v} + \nabla \bar{v}^T)$, with the drag obtained as the x component according to

$$F_x = \bar{F} \cdot \bar{e}_x = - \int_{|\bar{x}|=R} p n_x d\sigma + \int_{|\bar{x}|=R} \mu [(\nabla \bar{v} + \nabla \bar{v}^T) \cdot \bar{n}] \cdot \bar{e}_x d\sigma. \quad (6.38)$$

A quick look at (6.36) and (6.38) indicates that, as stated in (6.30), the drag force is indeed a function of ρ , U , R and μ , but also a function of P_∞ and g . This last dependences can be however eliminated by using the reduced pressure $p + \rho g z$ to incorporate the effect of gravity according to $-\nabla p - \rho g \bar{e}_z = -\nabla(p + \rho g z)$ and noting that the effect of the pressure enters in (6.36) and (6.38)

through the relative pressure differences, while the ambient level P_∞ is inconsequential. These observations suggest the use of the alternative variable $P' = p + \rho g z - P_\infty$ as a replacement for p . Rewriting (6.36) in terms of P' yields

$$\begin{aligned} \nabla \cdot \bar{v} &= 0 \\ \rho \bar{v} \cdot \nabla \bar{v} &= -\nabla P' + \mu \nabla^2 \bar{v} \end{aligned} \quad \left\{ \begin{array}{l} |\bar{x}| \rightarrow \infty : P' = 0, \bar{v} = U \bar{e}_x \\ |\bar{x}| = R : \bar{v} = 0, \end{array} \right. \quad (6.39)$$

which reveals that the distributions $P'(\bar{x})$ and $\bar{v}(\bar{x})$ are independent of P_∞ and g . The force (6.37) can be expressed in terms of these two variables to give¹

$$\bar{F} = - \int_{|\bar{x}|=R} P' \bar{n} d\sigma + \int_{|\bar{x}|=R} \mu (\nabla \bar{v} + \nabla \bar{v}^T) \cdot \bar{n} d\sigma + \int_{|\bar{x}|=R} \rho g z \bar{n} d\sigma. \quad (6.40)$$

The last term corresponds to the buoyancy force associated with Archimedes law, as can be seen by using the Gauss formula to yield

$$\int_{|\bar{x}|=R} \rho g z \bar{n} d\sigma = \int_V \nabla(\rho g z) dV = \rho g \frac{4}{3} \pi R^3 \bar{e}_z, \quad (6.41)$$

and therefore gives no contribution to the drag force. The value of F_x can be therefore computed according to

$$F_x = \bar{F} \cdot \bar{e}_x = - \int_{|\bar{x}|=R} P' n_x d\sigma + \int_{|\bar{x}|=R} \mu [(\nabla \bar{v} + \nabla \bar{v}^T) \cdot \bar{n}] \cdot \bar{e}_x d\sigma \quad (6.42)$$

in terms of $P'(\bar{x})$ and $\bar{v}(\bar{x})$, independent of P_∞ and g , thereby demonstrating that $F_x = F_x(R, U, \rho, \mu)$ is the correct parametric dependence, so that the final result obtained by application of the Π theorem should be (6.35) instead of (6.34).

Note that a dependence on the pressure level appears in general in problems involving gas motion, because the absolute value of the pressure enters in the problem through the equation of state $p = \rho R_g T$. For liquid motion, a dependence on gravity appears in problems involving nonplanar free surfaces separating the liquid from the air ambient, as occurs when waves are present. For instance, if the sphere above were moving close to the surface of the liquid (i.e., at a depth of the order of R), a boundary condition $p = p_a$ applies at the free surface $z_s = z_s(x, y)$, whose shape should be computed as part of the integration. The resulting drag force, although independent of p_a , is however a function of g (and also of the depth). The reader can verify that, in that case, using the reduced pressure in the place of p does not get rid of g in the formulation of the problem.

It is also convenient to note at this point that the selection of the subset of governing parameters with independent dimensions $a_1, \dots, a_k$ is not always unique. For instance, for the problem of the drag force on the sphere, one could have alternatively selected the subset U , R and μ , so that application of the Π theorem would have led to

$$\frac{F_x}{\mu U R} = f\left(\frac{\rho U R}{\mu}\right). \quad (6.43)$$

In principle, both expressions (6.35) and (6.43) are equally valid. The dimensionless force, which is different in the two cases, can be seen to be a function of the dimensionless parameter

$$Re = \frac{\rho U R}{\mu} \quad (6.44)$$

which is the Reynolds number of the problem.

¹Note that $\int_{|\bar{x}|=R} P_\infty \bar{n} d\sigma = 0$.

Nondimensionalization of the Navier-Stokes equations

An alternative to the Π theorem that allows us to identify the minimum number of parameters governing a given fluidmechanical problem consists in rewriting in dimensionless form the mathematical problem involved in its solution by introducing appropriately rescaled dependent and independent variables. The scales to be used depend on the initial and boundary conditions of the problem. For instance, for the motion over a sphere, the velocity U is the appropriate scale to define a dimensionless velocity $\bar{v}^* = \bar{v}/U$, while the radius of the sphere R introduces a natural length to scale the spatial coordinate, yielding $\bar{x}^* = \bar{x}/R$, $\nabla^* = R\nabla$ and $d\sigma^* = R^{-2}d\sigma$. In principle, it is not easy to find in this case a characteristic value for the pressure difference Δp to be used in scaling $P' = p + \rho gz - P_\infty$. This value, in fact, depends on the character of the fluid motion. To see this, it is convenient to estimate the order of magnitude of each one of the terms appearing in the momentum equation

$$\begin{aligned} \rho \bar{v} \cdot \nabla \bar{v} &= -\nabla P' + \mu \nabla^2 \bar{v}. \\ \rho U^2/R &\quad \Delta p/R \quad \mu U/R^2 \end{aligned} \tag{6.45}$$

To estimate the order of magnitude of the inertial term $O(\rho \bar{v} \cdot \nabla \bar{v}) \sim \rho U^2/R$ we have taken into account that the characteristic velocity is of order $O(\bar{v}) \sim U$ and that in the flow field we observe velocity differences of order U when traversing distances of order R , so that $O(\nabla \bar{v}) \sim U/R$. The estimate of the pressure gradient $O(\nabla P') \sim \Delta p/R$ is written with use made of the unknown characteristic value Δp of the pressure differences in the flow field. Similarly, the order of magnitude of the resultant of the viscous stresses per unit volume is $O(\mu \nabla^2 \bar{v}) \sim \mu U/R^2$.

The above estimates enable the relative importance of the different terms to be easily anticipated. We know from experience that the pressure gradient is almost always essential for the fluid motion², so that the remaining question is to clarify the relative importance of the convective acceleration $\rho \bar{v} \cdot \nabla \bar{v}$ and the resultant of the viscous forces $\mu \nabla^2 \bar{v}$ by comparing their orders of magnitude according to

$$\frac{O(\rho \bar{v} \cdot \nabla \bar{v})}{O(\mu \nabla^2 \bar{v})} = \frac{\rho U R}{\mu}. \tag{6.46}$$

As can be seen, the dimensionless number that measures the relative importance of inertial (convective) effects and viscosity is the so-called Reynolds number, Re , defined above in (6.44). Given the sphere radius, its velocity and the fluid properties, one may easily determine the value of Re . For the motion of fluids at $Re \sim O(1)$, acceleration and viscous stresses have comparable effects so that in principle one needs to retain all three terms in (6.45) for describing the solution. On the other hand, if $Re \gg 1$, the effect of viscosity can be anticipated to be negligible, so that (6.45) simplifies to the balance $\rho \bar{v} \cdot \nabla \bar{v} = -\nabla P'$ between acceleration and pressure forces. Conversely, if $Re \ll 1$ the flow has negligible acceleration, and the motion established responds to a local balance between pressure and viscous forces according to $0 = -\nabla P' + \mu \nabla^2 \bar{v}$.

These results can be used to evaluate the characteristic value of Δp depending on the value of Re . Thus, for fluid motion at large Reynolds numbers, using the condition $O(\rho \bar{v} \cdot \nabla \bar{v}) \sim O(\nabla P')$ gives $\Delta p \sim \rho U^2$, whereas for $Re \ll 1$ the value $\Delta p \sim \mu U/R$ follows from $O(\mu \nabla^2 \bar{v}) \sim O(\nabla P')$. Note that, for $Re \sim 1$ both estimates are equivalent. Clearly, the selection for Δp depends on

²Note that, if the pressure gradient were not important, then the mathematical problem (6.39) would involve three unknowns (the three components of the velocity) but four equations (continuity plus the three components of momentum) and would be therefore overdetermined.

the character of the problem. For $Re \gtrsim 1$ the appropriate choice is $p^* = (p + \rho gz - P_\infty)/(\rho U^2)$, reducing (6.40) to

$$\nabla^* \cdot \bar{v}^* = 0, \quad \bar{v}^* \cdot \nabla^* \bar{v}^* = -\nabla^* p^* + \frac{1}{Re} \nabla^{*2} \bar{v}^* \begin{cases} |\bar{x}^*| \rightarrow \infty : p^* = 0, \bar{v}^* = \bar{e}_x \\ |\bar{x}^*| = 1 : \bar{v}^* = 0 \end{cases} \quad (6.47)$$

and

$$\frac{F_x}{\rho U^2 R^2} = - \int_{|\bar{x}^*|=1} p^* \bar{n} \cdot \bar{e}_x d\sigma^* + \frac{1}{Re} \int_{|\bar{x}^*|=1} [(\nabla^* \bar{v}^* + \nabla^{*T} \bar{v}^*) \cdot \bar{n}] \cdot \bar{e}_x d\sigma^*. \quad (6.48)$$

As anticipated before, by writing the problem in dimensionless form we are able to reduce to the minimum the set of governing parameters that influence the solution. In this case, observation of (6.48) indicates that the value of $F_x/(\rho U^2 R^2)$ is only a function of the dimensionless parameter $Re = \rho U R / \mu$, exactly the same result (6.35) that we previously obtained by application of the Π theorem.

On the other hand, when $Re \lesssim 1$ it is more convenient to use $\Delta p \sim \mu U / R$ to scale the pressure variations $p^* = (p + \rho gz - P_\infty)/(\mu U / R)$, which reduces the problem (6.40) to the dimensionless alternative form

$$\nabla^* \cdot \bar{v}^* = 0, \quad Re \bar{v}^* \cdot \nabla^* \bar{v}^* = -\nabla^* p^* + \nabla^{*2} \bar{v}^* \begin{cases} |\bar{x}^*| \rightarrow \infty : p^* = 0, \bar{v}^* = \bar{e}_x \\ |\bar{x}^*| = 1 : \bar{v}^* = 0 \end{cases} \quad (6.49)$$

and

$$\frac{F_x}{\mu U R} = - \int_{|\bar{x}^*|=1} p^* \bar{n} \cdot \bar{e}_x d\sigma^* + \int_{|\bar{x}^*|=1} [(\nabla^* \bar{v}^* + \nabla^{*T} \bar{v}^*) \cdot \bar{n}] \cdot \bar{e}_x d\sigma^*. \quad (6.50)$$

In this case, the dimensionless problem indicates that $F_x/(\mu U R)$ is only a function of Re , the same result we obtained previously in (6.43).

Selection of parameters with independent dimensions

As discussed above, there is often a certain degree of arbitrariness when selecting the subset of parameters $a_1, \dots, a_k$, in that different choices are available. The selection must be guided by the general principle that the parameters selected must have a significant influence for the flow conditions considered. For instance, for the motion over a sphere, one can anticipate that the force will increase significantly with increasing values of R and U , so that it is reasonable to use these two parameters as part of the subset $a_1, \dots, a_k$. On the other hand, in cases where the viscosity is anticipated to play an important role, the third parameter should be μ , which leads to (6.43). On the contrary, if the viscosity is not foreseen as having a significant influence, then it is more appropriate to use ρ as the third parameter with independent dimensions, yielding (6.35) after simplification.

To anticipate the relative importance of a given parameter, we often have to use our intuition. In fluid mechanics, the precise knowledge of the mathematical framework describing fluid motion simplifies things. For instance, to see whether the viscosity is important in a given problem one could evaluate the Reynolds number, given in general by $Re = \rho U_c L_c / \mu$, where U_c and L_c are the characteristic values of the fluid velocity and flowfield size. If the scales are such that $Re \lesssim 1$, then we can anticipate that the viscosity will be important, so that μ is to be selected as part of the subset $a_1, \dots, a_k$, while discarding the density. Conversely, when $Re \gtrsim 1$ we expect the viscosity to have a secondary role, and the appropriate subset of parameters with independent dimensions should be (U_c, L_c, ρ) (with $U_c = U$ and $L_c = R$ for the flow over a sphere).

The selection is more justified for extreme values of Re . For instance, for $Re \gg 1$ one can conclude from (6.47) and (6.48) that the effect of viscosity will be entirely negligible, so that $F_x/(\rho U R^2)$ becomes independent of Re , which implies that (6.35) simplifies to give

$$\lim_{Re \gg 1} \frac{F_x}{\rho U^2 R^2} = C_1, \quad (6.51)$$

where C_1 is a constant. Similarly, in view of (6.49) and (6.50), it is clear that for $Re \ll 1$ the value of $F_x/(\mu U R)$ becomes independent of Re , yielding

$$\lim_{Re \ll 1} \frac{F_x}{\mu U R} = C_2, \quad (6.52)$$

as a simplification of (6.43) for this case of negligibly small Reynolds number. As can be seen, the parameter Re ceases to exert an influence on the solution when it takes on extreme values, provided that in the analysis we have made an appropriate choice for $a_1, \dots, a_k$, i.e., to analyze the limit $Re \gg 1$ we need to anticipate that the effect of viscosity is negligible in selecting $a_1, \dots, a_k = \rho, U$ and R , whereas to analyze the limit $Re \ll 1$ the dominant effect of the viscosity should be accounted for when selecting $a_1, \dots, a_k = \mu, U$ and R as an appropriate subset of governing parameters with independent dimensions.

The solution would not be independent of Re if the selection of $a_1, \dots, a_k$ is not appropriate. For instance, for small values of Re the limiting value of $f(Re)$ in (6.35) does not become independent of Re . Instead, one obtains

$$\lim_{Re \ll 1} \frac{F_x}{\rho U^2 R^2} = C_2 Re^{-1}, \quad (6.53)$$

as can be concluded in view of (6.52). Correspondingly, the limiting value of (6.43) for $Re \gg 1$ leads to

$$\lim_{Re \gg 1} \frac{F_x}{\mu U R} = C_1 Re, \quad (6.54)$$

as can be deduced from (6.51).

In view of these results, it is possible to make some general comments concerning the simplification of (6.25)

$$\Pi_0 = F(\Pi_1, \dots, \Pi_{n-k})$$

arising when one of the parameters $\Pi_1, \dots, \Pi_{n-k}$ takes on extreme values (very small or very large). If for instance $\Pi_1 \gg 1$, it often occurs that Π_0 no longer depends on this parameter, so that

$$\lim_{\Pi_1 \gg 1} \Pi_0(\Pi_1, \dots, \Pi_{n-k}) = f(\Pi_2, \dots, \Pi_{n-k}). \quad (6.55)$$

In this case, we reduce the number of parameters in one. Other times, the limiting value of Π_0 continues depending on the large (or small) parameter Π_1 , but the dependence takes the simplified form

$$\lim_{\Pi_1 \gg 1} \Pi_0(\Pi_1, \dots, \Pi_{n-k}) = \Pi_1^\lambda f(\Pi_2, \dots, \Pi_{n-k}), \quad (6.56)$$

where λ is a given number, to be determined. As we have seen above, an appropriate selection of $a_1, \dots, a_k$ often simplifies the treatment of limiting cases, leading to expressions of the form (6.55).

Physical similarity

Dimensional analysis serves in particular to guide the design of experiments. Often, it is not possible to reproduce exactly in the lab all of the conditions found in a given physical phenomenon of interest. For instance, if the size involved is too big, then the model for the experiments will need to be scaled down to a smaller size that we can handle in the lab. It is then appropriate to ask for the conditions that the experiment must satisfy to reproduce reliably the phenomenon occurring at the real large scale, and also how the experimental measurements must be adapted to yield the quantities of interest.

Let us consider again the moving sphere. In particular, assume we want to compute the drag force F_1 acting on a sphere of radius R_1 that moves with velocity U_1 in an oil of density ρ_1 and viscosity μ_1 . To obtain the result, we propose to carry out an experiment in a hydrodynamic tunnel. In designing the experiment, we need to take into account that the density and viscosity of water are different from those of the oil. Besides, the only spheres available have a radius $R_2 \neq R_1$.

To design the experiment, we shall use dimensional analysis. According to what we have seen above, the dependence of the force on the different parameters can be expressed in the form (6.35), indicating that the value of $F_x/(\rho U^2 R^2)$ is only a function of $Re = \rho U R / \mu$. In other words, if we get the Reynolds number in the experiment $Re_2 = \rho_2 U_2 R_2 / \mu_2$ to match exactly the value $Re_1 = \rho_1 U_1 R_1 / \mu_1$ corresponding to the real phenomenon, then the resulting values of $F_1/(\rho_1 U_1^2 R_1^2)$ and $F_2/(\rho_2 U_2^2 R_2^2)$ will also be exactly equal. With this information, we can now proceed to design the experiment.

For the laboratory experiment to be physically similar to the real phenomenon we need to impose the condition $Re_1 = Re_2$, which provides the value

$$U_2 = \frac{\mu_2}{\mu_1} \frac{\rho_1 R_1}{\rho_2 R_2} U_1. \quad (6.57)$$

to be used in the experiment. The resulting measurement in the lab gives the force F_2 acting on the experimental model. The condition $Re_1 = Re_2$ guarantees that

$$\frac{F_1}{\rho_1 U_1^2 R_1^2} = \frac{F_2}{\rho_2 U_2^2 R_2^2}, \quad (6.58)$$

which can be used finally to determine the value

$$F_1 = \frac{\rho_1 U_1^2 R_1^2}{\rho_2 U_2^2 R_2^2} F_2 \quad (6.59)$$

for the force acting on the sphere moving in oil.

The procedure can be generalized. Dimensional analysis shows that the parametric dependence of the unknown value of the physical quantity

$$a_0 = f(a_1, \dots, a_k, a_{k+1}, \dots, a_n),$$

where k is the maximum number of governing parameters with independent dimensions, can be expressed in the simplified form

$$\Pi_0 = F(\Pi_1, \dots, \Pi_{n-k}).$$

When designing an experiment, it is not necessary to reproduce exactly in the laboratory the values of $a_1, \dots, a_n$. Instead, for the experiment to reproduce reality, or more specifically, for the experiment to be physically similar to the real phenomenon, it is only necessary that the values of the different dimensionless parameters $\Pi_1, \dots, \Pi_{n-k}$ be equal in the experiment and in the real phenomenon. If physical similarity is satisfied, it is then possible to determine the value of a_0 in the real phenomenon from the measurement of a_0 in the lab by equating the corresponding values of Π_0 . Note that at times it is not possible to design the experiment in such a way that all of the governing parameters $\Pi_1, \dots, \Pi_{n-k}$ take on equal values. In that case, physical similarity is only partially satisfied. For the experiment to be successful, we need to ensure the equality of those parameters anticipated to be most influential for the problem at hand, while permitting that parameters of secondary importance take on different values.

Chapter 7

Viscous Flows

We address now liquid flows dominated by viscosity in slender configurations including pipes and thin films. We begin by considering the case of strictly unidirectional motion, including the steady solutions corresponding to Couette and Poiseuille flow along with the unsteady solutions of the Rayleigh and Stokes problems. The case of unsteady flow in a channel is used to introduce the concept of quasi-steady motion. Quasi-steady flow in pipes and channels of slowly varying cross-section is addressed next, including a careful analysis of the entrance region. Lubrication theory is addressed also, with a simple planar bearing used as illustration of the wedge effect that explains the lubricating capability of thin oil layers.

Strictly unidirectional liquid motion

Formulation of the problem

We begin by considering the flow of perfect liquids of constant viscosity when the boundary conditions are such that all streamlines are parallel to a given direction x . This is the case, for instance, of idealized flow in an infinitely long pipe of constant section. In a cartesian coordinate system, with y and z representing the transverse coordinates, the velocity vector reduces to $\bar{v} = v_x \bar{e}_x$ and the equation of continuity $\nabla \cdot \bar{v} = 0$ yields $\partial v_x / \partial x = 0$, indicating that

$$v_x = v_x(y, z, t). \quad (7.1)$$

The effect of gravity will be incorporated with use made of the reduced pressure $P = p + \rho g \mathcal{Z}$, with $\mathcal{Z}$ representing the vertical distance measured opposite to the gravity vector (different in general from the transverse coordinate z). With $\partial v_x / \partial x = v_y = v_z = 0$, the x component of the momentum equation simplifies to give

$$\rho \frac{\partial v_x}{\partial t} = -\frac{\partial P}{\partial x} + \mu \left(\frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right), \quad (7.2)$$

while the transverse components provide

$$0 = \frac{\partial P}{\partial y} = \frac{\partial P}{\partial z}. \quad (7.3)$$

As can be seen in (7.2) the nonlinear convective terms vanish, so that the resulting problem is linear. According to (7.3) the reduced pressure is independent of y and z , so that, in principle,

the streamwise loss of reduced pressure per unit length

$$P_l = -\frac{\partial P}{\partial x} \quad (7.4)$$

could be at the most a function of x and t . The result is however simpler, in that $\partial P_l / \partial x = 0$, as can be inferred from straightforward derivation of (7.2) with respect to x with $\partial v_x / \partial x = 0$. The value of $P_l(t)$ can be computed from the instantaneous measurement of $P(x, t)$ at two different sections x_1 and x_2 according to

$$P_l = \frac{P_1 - P_2}{x_2 - x_1}. \quad (7.5)$$

For strictly unidirectional flow the problem therefore reduces to that of integrating the parabolic equation

$$\rho \frac{\partial v_x}{\partial t} = P_l(t) + \mu \left(\frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right). \quad (7.6)$$

with initial conditions $v_x = v_i(y, z)$ at $t = 0$. If the fluid is confined by a cylinder of section $C(y, z) = 0$ moving with velocity $v_c(y, z, t)$, the nonslip condition yields $v_x = v_c(y, z, t)$ at $C(y, z) = 0$. For axisymmetric problems, it is convenient to express the transverse laplacian of (7.6) in cylindrical coordinates to give the alternative equation

$$\rho \frac{\partial v_x}{\partial t} = P_l(t) + \mu \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right) \quad (7.7)$$

for $v_x(r, t)$. In some problems, the pressure loss per unit length $P_l(t)$ is not given, and should be determined as part of the solution in terms of some other known quantity (e.g., the volume flux). We shall now apply the general formulation given above to a number of simple problems, beginning with some steady solutions

Introductory examples of steady planar flow

Consider the steady planar motion of a liquid in a two-dimensional channel formed between two parallel walls separated by a distance h . We begin by studying the flow that appears in the absence of pressure gradient when the walls move relative to each other. In a reference frame attached to the bottom wall the problem reduces to that of integrating

$$0 = \frac{\partial^2 v_x}{\partial y^2} \quad \begin{cases} y = 0 : & v_x = 0 \\ y = h : & v_x = V \end{cases} \quad (7.8)$$

to give the linear velocity profile, known as Couette solution,

$$v_x = Vy/h. \quad (7.9)$$

In terms of the velocity, one may for instance compute the volume flux in the channel $Q = \int_0^h v_x dy = Vh/2$ or the viscous-stress-tensor components $\tau'_{xx} = \tau'_{yy} = 0$ and $\tau'_{xy} = \mu V/h$. The viscous force per unit surface acting on the upper wall, $\bar{\tau}' \cdot \bar{n}$ with $\bar{n} = -\bar{e}_y$, is $\bar{\tau}' \cdot \bar{n} = -(\mu V/h)\bar{e}_x$, so that the power per unit surface needed to keep the upper wall in motion is $\mu V^2/h$. This energy input is converted within the fluid into internal energy by viscous dissipation, as can be seen by computing the viscous dissipation rate $\phi_v = \mu(\partial u / \partial y)^2 = \mu V^2/h^2$, which can be integrated

across the channel to give $\int \phi_v dy = \mu V^2/h$ as the amount of kinematic energy dissipated in the channel per unit length per unit time.

Another steady simple solution arises in the presence of a constant pressure gradient P_l with both walls assumed to be at rest. This is the so-called Hagen-Poiseuille solution, to be determined by integrating

$$0 = P_l + \mu \frac{\partial^2 v_x}{\partial y^2} \quad \begin{cases} y = 0 : & v_x = 0 \\ y = h : & v_x = 0 \end{cases} . \quad (7.10)$$

to give

$$v_x = \frac{1}{\mu} \frac{P_l}{2} y(h-y). \quad (7.11)$$

As before, in terms of the velocity, it is straightforward to compute the volume flux $Q = P_l h^3/(12\mu)$ or the viscous stresses on the walls $\bar{\tau}' \cdot \bar{n} = P_l h/2 \bar{e}_x$.

Note that, since the problem of strictly unidirectional motion is linear in v_x , one can use superposition of elementary solutions to obtain the velocity induced by the combined application of a pressure gradient P_l and a moving wall. For instance, if it is the upper wall that moves with velocity $V \bar{e}_x$, then the solution is obtained as the sum of (7.9) and (7.11) to yield

$$v_x = V \frac{y}{h} + \frac{1}{\mu} \frac{P_l}{2} y(h-y), \quad (7.12)$$

with corresponding volume flux and viscous stresses obtained also by addition of the two separate contributions.

Steady flow in a circular pipe

For steady liquid flow driven a constant pressure gradient along an infinitely long circular pipe of radius a the velocity is obtained by integrating

$$0 = P_l + \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right) \quad \begin{cases} r = 0 : & \partial v_x / \partial r = 0 \\ r = a : & v_x = 0 \end{cases} \quad (7.13)$$

to give the parabolic profile

$$v_x = \frac{P_l}{4\mu} (a^2 - r^2). \quad (7.14)$$

known as Poiseuille solution. In this case, the volume flux is determined as

$$Q = \int_0^a 2\pi r v_x dr = \frac{\pi}{8} \frac{P_l a^4}{\mu}, \quad (7.15)$$

while the viscous stress on the wall is

$$\tau_p = \bar{\tau}' \cdot \bar{n} = -\mu \left(\frac{\partial v_x}{\partial r} \right)_{r=a} = \frac{P_l a}{2}. \quad (7.16)$$

The motion established is the result of a balance between the reduced pressure gradient along the pipe and the friction with the surrounding wall associated with viscous forces. To see this balance more clearly, it is convenient to consider the control volume displayed in figure 7.1.

Momentum conservation requires that

$$\int_{\Sigma_c} \rho \bar{v} \bar{v} \cdot \bar{n} d\sigma = - \int_{\Sigma_c} P \bar{n} d\sigma + \int_{\Sigma_c} \bar{\tau} \cdot \bar{n} d\sigma, \quad (7.17)$$

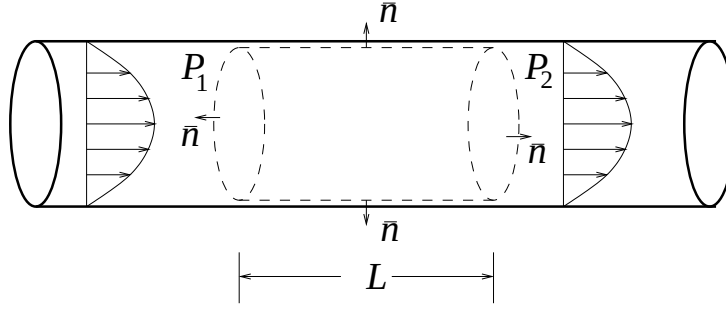


Figure 7.1: Fully developed laminar flow in a circular pipe.

where we have used Gauss formula to rewrite the contribution of gravity as a surface integral according to $\int_{V_c} \rho \bar{g} dV = - \int_{V_c} \nabla(\rho g \mathcal{Z}) dV = - \int_{\Sigma_c} \rho g \mathcal{Z} \bar{n} d\sigma$, yielding $- \int_{\Sigma_c} P \bar{n} d\sigma$ when added to the pressure force $- \int_{\Sigma_c} p \bar{n} d\sigma$, with $P = p + \rho g \mathcal{Z}$.

The first term in (7.17) is identically zero, because the velocity profile does not vary along the pipe. The two remaining terms provide $- \int_{\Sigma_c} P \bar{n} d\sigma = (P_1 - P_2) \pi a^2 \bar{e}_x$ and $\int_{\Sigma_c} \bar{\tau} \cdot \bar{n} d\sigma = -\tau_p 2\pi a L$, reducing (7.17) to

$$\tau_p = \frac{P_1 - P_2}{L} \frac{a}{2}, \quad (7.18)$$

which reproduces the result previously obtained in (7.16), with $P_l = (P_1 - P_2)/L$ as obtained with constant P_l from the values of the reduced pressure at two different sections.

Rayleigh's problem

As a first example of unsteady flow let us consider Rayleigh's solution for the flow induced by a suddenly accelerated wall. A semi-infinite space filled with liquid is bounded by a plane wall that remains at rest for $t < 0$. At $t = 0$, the wall is suddenly set in parallel motion with constant velocity V . Viscous stresses transport the momentum away from the wall, accelerating a layer of liquid of increasing thickness, where the velocity, with a single component $v_x(y, t)$ parallel to the wall, is a function of time and of the distance to the wall y . According to the transverse component of the momentum equation, $\partial P / \partial y = 0$, so that the reduced pressure everywhere in the flow field equals that encountered far from the wall, where $v_x = 0$ and $P = P_\infty$, yielding $P_l = -\partial P / \partial x = 0$. To determine $v_x(y, t)$ one therefore needs to integrate the problem

$$\rho \frac{\partial v_x}{\partial t} = \mu \frac{\partial^2 v_x}{\partial y^2} \begin{cases} t = 0 : & v_x = 0 \\ t > 0 : & \begin{cases} y = 0 : & v_x = V \\ y \rightarrow \infty : & v_x \rightarrow 0 \end{cases} \end{cases}, \quad (7.19)$$

Because of the absence of space and time scales, the resulting solution is selfsimilar, in that the velocity is a function of y and t through a single independent variable constructed as combination of both. In this case, this simplified dependence can be inferred by application of the Π theorem. For the analysis, it is convenient to rewrite (7.19) in terms of the dimensionless velocity $\tilde{v} = v_x / V$ to give

$$\rho \frac{\partial \tilde{v}}{\partial t} = \mu \frac{\partial^2 \tilde{v}}{\partial y^2} \begin{cases} t = 0 : & \tilde{v} = 0 \\ t > 0 : & \begin{cases} y = 0 : & \tilde{v} = 1 \\ y \rightarrow \infty : & \tilde{v} \rightarrow 0 \end{cases} \end{cases} \quad (7.20)$$

By inspection of the above problem one may conclude that the dependence of $\tilde{v}$ is given by

$$\frac{v_x}{V} = \tilde{v}(\rho, \mu, y, t) \quad (7.21)$$

Three of the four quantities (ρ, μ, y, t) have independent dimensions, indicating that we can reduce the number of governing parameters from $n = 4$ to $n - k = 1$. For instance, the selection of ρ , μ and t as the set of parameters with independent dimension reduces the parametric dependence to

$$\frac{v_x}{V} = \tilde{v}\left(\frac{y}{\sqrt{\nu t}}\right), \quad (7.22)$$

where $\nu = \mu/\rho$ represents the kinematic viscosity coefficient. As can be seen in (7.22), the dependence of v_x/V on y and t is embodied in a single independent variable $y/\sqrt{\nu t}$.

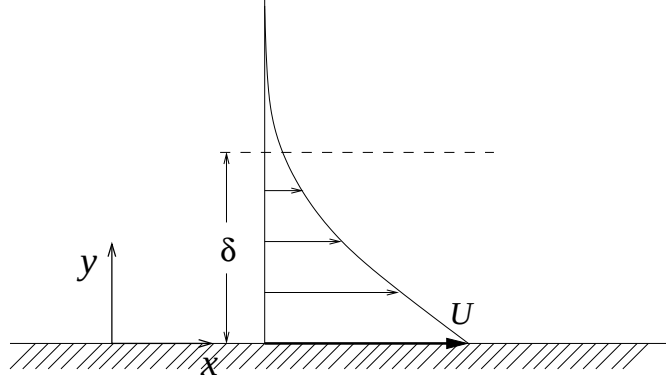


Figure 7.2: Variation with t of the velocity for Rayleigh's problem.

One may also arrive at (7.22) by a more intuitive reasoning based on the absence of characteristic length scales. As previously mentioned when discussing dimensional analysis, the characteristic scales used to express a given problem in dimensionless form are in general obtained from its initial and boundary conditions. For instance, for Rayleigh's problem the boundary condition at the wall naturally provides $v_x \sim V$ as a scale for the velocity. The boundary conditions in (7.19) do not provide, however, the value of the characteristic size of the flow field, given by the thickness δ of the layer where we find velocities comparable to that found at the wall. This quantity is in fact an increasing function of time, with a characteristic value $\delta(t) = \sqrt{\nu t}$ at a given time t determined by the condition that in the layer of moving fluid the acceleration of the fluid, of order $O(\rho \partial v_x / \partial t) \sim \rho V / t$, must be comparable to the resultant of the viscous stresses, of order $O(\mu \partial^2 v_x / \partial y^2) \sim \mu V / \delta^2$. Using δ to nondimensionalize the distance to the wall y leads to the similarity variable $\eta = y / \sqrt{\nu t}$, obtained also in (7.22) from application of the Π theorem. Note also that the balance between acceleration and viscous stresses $\rho V / t \sim \mu V / \delta^2$ can also be used to determine the characteristic time

$$t_v = \frac{\delta^2}{\mu / \rho} \quad (7.23)$$

required for shear stresses to set in motion a fluid particle located at a distance from the wall of order δ . This is the so-called viscous time, to be used below in characterizing the acceleration produced by viscous forces.

Introducing the change of variables $(y, t) \rightarrow \eta = y/\sqrt{\nu t}$, with $\partial/\partial t = -(2t)^{-1}\eta d/d\eta$ and $\partial^2/\partial y^2 = (\nu t)^{-1}d^2/d\eta^2$, reduces (7.20) to

$$\frac{d^2\tilde{v}}{d\eta^2} + \frac{\eta}{2} \frac{d\tilde{v}}{d\eta} = 0 \begin{cases} \eta = \infty : & \tilde{v} = 0 \\ \eta = 0 : & \tilde{v} = 1, \end{cases} \quad (7.24)$$

which can be integrated to give

$$\frac{v_x}{V} = 1 - \frac{\int_0^\eta \exp(-\bar{\eta}^2/4) d\bar{\eta}}{\int_0^\infty \exp(-\bar{\eta}^2/4) d\bar{\eta}} = \text{erfc}(\eta/2) \quad (7.25)$$

where $\text{erfc}(x) = (2/\sqrt{\pi}) \int_x^\infty e^{-z^2} dz$ is the complementary error function. From the expression (7.25) the reader can evaluate different quantities associated with the motion, including the volume flux driven by the moving wall $Q(t) = \int_0^\infty v_x dy$ and the viscous stresses on the wall $\bar{\tau}' \cdot \bar{n} = \mu(\partial v_x / \partial y)_{y=0}$.

Stokes flow near a wall

Consider now the periodic velocity induced in a liquid by the oscillatory motion of a plane wall with velocity $V \cos(\omega t)$. The equation with boundary conditions to be integrated in this case is

$$\rho \frac{\partial v_x}{\partial t} = \mu \frac{\partial^2 v_x}{\partial y^2} \begin{cases} y = 0 : & v_x = V \cos(\omega t) \\ y \rightarrow \infty : & v_x \rightarrow 0 \end{cases}. \quad (7.26)$$

No initial conditions are needed for this problem, because the solution sought is periodic. Since the wall velocity changes sign alternatively, appreciable motion with velocities of the order of V is only found within a layer of finite thickness δ , whereas negligible velocities are found for $y \gg \delta$. The characteristic value of δ follows from an order-of-magnitude analysis of (7.26). With ω^{-1} used as an estimate for the characteristic time for velocity variation, the balance between acceleration and viscous forces gives in this case $\rho V / \omega^{-1} \sim \mu V / \delta^2$, which can be solved to yield the characteristic thickness

$$\delta = \sqrt{\nu / \omega}. \quad (7.27)$$

To solve (7.26) it is convenient to introduce an intermediate complex variable v^* satisfying the alternative problem

$$\rho \frac{\partial v^*}{\partial t} = \mu \frac{\partial^2 v^*}{\partial y^2} \begin{cases} y = 0 : & v^* = V \exp(i\omega t) \\ y \rightarrow \infty : & v^* \rightarrow 0 \end{cases} \quad (7.28)$$

defined such that the velocity can be computed as the real part of v^* , i.e., $v_x = \text{Re}(v^*)$. In view of the boundary condition at the wall, one may anticipate that the solution of (7.28) is expressible in the form $v^* = V \exp(i\omega t) f(y)$. Substitution into (7.28) provides the reduced problem

$$\nu \frac{d^2 f}{dy^2} - i\omega f = 0 \begin{cases} y = 0 : & f = 1 \\ y \rightarrow \infty : & f \rightarrow 0 \end{cases} \quad (7.29)$$

which can be integrated to give

$$f = \exp\left(-\frac{1+i}{\sqrt{2}} \frac{y}{\sqrt{\nu/\omega}}\right). \quad (7.30)$$

Making use of $v_x = \text{Re}(v^*)$ finally gives

$$v_x = V \exp\left(-y/\sqrt{2\nu/\omega}\right) \cos\left(\omega t - y/\sqrt{2\nu/\omega}\right), \quad (7.31)$$

corresponding to a wave that propagates into the fluid with velocity $\sqrt{2\nu\omega}$, whose amplitude attenuates over a distance of the order $\sqrt{\nu/\omega}$.

As a straightforward extension of Stokes problem, one may consider the periodic motion induced in an infinitely long channel of thickness h whose lower wall moves with velocity $V \cos(\omega t)$. The problem to be solved in this case is

$$\underbrace{\rho \frac{\partial v_x}{\partial t}}_{\rho V/\omega^{-1}} = \underbrace{\mu \frac{\partial^2 v_x}{\partial y^2}}_{\mu V/h^2} \quad \begin{cases} y = 0 : & v_x = V \cos \omega t \\ y = h : & v_x = 0 \end{cases} \quad (7.32)$$

where anticipated orders of magnitude have been indicated for each term in the conservation equation. The relative importance of acceleration and viscous stresses is measured in this case by the dimensionless parameter $h^2\omega/\nu$ obtained by comparing the characteristic orders of magnitude indicated in (7.32).

For $h^2\omega/\nu$ of order unity, the values of $\rho \partial v_x / \partial t$ and $\mu \partial^2 v_x / \partial y^2$ have comparable magnitudes all across the channel. The reader can obtain the exact solution in this case by following the procedure used above, i.e., introducing a complex function v^* such that $v_x = \text{Re}(v^*)$, with simple solutions emerging for extreme values of $h^2\omega/\nu$. For instance, for oscillatory motion at sufficiently low frequencies such that $h^2\omega/\nu \ll 1$, the unsteady term is small, and can be neglected in (7.32) to give the reduced problem

$$0 = \mu \frac{\partial^2 v_x}{\partial y^2} \quad \begin{cases} y = 0 : & v_x = V \cos(\omega t) \\ y = h : & v_x = 0 \end{cases} \quad (7.33)$$

leading upon integration to

$$v_x = V \cos(\omega t) \frac{h-y}{h}. \quad (7.34)$$

As can be seen, the velocity is given at each instant of time by the steady Couette profile (7.9) corresponding to the instantaneous wall velocity $V \cos(\omega t)$. To interpret the result, note that this limit $h^2\omega/\nu \ll 1$ corresponds to values of the viscous time across the channel $t_v = h^2/\nu$ much smaller than the characteristic time for variation of the wall velocity, i.e.,

$$\frac{h^2/\nu}{\omega^{-1}} \ll 1. \quad (7.35)$$

Therefore, as the wall velocity changes, viscous stresses rapidly transport momentum across the channel, modifying almost instantaneously the velocity profile, which remains linear in the first approximation, as corresponds to the Couette solution dominated by viscosity (7.9). The solution is **quasi-steady**, in the sense that, although transient effects are negligible in the conservation equation, the result (7.34) varies slowly with time (in a time scale ω^{-1} much larger than the characteristic viscous time across the channel h^2/ν).

The opposite limit $h^2\omega/\nu \gg 1$ of high-frequency wall oscillations also leads to a simplified description. With viscous stresses being negligible across the channel, the simplified equation $\partial v_x / \partial t = 0$ is compatible with the boundary condition on the upper wall, thereby giving $v_x = 0$ in

the first approximation. This solution of stagnant flow does not satisfy, however, the boundary condition on the lower wall. To understand the result, note that the condition $h^2\omega/\nu \gg 1$ implies that the channel width h is much larger than the characteristic length of penetration of the viscous effects $\delta \sim (\nu/\omega)^{1/2}$, given in (7.27). As a result, in the limit $h \gg (\nu/\omega)^{1/2}$ the flow remains stagnant across the channel, except in a thin layer of thickness $(\nu/\omega)^{1/2}$ adjacent to the lower wall, where the solution is in the first approximation that given in (7.31), independent of h .

Quasisteady viscous flows in pipes and channels of slowly varying section

The previous Couette and Poiseuille quasi-steady solutions, corresponding to steady unidirectional flows, apply approximately to the description of slowly evolving flows in finite-length slender domains. One such application occurs in hydrodynamic lubrication, involving thin layers of viscous fluid separating solid surfaces in relative motion. Another example of practical interest is that of motion in a pipe of slowly varying section, schematically represented in figure 7.3. This second example will be used to illustrate the simplifications that arise as well as the solution procedure for these slowly-evolving viscous flows. Next, the methodology developed will be used in addressing hydrodynamic-lubrication flows.

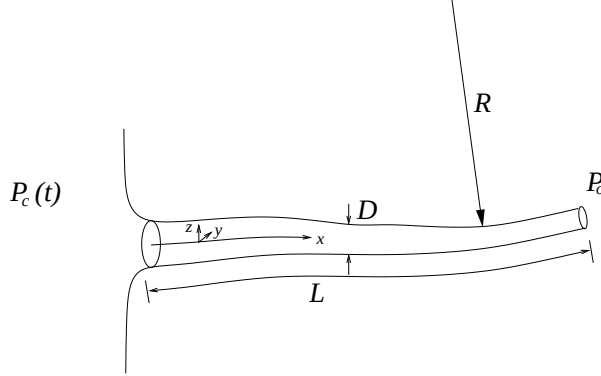


Figure 7.3: Discharge flow through a pipe of variable section.

Slender flows dominated by viscosity

Consider the discharge of a container of pressure $P_c(t)$ through a pipe. The diameter varies along the pipe slowly, in the sense that variations of the diameter of the order of its characteristic value D occur over distances $L \gg D$. The pipe is allowed to be curved, but only weakly, i.e., with associated curvature radii of order $R \gg D$. The description uses a set of curvilinear coordinates including the distance x along the pipe, defined by the curve joining the center points at each cross section, together with the corresponding transverse coordinates y and z . For weakly curved pipes, the conservation equations reduce, with small relative errors of order $D/R \ll 1$, to those

corresponding to cartesian coordinates

$$\underbrace{\frac{\partial v_x}{\partial x}}_{v_L/L} + \underbrace{\frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}}_{v_T/D} = 0 \quad (7.36)$$

$$\underbrace{\rho \frac{\partial v_x}{\partial t}}_{\rho v_L/t_o} + \underbrace{\rho \left(v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right)}_{\rho v_L^2/L} = - \underbrace{\frac{\partial P}{\partial x}}_{\Delta_L P/L} + \underbrace{\mu \frac{\partial^2 v_x}{\partial x^2}}_{\mu v_L/L^2} + \underbrace{\mu \frac{\partial^2 v_x}{\partial y^2} + \mu \frac{\partial^2 v_x}{\partial z^2}}_{\mu v_L/D^2} \quad (7.37)$$

$$\underbrace{\rho \frac{\partial v_y}{\partial t}}_{\rho v_T/t_o} + \underbrace{\rho \left(v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right)}_{\rho v_L v_T/L} = - \underbrace{\frac{\partial P}{\partial y}}_{\Delta_T P/D} + \underbrace{\mu \frac{\partial^2 v_y}{\partial x^2}}_{\mu v_T/L^2} + \underbrace{\mu \frac{\partial^2 v_y}{\partial y^2} + \mu \frac{\partial^2 v_y}{\partial z^2}}_{\mu v_T/D^2}. \quad (7.38)$$

For the order-of-magnitude discussion, we choose to write the y component of the momentum equation (7.38) as representative of the momentum balance in the transverse direction, the discussion being identical if use is made of the z component instead.

The order of magnitude is indicated below each term. In the notation, v_L and v_T denote the characteristic values of the longitudinal and transverse velocity components, $\Delta_L P$ is the characteristic pressure difference found along the pipe, of order $P_c - P_o$ for the specific problem at hand, and $\Delta_T P$ is the characteristic transverse pressure variation found across a given cross section. The characteristic time t_o for unsteady velocity variations in the pipe is associated with that characterizing the variation with time of the boundary conditions (note that, for constant boundary conditions, the flow in the pipe would be steady). If, for instance, the cause of the unsteady motion is the variation of the reduced pressure in the container, then t_o would be the time necessary for the feeding pressure to change by a relative amount of order unity.

The relative importance of the different physical phenomena in the pipe can be anticipated by evaluating the ratio of the different orders of magnitude. Some resulting features, common to all slender flows, follow from the condition $L \gg D$. For instance, the continuity equation provides

$$v_T \sim \frac{D}{L} v_L \ll v_L, \quad (7.39)$$

for the order of magnitude of the transverse velocity component, much smaller than the longitudinal component, indicating that because of the geometry of the pipe all streamlines are aligned. On the other hand, observation of the expressions for the viscous forces reveals that the contribution associated with longitudinal derivatives is always a factor $(D/L)^2$ smaller than that of the transverse derivatives, and can be correspondingly neglected for the description of slender flows.

For the flow to be dominated by viscosity, the resultant of the viscous forces has to be much larger than the local acceleration and also much larger than the convective acceleration. The first condition is satisfied if

$$\frac{O(\rho \partial \bar{v} / \partial t)}{O(\mu \nabla^2 \bar{v})} \sim \frac{D^2 / (\mu \rho)}{t_o} \ll 1, \quad (7.40)$$

whereas the second condition requires

$$\frac{O(\rho \bar{v} \cdot \nabla \bar{v})}{O(\mu \nabla^2 \bar{v})} \sim \frac{\rho v_L D}{\mu} \frac{D}{L} \ll 1. \quad (7.41)$$

The first condition (7.40) is analogous to that given in (7.35) for Stokes flow in a channel, that is, the viscous flow is quasi-steady when the characteristic viscous time across the pipe or channel is much smaller than the characteristic time of variation of the boundary conditions (ω^{-1} for the oscillating wall and t_o for the pipe with variable feeding pressure). Under those conditions, the velocity field adjusts almost automatically to the changing boundary conditions. The second condition (7.41) implies that the residence time in the pipe L/v_L is much larger than the local viscous time $D^2/(\mu/\rho)$, so that as the fluid flows along the pipe it adapts to the local balance between pressure and viscous forces without appreciable convective effects. As the criterion is written in (7.41), it requires that the product of the Reynolds number $\rho v_L D/\mu$ and the slenderness ratio D/L be small. Viscosity-dominated flows therefore appear in this case regardless of the value of the Reynolds numbers provided that the pipe is sufficiently slender for the condition (7.41) to hold.

If (7.40) and (7.41) are simultaneously satisfied, then in the first approximation (7.37) and (7.38) reduce to the balance

$$0 = - \underbrace{\frac{\partial P}{\partial x}}_{\Delta_L P/L} + \underbrace{\mu \frac{\partial^2 v_x}{\partial y^2} + \mu \frac{\partial^2 v_x}{\partial z^2}}_{\mu v_L/D^2} \quad (7.42)$$

$$0 = - \underbrace{\frac{\partial P}{\partial y}}_{\Delta_T P/D} + \underbrace{\mu \frac{\partial^2 v_y}{\partial y^2} + \mu \frac{\partial^2 v_y}{\partial z^2}}_{\mu v_T/D^2} \quad (7.43)$$

between pressure and viscous forces, which can be used to give

$$\Delta_L P \sim \mu v_L L/D^2 \quad (7.44)$$

and $\Delta_T P \sim \mu v_T/D$ for the characteristic values of the longitudinal and transverse pressure variations. The former are significantly larger, as can be seen by writing the ratio

$$\frac{\Delta_T P}{\Delta_L P} \sim \frac{D^2}{L^2} \ll 1. \quad (7.45)$$

Again, this is a result of the slenderness of the flow that can be seen to apply whether or not the flow is dominated by viscosity; if the streamlines are aligned, the associated transverse pressure variations are necessarily very small, because larger pressure variations would lead to significant transverse acceleration, destroying the alignment. These small pressure variations will be neglected in the following description, for which the reduced pressure will be a function of x and t , but will be taken as uniform across any given cross section. Correspondingly, with this approximation $P(x, t)$, the y and z components of the momentum equation will not be needed at leading order in the description. The problem therefore reduces to the integration of

$$0 = -\frac{\partial P}{\partial x} + \mu \left(\frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right) \quad (7.46)$$

with $P(x, t)$, to be used for the analysis of pipe and channel flows.

Solution for the flow along a pipe of circular cross section

For a pipe of circular cross section, as that depicted in figure 7.3, the problem is more conveniently written in cylindrical coordinates to give

$$0 = -\frac{\partial P}{\partial x} + \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right). \quad (7.47)$$

The pipe geometry is defined by its variable radius $a(x)$, assumed to be known in the description. Integrating (7.47) at a given section x and at a given time t with boundary conditions $v_x = 0$ at $r = a$ and with the symmetry condition $\partial v_x / \partial r = 0$ at $r = 0$ yields as solution the Poiseuille parabolic profile (7.48),

$$v_x = \frac{P_l}{4\mu} (a^2 - r^2), \quad (7.48)$$

corresponding to the local radius $a(x)$ and local instantaneous value of the reduced pressure gradient $P_l(x, t) = -\partial P / \partial x$. Straightforward integration across the pipe gives

$$Q(t) = \int_0^a 2\pi r u dr = -\frac{\pi a^4}{8\mu} \frac{\partial P}{\partial x}, \quad (7.49)$$

for the variation with time of the volume flux.

To close the problem, we need to integrate (7.49) to determine the volume flux $Q(t)$ and the reduced-pressure variation along the pipe $P(x, t)$ as a function of the its values at the entrance section $P(0, t) = P_e(t)$ and outlet section $P(L, t) = P_o$, the latter obtained from the known pressure surrounding the pipe exit, where the liquid discharges as a jet with quasi-parallel streamlines. The value of P_e is however not known a priori, since it depends on the characteristics of the flow at the entrance (we shall demonstrate below that, in the first approximation, one may take $P_e = P_c$).

Integrating once (7.49) yields

$$P_e(t) - P(x, t) = \frac{8}{\pi} \mu Q \int_0^x \frac{1}{a^4(x)} dx, \quad (7.50)$$

after imposing the condition $P = P_e$ at $x = 0$. The condition $P = P_o$ at $x = L$ provides

$$Q(t) = \frac{\pi}{8} \frac{P_e(t) - P_o}{\mu} \frac{1}{\int_0^L \frac{1}{a^4(x)} dx}. \quad (7.51)$$

for the volume flux. If the geometry $a(x)$ and the entrance pressure $P_e(t)$ were known, then (7.51) would determine the volume flux, while (7.50) could be used to obtain the pressure along the pipe. Using $Q(t)$ in (7.49) would then yield P_l , to be finally substituted into (7.48) for the velocity profile at each section. For pipes of constant cross section, it is easy to prove from (7.49) that the associated pressure gradient is uniform, i.e., $P_l(t)$, and can be therefore determines simply as $P_l = (P_e - P_o)/L$ as a function of the pressure at the entrance and outlet sections.

The entrance region

Unlike the motion at the outlet, where the pipe flow discharges at the ambient as a jet, at the entrance there exists a region of nonslender flow where the fluid approaches the pipe from all

directions. The entrance region where we find velocities of the order of v_L extends to distances of order D into the container (note that straightforward application of mass conservation reveals that the velocity decays in the container for increasing distances ℓ from the pipe according to $v \sim (D/\ell)^2 v_L$). To determine the velocity field one needs to integrate in this region the full conservation equations

$$\nabla \cdot \bar{v} = 0 \quad (7.52)$$

$$\underbrace{\rho \frac{\partial \bar{v}}{\partial t}}_{\rho v_L / t_o} + \underbrace{\rho \bar{v} \cdot \nabla \bar{v}}_{\rho v_L^2 / D} = - \underbrace{\nabla P}_{\Delta_E P / D} + \underbrace{\mu \nabla^2 \bar{v}}_{\mu v_L / D^2} \quad (7.53)$$

with appropriate initial and boundary conditions (e.g., nonslip condition at the walls). We have given below each term its anticipated order of magnitude, with $\Delta_E P$ representing the characteristic pressure loss across this entrance region.

If the motion in the pipe is dominated by viscosity, then the two conditions (7.40) and (7.41) are simultaneously satisfied. The first condition guarantees that the flow in the entrance region is also quasi-steady, since $O(\rho \partial \bar{v} / \partial t) / O(\rho \bar{v} \cdot \nabla \bar{v}) \sim (D^2 / \nu) / t_o \ll 1$ in (7.53). On the other hand, the relative importance of the convection term for the nonslender flow at the entrance is measured by the Reynolds number, as can be seen by writing from (7.53) the order-of-magnitude ratio

$$\frac{O(\rho \bar{v} \cdot \nabla \bar{v})}{O(\mu \nabla^2 \bar{v})} = \frac{\rho v_L D}{\mu}. \quad (7.54)$$

Since the condition (7.41) for the pipe flow can be satisfied both for small and large values of $\rho v_L D / \mu$, both possibilities need to be contemplated, with associated entrance flows sketched in figure 7.4.

If the Reynolds number $\rho v_L D / \mu$ is small, the momentum equation in the entrance region reduces to $0 = -\nabla P + \mu \nabla^2 \bar{v}$. The flow is seen to evolve rapidly in the entrance region to reach the Poiseuille profile at distances of order D from the entrance. The associated pressure loss in this region is of order

$$\Delta_E P \sim \mu v_L / D, \quad (7.55)$$

as can be seen from the balance between viscous and pressure forces in (7.53). If the Reynolds number is of order unity, convective terms should be also retained in (7.53) for the description of the entrance region, which would be qualitatively similar, with associated pressure losses given also by (7.55). This value is to be compared with the characteristic pressure loss along the pipe, given in (7.44) to give in this case

$$\frac{\Delta_E P}{\Delta_L P} \sim \frac{D}{L} \ll 1. \quad (7.56)$$

Clearly, the associated pressure loss in the entrance region is very small, so that in the first approximation one may neglect it and use $P_e = P_c$ in (7.50) and (7.51). Note that the errors introduced in the computation when this approximation is used are expected to be of order $D/L \ll 1$, very small for sufficiently long pipes.

The flow structure at the entrance is different when the Reynolds number $\rho v_L D / \mu$ is large. Viscosity is negligible at distances of order D from the entrance section, where the momentum equation reduces to $\rho \bar{v} \cdot \nabla \bar{v} = -\nabla P$. This equation can be projected along any given streamline to give Bernoulli's equation

$$P + \rho |\bar{v}|^2 / 2 = P_c. \quad (7.57)$$

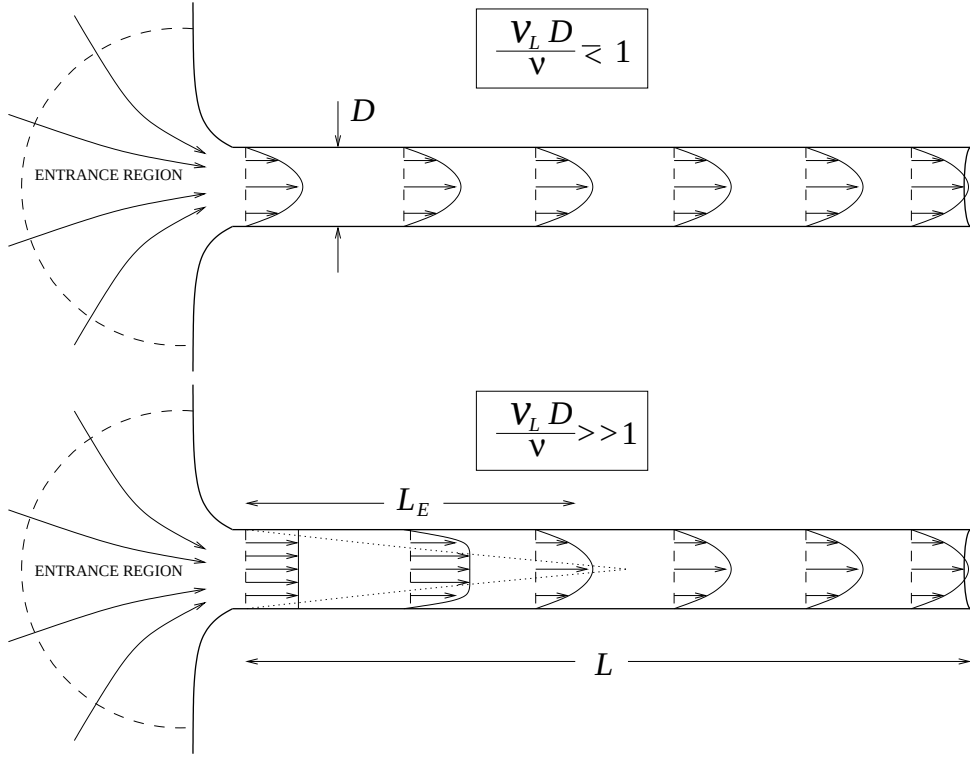


Figure 7.4: The entrance region in pipe flows dominated by viscosity.

Once the streamlines become aligned at the entrance of the pipe the pressure difference between different streamlines vanish, so that (7.57) provides $|\bar{v}|(0) = \sqrt{2[P_c - P(0)]/\rho}$ for the velocity at the entrance section. As can be seen, in the absence of viscous effect, the resulting velocity profile is uniform. According to (7.57), the pressure drop between the container and this entrance region is in this case

$$P_c - P(0) \sim \rho v_L^2. \quad (7.58)$$

Clearly, this solution does not satisfy the nonslip boundary condition at the wall, so that we can anticipate that in the vicinity of the wall there exists a thin layer (the so-called boundary layer), where the effect of viscosity is nonnegligible. Across this layer the velocity evolves from the value $|\bar{v}| = \sqrt{2[P_c - P(0)]/\rho}$ corresponding to the uniform profile to vanish at the wall. The thickness of this boundary layer increases downstream, until the boundary layer eventually occupies the whole section of the pipe, which occurs at streamwise distances from the entrance of order $L_E = \rho v_L D^2 / \mu$, with the velocity profile approaching the Poiseuille parabolic shape (7.48). Farther downstream, viscosity dominates the fluid motion, so that the velocity maintains the Poiseuille profile along the stretch $L_E \lesssim x \leq L$, adapting its shape to the local radius as dictated by (7.48).

Simple estimates can be used to provide the order of magnitude of the characteristic length L_E for the transition region from uniform profile to Poiseuille profile. At the entrance, the flow is given by a balance between convective acceleration and pressure forces, whereas downstream from the transition region, the motion is given by a balance between pressure and viscous forces.

Clearly, in the transition region all three contributions must be comparable according to

$$\underbrace{\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right)}_{\rho v_L^2 / L_E} = - \underbrace{\frac{\partial P}{\partial x}}_{\Delta_E P / L_E} + \underbrace{\mu \left(\frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)}_{\mu v_L / D^2}. \quad (7.59)$$

thereby providing

$$L_E = \frac{\rho v_L D^2}{\mu} \quad (7.60)$$

and

$$\Delta_E P = \rho v_L^2 \quad (7.61)$$

for the characteristic transition length and its associated pressure loss. The comparison of L_E with the total length of the pipe yields

$$\frac{L_E}{L} = \frac{\rho v_L D}{\mu} \frac{D}{L} \quad (7.62)$$

which is very small, according to (7.41). On the other hand, the pressure loss $\Delta_E P$ across the transition stretch is seen to be comparable to that found between the container the the entrance section, given in (7.58). We may therefore conclude that, for large values of the Reynolds number $\rho v_L D / \mu$, the total pressure loss between the container and the first pipe section $x \sim L_E$ where the Poiseuille solution (7.48) applies is of order $\Delta_E P \sim \rho v_L^2$, to be compared with the pressure loss along the pipe, given in (7.44). The comparison gives

$$\frac{\Delta_E P}{\Delta_L P} \sim \frac{\rho v_L D}{\mu} \frac{D}{L}, \quad (7.63)$$

which is negligibly small, according to (7.41). We see, therefore, that also in this case of large Reynolds numbers the equation $P_e = P_c$ should be used in the first approximation for the reduced pressure at the entrance of the pipe when evaluating (7.50) and (7.51).

Note that, according to the above discussion, the relative errors in using the approximate value $P_e = P_c$ can be expected to be of order $D/L \ll 1$ if the Reynolds number at the entrance $\rho v_L D / \mu$ is of order unity or smaller and of order $(\rho v_L D / \mu)(D/L) \ll 1$ if $\rho v_L D / \mu \gg 1$. Also worth noting is that the localized pressure losses emerging in joints, contractions, expansions, elbows, etc, are comparable to those found in the entrance region. Consequently, these secondary losses can always be neglected in computations of pipe flows dominated by viscosity, that is, whenever the two conditions $L \gg D$ and (7.41) are simultaneously satisfied.

Hydrodynamic lubrication: the wedge effect

When two solid surfaces in relative motion come into contact there appear friction forces that dissipate a large amount of energy, giving rise to a significant temperature increase that may damage the solid materials. This can be avoided by introducing a thin layer of lubricating fluid between the two solid surfaces. The fluid motion produces shear stresses on the solid surfaces that are much smaller than the associated pressure differences induced, responsible for keeping the two surfaces apart. This principle is used for instance in fluid bearings. Hydrodynamic lubrication is also responsible for the phenomenon of hydroplaning of road-vehicle tires and explains the behavior of the synovial fluid in our articulating joints.

To understand the basic principles of hydrodynamic lubrication let us consider a planar bearing of length L in relative motion at velocity $V(t)$ with respect to a horizontal wall, as shown in figure (7.5). Between the bearing and the plane wall there exists a thin channel of thickness $h(x)$ full of a fluid. For this application, we consider that the pressure takes on both sides of the channel equal values $P = P_o$.

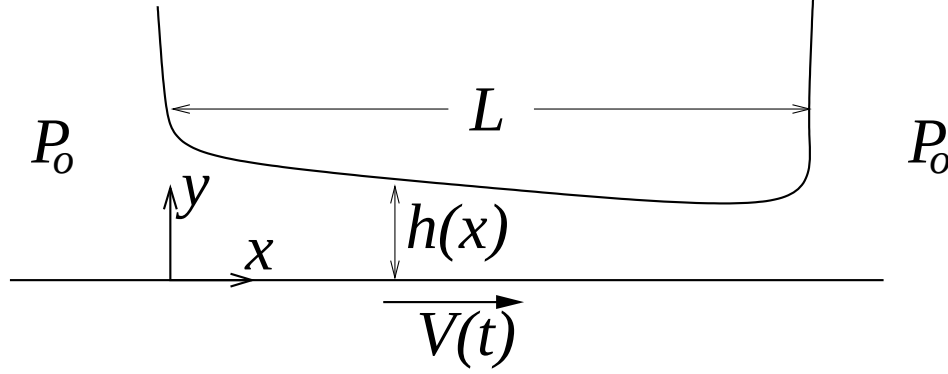


Figure 7.5: Plane bearing.

In a reference frame moving with the bearing, with x and y denoting, respectively, the distance from the channel entrance and the distance to the horizontal wall, the equations of motion for this planar case are

$$\underbrace{\frac{\partial v_x}{\partial x}}_{V_c/L} + \underbrace{\frac{\partial v_y}{\partial y}}_{v_T/h_c} = 0 \quad (7.64)$$

$$\underbrace{\rho \frac{\partial v_x}{\partial t}}_{\rho V_c/t_o} + \underbrace{\rho \left(v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} \right)}_{\rho V_c^2/L} = - \underbrace{\frac{\partial P}{\partial x}}_{\Delta_L P/L} + \underbrace{\mu \frac{\partial^2 v_x}{\partial x^2}}_{\mu V_c/L^2} + \underbrace{\mu \frac{\partial^2 v_x}{\partial y^2}}_{\mu V_c/h_c^2} \quad (7.65)$$

$$\underbrace{\rho \frac{\partial v_y}{\partial t}}_{\rho v_T/t_o} + \underbrace{\rho \left(v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} \right)}_{\rho V_c v_T/L} = - \underbrace{\frac{\partial P}{\partial y}}_{\Delta_T P/h_c} + \underbrace{\mu \frac{\partial^2 v_y}{\partial x^2}}_{\mu v_T/L^2} + \underbrace{\mu \frac{\partial^2 v_y}{\partial y^2}}_{\mu v_T/h_c^2}. \quad (7.66)$$

In the estimates for the orders of magnitude of each term, V_c and h_c are the characteristic values of $V(t)$ and $h(x)$, and t_o if the characteristic time of variation of $V(t)$. Following the discussion used above in simplifying (7.36)–(7.38), one may easily conclude that if the channel is slender (i.e., $h_c \ll L$) and the motion satisfies the two conditions,

$$\frac{\rho h_c^2/\mu}{t_o} \ll 1, \quad (7.67)$$

and

$$\frac{\rho V_c h_c}{\mu} \frac{h_c}{L} \ll 1, \quad (7.68)$$

equivalent to those given in (7.40) and (7.41) for the pipe flow, then the flow is quasi-steady and is dominated by viscosity, with negligible transverse pressure variations, so that $P(x, t)$. The

streamwise velocity is given by integration of

$$0 = - \underbrace{\frac{\partial P}{\partial x}}_{\Delta_L P/L} + \underbrace{\mu \frac{\partial^2 v_x}{\partial y^2}}_{\mu V_c/h_c^2}, \quad (7.69)$$

the two-dimensional counterpart of (7.46), with boundary conditions $v_x = V(t)$ at $y = 0$ and $v_x = 0$ at $y = h$, yielding

$$v_x = \frac{P_l}{2\mu}y(h-y) + V\frac{h-y}{h}, \quad (7.70)$$

with associated volume flux (per unit width of channel) given by

$$Q = \int_0^h v_x dy = \frac{P_l h^3}{12\mu} + \frac{Vh}{2}, \quad (7.71)$$

where $P_l = -\partial P/\partial x = P_l(x, t)$.

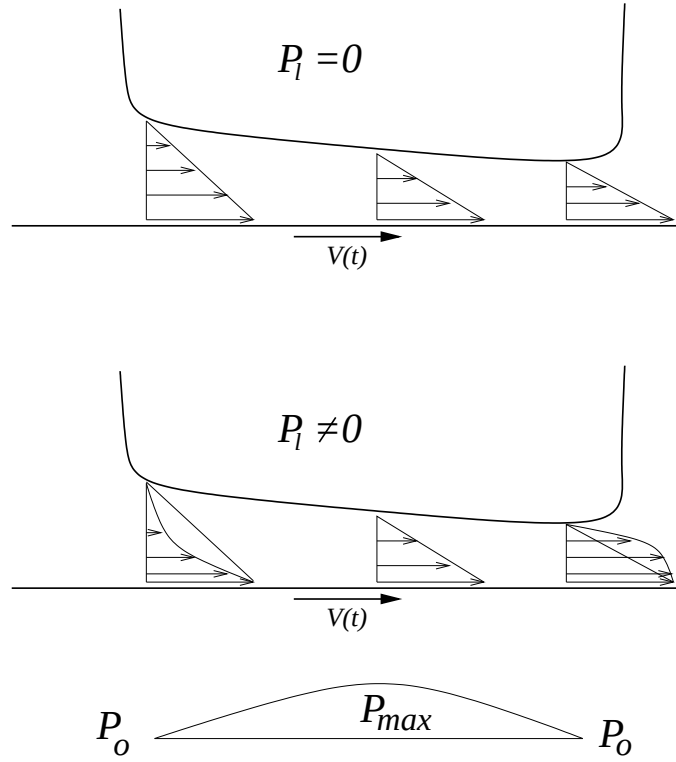


Figure 7.6: Schematic representation of the wedge effect including the velocity distributions emerging with $P_l = 0$ and with $P_l \neq 0$.

The volume flux is a function of t , independent of x , as can be seen by integrating the continuity equation (7.64) across a given section to give

$$\int_0^h \frac{\partial v_x}{\partial x} dy + \int_0^h \frac{\partial v_y}{\partial y} dy = \frac{\partial}{\partial x} \left(\int_0^{h(x)} v_x dy \right) + v_y(y=h) - v_y(y=0) = \frac{\partial}{\partial x} Q = 0, \quad (7.72)$$

The resulting value of $Q(t)$ is the sum of the Poiseuille contribution $P_l h^3/(12\mu)$, associated with the unknown value of $P_l(x, t)$, and the Couette contribution $Vh/2$, which varies linearly proportional to the local height. If the pressure gradient P_l were equal to zero, then Q would change along the pipe according to $Q(x, t) = V(t)h(x)/2$ as shown in the upper plot of figure 7.6, thereby violating continuity, since Q is only a function of t . As a consequence, although at both ends of the channel P takes on the same value $P = P_o$, pressure variations must necessarily emerge within the channel whenever h varies with x , as required to keep Q uniform for $0 < x < L$. For the planar bearing of decreasing h shown in figure 7.6, overpressures $P - P_o > 0$ appear within the liquid film, reaching a maximum value at an intermediate location $x = x_{max}$ where $P_l = 0$. A negative value $P_l < 0$ exists for $x < x_{max}$, and the associated reverse Poiseuille flow reduces the volume flux for $0 < x < x_{max}$ from that found with $P_l = 0$, as shown in figure 7.6. Conversely, for $x_{max} < x < L$ the positive value of $P_l > 0$ induces a positive Poiseuille flow additional to the existing Couette flow. The distribution of pressure is such that the resulting value of Q , given in (7.71), is uniform all along the channel.

To obtain $P(x, t)$, we write (7.71) as a differential equation for the pressure

$$\frac{\partial P}{\partial x} = \frac{6\mu V}{h^2} - \frac{12\mu Q}{h^3}. \quad (7.73)$$

As shown above, the pressure variations in the region near the entrance of the channel can be neglected in the first approximation, so that we can use the value $P = P_o$ at $x = 0$ in integrating (7.73) to give

$$P - P_o = 6\mu V \int_0^x h^{-2} dx - 12\mu Q \int_0^x h^{-3} dx. \quad (7.74)$$

Evaluating this expression at $x = L$, where the pressure recovers the value $P = P_o$, yields

$$Q = \frac{V \int_0^L h^{-2} dx}{2 \int_0^L h^{-3} dx} \quad (7.75)$$

for the volume flux $Q(t)$ as a function of the instantaneous velocity $V(t)$, thereby closing the problem; for a known value of Q , the pressure distribution is obtained from (7.74) with accompanying pressure gradient (7.73), finally determining the velocity profile (7.70) at each section x . The results can be employed to compute the forces acting on the bearing and on the wall. For instance, the force on the bottom wall is given in this case by

$$\bar{F} = - \int_0^L (P - P_o) dx \bar{e}_y + \int_0^L \mu (\partial v_x / \partial y)|_{y=0} dx \bar{e}_x. \quad (7.76)$$

According to (7.69) the overpressures $P - P_o$ in the channel are of order $\mu V_c L / h_c^2$, a result that can be also inferred from (7.74), giving a pressure force normal to the solid surfaces of order $\int_0^L (P - P_o) dx \sim \mu V_c L^2 / h_c^2$, while the corresponding tangential viscous forces are of order $\int_0^L \mu (\partial v_x / \partial y)|_{y=0} dx \sim \mu V_c L / h_c$. This is the essential feature that makes hydrodynamic lubrication so useful: the pressure forces keeping the solid surfaces apart are a factor L/h_c larger than the viscous forces opposing the relative motion.

Chapter 8

Ideal Flows

The effects of molecular transport (viscous stresses and thermal heat conduction) are found to be negligible in most of the flow field when the fluid motion takes place at large Reynolds numbers. Under those conditions, the Navier-Stokes equations reduce to the so-called Euler equations, which further simplify under conditions of practical interest (quasi-steady motion, unidirectional flow, ...).

Introductory remarks

The Navier-Stokes equations

As seen previously, to solve a given flow problem, the conservation equations of continuity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{v}) = 0, \quad (8.1)$$

momentum

$$\rho \frac{D\bar{v}}{Dt} = -\nabla p + \nabla \cdot \bar{\tau}' + \rho \bar{f}_m, \quad (8.2)$$

and energy

$$\rho \frac{De}{Dt} = -p \nabla \cdot \bar{v} + \bar{\tau}' : \nabla \bar{v} - \nabla \cdot \bar{q} + Q_c + Q_r, \quad (8.3)$$

must be supplemented with the equations of state

$$\rho = \rho(p, T) \quad \text{and} \quad e = e(p, T), \quad (8.4)$$

together with the constitutive equations

$$\bar{\tau}' = \mu(\nabla \bar{v} + \nabla \bar{v}^T) + (\mu_B - \frac{2}{3}\mu)(\nabla \cdot \bar{v})\bar{\mathbf{I}} \quad \text{and} \quad \bar{q} = -k \nabla T \quad (8.5)$$

and associated state functions

$$\mu = \mu(T), \quad \mu_B = \mu_B(T) \quad \text{and} \quad k = k(T). \quad (8.6)$$

In particular, the state equations (4.35) reduce to

$$\rho = \rho_o \quad \text{and} \quad e = e_o + cT \quad (8.7)$$

for a perfect liquid and to

$$\rho = \frac{p}{R_g T} \quad \text{and} \quad e = e_o + c_v T \quad (8.8)$$

for a perfect gas. The energy equation (8.3) can be alternatively expressed in terms of the enthalpy $h = e + p/\rho$ to give

$$\rho \frac{Dh}{Dt} - \frac{Dp}{Dt} = \bar{\tau}' : \nabla \bar{v} - \nabla \cdot \bar{q} + Q_c + Q_r, \quad (8.9)$$

and also in terms of the entropy, defined from $Tds = dh - dp/\rho$, yielding

$$\rho T \frac{Ds}{Dt} = \bar{\tau}' : \nabla \bar{v} + \nabla \cdot (k \nabla T) + Q_c + Q_r. \quad (8.10)$$

The thermal equations of state can be written for the enthalpy and entropy as

$$h = e_o + cT + \frac{p}{\rho_o} \quad \text{and} \quad \frac{s - s_o}{c} = \ln T \quad (8.11)$$

for a perfect liquid and as

$$h = e_o + c_p T \quad \text{and} \quad \frac{s - s_o}{c_v} = \ln \left(\frac{p}{\rho^\gamma} \right) \quad (8.12)$$

for a perfect gas, where c_v and c_p are the specific heat at constant volume and at constant pressure for the gas, with $\gamma = c_p/c_v$. According to (8.11) and (8.12), a flow is effectively isentropic when the entropy variations that appear Δs are much smaller than the corresponding specific heat. Also of interest is that, for a liquid, isentropic flow means isothermal flow, whereas according to (8.12) for gases one obtains $p/\rho^\gamma = \text{constant}$ when $\Delta s \ll c_v$.

The velocity of sound

Besides the enthalpy and the entropy, another thermodynamic quantity of interest in fluid-flow analysis is the velocity of sound, a , which is the velocity at which the small pressure perturbations propagate in a compressible medium at rest, to be computed according to

$$a^2 = \left(\frac{\partial p}{\partial \rho} \right)_s. \quad (8.13)$$

For a perfect liquid, with ρ taken as constant, the corresponding velocity of sound is infinite. For a perfect gas, straightforward differentiation of the second equation in (8.12) leads to

$$a = (\gamma p / \rho)^{1/2} = (\gamma R_g T)^{1/2}, \quad (8.14)$$

indicating that the velocity of sound is a function of the temperature. The reader can verify that for air at $T = 300$ K the above expression yields $a \simeq 340$ m/s.

The Mach number

Compressibility effects associated with the fluid motion are characterized by the Mach number

$$M = \frac{v}{a}, \quad (8.15)$$

defined as the ratio of the velocity magnitude $v = |\bar{v}|$ to the velocity of sound a . Flows can be subsonic, sonic or supersonic if the value of the Mach number is $M < 1$, $M = 1$, or $M > 1$, respectively. Since the perturbations move with respect to the flow with the velocity of sound, subsonic and supersonic flow show markedly different behaviors, as sketched in Fig. 8. As can be seen, a perturbation in a subsonic flow ends up affecting all of the flow field, because it propagates upstream sufficiently fast to overcome convection. In supersonic flow, however, convection sweeps downstream the perturbation, which affects only a conical region (termed the “Mach cone”) of semiangle α such that $\sin(\alpha) = 1/M$.

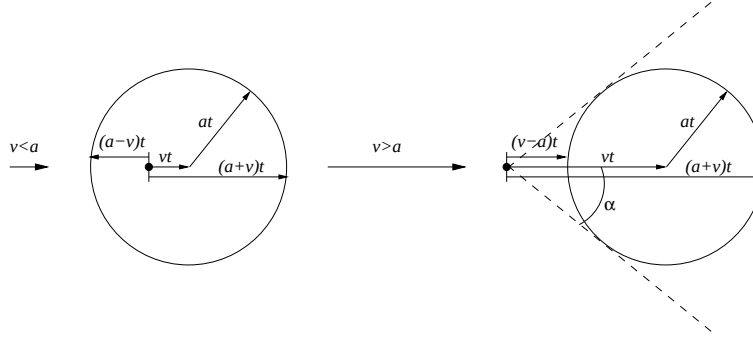


Figure 8.1: Perturbation regions for subsonic and supersonic flows.

Steady gas flow around a sphere

To compute a given flow field one needs to integrate (8.1) and (8.2) together with one of the alternative forms of the energy equation - i.e., (8.3), (8.9), or (8.10) - with appropriate initial and boundary conditions. To begin to introduce some of the concepts to be studied in the following chapters, it is of interest to consider, as an illustrative example, the steady buoyancy-free motion of a gas around a sphere of radius R with negligible radiation and chemical reaction. The ambient conditions in this case are given by $\bar{v} = U_\infty \bar{e}_x$, $T = T_\infty$, and $p = p_\infty$ whereas on the sphere surface the temperature is given by $T = T_w$.

In this case, the problem reduces to that of integrating

$$\nabla \cdot (\rho \bar{v}) = 0 \quad (8.16)$$

$$\underbrace{\rho \bar{v} \cdot \nabla \bar{v}}_{\rho_\infty U_\infty^2 / R} = \underbrace{-\nabla p}_{\Delta p / R} + \underbrace{\nabla \cdot \bar{\tau}'}_{\mu U_\infty / R^2} \quad (8.17)$$

$$\underbrace{\rho T \bar{v} \cdot \nabla s}_{\rho_\infty T_\infty U_\infty \Delta s / R} = \underbrace{\bar{\tau}' : \nabla \bar{v}}_{\mu U_\infty^2 / R^2} + \underbrace{\nabla \cdot (k \nabla T)}_{k \Delta T / R^2} \quad (8.18)$$

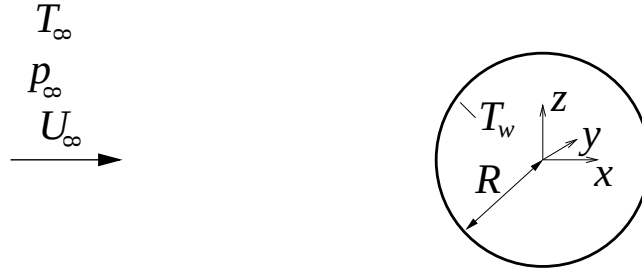


Figure 8.2: Steady gas motion around a sphere.

supplemented with the constitutive law for $\bar{\tau}'$ given in (8.5) and with the equations of state

$$\rho = \frac{p}{R_g T} \quad \text{and} \quad \frac{s - s_\infty}{c_v} = \ln \left(\frac{p/p_\infty}{(\rho/\rho_\infty)^\gamma} \right) \quad (8.19)$$

and subject to the boundary conditions

$$\begin{cases} |\bar{x}| = R : & \bar{v} = 0, \quad T = T_w \\ |\bar{x}| \gg R : & \bar{v} = U_\infty \bar{e}_x, \quad T = T_\infty, \quad p = p_\infty. \end{cases} \quad (8.20)$$

Note that, for convenience, the thermal equation of state in (8.19) has been written using the ambient conditions for the arbitrary reference state.

The estimated orders of magnitude are given below each term in the momentum and energy equations. The character of the solution depends on the value of the Reynolds number, $Re = \rho_\infty U_\infty R / \mu$, which measures the ratio

$$Re = \frac{O(\rho \bar{v} \cdot \nabla \bar{v})}{O(\bar{\tau}')} = \frac{\rho_\infty U_\infty R}{\mu}. \quad (8.21)$$

When $Re \gg 1$, the viscous forces are found to be negligible in most of the flow field. The simplified balance equation that appears can be used to estimate the characteristic value of the spatial pressure differences appearing across the flow field according to

$$\underbrace{\rho \bar{v} \cdot \nabla \bar{v}}_{\rho_\infty U_\infty^2 / R} \sim \underbrace{-\nabla p}_{\Delta p / R} \rightarrow \Delta p = \rho_\infty U_\infty^2. \quad (8.22)$$

It is of interest to compare these differences with the ambient value p_∞ according to

$$\frac{\Delta p}{p_\infty} = \frac{\rho_\infty U_\infty^2}{p_\infty} = \gamma \left(\frac{U_\infty}{a_\infty} \right)^2 = \gamma M_\infty^2. \quad (8.23)$$

As can be seen, the compressibility ratio $\Delta p / p_\infty$ is proportional to the square of the Mach number

$$M_\infty = U_\infty / a_\infty. \quad (8.24)$$

Appreciable compressibility effects appear in flows with $M_\infty \gtrsim 1$, whereas in flows with $M_\infty \ll 1$ the resulting spatial pressure differences are much smaller than the ambient pressure. In the latter case, one may rewrite the equation of state given in (8.19) in the simplified form $p_\infty = \rho R_g T$.

According to the energy equation (8.18) the entropy of a fluid particle crossing the flow field changes due to the effects of viscous dissipation and thermal heat conduction. The corresponding characteristic entropy increase due to each separate effect, $(s - s_\infty)_\phi$ and $(s - s_\infty)_k$, can be estimated by order-of-magnitude comparisons according to

$$\underbrace{\rho T \bar{v} \cdot \nabla s}_{\rho_\infty T_\infty U_\infty (s - s_\infty)_\phi / R} \sim \underbrace{\bar{\tau}' : \nabla \bar{v}}_{\mu U_\infty^2 / R^2} \rightarrow \frac{(s - s_\infty)_\phi}{c_v} = \gamma(\gamma - 1) \frac{M_\infty^2}{Re} \quad (8.25)$$

and

$$\underbrace{\rho T \bar{v} \cdot \nabla s}_{\rho_\infty T_\infty U_\infty (s - s_\infty)_k / R} \sim \underbrace{\nabla \cdot (k \nabla T)}_{k \Delta T / R^2} \rightarrow \frac{(s - s_\infty)_k}{c_v} = \frac{\gamma}{Pr} \frac{\Delta T}{T_\infty} \frac{1}{Re}. \quad (8.26)$$

where $Pr = \mu c_p / k$ is the Prandtl number ($Pr = 0.7$ for air). The above expressions clearly indicate that when the Reynolds number is large the resulting flow is effectively isentropic, thereby reducing (8.18) to $s = s_\infty$.

The Euler equations

According to the previous discussion, the effects of molecular transport, including viscous forces, heat conduction and also viscous dissipation, can be neglected in the first approximation whenever the condition $Re \gg 1$ is satisfied, where $Re = \rho_c v_c L_c / \mu_c$ is the Reynolds number of the problem, defined from the characteristic values of the density and viscosity, ρ_c and μ_c , velocity v_c , and length L_c ¹. The simplified set of conservation equations, called the Euler equations, is given by

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{v}) = 0, \quad (8.27)$$

$$\rho \left(\frac{\partial \bar{v}}{\partial t} + \bar{v} \cdot \nabla \bar{v} \right) = \rho \frac{D \bar{v}}{Dt} = -\nabla p + \rho \bar{f}_m, \quad (8.28)$$

$$\rho T \left(\frac{\partial s}{\partial t} + \bar{v} \cdot \nabla s \right) = \rho T \frac{Ds}{Dt} = Q_c + Q_r. \quad (8.29)$$

For the integration, one needs to provide initial conditions, given by the initial values of the velocity and thermodynamic state at every point of the integration domain. Because of the absence of the second-order spatial derivatives, the Euler equations cannot however satisfy all of the boundary conditions. For the aerodynamic problem of flow around a body the subset of boundary conditions to be satisfied by the Euler equations (8.27)–(8.29) include those far from the body, i.e., $\bar{v} = U_\infty \bar{e}_x$, $p = p_\infty$, and $T = T_\infty$ at $|\bar{x}| \gg R$. On the body surface, however, we cannot impose either the temperature or the velocity, so that integration of (8.27)–(8.29) provides in general a solution with $T \neq T_w$ and $\bar{v} \neq 0$ on the body surface $|\bar{x}| = R$. For the

¹Note that, regardless of the value of the Reynolds number, molecular transport is also negligible in unsteady flows when the condition

$$\frac{\rho_c L_c^2}{\mu_c t_c} \gg 1$$

is satisfied, where t_c is the characteristic time of flow variation. Under those conditions, the molecular-transport terms are found to be negligible compared with the unsteady terms, so that, for instance, $\rho \partial \bar{v} / \partial t \gg \nabla \cdot \bar{\tau}'$ in the momentum conservation equation.

velocity, the presence of the pressure forces in the momentum equation prevents the fluid from penetrating the solid surface, so that the resulting velocity component normal to the wall is strictly zero, that is, for integrating (8.27)–(8.29) we must impose $\bar{v} \cdot \bar{n} = 0$ on the body surface. If both radiation and chemical reaction are absent, as occurs in many practical applications, then the energy equation (8.29) reduces to

$$\frac{Ds}{Dt} = 0, \quad (8.30)$$

stating that the entropy of each fluid particle remains constant in the flow evolution. When (8.30) is satisfied, we say that the flow is **isentropic**. If at the initial instant all fluid particles have the same value of s , then in the subsequent evolution the entropy will remain uniform in the flow field, so that $\nabla s = 0$; when this happens, the flow is said to be **homentropic**. Radiation and chemical reaction are not further considered in the following development, for which (8.30) will be used in place of (8.29).

Quasi-steady flow

For a flow with characteristic length L_c , characteristic velocity v_c , and characteristic time of flow variation t_c , the importance of the unsteady terms in the conservation equations is measured through the so-called Strouhal number St , which compares the orders of magnitude of the unsteady and convective terms according to

$$St = \frac{O(\partial \rho / \partial t)}{O(\nabla \cdot (\rho \bar{v}))} = \frac{O(\partial \bar{v} / \partial t)}{O(\bar{v} \cdot \nabla \bar{v})} = \frac{O(\partial s / \partial t)}{O(\bar{v} \cdot \nabla s)} = \frac{L_c / v_c}{t_c}. \quad (8.31)$$

For a given problem, the values of L_c , v_c , and t_c can be estimated from the boundary conditions. For instance, for flow of a fluid stream of variable velocity $U_\infty(t)$ (e.g., $U_\infty = U_o \sin \Omega t$) around a body (e.g., a sphere), the value of L_c will be given by the characteristic body size (e.g., the sphere radius R), whereas the values of v_c and t_c will be the characteristic free-stream velocity (e.g., its mean value $v_c = U_o$) and the characteristic time of free-stream velocity variation (e.g., $t_c = \Omega^{-1}$).

Flows with $St \ll 1$ are said to be **quasi-steady**, in that, for their description, one may in the first approximation neglect the time derivatives in the conservation equations, which therefore become

$$\nabla \cdot (\rho \bar{v}) = 0, \quad (8.32)$$

$$\rho \bar{v} \cdot \nabla \bar{v} = -\nabla p + \rho \bar{f}_m, \quad (8.33)$$

$$\bar{v} \cdot \nabla s = 0. \quad (8.34)$$

The flow is termed quasi-steady, as opposed to strictly steady, because the resulting solution depends on the time through the slowly-varying boundary conditions. For instance, for the sphere example with variable free-stream velocity, when $St = R\Omega/U_o \ll 1$ the instantaneous velocity field around the sphere $\bar{v}(\bar{x}, t)$ will be that corresponding to steady motion with free-stream velocity $U_\infty = U_o \sin \Omega t$. Note that the condition $St \ll 1$ implies that the fluid-particle residence time L_c/v_c is much smaller than the time required to change appreciably the boundary conditions, so that the boundary conditions remain almost constant during the time that it takes each fluid particle to cross the flow field. As a result, each individual fluid particle behaves as though the flow were steady, with the values of the boundary conditions corresponding to the instant of time at which they cross the flow field.

Bernoulli's equation (revisited)

When the mass forces are conservative, i.e., $\bar{f}_m = -\nabla U$, we may rewrite (8.33) in the form

$$\bar{v} \cdot \nabla \bar{v} = -\frac{\nabla p}{\rho} - \nabla U. \quad (8.35)$$

The projection of this vector equation along a given stream line, obtained by scalar multiplication by $\bar{v}/v$ with use of the identity $\bar{v} \cdot \nabla \bar{v} = \nabla(v^2/2) - \bar{v} \wedge (\nabla \wedge \bar{v})$, yields

$$\frac{\partial}{\partial l} (v^2/2 + U) + \frac{1}{\rho} \frac{\partial p}{\partial l} = 0, \quad (8.36)$$

where

$$\frac{\partial}{\partial l} = \frac{\bar{v}}{v} \cdot \nabla \quad (8.37)$$

denotes the derivative along stream lines. This expression further simplifies to

$$\frac{\partial}{\partial l} \left(\frac{p}{\rho_o} + \frac{v^2}{2} + U \right) = 0 \quad (8.38)$$

for constant-density flows, indicating that, **for quasi-steady ideal flow of perfect liquids** the quantity

$$p + \rho_o v^2/2 + \rho_o U = C_l \quad (8.39)$$

remains constant along a given stream line, with a value C_l that may be different for different stream lines and may also vary slowly with time. Note that for the application of (8.39) the fluid must be a **perfect liquid** and the flow must be **ideal** (i.e., large values of the Reynolds number $Re \gg 1$) and with **negligible time derivatives** (i.e., strictly steady flow or quasi-steady flow with $St \ll 1$). Also of interest is that, if gravity is the only mass force acting on the fluid, then (8.39) takes the familiar form

$$p + \rho_o v^2/2 + \rho_o g \mathcal{Z} = C_l \quad (8.40)$$

with $\mathcal{Z}$ representing the vertical distance measured opposite to the gravity vector $\bar{f}_m = \bar{g}$ and $g = |\bar{g}|$. When no mass forces are present, (8.39) further reduces to give

$$p + \rho_o v^2/2 = C_l = p_0, \quad (8.41)$$

where the constant p_0 is called **stagnation pressure**.

This last equation finds application in the analysis of Pitot tubes, which are used to measure the velocity of a fluid stream. The device, sketched below in Fig. 8, consists of a slender tube which is placed aligned with the fluid stream, whose values of pressure and velocity are p_∞ and U_∞ . Two different pressure readings are taken at two different orifices, internally connected to two different pressure gauges. The static pressure p_∞ is measured through a lateral orifice, where the flow is hardly perturbed. A second orifice is placed at the tube pointed end. The ideal quasisteady deceleration of the flow along the streamline ending at the orifice yields $p = p_0 = p_\infty + \rho U_\infty^2/2$ there, as follows from (8.41). It is then straightforward to determine the flow velocity

$$U_\infty = [2(p_0 - p_\infty)/\rho]^{1/2} \quad (8.42)$$

from the two pressure readings p_∞ and p_0 .

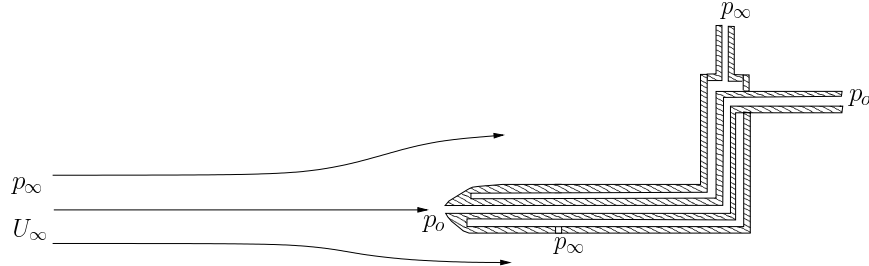


Figure 8.3: Pitot tube.

Steady ideal flow of a gas: stagnation flow properties

According to (8.34),

$$\frac{\partial s}{\partial l} = 0 \quad (8.43)$$

indicating that, in quasi-steady ideal flows, the entropy is constant along stream lines, with a value

$$s = s_0, \quad (8.44)$$

that may be different for different stream lines and may also vary slowly with time. This last result can be used to simplify the description of steady gas flow at large Reynolds numbers. Using the definition of entropy $Tds = dh - dp/\rho$ along with (8.43) yields

$$\frac{\partial h}{\partial l} = \frac{1}{\rho} \frac{\partial p}{\partial l}, \quad (8.45)$$

which can be substituted into (8.36) to give

$$\frac{\partial}{\partial l} \left(h + \frac{v^2}{2} + U \right) = 0. \quad (8.46)$$

In many gas-flow applications mass forces have a negligible effect on the gas motion, because the velocity is sufficiently large for the associated Froude number to satisfy

$$Fr = \frac{O(\rho \bar{v} \cdot \nabla \bar{v})}{O(\rho \bar{f}_m)} = \frac{v_c^2}{L_c f_{m_c}} \gg 1. \quad (8.47)$$

When this condition is satisfied, then (8.46) indicates that

$$h + \frac{v^2}{2} = h_0, \quad (8.48)$$

is constant along a given stream line. For known values of s_0 and h_0 , the equations (8.44) and (8.48) together with the equations of state

$$T = T(s, h), \quad p = p(s, h), \quad \rho = \rho(s, h), \quad \text{and} \quad a = a(s, h) \quad (8.49)$$

enable the computation of the values of s , h , T , p , ρ , and a in terms of the local value of the flow velocity v . The constants s_0 and h_0 together with the corresponding values T_0 , p_0 , ρ_0 , and

a_0 , obtained by substituting the values s_0 and h_0 in the equations of state (8.49), are called stagnation flow properties. The explicit form of the equations of state for a perfect gas, given above in (8.8), (8.12), and (8.14), can be used to express the different stagnation flow properties in terms of the local Mach number according to

$$\frac{h_0}{h} = \frac{T_0}{T} = \left(\frac{a_0}{a}\right)^2 = \left(\frac{p_0}{p}\right)^{(\gamma-1)/\gamma} = \left(\frac{\rho_0}{\rho}\right)^{\gamma-1} = 1 + \frac{\gamma-1}{2}M^2. \quad (8.50)$$

For **steady** (or quasi-steady) **ideal gas flow** with **negligible body forces**, the values of s_0 , h_0 , T_0 , a_0 , p_0 , and ρ_0 remain constant along a given stream line and (8.50) can be employed to compute h , T , a , p , and ρ as a function of the local value of the Mach number M . For $M \ll 1$, equation (8.50) indicates that the gas moves with small variations of the temperature, density, and pressure. The latter can be computed by expanding

$$\frac{p_0}{p} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\gamma/(\gamma-1)} \quad (8.51)$$

for $M \ll 1$ to give

$$p + \rho_0 v^2/2 = p_0, \quad (8.52)$$

to be compared with (8.41), thereby illustrating the similarities between gas motion at low Mach numbers and liquid motion.

Ideal flows in pipes

The ideal-flow quasisteady results (8.39), relating the velocity magnitude with the pressure of a liquid, and (8.50), relating the local Mach number with the thermodynamic variables for gas flow, are particularly useful in connection with ducted flows, for which the velocity direction is known a priori. We begin by considering the motion of a perfect liquid, following the discussion of viscous flow in pipes presented in the previous chapter. Steady gas flow in pipes is addressed next, including as an example flow in a convergent nozzle.

Quasisteady ideal liquid flow in pipes.

We address again the discharge of a container of reduced pressure $P_c(t)$ through a slowly-varying pipe of length L and characteristic transverse size $D \ll L$, given for instance by $D = \sqrt{A(L)}$ in terms of area $A(L)$ of the outlet cross section. The equations that describe the flow, written in terms of the reduced pressure $P = p + \rho_o U$ with the anticipated orders of magnitude given below each term, are given by (see the discussion given in the preceeding chapter for details)

$$\underbrace{\frac{\partial v_x}{\partial x}}_{v_L/L} + \underbrace{\frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}}_{v_T/D} = 0 \quad (8.53)$$

$$\underbrace{\rho_o \frac{\partial v_x}{\partial t}}_{\rho_o v_L/t_c} + \underbrace{\rho_o \left(v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right)}_{\rho_o v_L^2/L} = - \underbrace{\frac{\partial P}{\partial x}}_{\Delta_L P/L} + \underbrace{\mu \frac{\partial^2 v_x}{\partial x^2}}_{\mu v_L/L^2} + \underbrace{\mu \frac{\partial^2 v_x}{\partial y^2} + \mu \frac{\partial^2 v_x}{\partial z^2}}_{\mu v_L/D^2} \quad (8.54)$$

$$\underbrace{\rho_o \frac{\partial v_y}{\partial t}}_{\rho_o v_T/t_c} + \underbrace{\rho_o \left(v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right)}_{\rho_o v_L v_T/L} = - \underbrace{\frac{\partial P}{\partial y}}_{\Delta_T P/D} + \underbrace{\mu \frac{\partial^2 v_y}{\partial x^2}}_{\mu v_T/L^2} + \underbrace{\mu \frac{\partial^2 v_y}{\partial y^2} + \mu \frac{\partial^2 v_y}{\partial z^2}}_{\mu v_T/D^2}. \quad (8.55)$$

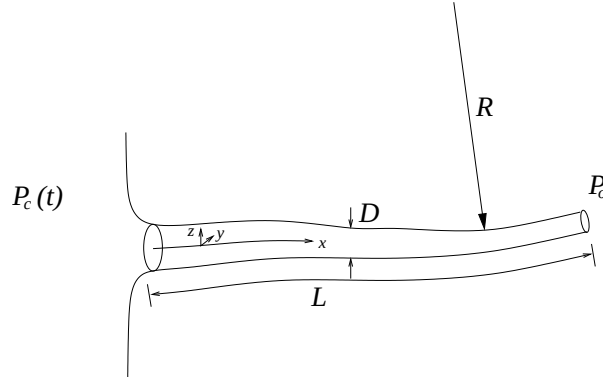


Figure 8.4: Discharge flow through a pipe of variable section.

Because of the pipe geometry, the streamlines are aligned, with

$$v_T \sim \frac{D}{L} v_L \ll v_L, \quad (8.56)$$

as follows from (8.53), so that $|\bar{v}| = v \simeq v_x$.

The relative importance of viscous forces can be measured by comparing its order of magnitude to that of convection, yielding

$$\frac{O(\rho_o \bar{v} \cdot \nabla \bar{v})}{O(\mu \nabla^2 \bar{v})} \sim \frac{\rho_o v_L D}{\mu} \frac{D}{L}. \quad (8.57)$$

Clearly, for this case of pipe flow, the liquid behaves as ideal if the condition

$$\frac{\rho_o v_L D}{\mu} \frac{D}{L} \gg 1 \quad (8.58)$$

which implies that the pipe length be much shorter than the length of the entrance region

$$L \ll L_E = \frac{\rho_o v_L D}{\mu} D \quad (8.59)$$

required for viscous effects to act across the pipe (see the bottom plot in Fig. 7.4).

With viscous terms being negligible, the balance between convection and pressure forces can be used to give

$$\Delta_L P \sim \rho_o v_L^2 \quad (8.60)$$

and $\Delta_T P \sim \rho_o v_L v_T D/L \sim \rho_o v_L^2 (D/L)^2$, so that

$$\frac{\Delta_T P}{\Delta_L P} \sim \frac{D^2}{L^2} \ll 1 \quad \rightarrow \quad P = P(x), \quad (8.61)$$

indicating that pressure differences across a given pipe section are negligibly small and can be therefore neglected in the first approximation.

Attention will be restricted to the analysis of quasisteady flows satisfying

$$St = \frac{L/v_L}{t_c} \ll 1, \quad (8.62)$$

for which the momentum equation reduces to

$$\rho_o \bar{v} \cdot \nabla \bar{v} = -\nabla P. \quad (8.63)$$

Projection along any given streamline leads to (see above the derivation of (8.39))

$$P + \rho_o v^2/2 = p + \rho_o U + \rho_o v^2/2 = C_l. \quad (8.64)$$

The constant C_l can be evaluated by following the streamline upstream into the container, where $\bar{v} \rightarrow 0$ and $p + \rho_o U = P_c$, so that one obtains

$$P + \rho_o v^2/2 = P_c. \quad (8.65)$$

Since $C_l = P_c$ for all streamlines and since according to (8.61) the pressure $P(x)$ is uniform across any given pipe section, the above equation indicates that the velocity profile is also necessarily uniform, i.e., $v = v(x)$. Evaluating (8.65) at the outlet section, where the reduced pressure takes a known value P_o , yields $v_o = [2(P_c - P_o)/\rho_o]^{1/2}$, which can be used to write the discharge volume flux

$$Q = A(L)[2(P_c - P_o)/\rho_o]^{1/2} = v(x)A(x). \quad (8.66)$$

This last equation can be used to determine the velocity distribution $v(x)$ along the pipe, which in turn yields $P(x)$ when substituted into (8.65).

Quasisteady ideal gas flow in pipes

If the conditions (8.58) and (8.62) are satisfied for gas flow in a pipe of length L and characteristic transverse size D (with ρ_o indicating in (8.58) the characteristic value of the gas density), then the flow is ideal and quasi-steady. We shall consider, in particular, configurations for which the mass forces have a negligible effect on the gas motion, as occurs in many applications where the condition (8.47) holds. As seen above, the energy and momentum equations can be projected along a given stream line to give

$$\frac{\partial s}{\partial l} = 0 \quad \text{and} \quad \frac{\partial}{\partial l} \left(h + \frac{v^2}{2} \right) = 0, \quad (8.67)$$

indicating that the entropy s and the sum $h + v^2/2$ remain constant, i.e.,

$$s = s_0, \quad (8.68)$$

and

$$h + \frac{v^2}{2} = h_0. \quad (8.69)$$

Evaluating the above expressions by following the stream line upstream into the container yields $s_0 = s_c$ and $h_0 = h_c$. On the other hand, the arguments leading to (8.61) provide in this case of gas flow with no mass forces the condition that the pressure remains uniform across each pipe cross section. With $s = s_c$ everywhere in the pipe and with $p = p(x)$, it is clear that all other thermodynamic variables (e.g., $h = h(s, p)$, $T = T(s, p)$, $\rho = \rho(s, p), \dots$) are also only a function of x . For instance, $h(s, p) = h(s_c, p(x)) = h(x)$. This last result can be used in (8.69) to demonstrate that $v = v(x)$.

In view of the above considerations, the solution can be determined from (8.68) and (8.69) together with

$$G = \rho(x)v(x)A(x), \quad (8.70)$$

where G is the constant mass flow rate circulating in the pipe. Equations (8.68) and (8.69) can be written with use made of the equations of state in terms of the different stagnation flow variables, according to

$$\frac{h_0}{h(x)} = \frac{T_0}{T(x)} = \left(\frac{a_0}{a(x)}\right)^2 = \left(\frac{p_0}{p(x)}\right)^{(\gamma-1)/\gamma} = \left(\frac{\rho_0}{\rho(x)}\right)^{\gamma-1} = 1 + \frac{\gamma-1}{2}M(x)^2, \quad (8.71)$$

where $h_0 = h_c$, $T_0 = T_c$, $a_0 = a_c$, $p_0 = p_c$, and $\rho_0 = \rho_c$ can be evaluated from the container properties. Similarly, it is convenient to rewrite (8.70) in term of the local Mach number $M(x)$ to give

$$G = \rho_0 a_0 A(x) M(x) \left(1 + \frac{\gamma-1}{2}M(x)^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}}. \quad (8.72)$$

Subsonic and supersonic flow in pipes

It is of interest to comment on the effect of the Mach number on the character of the fluid motion. For liquid flow in a pipe of varying cross section, our intuition tells us that as the cross section decreases the velocity must increase to accomodate the constant volume flux, while the corresponding reduced pressure must decrease to provide the needed acceleration, in agreement with (8.65) and (8.66). Things are not always quite so intuitive for gas flow, for which the resulting behavior depends on whether the flow is subsonic or supersonic. To see how changes in cross-section area affect the acceleration, we begin by differentiating (8.70) to give

$$\frac{1}{\rho} \frac{d\rho}{dx} + \frac{1}{v} \frac{dv}{dx} = -\frac{1}{A} \frac{dA}{dx}. \quad (8.73)$$

On the other hand, for a perfect gas, the condition of isentropic flow (8.68) gives

$$\frac{p}{\rho^\gamma} = \frac{p_0}{\rho_0^\gamma}, \quad (8.74)$$

providing

$$\frac{1}{\rho} \frac{d\rho}{dx} = \frac{1}{a^2} \frac{1}{\rho} \frac{dp}{dx} \quad (8.75)$$

upon differentiation, with $a^2 = \gamma p / \rho$. Since

$$\frac{1}{\rho} \frac{dp}{dx} = \frac{dh}{dx} \quad (8.76)$$

as follows from the condition of isentropic flow $Tds = dh - dp/\rho = 0$ and with

$$\frac{dh}{dx} = -v \frac{dv}{dx}, \quad (8.77)$$

obtained by differentiating (8.69), (8.73) finally yields

$$(1 - M^2) \frac{1}{v} \frac{dv}{dx} = -\frac{1}{A} \frac{dA}{dx}. \quad (8.78)$$

For subsonic flow ($M < 1$) along a convergent nozzle ($dA/dx < 0$) the velocity increases, as can be seen from (8.78), and the enthalpy, temperature, pressure, density, and velocity of sound decrease, as seen in (8.71) and (8.77). With v increasing and a decreasing the associated Mach number $M = v/a$ continuously increases as the cross section decreases. Because of the presence of the factor $(1 - M^2)$ in (8.78) the opposite behavior is found for supersonic flow in a convergent nozzle, that is, the flow decelerates and the Mach number decreases, while h , T , p , ρ , and a increase. It is therefore not possible to accelerate a subsonic stream beyond sonic conditions with use made of a convergent nozzle.

Observation of (8.78) reveals that a divergent nozzle ($dA/dx > 0$) is required to accelerate a supersonic flow. It also indicates that, if sonic conditions $M = 1$ are reached at a given section x , at that location the cross-section area must satisfy $dA/dx = 0$. These findings suggest that supersonic flow can be therefore achieved by utilizing a convergent-divergent nozzle so that the flow accelerates through the converging stretch to reach sonic conditions at the throat (the section of minimum area), further accelerating downstream into the supersonic regime in the diverging stretch. We shall see that, depending on the conditions, the flow in convergent-divergent nozzles may include a number of complicating features, including shock and expansion waves, to be treated in a later chapter.

Critical magnitudes

For steady gas flow in a pipe (8.71) provides the variation of the different thermodynamic variables with M . It can be used to determine, for instance, their critical values

$$\frac{h_0}{h^*} = \frac{T_0}{T^*} = \left(\frac{a_0}{a^*}\right)^2 = \left(\frac{p_0}{p^*}\right)^{(\gamma-1)/\gamma} = \left(\frac{\rho_0}{\rho^*}\right)^{\gamma-1} = \frac{\gamma+1}{2}. \quad (8.79)$$

found at a section where the flow is sonic. At that critical section, (8.72) provides

$$G = G^* = \rho_0 a_0 A^* \left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma+1}{2(\gamma-1)}}, \quad (8.80)$$

which can be written in the form

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}} \quad (8.81)$$

thereby relating the values of A and M found at a given section with the area A^* of the section where sonic conditions would be attained.

Flow in a convergent nozzle

We are now in a position to analyze the discharge of a gas container with known gas properties (p_c , ρ_c , T_c , a_c , ...) through a convergent nozzle of minimum cross section $A(L)$ at the outlet $x = L$ into a surrounding gas atmosphere at ambient pressure p_a . The solution, depending on the thermodynamic properties in the container and on the ambient pressure p_a , can be determined with use made of (8.71) and (8.72) with $T_0 = T_c$, $a_0 = a_c$, $p_0 = p_c$, and $\rho_0 = \rho_c$. An outflow is established whenever $p_a < p_c$. The flow is strongly subsonic for small pressure differences $p_c - p_a \ll p_c$, when the gas discharges into the outer atmosphere as a low-Mach-number jet with

pressure $p(L) = p_a$ at the exit section $x = L$. This last condition enables the Mach number $M(L)$ to be computed from (8.71) according to

$$\left(\frac{p_c}{p_a}\right)^{(\gamma-1)/\gamma} = 1 + \frac{\gamma-1}{2}M(L)^2 \quad \rightarrow \quad M(L) = \left(\frac{2}{\gamma-1}\right)^{1/2} \left[\left(\frac{p_c}{p_a}\right)^{(\gamma-1)/\gamma} - 1\right]^{1/2}, \quad (8.82)$$

to be substituted into (8.72) to determine the gas flow rate through the nozzle

$$G = \rho_c a_c A(L) \left(\frac{2}{\gamma-1}\right)^{1/2} \left(\frac{p_c}{p_a}\right)^{-\frac{\gamma+1}{2\gamma}} \left[\left(\frac{p_c}{p_a}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]^{1/2}. \quad (8.83)$$

As the value of p_c/p_a increases so does the value of $M(L)$ given by (8.82) and also the value of G given by (8.83), with variations given in Fig. 8.5 for $\gamma = 1.4$. Using the value of G , computed for a given value of p_c/p_a from (8.83), in (8.72) enables the Mach number distribution to be obtained along the nozzle, whereas (8.71) provides the accompanying distributions of pressure, density, and temperature. Sample schematic profiles are given in Fig. 8.6.

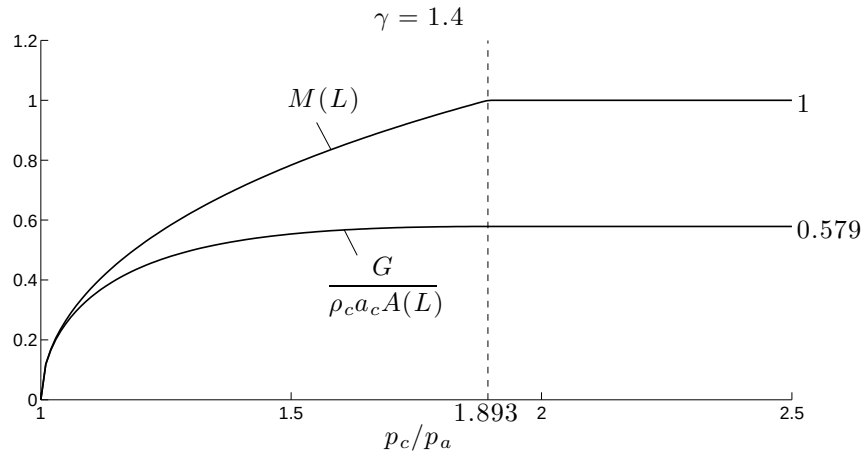


Figure 8.5: The variation of the exit Mach number $M(L)$ and mass flow rate G with p_c/p_a as obtained from (8.82) and (8.83) for $\gamma = 1.4$.

As discussed above, the Mach number at the exit section increases for increasing values of p_c/p_a . The jet solution with $p(L) = p_a$ and the computation procedure described above continue to hold as long as p_a remains above a critical choking value p_{CH} at which the flow reaches sonic conditions at the outlet section. This choking value is determined from (8.82) with $M(L) = 1$ as

$$\frac{p_c}{p_{CH}} = \left(\frac{\gamma+1}{2}\right)^{\gamma/(\gamma-1)} \quad (8.84)$$

giving $p_c/p_{CH} \simeq 1.893$ for $\gamma = 1.4$ ($p_{CH} \simeq 0.53p_c$). For these choking conditions the mass flow rate reaches the critical value G^*

$$G^* = \rho_c a_c A(L) \left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma+1}{2(\gamma-1)}}, \quad (8.85)$$

given in (8.80). The flow cannot accelerate past sonic conditions in the convergent nozzle by decreasing the ambient pressure. Instead, the solution becomes choked, that is, for values of $p_a < p_{CH}$ the flow is sonic at the exit, and becomes independent of the outer pressure². In particular, the mass flow rate, which for $p_a > p_{CH}$ increases for increasing values of p_c/p_a , remains equal to G^* , independent of p_a , for $p_a < p_{CH}$. In the choked flow that emerges for $p_c/p_a > [(\gamma + 1)/2]^{\gamma/(\gamma-1)}$, the condition $M(L) = 1$ therefore replaces $p(L) = p_a$. The pressure at the exit section is $p(L) = p_{CH} > p_a$, so that the jet flow with parallel streamlines found downstream from the pipe outlet for $p_a > p_{CH}$ is replaced for $p_a < p_{CH}$ by a complicated flow pattern including expansion waves that deflect the streamlines outwards.

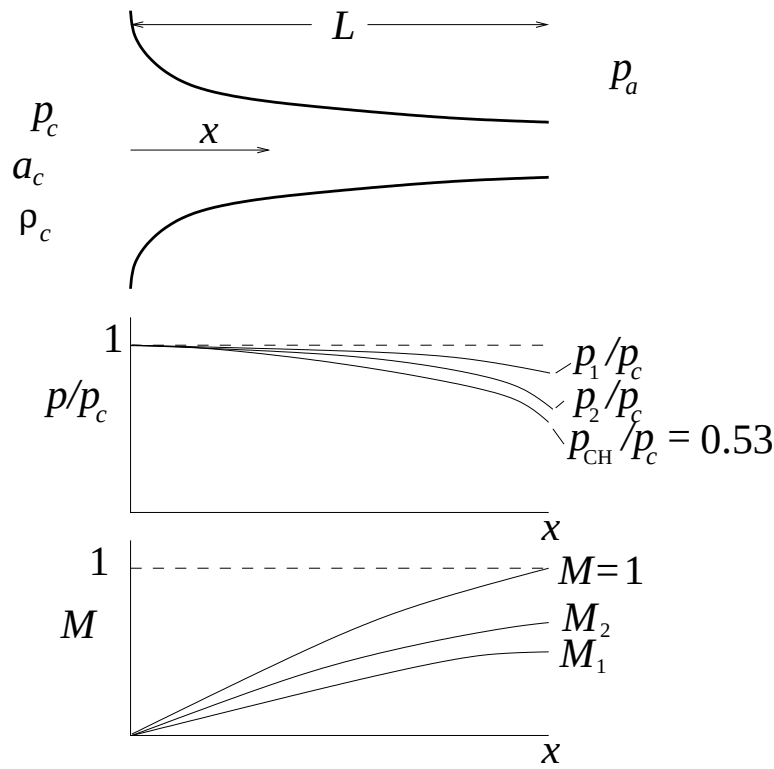
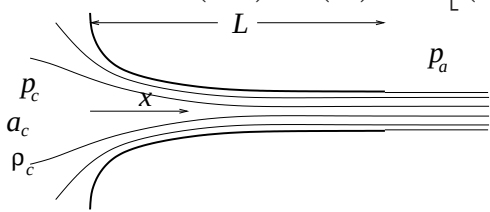
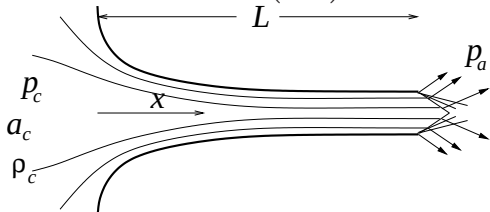


Figure 8.6: Pressure and Mach number distributions along a nozzle for different values of p_a/p_c .

In summary, the discharge of a container of pressure p_c into the atmosphere, at pressure p_a , through a convergent nozzle of minimum cross section $A(L)$ depends on the value of p_c/p_a , so that

²Note that the decrease of the ambient pressure below p_{CH} cannot be “felt” within the nozzle, because the flow at the exit is already sonic, and therefore the perturbations cannot travel upstream, so that for $p_a < p_{CH}$ the flow in the nozzle becomes independent of p_a .

If $\frac{p_c}{p_a} \leq \left(\frac{\gamma+1}{2}\right)^{\gamma/(\gamma-1)}$:	$p(L) = p_a$ $M(L) = \left(\frac{2}{\gamma-1}\right)^{1/2} \left[\left(\frac{p_c}{p_a}\right)^{(\gamma-1)/\gamma} - 1 \right]^{1/2} \leq 1$ $G = \rho_c a_c A(L) \left(\frac{2}{\gamma-1}\right)^{1/2} \left(\frac{p_c}{p_a}\right)^{-\frac{\gamma+1}{2\gamma}} \left[\left(\frac{p_c}{p_a}\right)^{\frac{\gamma-1}{\gamma}} - 1 \right]^{1/2}$ 
If $\frac{p_c}{p_a} > \left(\frac{\gamma+1}{2}\right)^{\gamma/(\gamma-1)}$:	$M(L) = 1$ $p(L) = p_{CH} = \left(\frac{\gamma+1}{2}\right)^{-\gamma/(\gamma-1)} p_c > p_a$ $G = G^* = \rho_c a_c A(L) \left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$ 

Discharge and charge processes in gas containers

The gas flow in nozzles studied above enters for instance when addressing the discharge and charge processes of gas containers. We begin by considering the discharge of a gas container of volume V through a convergent nozzle of exit cross section A and length $L \sim \sqrt{A}$ ³. As occurs in most applications, the nozzle is assumed to be small, in that its characteristic size $A^{1/2}$ is much smaller than the characteristic container size $V^{1/3}$. This disparity of length scales enables the description to be simplified, as explained below.

Two different regions can be identified. Region (I) includes the interior of the nozzle and also a small portion of the container upstream from the nozzle entrance, of characteristic size $A^{1/2}$, where the flow velocity is comparable to that found at the nozzle exit v_e . We shall treat discharge processes for which the Reynolds number in this region is much larger than unity,

$$Re_e = \frac{\rho_e v_e A^{1/2}}{\mu_e} \gg 1, \quad (8.86)$$

with the values of the flow variables at the nozzle exit (denoted by the subscript e) used as characteristic values in the nozzle. Following the discussion of flow in nozzles, we anticipate that the associated exit Mach number M_e takes in general values of order unity or smaller ($M_e = 1$ when the flow in the convergent nozzle is choked).

³As an exercise, the reader might repeat the order-of-magnitude analysis given below for the case of slender nozzles with length $L \gg A^{1/2}$.

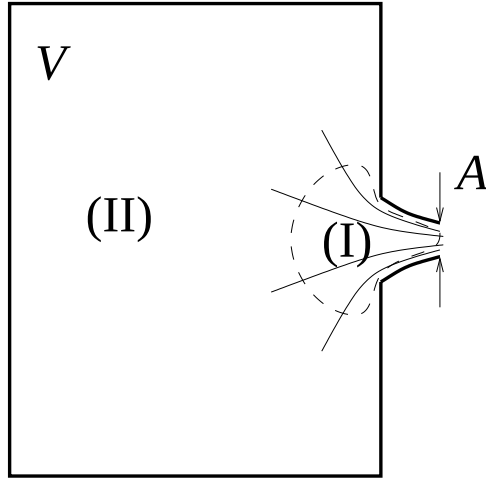


Figure 8.7: Characteristic regions to be considered in the analysis of the discharge of a gas container.

Region (II) includes the rest of the container, where the flow velocity is very small, as can be seen by applying continuity to give

$$\rho_c v_c V^{2/3} \sim \rho_e v_e A, \quad (8.87)$$

where the subscript c denotes the characteristic flow variables in the container. From (8.71) with $M_e \sim O(1)$ we can anticipate that $\rho_c \sim \rho_e$ (and also that $p_c \sim p_e$, $T_c \sim T_e$, $a_c \sim a_e$, ...), so that the above equation yields

$$\frac{v_c}{v_e} \sim \frac{A}{V^{2/3}} \ll 1. \quad (8.88)$$

The Strouhal number for the flow in the container is of order unity

$$St = \frac{O(\partial/\partial t)}{O(\bar{v} \cdot \nabla)} = \frac{V^{1/3}/v_c}{t_d} \sim O(1) \quad (8.89)$$

because the discharge time $t_d \sim (\rho_c V)/(\rho_e v_e A)$ is comparable to the residence time $V^{1/3}/v_c$. In the nozzle region, however, the associated Strouhal number is small,

$$St = \frac{A^{1/2}/v_e}{t_d} \sim \frac{A^{3/2}}{V} \ll 1, \quad (8.90)$$

so that the flow is quasisteady.

On the other hand, the Mach number for the flow in the container is

$$M_c \sim \frac{v_c}{a_c} \sim M_e \frac{A}{V^{2/3}} \ll 1, \quad (8.91)$$

whereas the Reynolds number is given by

$$Re_c = \frac{\rho_c v_c V^{1/3}}{\mu_c} \sim \frac{v_c V^{1/3}}{v_e A^{1/2}} Re_e \sim \frac{A^{1/2}}{V^{1/3}} Re_e. \quad (8.92)$$

As can be seen, the Reynolds number in the container may take large or small values depending on the flow conditions and on the nozzle-to-container size ratio. In most applications, the Reynolds number for the nozzle flow is sufficiently large for the condition $Re_c \gg 1$ to hold. In this case, the associated spatial pressure variations can be obtained from the balance $\rho \bar{v} \cdot \nabla \bar{v} \sim \nabla p$ to give

$$Re_c \gg 1 \quad \rightarrow \quad \Delta_c p \sim \rho_c v_c^2 \quad \rightarrow \quad \frac{\Delta_c p}{p_c} \sim \frac{v_c^2}{p_c / \rho_c} \sim M_c^2 \sim M_e^2 \frac{A^2}{V^{4/3}} \ll 1, \quad (8.93)$$

because M_e is at the most of order unity. On the other hand, in the opposite limit of very large containers such that $Re_c \ll 1$ the momentum balance $\nabla p \sim \nabla \cdot \bar{\tau}'$ yields

$$Re_c \ll 1 \quad \rightarrow \quad \Delta_c p \sim \mu_c v_c / V^{1/3} \quad \rightarrow \quad \frac{\Delta_c p}{p_c} \sim \frac{M_e^2 A^{3/2}}{Re_e V} \ll 1. \quad (8.94)$$

As can be seen in (8.93) and (8.94), regardless of the container Reynolds number, the associated pressure variations in the container are negligible, so that in the first approximation one may assume a uniform distribution $p_c(t)$ when describing the discharge process. Note that the pressure variations in the nozzle, of order $\Delta_e p \sim \rho_e v_e^2$ because $Re_e \gg 1$, are such that $\Delta_e p / p_e \sim M_e^2 \sim O(1)$.

The energy equation can be used to estimate the entropy changes occurring as the fluid particle travels across the container during the discharge process according to

$$\underbrace{\rho T \frac{Ds}{Dt}}_{\rho_c T_c \Delta s / t_d} = \underbrace{\bar{\tau}' : \nabla \bar{v}}_{\mu_c v_c^2 / V^{2/3}} + \underbrace{\nabla \cdot (k \nabla T)}_{k_c \Delta_c T / V^{2/3}}. \quad (8.95)$$

It can be seen that viscous dissipation leads to negligibly small entropy variations

$$\frac{\Delta_\phi s}{c_v} \sim \frac{M_e^2 A^{3/2}}{Re_e V} \ll 1, \quad (8.96)$$

whereas heat conduction produces entropy changes that depend on the container Reynolds number

$$\frac{\Delta_k s}{c_v} \sim \frac{\gamma}{Pr} \frac{\Delta T}{T_c} \frac{1}{Re_e}, \quad (8.97)$$

leading to two limiting behaviors. If $Re_c \gg 1$, then $\Delta s / c_v \ll 1$, and the evolution of the container gas is isentropic, i.e.,

$$\frac{Ds}{Dt} = 0. \quad (8.98)$$

As a result, the entropy in the container remains constant during the discharge process, equal to its initial value

$$s_c = s_i. \quad (8.99)$$

If $Re_c \ll 1$, on the other hand, since $\Delta_k s / c_v$ can never exceed an order-unity value, (8.97) leads to $\Delta T / T_c \ll 1$, indicating that the gas in the container is isothermal, with a temperature equal to that of the wall

$$T_c = T_w. \quad (8.100)$$

An alternative way to understand this last result involves the writing of (8.97) in the form

$$\frac{\Delta_k s}{c_v} \sim \gamma \left(\frac{(\alpha t_d)^{1/2}}{V^{1/3}} \right)^2 \frac{\Delta T}{T_c}, \quad (8.101)$$

in terms of the thermal diffusivity $\alpha = k_c/(\rho_c c_p)$, with $(\alpha t_d)^{1/2}$ being the thermal penetration depth (the thickness of the layer where heat conduction from the wall modifies the temperature after a time t_d). If $(\alpha t_d)^{1/2} \ll V^{1/3}$, corresponding to the limit $Re_c \gg 1$, heat conduction affects only a thin layer of gas around the container wall, thereby modifying its temperature, while the flow remains isentropic in most of the container interior. In the opposite limit $(\alpha t_d)^{1/2} \gg V^{1/3}$ (i.e., $Re_c \ll 1$), heat conduction is dominant, and ensures that the temperature remains everywhere equal to that found on the container wall. Note that, regardless of the value of Re_c , for containers with insulated walls, the heat conducted from the wall into the container would be negligible, resulting in an isentropic evolution.

Since the pressure is uniform in the container and either the entropy (for $Re_c \gg 1$ or for containers with insulated walls) or the temperature (for $Re_c \ll 1$) are constant, it turns out that all thermodynamic variables in the container are uniform during the discharge process (i.e., $\rho_c = \rho_c(t)$, $h_c = h_c(t)$, $a_c = a_c(t)$, $\dots$) in the two limiting cases considered. Nonuniform distributions should be accounted for when $Re_c \sim O(1)$, a case seldom encountered in applications.

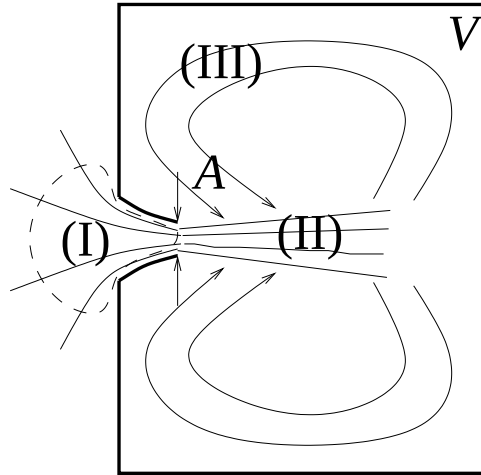


Figure 8.8: Characteristic regions to be considered in the analysis of the charge of a gas container.

The analysis of the charging process is more complicated and involves consideration of three different regions. Besides the quasisteady inviscid nozzle region (I), there is a slender jet region (II) extending into the container, where the velocity is also of order v_e . The jet decelerates as it mixes with the container gas, in a nonisentropic process, leading to jet-penetration distances of order $Re_e A^{1/2}$ for the laminar case and of the order of a few tens of diameters for the case of turbulent jets. Outside the jet, a region of slow recirculating flow appears, as induced by the jet entrainment. The recirculating motion, associated with small values of $M_c \ll 1$ and thereby inducing negligibly small spatial pressure differences, promotes mixing in the container, so that the assumption of uniform thermodynamic properties holds (although the evolution is never isentropic).

Ideal fluid machines

For the analysis of ideal flow in pipes, we have used continuity written in integral form together with the momentum and energy differential equations projected along streamlines. Likewise, an appropriate combination of differential and integral conservation equations will be used below to analyze closed fluid machines, such as pumps, compressors and turbines, providing the equations that relate the jumps in flow properties occurring as the fluid stream crosses the machine.

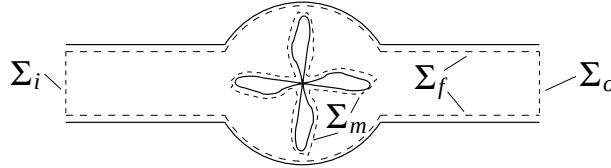


Figure 8.9: Schematic representation of an ideal fluid machine.

Figure 8.9 shows a schematic representation of a fluid machine, including inlet and outlet sections Σ_o and Σ_i for the internal stream, which is bounded by a surrounding fixed casing Σ_f and by the internal moving parts (typically, rotating blades), with surface Σ_m . The motion takes place at large values of the Reynolds number, so that the energy equation reduces to

$$\frac{Ds}{Dt} = 0, \quad (8.102)$$

indicating that each fluid particle conserves the entropy. Assuming uniform entropy at the inlet section yields

$$s = s_i \quad (8.103)$$

for the flow inside the machine. Molecular transport effects (viscous forces and heat conduction) can be expected to be nonnegligible in boundary layers adjacent to the machine walls, so that the condition (8.103) of homentropic flow, to be used below for the analysis of ideal fluid machines, should be slightly modified for increased accuracy in realistic applications.

Because of the presence of rotating parts, the fluid motion is intrinsically unsteady, so that, although the flow is inviscid, the value of $h + v^2/2 + U$ is not conserved along streamlines. To analyze the fluid motion it is convenient to use a laboratory reference frame and apply the continuity and energy equations in integral form to a control volume bounded by the machine walls and the inlet and outlet sections, as shown in Fig. 8.9. In particular, continuity reads

$$\frac{d}{dt} \int_{V_c(t)} \rho dV + \int_{\Sigma_c(t)} \rho(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = 0. \quad (8.104)$$

Although the motion is unsteady, the variation with time of the amount of mass contained in the control volume is only due to changes in the operating regime. Therefore, for steady operation of turbomachines, this first term is entirely negligible. It is also negligible with variable operating conditions when, as typically occurs in applications, the time for variation of the regime t_c is much larger than the characteristic residence time in the machine $V/(Av_c)$, so that

$$\frac{\frac{d}{dt} \int_{V_c(t)} \rho dV}{\int_{\Sigma_c(t)} \rho(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma} \sim \frac{V/(Av_c)}{t_c} \ll 1. \quad (8.105)$$

Under those conditions, (8.104) reduces to

$$-\int_{\Sigma_i} \rho \bar{v} \cdot \bar{n} d\sigma = \int_{\Sigma_o} \rho \bar{v} \cdot \bar{n} d\sigma = G(t), \quad (8.106)$$

where G represents the mass flow rate circulating through the machine, a function of the operating regime that may, therefore, change slowly with time.

For the analysis, it is convenient to write the energy equation

$$\begin{aligned} \frac{d}{dt} \left[\int_{V_c(t)} \rho(e + v^2/2) dV \right] + \int_{\Sigma_c(t)} \rho(e + v^2/2)(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma &= - \int_{\Sigma_c(t)} p \bar{v} \cdot \bar{n} d\sigma \\ &+ \int_{\Sigma_c(t)} \bar{v} \cdot \bar{\tau}' \cdot \bar{n} d\sigma + \int_{V_c(t)} \rho \bar{f}_m \cdot \bar{v} dV + \int_{\Sigma_c(t)} k \nabla T \cdot \bar{n} d\sigma \end{aligned} \quad (8.107)$$

with the term $\int_{V_c(t)} \rho \bar{f}_m \cdot \bar{v} dV$ for the work exerted by external forces expressed in a different way. To that end, we assume that the mass forces are conservative and steady, so that $\bar{f}_m = -\nabla U$ with $\partial U / \partial t = 0$ and we can write

$$\rho \bar{v} \cdot \bar{f}_m = -\frac{\partial}{\partial t}(\rho U) - \nabla \cdot (\rho \bar{v} U) \quad (8.108)$$

and

$$\int_{V_c(t)} \rho \bar{f}_m \cdot \bar{v} dV = - \int_{V_c(t)} \frac{\partial}{\partial t}(\rho U) dV - \int_{\Sigma_c(t)} \rho U \bar{v} \cdot \bar{n} d\sigma. \quad (8.109)$$

On the other hand, application of Reynolds transport theorem to a variable control volume yields

$$\frac{d}{dt} \int_{V_c(t)} \rho U dV = \int_{V_c(t)} \frac{\partial}{\partial t}(\rho U) dV + \int_{\Sigma_c(t)} \rho U \bar{v}_c \cdot \bar{n} d\sigma, \quad (8.110)$$

which can be combined with (8.109) to give

$$\int_{V_c(t)} \rho \bar{f}_m \cdot \bar{v} dV = -\frac{d}{dt} \int_{V_c(t)} \rho U dV - \int_{\Sigma_c(t)} \rho U (\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma. \quad (8.111)$$

Substituting this last result into (8.107) finally provides

$$\begin{aligned} \frac{d}{dt} \left[\int_{V_c(t)} \rho(e + v^2/2 + U) dV \right] + \int_{\Sigma_c(t)} \rho(e + v^2/2 + U)(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma &= \\ &- \int_{\Sigma_c(t)} p \bar{v} \cdot \bar{n} d\sigma + \int_{\Sigma_c(t)} \bar{v} \cdot \bar{\tau}' \cdot \bar{n} d\sigma + \int_{\Sigma_c(t)} k \nabla T \cdot \bar{n} d\sigma \end{aligned} \quad (8.112)$$

as an alternative form of the energy equation that is particularly useful for the analysis of turbomachinery.

The first term above is negligible for quasi-steady operating conditions, the case considered here, while the convective flux of energy is zero across all internal walls Σ_f and Σ_m . For large values of the Reynolds number, the work due to viscous forces and the heat added by conduction are

negligible everywhere. The work of pressure forces on the fixed internal walls Σ_f is zero, while that on the surface of the moving parts is

$$W = - \int_{\Sigma_m} p \bar{v} \cdot \bar{n} d\sigma, \quad (8.113)$$

where W is the power of the turbomachine, defined to be positive when the work is exerted on the fluid stream. Finally, the energy balance reduces to

$$\int_{\Sigma_i + \Sigma_o} \rho(e + v^2/2 + U) \bar{v} \cdot \bar{n} d\sigma = - \int_{\Sigma_i + \Sigma_o} p \bar{v} \cdot \bar{n} d\sigma + W, \quad (8.114)$$

which can alternatively be written in terms of the enthalpy to give

$$\int_{\Sigma_i + \Sigma_o} \rho(h + v^2/2 + U) \bar{v} \cdot \bar{n} d\sigma = W. \quad (8.115)$$

Equations (8.106) y (8.115) admit simplified versions when the conditions at the inlet and outlet sections are uniform, yielding

$$G = \rho_i v_i A_i = \rho_o v_o A_o \quad (8.116)$$

and

$$G[(h + v^2/2 + U)_o - (h + v^2/2 + U)_i] = W, \quad (8.117)$$

where A_i and A_o denote the surface area of the inlet and outlet sections. In pumps and compressors the inequality $(h + v^2/2 + U)_o > (h + v^2/2 + U)_i$ is verified, whereas the opposite is found in turbines. Equations (8.103), (8.116), and (8.117) give the solution for the motion in ideal turbomachinery under quasi-steady operating conditions.

For pumps and hydraulic turbines the equations reduce to

$$T = T_i = T_o \quad (8.118)$$

$$W = Q[(p_o + \rho v_o^2/2 + \rho U_o) - (p_i + \rho v_i^2/2 + \rho U_i)] \quad (8.119)$$

$$Q = G/\rho = v_o A_o = v_i A_i, \quad (8.120)$$

where Q is the volume flow rate. As can be seen, the power of a pump is used to increase the pressure, the potential energy and the kinetic energy of the liquid stream. Note that while the change in kinetic energy is significant in many cases, and needs to be retained for a correct analysis of hydraulic turbomachinery, in most situations the value of $\rho(U_o - U_i)$ is only a small fraction of W/Q and can be correspondingly neglected in (8.119).

For thermal turbomachinery (8.103), (8.116), and (8.117) can be written with mass forces neglected to give

$$\frac{h_o}{h_i} = \left(\frac{p_o}{p_i} \right)^{(\gamma-1)/\gamma} = \left(\frac{\rho_o}{\rho_i} \right)^{\gamma-1} \quad (8.121)$$

$$W = G[(h_o + v_o^2/2) - (h_i + v_i^2/2)] \quad (8.122)$$

$$G = \rho_o v_o A_o = \rho_i v_i A_i. \quad (8.123)$$

When the Mach number is small at the inlet and outlet sections (8.122) can be further simplified to give

$$\frac{W}{G h_i} = \frac{h_o}{h_i} - 1 = \left(\frac{p_o}{p_i} \right)^{(\gamma-1)/\gamma} - 1. \quad (8.124)$$

Chapter 9

Introduction to Potential Flow

The concept of vorticity

$$\bar{\omega} = \nabla \wedge \bar{v} \quad (9.1)$$

was introduced previously when analyzing the kinematics of fluid motion. Physically, it represents twice the angular velocity with which the fluid particle rotates as a solid body. The present chapter deals with irrotational (or potential) flows, for which the vorticity is zero at every point in the flow field. For simplicity in the presentation, attention will be restricted to incompressible flow, which applies to the motion of liquids and also of gas at low Mach numbers. Complex variable will be seen to facilitate the description of plane flows. The analysis of motion around a circular cylinder will allow us to introduce some aerodynamic concepts including the D'Alembert's paradox and the Kutta-Joukowski formula.

The vorticity equation

A conservation equation for the vorticity can be derived from the momentum equation

$$\frac{\partial \bar{v}}{\partial t} + \nabla(|\bar{v}|^2/2) - \bar{v} \wedge (\nabla \wedge \bar{v}) = -\frac{1}{\rho} \nabla p + \frac{1}{\rho} \nabla \cdot \bar{\tau}' + \bar{f}_m. \quad (9.2)$$

For simplicity, attention will be restricted to incompressible flows with conservative mass forces and constant viscosity, for which the above equation takes the form

$$\frac{\partial \bar{v}}{\partial t} + \nabla(|\bar{v}|^2/2) - \bar{v} \wedge \bar{\omega} = -\nabla \left(\frac{p}{\rho} + U \right) + \nu \nabla^2 \bar{v}. \quad (9.3)$$

Since the curl of a gradient is identically zero, taking the curl of this last equation leads to

$$\frac{D\bar{\omega}}{Dt} = \frac{\partial \bar{\omega}}{\partial t} + \bar{v} \cdot \nabla \bar{\omega} = \bar{\omega} \cdot \nabla \bar{v} + \nu \nabla^2 \bar{\omega}. \quad (9.4)$$

In writing the term $\nabla \wedge (\bar{v} \wedge \bar{\omega})$ we have used the vector identity

$$\nabla \wedge (\bar{v} \wedge \bar{\omega}) = \bar{v}(\nabla \cdot \bar{\omega}) - \bar{\omega}(\nabla \cdot \bar{v}) - \bar{v} \cdot \nabla \bar{\omega} + \bar{\omega} \cdot \nabla \bar{v} \quad (9.5)$$

where the first two terms vanish because of the additional condition that the divergence of a curl is identically zero (i.e., $\nabla \cdot \bar{\omega} = 0$) and the fact that, for an incompressible fluid, the velocity

field is solenoidal, that is, $\nabla \cdot \bar{v} = 0$. Equation (9.4) indicates that for an incompressible fluid the variation of the vorticity following a given fluid particle is due to two different contributions. The term $\nu \nabla^2 \bar{\omega}$ represents the diffusion of vorticity due to molecular transport effects (remember that molecular transport was seen to be responsible for the appearance of viscous forces in the momentum equation and of heat conduction in the energy equation). For a given flow with characteristic length and velocity scales L_c and v_c , the relative importance of convective and diffusive transport of vorticity is measured by the Reynolds number, according to

$$\frac{O(\bar{v} \cdot \nabla \bar{\omega})}{O(\nu \nabla^2 \bar{\omega})} \sim \frac{v_c L_c}{\nu}, \quad (9.6)$$

so that vorticity diffusion is negligible for large values of the Reynolds number $v_c L_c / \nu \gg 1$. The term $\bar{\omega} \cdot \nabla \bar{v}$, on the other hand, represents the production or destruction of vorticity by stretching or turning the vortex lines (lines that are tangent to the vorticity vector $\bar{\omega}$ at each point). In planar flows, for which $\bar{\omega}$ is perpendicular to the plane of motion, this term can be seen to vanish. It is of interest that pressure and gravity forces do not appear in (9.4), the physical reason being that these forces act through the center of mass of the fluid particle and therefore cannot change its angular velocity.

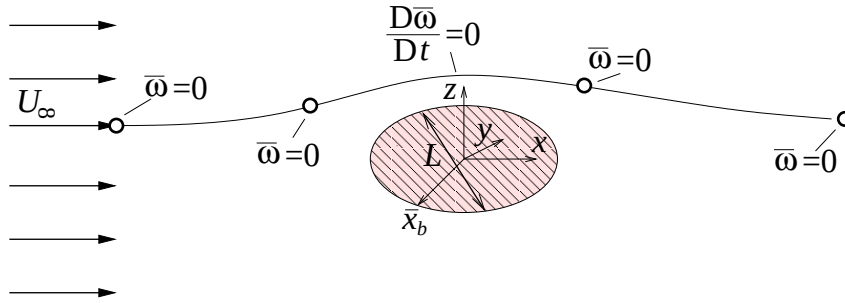


Figure 9.1: Vorticity following a fluid particle in ideal flow about a general three-dimensional body.

Equation (9.4) determines the vorticity of a given incompressible flow. It can be used in particular to demonstrate that the vorticity is zero in some problems of practical interest, an example being the ideal flow about a body, the central problem of interest in aerodynamics. Consider a uniform stream of velocity $\bar{v} = U_\infty \bar{e}_x$ flowing about a body of characteristic size L , as sketched in Fig. 9.1. The evolution of the vorticity of a given fluid particle is determined by (9.4). If the Reynolds number $U_\infty L / \nu$ is large, the diffusion of vorticity is negligible, and the equation reduces to

$$\frac{D\bar{\omega}}{Dt} = \bar{\omega} \cdot \nabla \bar{v}. \quad (9.7)$$

Since the free-stream vorticity is zero, because $\nabla \wedge (U_\infty \bar{e}_x) = 0$, the evolution of the vorticity for a fluid particle that enters the flow field from the left boundary is obtained by integration of (9.7) with initial condition $\bar{\omega} = 0$, giving $\bar{\omega} = 0$ at any given instant of time. Since that is true for any fluid particle, it is clear that the resulting flow is irrotational, i.e.,

$$\bar{\omega} = \nabla \wedge \bar{v} = 0. \quad (9.8)$$

In assessing the validity of the previous equation, one should note that at large Reynolds numbers vorticity appears confined in a thin boundary layer around the body where diffusion of vorticity from the wall, where it is created due to viscous effects, is significant. If the boundary layer remains attached to the body, as occurs in aerodynamic streamlined bodies, then (9.8) holds everywhere in the flow field outside the boundary layer. Often, however, the boundary layer over blunt bodies separates at a given location, ejecting vorticity into the flow field in the form of a concentrated vortex layer (also called mixing layer) bounding a wake, which is unsteady in many practical situations. The presence of concentrated vorticity invalidates the existence of uniformly-valid potential flow solutions whenever boundary-layer separation occurs.

Potential flow

A flow (viscous or inviscid, compressible or incompressible, steady or unsteady) is said to be potential or irrotational when the vorticity is identically zero at every point in the flow field. This condition (9.8) simplifies the flow description significantly, because it is known from vector analysis that an irrotational vector field can be expressed as a gradient of a scalar potential function. In connection with irrotational flows, we therefore introduce the velocity potential $\phi(\vec{x}, t)$ such that, if (9.8) holds everywhere in the flow field, then the velocity is given by

$$\vec{v} = \nabla\phi, \quad (9.9)$$

where ϕ may have an arbitrary function of time added without a change in $\vec{v}$.

The velocity potential can be used for the description of compressible and incompressible irrotational fluid motion. For the case of incompressible fluid motion treated here, the associated conservation equation to be satisfied by the velocity potential is particularly simple, in that substitution of (9.9) into the corresponding continuity equation

$$\nabla \cdot \vec{v} = 0 \quad (9.10)$$

yields the well-known Laplace equation

$$\nabla^2\phi = 0. \quad (9.11)$$

Once the potential is obtained by integration of (9.11), the associated pressure field can be determined from (9.3) written with $\vec{\omega} = 0$ ¹ in the form

$$\nabla \left(\frac{\partial\phi}{\partial t} + \frac{p}{\rho} + \frac{1}{2}\nabla\phi \cdot \nabla\phi + U \right) = 0, \quad (9.12)$$

which can be integrated to give the generalized Bernoulli's equation

$$\frac{\partial\phi}{\partial t} + \frac{p}{\rho} + \frac{1}{2}\nabla\phi \cdot \nabla\phi + U = F(t), \quad (9.13)$$

where the function $F(t)$ is to be determined from boundary information.

¹For incompressible flow with constant viscosity, $\nabla \cdot \vec{\tau} = \mu\nabla^2\vec{v} = -\mu\nabla \wedge (\nabla \wedge \vec{v})$, so that viscous forces are identically zero if the flow is irrotational.

As an illustrative example, let us consider the incompressible aerodynamic problem sketched in Fig. 9.1. Since the flow is irrotational, the velocity field is given by (9.9), where the velocity potential is determined by integration of (9.11) with boundary conditions

$$\begin{cases} |\bar{x}| \gg L : & \phi = U_\infty x \\ \bar{x} = \bar{x}_b : & \bar{n} \cdot \nabla \phi = 0 \end{cases} \quad (9.14)$$

corresponding to the free-stream velocity $\bar{v} = U_\infty \bar{e}_x$ far from the body and the condition of zero normal velocity $\bar{v} \cdot \bar{n} = 0$ on the body surface $\bar{x} = \bar{x}_b$. The corresponding pressure field is given by (9.13), yielding for instance

$$\frac{p}{\rho} + g\mathcal{Z} + \frac{1}{2} \nabla \phi \cdot \nabla \phi = \frac{P_\infty}{\rho} + \frac{1}{2} U_\infty^2 \quad (9.15)$$

for steady motion with gravity forces included, with $P_\infty = (p + \rho g\mathcal{Z})_\infty$ being the value of the reduced pressure far from the body.

The description given by (9.11) and (9.13) applies in general to irrotational flows of perfect liquids, but also to other fluids, as long as they behave as effectively incompressible for the flow conditions considered. That would be the case of ideal air flow about a body when the Mach number of the approaching free stream U_∞/a_∞ is sufficiently small. To see it, note that the condition of large Reynolds number $U_\infty L/\nu \gg 1$ implies that the flow is isentropic. Integration of

$$\frac{Ds}{Dt} = c_v \frac{D}{Dt} [\ln(p/\rho^\gamma)] = 0 \quad (9.16)$$

for a given fluid particle yields

$$\frac{p}{\rho^\gamma} = \frac{p_\infty}{\rho_\infty^\gamma} \quad (9.17)$$

if the conditions are uniform in the free stream. On the other hand, for large Reynolds numbers, the spatial pressure differences Δp can be estimated from the balance $\rho \bar{v} \cdot \nabla \bar{v} \sim -\nabla p$ to give

$$\Delta p/p_\infty \sim \rho_\infty U_\infty^2/p_\infty \sim \gamma (U_\infty/a_\infty)^2. \quad (9.18)$$

Therefore, for air flows where the free-stream velocity is much smaller than the velocity of sound (i.e., low-Mach-number flows), the relative spatial pressure differences are of order $(U_\infty/a_\infty)^2 \ll 1$, giving also negligibly small density differences $\Delta \rho/\rho_\infty \sim (U_\infty/a_\infty)^2 \ll 1$, according to (9.17). Under those conditions, the irrotational velocity and pressure fields of the air flow are determined in the first approximation by the incompressible equations (9.11) and (9.13), with errors, due to compressibility, that would be of order $(U_\infty/a_\infty)^2 \ll 1$.

Planar potential flows

Incompressible potential flows, determined by (9.11) and (9.13), are relatively easy to analyze, because there is a vast amount of mathematical information about Laplace equation and its solutions. A particularly simple situation arises when the flow field is planar, with a velocity that can be expressed in cartesian coordinates according to

$$\bar{v} = v_x \bar{e}_x + v_y \bar{e}_y. \quad (9.19)$$

The condition of irrotational motion (9.8) leads to

$$\omega_z = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = 0 \quad (9.20)$$

while continuity provides

$$\nabla \cdot \bar{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0. \quad (9.21)$$

Because the flow is irrotational, there exists a velocity potential such that

$$v_x = \frac{\partial \phi}{\partial x} \quad \text{and} \quad v_y = \frac{\partial \phi}{\partial y} \quad (9.22)$$

which, since the flow is incompressible, satisfies

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0, \quad (9.23)$$

obtained by substituting (9.22) into (9.21). On the other hand, for a planar incompressible flow we can define a stream function ψ such that ²

$$v_x = \frac{\partial \psi}{\partial y} \quad \text{and} \quad v_y = -\frac{\partial \psi}{\partial x}, \quad (9.24)$$

which, since the flow is irrotational, satisfies

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0, \quad (9.25)$$

obtained by substituting (9.24) into (9.20).

The flow can be analyzed by introducing the so-called complex potential

$$f(z) = \phi(x, y) + i\psi(x, y), \quad (9.26)$$

defined to be an *analytic function* ³ in the complex plane $z = x + iy = re^{i\theta}$. The definition suits perfectly our needs, because if $f(z)$ is analytic, then its real and imaginary parts satisfy Laplace equation (i.e., equations (9.23) and (9.25)) along with the Cauchy-Riemann conditions

$$v_x = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} \quad \text{and} \quad v_y = \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}. \quad (9.29)$$

²The definition and properties of the stream function are found in Chapter 2.

³A function of complex variable

$$f(z) = \mathcal{R}(x, y) + i\mathcal{I}(x, y) \quad (9.27)$$

is said to be analytic in a region if it is differentiable at every point of that region, that is, at every point the partial derivatives $\partial \mathcal{R}/\partial x$, $\partial \mathcal{R}/\partial y$, $\partial \mathcal{I}/\partial x$, and $\partial \mathcal{I}/\partial y$, are continuous and satisfy the so-called Cauchy-Riemann equations

$$\frac{\partial \mathcal{R}}{\partial x} = \frac{\partial \mathcal{I}}{\partial y} \quad \text{and} \quad \frac{\partial \mathcal{R}}{\partial y} = -\frac{\partial \mathcal{I}}{\partial x}, \quad (9.28)$$

which guarantee that the value of df/dz is independent of the direction in which it is calculated. If a function is analytic, then it can be demonstrated that its real and imaginary parts satisfy Laplace equation.

In other words, for every analytic function $f(z)$ the real part is automatically a valid velocity potential and the imaginary part is a valid stream function. The derivative of the complex potential is the so-called complex velocity

$$\frac{df}{dz} = v_x - \imath v_y = (v_r - \imath v_\theta)e^{-\imath\theta}, \quad (9.30)$$

as can be seen by straightforward derivation of (9.27) (for instance, in the direction $z = x$ or $z = \imath y$).

An approach to be followed below in investigating planar potential flows involves consideration of simple analytic functions that represent elementary flows of interest, such as sources, sinks, or vortices. More complex flows can be constructed by superposition of these elementary solutions, a procedure that is always admissible because the Laplace equation is linear. This approach, albeit indirect, provides interesting results for some flows of interest and avoids difficulties of solving partial differential equations. Note that, while the velocity field obtained by adding two different complex potentials is the sum of the associated velocity fields, the pressure is not, because the corresponding equation (9.13) is nonlinear. Below, we proceed to introduce some elementary complex potentials of interest and use them subsequently for the analysis of flow over a circular cylinder.

Simple analytic functions

The simplest analytic function is proportional to z , $f(z) = cz$, with a proportionality constant that can be selected to give

$$f(z) = U_\infty e^{-\imath\alpha} z. \quad (9.31)$$

The above complex potential represents a **uniform stream** of velocity U_∞ inclined at an angle α to the x axis, as shown in Fig. 9.2. Taking the derivative of (9.31) to compute the complex velocity (9.30) yields

$$\frac{df}{dz} = U_\infty \cos \alpha - \imath U_\infty \sin \alpha, \quad (9.32)$$

so that

$$v_x = U_\infty \cos \alpha \quad \text{and} \quad v_y = U_\infty \sin \alpha. \quad (9.33)$$

On the other hand, the velocity potential and the stream function for the flow are

$$\phi = U_\infty(x \cos \alpha + y \sin \alpha) \quad \text{and} \quad \psi = U_\infty(y \cos \alpha - x \sin \alpha), \quad (9.34)$$

as obtained from taking the real and imaginary parts of $f(z)$, with the latter yielding $y = x \tan \alpha + \text{constant}$ for the equations of the streamlines (straight lines with inclination α with respect to the horizontal line).

The complex potential for a **source** of strength Q centered at the origin is given by

$$f(z) = \frac{Q}{2\pi} \ln(z), \quad (9.35)$$

with $\ln(z) = \ln(r) + \imath\theta$ taken as the principal part corresponding to $0 < \theta < 2\pi$. Taking the derivative of (9.35) yields

$$\frac{df}{dz} = \frac{Q}{2\pi z} = \frac{Q}{2\pi r} e^{-\imath\theta} \quad (9.36)$$

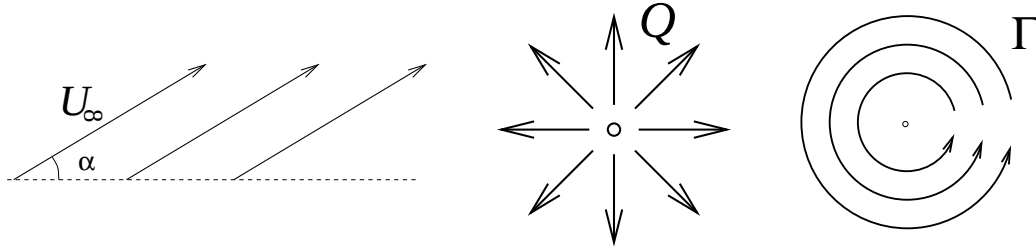


Figure 9.2: Uniform flow, source, and vortex elementary solutions.

for the complex velocity, so that

$$v_r = \frac{Q}{2\pi r} \quad \text{and} \quad v_\theta = 0, \quad (9.37)$$

as follows from comparison with (9.30). The volume flux (per unit length perpendicular to the plane of motion) across a circle of radius r is given by $\int_0^{2\pi} v_r r d\theta = Q$, equal to the strength of the source. It is straightforward to generalize (9.35) for a source centered at $z = z_o$ by writing

$$f(z) = \frac{Q}{2\pi} \ln(z - z_o), \quad (9.38)$$

with Q replaced with $-Q$ to represent a sink.

The complex potential for a **vortex** of strength Γ centered at the origin is given by

$$f(z) = -\frac{\Gamma i}{2\pi} \ln(z), \quad (9.39)$$

with corresponding velocity components

$$v_r = 0 \quad \text{and} \quad v_\theta = \frac{\Gamma}{2\pi r}, \quad (9.40)$$

obtained by derivation. The positive (counterclockwise) circulation associated with this flow can be computed to be $\int_0^{2\pi} v_\theta r d\theta = \Gamma$, equal to the strength of the vortex. Note that for vortices with clockwise (negative) circulation, associated with lifting bodies, one should use a plus sign on the right-hand-side of (9.39). A general vortex of circulation Γ centered at $z = z_o$ is in general given by

$$f(z) = -\frac{\Gamma i}{2\pi} \ln(z - z_o), \quad (9.41)$$

Another elementary solution of interest is the **doublet**. It can be seen as the limiting solution for the flow induced by a source and a sink of equal strengths Q located at an infinitesimally small distance $2a$

$$f(z) = \frac{Q}{2\pi} \ln(z + a) - \frac{Q}{2\pi} \ln(z - a) = \frac{Q}{2\pi} [\ln(1 + a/z) - \ln(1 - a/z)]. \quad (9.42)$$

The solution at distances $z \gg a$ is given in the first approximation by

$$f(z) = \frac{M}{z}, \quad (9.43)$$

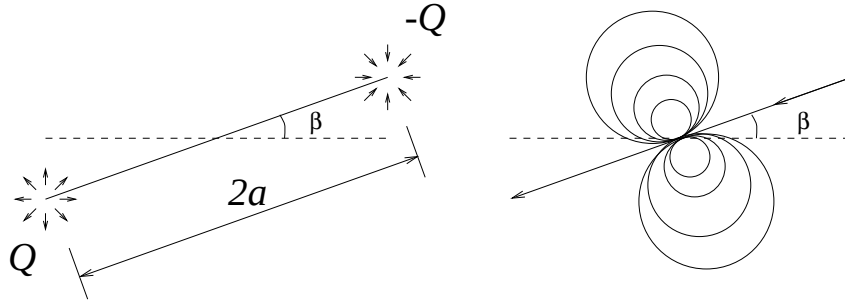


Figure 9.3: The doublet as limiting superposition of a source and a sink.

where $M = Qa/\pi$ is the strength of the doublet. It can be seen that the corresponding streamlines are circles of radius R centered at $z = \pm Ri$. The solution can be generalized to doublets centered at $z = z_o$ with inclination β according to

$$f(z) = \frac{Me^{i\beta}}{z}. \quad (9.44)$$

The reader can verify that the doublet can be also obtained as limiting solution for two close vortices with opposite circulation.

The above elementary analytic functions, summarized in the table below, can be combined linearly to generate more complex flows by superposition. For instance, the complex potential for the flow over a Rankine oval can be obtained by superposing a uniform stream with a source located at $z = -a$ and a sink of equal strength located at $z = a$, yielding $f(z) = U_\infty z + Q \ln(z + a)/(2\pi) - Q \ln(z - a)/(2\pi)$.

Elementary flow	Complex potential $f(z)$
Uniform flow with inclination α	$U_\infty e^{-i\alpha} z$
Source with volume rate Q	$Q \ln(z - z_o)/(2\pi)$
Vortex with circulation Γ	$-i\Gamma \ln(z - z_o)/(2\pi)$
Doublet of strength $Me^{i\beta}$	$Me^{i\beta}/(z - z_o)$

Circular cylinder without circulation

Consider the superposition of a uniform flow of velocity U_∞ and a doublet located at the origin with strength $U_\infty a^2$. The associated complex potential

$$f(z) = U_\infty \left(z + \frac{a^2}{z} \right) \quad (9.45)$$

corresponds to the irrotational flow of a uniform stream about a circular cylinder of radius a . To prove it, note that the stream function $\psi = U_\infty y[1 - a^2/(x^2 + y^2)]$ obtained as the imaginary part of (9.45) vanishes on the cylinder surface, which is therefore a stream line of the flow, that is, the condition of zero normal velocity $\bar{v} \cdot \bar{n} = 0$ is satisfied, whereas far from the body the

velocity approaches the free stream solution $\bar{v} = U_\infty \bar{e}_x$, as can be seen by evaluating the complex velocity

$$\frac{df}{dz} = U_\infty \left(1 - \frac{a^2}{z^2} \right) \quad (9.46)$$

as $z \gg a$. The velocity field, given in cylindrical coordinates by

$$v_r = U_\infty \cos(\theta) \left(1 - \frac{a^2}{r^2} \right) \quad \text{and} \quad v_\theta = -U_\infty \sin(\theta) \left(1 + \frac{a^2}{r^2} \right) \quad (9.47)$$

can be used in (9.13) to give the corresponding pressure distribution

$$p - p_\infty = \frac{1}{2} \rho U_\infty^2 \left(1 - \frac{v_r^2 + v_\theta^2}{U_\infty^2} \right) \quad (9.48)$$

everywhere in the flow field.

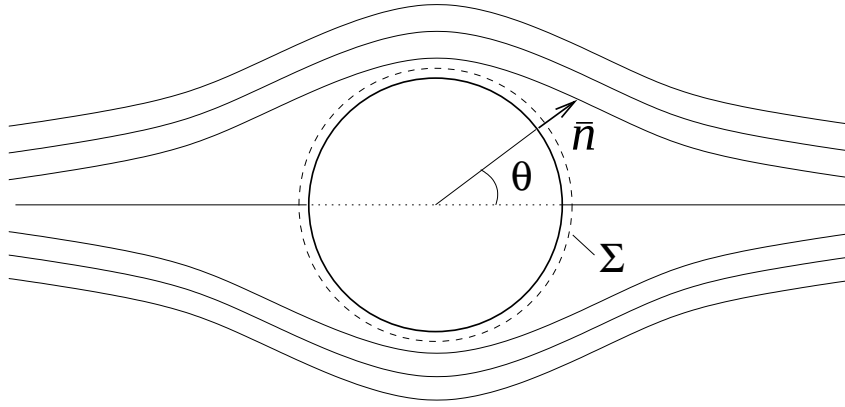


Figure 9.4: Flow about a circular cylinder.

The flow exhibits two stagnation points, located at $z = \pm a$, as can be seen from (9.46). The velocity distribution on the surface of the cylinder is given by

$$v_\theta(\theta) = -2U_\infty \sin(\theta), \quad (9.49)$$

so that the corresponding pressure distribution becomes

$$p - p_\infty = \frac{1}{2} \rho U_\infty^2 [1 - 4 \sin^2(\theta)]. \quad (9.50)$$

For large Reynolds numbers one may neglect viscous forces in the first approximation when computing the force acting on the cylinder $\bar{F} = - \int_\Sigma (p - p_\infty) \bar{n} d\sigma$. The transverse component $L = - \int_\Sigma (p - p_\infty) n_y d\sigma$, called lift, is given by

$$L = - \int_0^{2\pi} (p - p_\infty) \sin(\theta) a d\theta = - \frac{1}{2} \rho U_\infty^2 a \int_0^{2\pi} [1 - 4 \sin^2(\theta)] \sin(\theta) d\theta = 0, \quad (9.51)$$

a result that could have been anticipated in view of the symmetry of the problem. Somewhat more surprising is the result to be obtained for the drag force $D = - \int_\Sigma (p - p_\infty) n_x d\sigma$ (the streamwise component of the force acting on the cylinder), which can be computed according to

$$D = - \int_0^{2\pi} (p - p_\infty) \cos(\theta) a d\theta = - \frac{1}{2} \rho U_\infty^2 a \int_0^{2\pi} [1 - 4 \sin^2(\theta)] \cos(\theta) d\theta = 0. \quad (9.52)$$

This last result, obtained here for the circular cylinder, is a manifestation of the so-called *D'Alembert's paradox*: the steady motion of solid bodies moving in an incompressible ideal fluid results in zero drag. The resolution of this paradox is due to Prandtl, who introduced the concept of boundary layer, as explained below.

Circular cylinder with circulation

The potential solution given in (9.45) is unique only if one imposes the additional condition that the circulation around the cylinder be zero. In other words, the solution

$$f(z) = U_{\infty} \left(z + \frac{a^2}{z} \right) + \frac{i\Gamma}{2\pi} \ln \left(\frac{z}{a} \right), \quad (9.53)$$

obtained by superposing a vortex of strength $-\Gamma$ located at the center of the cylinder, is also a valid solution, in that $r = a$ is still a stream line and one still recovers for $z \gg a$ the free-stream velocity $\bar{v} = U_{\infty} \bar{e}_x$. Although it is not necessary in the following, in defining (9.53) we have added an arbitrary constant $i\Gamma \ln(a)/(2\pi)$, so that the constant value of ψ on the cylinder surface is still given by $\psi = 0$. The complex velocity

$$\frac{df}{dz} = U_{\infty} \left(1 - \frac{a^2}{z^2} \right) + \frac{i\Gamma}{2\pi z}, \quad (9.54)$$

is modified by the presence of the vortex, yielding

$$\frac{z}{a} = \pm(1 - B^2)^{1/2} - iB \quad (9.55)$$

for the location of the stagnation points, where the value of the parameter $B = \Gamma/(4\pi a U_{\infty})$ determines the structure of the flow field according to the schematic shown in Fig. 9.5.

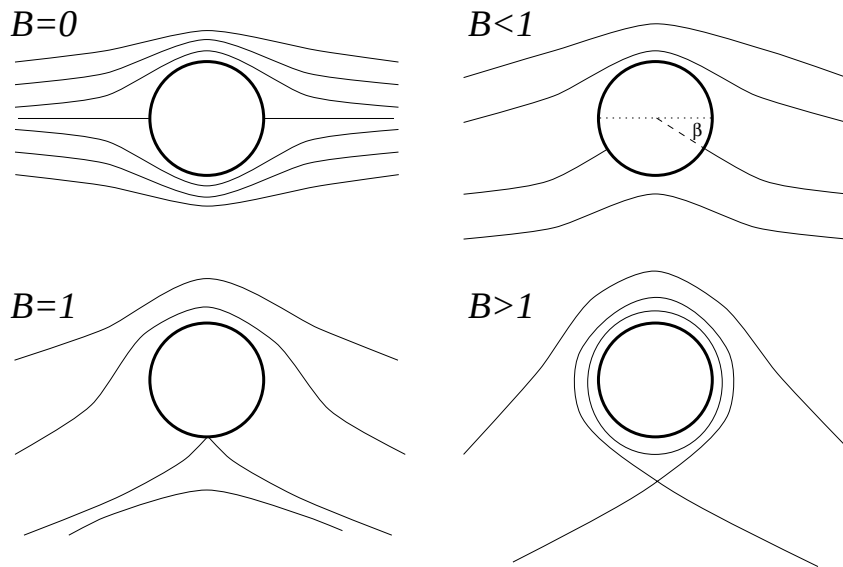


Figure 9.5: Flow about a circular cylinder with circulation.

If $B < 1$ both stagnation points are located on the surface of the cylinder, at $z/a = \pm \cos(\beta) - i \sin(\theta)$, where $\sin \beta = B$. For $B = 1$ both points collapse at a single location $z = -ia$, whereas for $B > 1$ there appears a single stagnation point outside the cylinder, given by $z/a = -(B + (B^2 - 1)^{1/2})$. The modified velocity and pressure distributions on the cylinder are given by

$$v_\theta(\theta) = -2U_\infty \sin(\theta) - \frac{\Gamma}{2\pi a}, \quad (9.56)$$

and

$$p - p_\infty = \frac{1}{2}\rho U_\infty^2 \left[1 - \left(\frac{\Gamma}{2\pi a U_\infty} \right)^2 - 4\sin^2(\theta) - \frac{2\Gamma \sin(\theta)}{\pi a U_\infty} \right]. \quad (9.57)$$

Note that the pressure still exhibits symmetry about the y axis, i.e., $p(\theta) = p(\pi - \theta)$, so that the resulting force has a zero drag component $D = -\int_0^{2\pi} (p - p_\infty) \cos(\theta) a d\theta = 0$. However, the presence of the vortex destroys the symmetry about the x axis, because it accelerates (decelerates) the flow on the upper (lower) half of the cylinder, reducing (increasing) the pressure there. As a result, a nonzero lift force appears, to be computed from the first equation in (9.51) to give the so-called *Kutta-Joukowski formula*

$$L = \rho U_\infty \Gamma. \quad (9.58)$$

The value of the circulation Γ therefore determines the location of the stagnation points and the lift force acting on the cylinder.

Chapter 10

Boundary Layer Flows

Introduction

We have seen previously, in connection with the problem of external flow about a body of characteristic size L_c , that the effects of viscous forces and heat conduction are negligible in most of the flow field when the Reynolds number $\rho UL_c/\mu$ is much larger than unity, with U representing the free-stream velocity and ρ and μ being the characteristic values of the density and viscosity of the fluid. Under such conditions, we say that the flow is inviscid or ideal. We have also seen that, for flow in pipes of characteristic transverse size D much smaller than their length L , the condition for the flow to be ideal involves the product of the Reynolds number and the slenderness ratio according to $(\rho UD/\mu)(D/L) \gg 1$.

Ideal flows are described by integration of the Euler equations, obtained from the complete Navier-Stokes equations by neglecting the terms associated with molecular transport. Since the missing terms involve the only second-order spatial derivatives present in the description, the simplified equations cannot satisfy all of the boundary conditions. As a result, if one wants the solution of the Euler equations to represent adequately a given large-Reynolds-number flow, we need to consider the existence of discontinuity surfaces (boundary layers, vortex sheets, and mixing layers) across which the temperature and/or velocity undergo a discontinuous jump.

For instance, when the flow is limited by a solid wall, in integrating the Euler equations we replace the non-slip condition $\bar{v} = 0$ with the condition of zero normal velocity component $\bar{v} \cdot \bar{n} = 0$. Hence, the resulting solution has in general a nonzero component tangent to the wall v_e . Since the velocity on the wall surface has to vanish, for the solution of the Euler equations to provide a realistic representation of the flow we have to permit the existence of a discontinuity surface on the wall, across which the velocity suddenly jumps from the external value v_e determined from the solution of the Euler equations to zero. In reality, the surface is rather a layer of small characteristic thickness δ , such that $\delta \ll D$ for flow in pipes and $\delta \ll L_c$ for flow around bodies. Inside this **boundary layer** viscosity has a nonnegligible effect and cause the velocity to evolve continuously from the external value v_e to vanish at the wall.

Boundary-layer characteristics

Consider as an introductory example the flow at the entrance of the aerodynamic intake of characteristic size L shown in Fig. 10.1, whose lower wall is assumed to be planar. For simplicity, we shall consider the fluid to be incompressible, an excellent approximation when the Mach

number of the incoming stream is much smaller than unity. The Reynolds number of the flow $Re = \rho U_\infty L / \mu$ is constructed from the characteristic flow-field size L , the fluid density and viscosity- assumed to be constant- and the characteristic value of the velocity- taken to be that of the approaching stream U_∞ . In typical applications, $Re \gg 1$, and the flow can be assumed to be ideal in the first approximation. Integration of the Euler equations, including a condition of zero normal velocity on the walls, provides the velocity and pressure fields everywhere, including a nonzero tangential component on the walls, given by $v_e(x, t)$ on the flat lower wall (with $v_e(x, t) \sim U_\infty$), where the accompanying pressure distribution is given $p_e(x, t)$. The plane wall is a stream line for the Euler flow, along which the momentum equation reduces to

$$\rho \frac{\partial v_e}{\partial t} + \rho v_e \frac{\partial v_e}{\partial x} = - \frac{\partial p_e}{\partial x}. \quad (10.1)$$

This equation serves to relate the variations of p_e and v_e , and simplifies to Bernoulli's equation

$$\frac{d}{dx}(p_e + \rho v_e^2 / 2) = 0 \quad (10.2)$$

when the motion is steady.

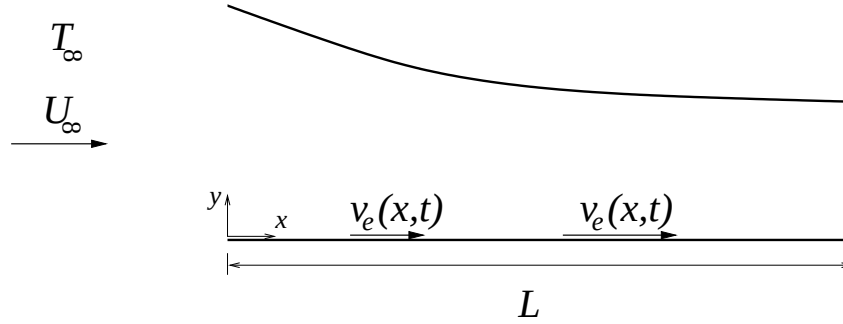


Figure 10.1: Flow at the entrance of an aerodynamic intake.

As previously anticipated, in the immediate vicinity of the wall there exists a thin layer - termed boundary layer- where the effect of viscosity is important. Since the thickness δ of this viscous layer is seen to be very small compared with L , with the scales of the outer flow, the boundary layer is seen in the first approximation as a discontinuity surface. To describe the continuous evolution of the velocity within the boundary layer from the value v_e found at the outer edge of the boundary layer to vanish at the wall, we need to write the conservation equations that govern the viscous solution near the wall, a development that provides the simplified form of these equations along with the characteristic scales of the boundary-layer flow, as shown below. Using the cartesian coordinate system indicated in Fig. 10.1, the continuity and momentum equations take the form

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \quad (10.3)$$

$$\frac{U_\infty}{L} \quad \frac{V}{\delta}$$

$$\rho \frac{\partial v_x}{\partial t} + \rho v_x \frac{\partial v_x}{\partial x} + \rho v_y \frac{\partial v_x}{\partial y} = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} \right) \quad (10.4)$$

$$\begin{array}{cccccc} \rho \frac{U_\infty}{t_o} & \rho \frac{U_\infty^2}{L} & \rho \frac{U_\infty V}{\delta} & \frac{\Delta_x p}{L} & \mu \frac{U_\infty}{L^2} & \mu \frac{U_\infty}{\delta^2} \end{array}$$

$$\rho \frac{\partial v_y}{\partial t} + \rho v_x \frac{\partial v_y}{\partial x} + \rho v_y \frac{\partial v_y}{\partial y} = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} \right) \quad (10.5)$$

$$\begin{array}{cccccc} \rho \frac{V}{t_o} & \rho \frac{U_\infty V}{L} & \rho \frac{V^2}{\delta} & \frac{\Delta_y p}{\delta} & \mu \frac{V}{L^2} & \mu \frac{V}{\delta^2} \end{array}$$

We are writing below each term its anticipated order of magnitude, with δ and V representing the unknown characteristic values of the boundary-layer thickness and transverse velocity. Since the velocity v_e on the outer edge of the boundary layer is of order U_∞ , the longitudinal velocity v_x within the boundary and its variations can be expected to be also of order U_∞ . If the velocity of the approaching stream varies with time, the resulting solution will be unsteady, so that $v_e(x, t)$. The characteristic time t_o for variation of U_∞ will be also the characteristic time scale for the time derivatives of the boundary-layer flow, as indicated above.

The order-of-magnitude analysis of the different terms indicates that:

- The transverse velocity in the boundary layer is given by

$$V \sim (\delta/L)U_\infty, \quad (10.6)$$

as can be obtained from (10.3). Anticipating that $\delta \ll L$, we conclude that $v_y \sim V \ll U_\infty$

- The relative importance of the unsteady term depends on the Strouhal number of the flow $St = (L/U_\infty)/t_o$. In particular, for flows with $St \ll 1$ the flow will be quasisteady.
- The longitudinal derivatives give a negligible contribution to the viscous forces because, if $\delta \ll L$, then

$$\frac{\partial^2 v_x}{\partial x^2} \ll \frac{\partial^2 v_x}{\partial y^2} \quad \text{and} \quad \frac{\partial^2 v_y}{\partial x^2} \ll \frac{\partial^2 v_y}{\partial y^2} \quad (10.7)$$

- The boundary-layer thickness is given by

$$\delta = \sqrt{\frac{\mu L}{\rho U_\infty}} = Re^{-1/2} L \ll L, \quad (10.8)$$

as follows from the comparison of convective and viscous terms in either component of the momentum equation

- The longitudinal pressure variations along the boundary layer are of order

$$\Delta_x p \sim \rho U_\infty^2, \quad (10.9)$$

whereas the transverse variations are much smaller, of order

$$\Delta_y p \sim (\delta/L)^2 \rho U_\infty^2 \sim Re^{-1} \Delta_x p, \quad (10.10)$$

as follows from (10.5).

This last observation leads to a significant simplification of the boundary-layer description. Since $\Delta_y p$ is negligibly small, to compute the velocity field one may neglect completely the small transverse variations of the pressure, which amounts to replacing in the integration the conservation equation (10.5) with the condition

$$\frac{\partial p}{\partial y} = 0, \quad (10.11)$$

so that in the first approximation $p(x, y, t) = p_e(x, t)$.

The fact that $\Delta_y p$ is negligibly small is an important result with practical implications. In particular, it means that the pressure felt on the wall is that existing just outside the boundary layer, which can be computed from integration of the Euler equations. As a result, if the boundary layer remains attached to the wall, as occurs in aerodynamic bodies when their shape is correctly designed, the force acting on the body can be computed in the first approximation according to $\bar{F}_p = - \int_{\Sigma} p_e \bar{n} d\sigma$. The solution of the boundary-layer flow would be needed to guarantee that the boundary layer remains attached, and also to compute the viscous force acting on the wall and the heat conducted from the external stream to the body surface.

Equations and boundary conditions

To resolve the velocity field in the boundary layer one needs to integrate the continuity equation

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \quad (10.12)$$

together with the momentum equation

$$\rho \frac{\partial v_x}{\partial t} + \rho v_x \frac{\partial v_x}{\partial x} + \rho v_y \frac{\partial v_x}{\partial y} = - \frac{\partial p_e}{\partial x} + \mu \frac{\partial^2 v_x}{\partial y^2}, \quad (10.13)$$

written with the simplification $p(x, y, t) = p_e(x, t)$ and with the longitudinal derivatives neglected in the viscous force. Note that, once the velocity field is computed by integrating the above two equations, one could make use of (10.5) to determine the small pressure variations appearing across the boundary layer.

If the problem is unsteady, as occurs in the example of Fig. 10.1 when the velocity $U_{\infty}(t)$ of the outer stream changes with time, one needs to provide the initial velocity distribution in the boundary layer, i.e., $v_x = v_{x_i}(x, y)$ at $t = 0$. The boundary conditions for the problem include the nonslip condition on the wall surface and the external velocity in the outer inviscid stream found outside the boundary layer

$$\begin{aligned} y = 0 : \quad & v_x = v_y = 0 \\ y \rightarrow \infty (y \gg \delta) : \quad & v_x = v_e(x, t) \end{aligned} \quad (10.14)$$

for $x > 0$. Besides, we need to specify the longitudinal velocity distribution $v_x(y, t)$ at $x = 0$, where the boundary layer begins to develop, obtained from the solution of the Euler equations. Note that, since v_x approaches v_e when $y \gg \delta$, the momentum equation (10.13) simplifies to (10.1) in the external stream, as corresponds to the projection of Euler momentum equation along a line parallel to the wall.

Also of interest is that, since in the boundary-layer equations the only derivative of the transverse velocity component is $\partial v_y / \partial y$, we can only impose the value of v_y at $y = 0$, whereas the transverse

velocity outside the boundary layer $v_y(y \rightarrow \infty) = v_{y\infty}$ is determined as part of the integration. According to the estimate given above in (10.6), this velocity component is much smaller than the streamwise component $v_e \sim U_\infty$, and therefore constitutes a small correction to the outer velocity field, obtained from the Euler equations. From the view point of the external flow, the effect of the boundary layer amounts to a small displacement of the wall, so that the velocity is not parallel to $y = 0$ but exhibits instead a small transverse component. Since $v_{y\infty} \ll U_\infty$, this is a higher-order effect that can be neglected in the first approximation when integrating the Euler equations for the outer inviscid problem.

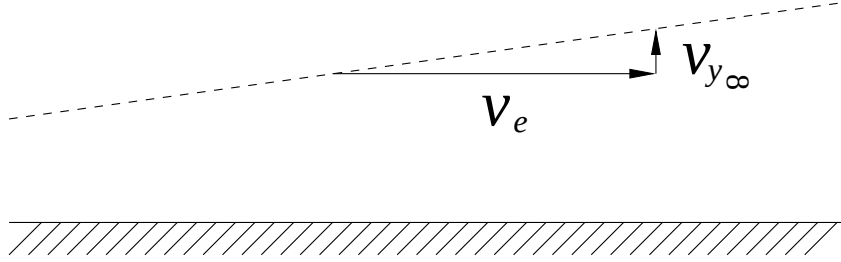


Figure 10.2: Velocity components at the edge of the boundary layer.

Boundary-layer thickness

At a given location x , the integration of the boundary-layer equations provides the velocity distribution v_x across the boundary layer, giving a profile that increases from the value $v_x = 0$ at $y = 0$ to approach asymptotically the value $v_x = v_e$ as $y \rightarrow \infty$. The resulting boundary-layer thickness is not uniquely defined from this result. One could for instance define δ as the transverse location where the velocity reaches $v_x = 0.95v_e$ or $v_x = 0.99v_e$. This arbitrariness can be removed by introducing definitions for the boundary-layer thickness that are directly related to physical quantities of interest, namely, mass and momentum flux.

Thus, we define the **displacement thickness** δ^* as the distance that we need to displace the wall towards the fluid for the mass flux of the unperturbed outer stream to be that found in reality in the presence of the boundary layer, as sketched in Fig. 10.3. For an incompressible fluid, this condition reduces to

$$\int_{\delta^*}^{\infty} v_e dy = \int_0^{\infty} v_x dy$$

which leads to

$$\delta^* = \int_0^{\infty} \left(1 - \frac{v_x}{v_e}\right) dy. \quad (10.15)$$

Similarly, we define the **momentum thickness** θ as the additional displacement required for the momentum flux of the unperturbed outer stream to equal that associated with the boundary layer, according to

$$\int_{\delta^*+\theta}^{\infty} v_e^2 dy = \int_0^{\infty} v_x^2 dy.$$

Using (10.15) and solving for θ yields the explicit expression

$$\theta = \int_0^\infty \frac{v_x}{v_e} \left(1 - \frac{v_x}{v_e}\right) dy. \quad (10.16)$$

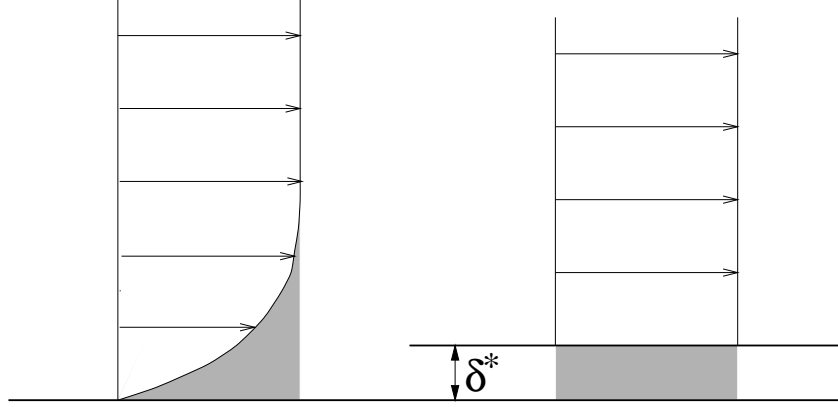


Figure 10.3: Displacement thickness.

Blasius solution

The incompressible boundary-layer problem presented above has simple solutions in just a few cases, one of them being the steady boundary layer developing over a semi-infinite flat plate aligned with an incident uniform stream of velocity U_∞ . In the absence of viscosity, the presence of the infinitely thin plate does not perturb the flow. In other words, the solution of the incompressible Euler equations subject to the boundary conditions $\bar{v} = U_\infty \bar{e}_x$ far from the wall and $\bar{v} \cdot \bar{n} = v_y = 0$ at $y = 0$ for $x > 0$ reduces to $\bar{v} = U_\infty \bar{e}_x$ and $p = p_\infty$, yielding $v_e = U_\infty$ and $p_e = p_\infty$ for the solution outside the boundary layer. The corresponding equation that describe the development of the boundary layer reduce in this case to

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \quad (10.17)$$

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \frac{\partial^2 v_x}{\partial y^2} \quad (10.18)$$

with boundary conditions

$$\begin{aligned} x = 0, \quad 0 < y < \infty : v_x &= U_\infty \\ x > 0 : \quad \begin{cases} y = 0 : v_x = v_y = 0 \\ y \rightarrow \infty : v_x = U_\infty. \end{cases} \end{aligned} \quad (10.19)$$

Because of the absence of length scales, the problem admits a selfsimilar solution, to be described by scaling the distance to the wall with the characteristic value of the boundary-layer thickness

$$\delta(x) \sim \sqrt{\frac{\nu x}{U_\infty}}, \quad (10.20)$$

obtained at a given downstream location x from the condition that inertial and viscous forces are comparable in the boundary layer. Introducing this result into the continuity equation yields

$$v_y \sim U_\infty \frac{\delta}{x} \sim \sqrt{\frac{\nu U_\infty}{x}} \quad (10.21)$$

as an estimate for the transverse velocity component.

The problem can be described by introducing a selfsimilar coordinate

$$\eta = \frac{y}{(\nu x / U_\infty)^{1/2}} \quad (10.22)$$

along with a selfsimilar stream function

$$\psi = (\nu U_\infty x)^{1/2} F(\eta), \quad (10.23)$$

defined such that

$$v_x = U_\infty F'(\eta) \quad \text{and} \quad v_y = \frac{1}{2} \sqrt{\frac{\nu U_\infty}{x}} [\eta F'(\eta) - F(\eta)], \quad (10.24)$$

where the prime denotes here differentiation with respect to the variable η . In terms of these variables, the problem reduces to that of integrating the so-called Blasius problem

$$F''' + \frac{1}{2} F F'' = 0, \quad F(0) = F'(0) = F'(\infty) - 1 = 0. \quad (10.25)$$

whose solution is represented below in Fig. 10.4. The numerical integration provides, in particular, the numerical values of

$$F''(0) = 0.332 \quad \text{and} \quad [\eta - F(\eta)]_{\eta \rightarrow \infty} = 1.721 \quad (10.26)$$

to be used below.

For the Blasius boundary layer, the displacement thickness can be computed according to

$$\delta^*(x) = \int_0^\infty \left(1 - \frac{v_x}{U_\infty}\right) dy = \sqrt{\frac{\nu x}{U_\infty}} \int_0^\infty (1 - F') d\eta = \sqrt{\frac{\nu x}{U_\infty}} [\eta - F(\eta)]_{\eta \rightarrow \infty} = 1.721 \sqrt{\frac{\nu x}{U_\infty}}, \quad (10.27)$$

while the momentum thickness is given by

$$\theta(x) = \int_0^\infty \frac{v_x}{U_\infty} \left(1 - \frac{v_x}{U_\infty}\right) dy = \sqrt{\frac{\nu x}{U_\infty}} \int_0^\infty F'(1 - F') d\eta = \sqrt{\frac{\nu x}{U_\infty}} 2F''(0) = 0.664 \sqrt{\frac{\nu x}{U_\infty}}. \quad (10.28)$$

On the other hand, the velocity gradient on the plate is

$$\left. \frac{\partial v_x}{\partial y} \right|_{y=0} = U_\infty \sqrt{\frac{U_\infty}{\nu x}} F''(0) = 0.332 U_\infty \sqrt{\frac{U_\infty}{\nu x}}. \quad (10.29)$$

The only force acting on the plate is that due to viscous stresses

$$\tau_w = \mu \left. \frac{\partial v_x}{\partial y} \right|_{y=0} = 0.332 \mu U_\infty \sqrt{\frac{U_\infty}{\nu x}}, \quad (10.30)$$

yielding

$$F_x = \int_0^x \tau_w dx = 0.664 \mu U_\infty \sqrt{\frac{U_\infty x}{\nu}} \quad (10.31)$$

for the force acting on the plate stretch $0 - x$.

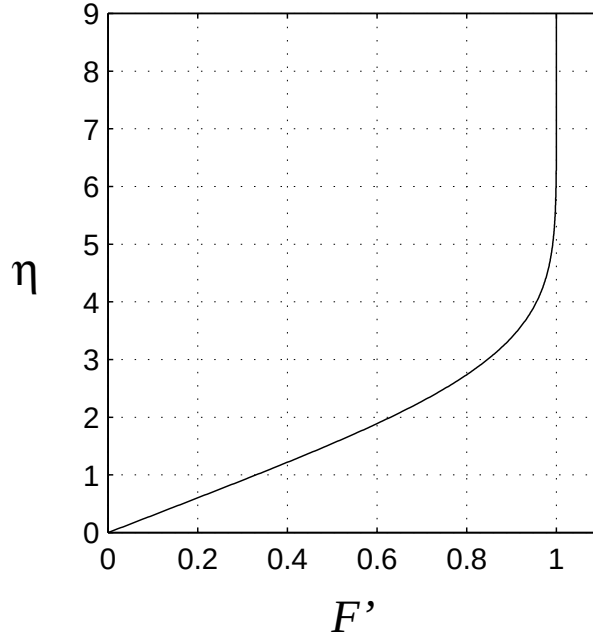


Figure 10.4: Blasius solution for the boundary-layer over a flat plate.

Von Karman boundary-layer integral equation

Besides the case of constant external velocity $v_e = U_\infty$ corresponding to Blasius flow over a flat plate, there exist only a few exact solutions of the boundary-layer problem. In general, numerical computation is needed to integrate the boundary-layer equations (10.12) and (10.13). Simplified methods aimed at providing approximate solutions have been derived over the years. Most of the methods are based on the integral evolution equation developed by Theodore Von-Karman, which is presented below for the case of steady incompressible boundary layers.

The development begins by writing the steady boundary-layer momentum equations with use made of $dp_e/dx = -\rho v_e dv_e/dx$ in the form

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = v_e \frac{\partial v_e}{\partial x} + \nu \frac{\partial^2 v_x}{\partial y^2} \quad (10.32)$$

and integrating across the boundary layer to give

$$\int_0^\infty \left(v_x \frac{\partial v_x}{\partial x} - v_e \frac{\partial v_e}{\partial x} + v_y \frac{\partial v_x}{\partial y} \right) dy = \nu \int_0^\infty \frac{\partial^2 v_x}{\partial y^2} dy. \quad (10.33)$$

Integration of the viscous forces yields

$$\nu \int_0^\infty \frac{\partial^2 v_x}{\partial y^2} dy = \nu \left. \frac{\partial v_x}{\partial y} \right|_{y=0}^{y=\infty} = -\nu \left. \frac{\partial v_x}{\partial y} \right|_{y=0} = -\frac{\tau_w}{\rho}$$

where $\tau_w(x)$ denotes the value of the viscous stress on the wall. The transverse convection term is integrated by parts with use made of the continuity equation

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \quad (10.34)$$

to give

$$\int_0^\infty v_y \frac{\partial v_x}{\partial y} dy = v_x v_y \Big|_{y=0}^{y=\infty} - \int_0^\infty v_x \frac{\partial v_y}{\partial y} dy = v_x v_y \Big|_{y=\infty} + \int_0^\infty v_x \frac{\partial v_x}{\partial x} dy.$$

The transverse velocity component v_y can be expressed by integration of the continuity equation (10.34) in the form

$$v_y(y) = v_w - \int_0^y \frac{\partial v_x}{\partial x} dy.$$

To enable the description of boundary layers developing over porous walls with suction or blowing, we allow for a nonzero value of the transverse velocity at the wall $v_y(y=0) = v_w$, yielding

$$v_x v_y \Big|_{y=\infty} = v_e v_w - v_e \int_0^\infty \frac{\partial v_x}{\partial x} dy$$

so that

$$\int_0^\infty v_y \frac{\partial v_x}{\partial y} dy = v_e v_w - \int_0^\infty (v_e - v_x) \frac{\partial v_x}{\partial x} dy.$$

Collecting the different terms appearing in (10.33) and changing the sign leads to

$$\int_0^\infty \left(v_e \frac{\partial v_e}{\partial x} - v_x \frac{\partial v_x}{\partial x} + (v_e - v_x) \frac{\partial v_x}{\partial x} \right) dy = \frac{\tau_w}{\rho} + v_e v_w \quad (10.35)$$

Writing the left-hand-side integrals with use made of the definitions given above for the displacement thickness and momentum thickness

$$v_e \delta^* = \int_0^\infty (v_e - v_x) dy$$

$$v_e^2 \theta(x) = \int_0^\infty v_x (v_e - v_x) dy$$

finally yields the so-called Von-Karman integral equation

$$\frac{\partial}{\partial x} (v_e^2 \theta) + v_e \frac{\partial v_e}{\partial x} \delta^* = \frac{\tau_w}{\rho} + v_e v_w. \quad (10.36)$$

It is important to note that the above equation contains no additional simplifications beyond those already present in (10.12) and (10.13). The velocity profile $v_x(x, y)$, not present explicitly, enters in the equation through the values of δ^* and θ . To use (10.36) in predicting the streamwise evolution of the boundary layer, we need to introduce a presumed shape for the velocity profile, as shown in the following section.

The Karman-Pohlhausen method

To use (10.36) one may begin by assuming that the velocity profile has a selfsimilar shape defined by the function

$$\frac{v_x}{v_e(x)} = f(\zeta), \quad (10.37)$$

where $\zeta = y/\delta(x)$ and $\delta(x)$ is a function representing the boundary-layer thickness. In terms of this function f one may easily calculate the values of $\tau_w = C_1 \mu v_e / \delta$, $\theta = C_2 \delta$ and $\delta^* = C_3 \delta$,

involving constant factors $C_1 = (df/d\zeta)_{\zeta=0}$, $C_2 = \int_0^\infty f(1-f)d\zeta$ and $C_3 = \int_0^\infty (1-f)d\zeta$. Introducing the above expressions into (10.36) yields

$$C_2 \frac{\partial}{\partial x} (v_e^2 \delta) + C_3 v_e \frac{\partial v_e}{\partial x} \delta = C_1 \nu v_e / \delta + v_e v_w \quad (10.38)$$

which determines the evolution of $\delta(x)$ from a given initial value.

The accuracy of the result depends on the selection of the function $f(\zeta)$. Realistic profiles should satisfy a number of conditions stemming for instance from:

1. Nonslip condition on the wall surface,

$$v_x(y=0) = 0$$

2. Momentum equation (10.32) evaluated at $y = 0$,

$$v_w \frac{\partial v_x}{\partial y} \Big|_{y=0} = v_e \frac{\partial v_e}{\partial x} + \nu \frac{\partial^2 v_x}{\partial y^2} \Big|_{y=0}$$

3. Matching with the external solution outside the boundary layer

$$v_x - v_e = \frac{\partial v_x}{\partial y} = \frac{\partial^2 v_x}{\partial y^2} = \frac{\partial^3 v_x}{\partial y^3} = \dots = 0$$

as $y \gg \delta$ (or sometimes at $y = \delta$, if δ is taken as the exact value of the boundary-layer thickness).

As an example, consider the selfsimilar velocity profile given by the third-order polynomial

$$\frac{v_x}{v_e} = a + b\zeta + c\zeta^2 + d\zeta^3,$$

for $y \leq \delta$ ($\zeta \leq 1$) and $v_x/v_e = 1$ for $y > \delta$ ($\zeta > 1$). To select the values of the constants a , b , c and d we may choose to satisfy the two conditions given above at the wall surface together with the first two conditions associated with matching with the outer profile. If, for instance, the external velocity is constant and $v_w = 0$, as occurs for Blasius flow, one obtains

$$\frac{v_x}{U_\infty} = \frac{3}{2}\zeta - \frac{1}{2}\zeta^3,$$

which can be used to compute

$$\delta^* = \delta \int_0^1 \left(1 - \frac{v_x}{U_\infty}\right) d\zeta = \frac{3}{8}\delta,$$

$$\theta = \delta \int_0^1 \frac{v_x}{U_\infty} \left(1 - \frac{v_x}{U_\infty}\right) d\zeta = \frac{39}{280}\delta,$$

and

$$\frac{\tau_w}{\rho} = \frac{\nu}{\delta} \frac{dv_x}{d\zeta} \Big|_{\zeta=0} = \frac{3}{2} \frac{\nu U_\infty}{\delta}.$$

Introducing these expressions into (10.36) leads to the equation

$$\frac{39}{280}U_\infty^2 \frac{d\delta}{dx} = \frac{3}{2} \frac{\nu U_\infty}{\delta},$$

which can be integrated with initial condition $\delta(x=0) = 0$ to give

$$\delta(x) = \sqrt{\frac{280}{13}} \sqrt{\frac{\nu x}{U_\infty}} = 4.64 \sqrt{\frac{\nu x}{U_\infty}}$$

for the evolution of the boundary-layer thickness. Use of this last equation provides

$$\delta^* = 1.74 \sqrt{\frac{\nu x}{U_\infty}}, \quad \theta = 0.646 \sqrt{\frac{\nu x}{U_\infty}}, \quad \text{and} \quad \tau_w = 0.323 \mu U_\infty \sqrt{\frac{U_\infty}{\nu x}}$$

to be compared with the exact results given above in (10.27), (10.28), and (10.30).

Boundary-layer separation

The velocity profile in the mixing layer is associated with a concentrated vorticity distribution, given by $\omega_z \simeq -\partial v_x / \partial y$. Therefore, the validity of the potential flow solution for the flow about a solid body, studied in the previous chapter, requires that the boundary layer remains attached to the wall, which occurs for instance in airfoils at moderately small values of the angle of attack. In blunt bodies, such as the circular cylinder studied previously, the boundary layer separates. A vortex sheet of concentrated vorticity emerges from the separation point, thereby invalidating the potential flow solution. The resulting wake is typically unsteady, which further complicates the computation of the resulting flow. Some general comments about the sequence of events leading to separation and the associated required conditions are given below.

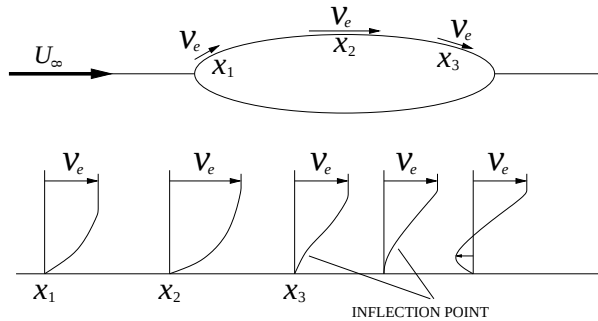


Figure 10.5: Flow over a blunt body.

Consider the potential solution for the flow about a blunt body, sketched in Fig. 10.5, along with the resulting boundary-layer velocity profiles. According to Bernoulli's equation, the peak pressure on the body surface is reached at the front and rear stagnation points. Consider the evolution as one moves along the surface for increasing values of the distance x from the forward stagnation point. As the velocity v_e increases, the corresponding pressure decreases, producing a favorable pressure gradient ($dp_e/dx < 0$) at the x_1 location in the figure. The maximum velocity and, therefore, the minimum pressure are reached at a given point x_2 , and for $x > x_2$

the potential flow decelerates, giving rise to a adverse pressure gradient, which acts, for instance, at x_3 .

To understand how the pressure gradient interacts with the boundary layer and may lead to its separation, we need to recall that the transverse pressure variation across the boundary layer is negligible, so that all fluid particles in the mixing layer, regardless of the distance from the wall, are subject to the same value of the pressure gradient dp_e/dx . For $x < x_2$, the velocity across the boundary layer increases due to the favorable pressure gradient. Conversely, the flow decelerates for $x > x_2$ under the action of the adverse pressure gradient. Although the pressure gradient acts uniformly across the boundary layer, the resulting velocity decrement

$$\Delta v_x \sim \frac{\Delta_x p_e}{\rho v_x}. \quad (10.39)$$

is larger where the velocity is smaller, as can be seen from (10.13). Therefore, the fluid near the wall is very sensitive to the adverse pressure gradient, which reduces progressively the velocity gradient $\partial v_x / \partial y$ at the wall. If the adverse pressure gradient acts for a sufficiently long time, the velocity in the wall vicinity ends up being negative, generating a region of reverse flow, indicative of the separation of the boundary layer. Once separation takes place, the boundary-layer solution no longer represents accurately the velocity near the wall. The separated boundary layer becomes a vortex sheet that modifies the outer solution, which is no longer potential.

The condition of zero shear stress $\partial v_x / \partial y = 0$ at the wall determines the location of the separation point. When it occurs, the associated velocity profile includes necessarily an inflection point at a given distance from the wall, separating a near-wall region where $\partial^2 v_x / \partial y^2 > 0$ from a far region where $\partial^2 v_x / \partial y^2 < 0$. Since $\partial^2 v_x / \partial y^2 > 0$ at the wall, where the momentum equation (10.13) reduces to

$$\frac{\partial p_e}{\partial x} = \mu \frac{\partial^2 v_x}{\partial y^2}, \quad (10.40)$$

it follows that separation may only occur in regions where an adverse pressure gradient $\partial p_e / \partial x > 0$ exists, in agreement with the physical arguments set forth previously. The design of aerodynamic bodies takes this finding into account by seeking body shapes such that the maximum velocity on the body surface is reached as far downstream as possible, so as to reduce the spatial extent of the region of adverse pressure gradient. The computation may combine potential flow theory, to determine the distributions of v_e and p_e over the body, with integrations of the boundary-layer equations (10.12) and (10.13), to predict whether the resulting boundary layer actually remains attached. Boundary-layer separation is indicated in the integrations of the parabolic problem (10.12) and (10.13) by the development of a singularity in the solution, including an infinitely large transverse velocity at a given x location, a behavior associated with the appearance of reverse flow, shown in Fig. 10.5.

Thwaites method

An approximate simple predictive method for boundary-layer separation was derived by Thwaites (1949). The development begins by introducing the non-dimensional factors

$$H = \frac{\delta^*}{\theta} \quad \text{and} \quad T = \frac{\tau_w}{\mu v_e / \theta}, \quad (10.41)$$

which depend on the shape of the boundary-layer velocity profile, giving for instance $H = 2.59$ and $T = 0.220$ for Blasius solution. Rewriting the integral equation (10.36) in terms of the these non-dimensional factors gives

$$v_e \frac{d}{dx} \left(\frac{\theta^2}{\nu} \right) + 2(2 + H) \frac{dv_e}{dx} \frac{\theta^2}{\nu} - 2T = 0, \quad (10.42)$$

for the case of a non-permeable wall with $v_w = 0$. By collecting a large number of numerical and experimental data, Thwaites demonstrated that the factors H and T can be expressed in general as a function of the parameter

$$\lambda = \frac{dv_e}{dx} \frac{\theta^2}{\nu}, \quad (10.43)$$

with resulting correlations, shown in Fig. 10.6, given by

$$T = (\lambda + 0.09)^{0.62} \quad (10.44)$$

and

$$2T - 2(H + 2) \frac{dv_e}{dx} \frac{\theta^2}{\nu} = 2T - 2(H + 2)\lambda = 0.45 - 6.0\lambda \quad (10.45)$$

Note in particular that, according to (10.44), boundary-layer separation ($\tau_w = 0$) occurs for $\lambda = -0.09$, a negative value that can be seen from (10.43) to be associated with adverse pressure gradients, i.e., $dv_e/dx < 0$.

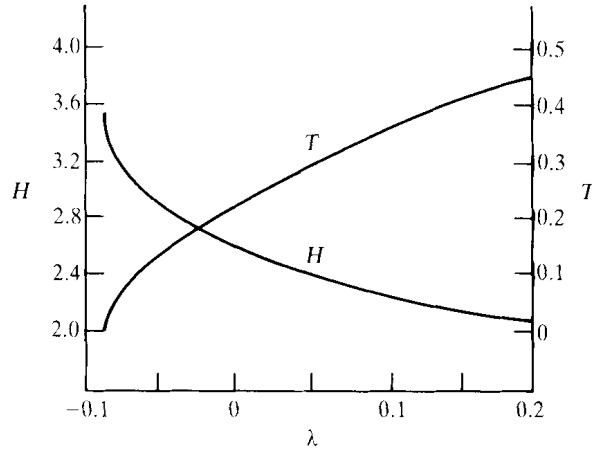


Figure 10.6: The variation with λ of Thwaites' parameters H and T .

Substituting (10.45) into (10.42) and integrating with initial condition $\theta = \theta_o$ at $x = x_o$ gives

$$\frac{\theta^2}{\nu} = \frac{\theta_o^2}{\nu} \left[\frac{v_e(x_o)}{v_e(x)} \right]^6 + \frac{0.45}{v_e^6(x)} \int_{x_o}^x v_e^5(\bar{x}) d\bar{x} \quad (10.46)$$

for the variation of $\theta(x)$ in terms of the distribution of external velocity $v_e(x)$. Once $\theta(x)$ is computed, the value of λ follows from (10.43), which can be used in (10.44) and (10.45) to give T and H , with corresponding values of δ^* and τ_w obtained from the definitions (10.41).

The use of Euler equations to determine $v_e(x)$ together with (10.43) and (10.46) provides a simple method to compute the evolution of $\lambda(x)$ as the boundary layer develops. If the resulting value of λ is everywhere larger than the critical value $\lambda = -0.09$, the boundary layer can be expected to remain attached, whereas if λ falls below this critical value at a given location, separation would occur. This approximate criterion for boundary-layer separation is subject to the validity of the experimental correlations (10.45) y (10.44). Note that boundary-layer separation would affect the external flow, so that, when separation occurs, the actual separation point does not necessarily coincide with that predicted by Thwaites method when the potential-flow solution is used for $v_e(x)$.

For an illustrative application of the method, consider the external-velocity distribution

$$\frac{v_e(x)}{U_o} = \left(\frac{x}{x_o} \right)^\alpha, \quad (10.47)$$

such that

$$\frac{dv_e(x)}{dx} = \alpha \frac{U_o}{x_o} \left(\frac{x}{x_o} \right)^{\alpha-1}. \quad (10.48)$$

This family of potential solutions include favorable pressure gradients ($\alpha > 0$), negative pressure gradients ($\alpha < 0$), and zero pressure gradients ($\alpha = 0$), the latter corresponding to the case of constant external velocity found in Blasius solution over a flat plate. Making use of (10.43) and (10.46) leads to

$$\frac{\theta^2}{\nu} = \frac{x_o}{U_o} \left(\frac{x}{x_o} \right)^{1-\alpha} \left[\frac{0.45}{1+5\alpha} + \left(\frac{U_o \theta_o^2}{\nu x_o} - \frac{0.45}{1+5\alpha} \right) \left(\frac{x}{x_o} \right)^{-(1+5\alpha)} \right] \quad (10.49)$$

and

$$\lambda = \alpha \left[\frac{0.45}{1+5\alpha} + \left(\frac{U_o \theta_o^2}{\nu x_o} - \frac{0.45}{1+5\alpha} \right) \left(\frac{x}{x_o} \right)^{-(1+5\alpha)} \right]. \quad (10.50)$$

When $\alpha > -0.2$ (either favorable or slightly adverse pressure gradients), the solution becomes independent of the initial conditions for $x \gg x_o$, where the above expressions reduce to

$$\frac{\theta^2}{\nu} = \frac{0.45}{1+5\alpha} \frac{x_o}{U_o} \left(\frac{x}{x_o} \right)^{1-\alpha} \quad (10.51)$$

and

$$\lambda = \frac{0.45\alpha}{1+5\alpha}. \quad (10.52)$$

Different cases of practical interest can be considered within the range of validity ($\alpha > -0.2$) of these two expressions.

- If $\alpha > 1$ (increasing velocity gradients dv_e/dx), the boundary-layer thickness decreases with x . This is seen to occur, for instance, in boundary layers over walls of strongly convergent pipes.
- If $\alpha = 1$ (constant velocity gradients dv_e/dx) the boundary-layer thickness remains constant. This is seen to occur in the vicinity of stagnation points, where the outer velocity is given by $v_e = Ax$ in terms of the strain rate A .

- For $\alpha < 1$ the boundary layer thickens as it develops downstream. For $\alpha = 0$, the case corresponding to uniform flow over a semi-infinite flat plate, one obtains

$$\frac{\theta^2}{\nu} = \frac{0.45x}{U_o}$$

and

$$\lambda = 0.$$

The latter can be used in (10.44) and (10.45) to yield $T = 0.225$ and $H = 2.55$, so that

$$\delta^* = 1.71\sqrt{\frac{\nu x}{U_o}}, \quad \theta = 0.671\sqrt{\frac{\nu x}{U_o}}, \quad \text{and} \quad \tau_w = 0.335\mu U_o\sqrt{\frac{U_o}{\nu x}},$$

to be compared with the exact results given above in (10.27), (10.28), and (10.30).

- Values of $\alpha < 0$ correspond to adverse pressure gradients. Equating (10.52) to $\lambda = -0.09$ gives the critical value $\alpha = -0.1$ below which boundary-layer separation is predicted to occur. As can be seen, according to Thwaites method, the boundary layer may withstand slightly adverse pressure gradients, corresponding to $\alpha > -0.1$. This indicates, for instance, that the shape of flow diffusers (divergent pipes) must be designed so as to have a gentle streamwise increase in cross-section, if boundary-layer separation is to be prevented.

Thermal boundary layer

Let us now address the computation of the temperature distribution for the flow shown in Fig. 10.1. We have seen previously that, if the Reynolds number $Re = \rho U_\infty L / \mu$ is large, then both heat conduction and viscous dissipation are negligible in the first approximation, thereby reducing the energy equation to the condition of isentropic motion

$$\frac{Ds}{Dt} = 0. \quad (10.53)$$

For incompressible flows, with $Tds = cdT$, this last condition reduces to $DT/Dt = 0$, indicating that each fluid particle moves with constant temperature, equal to its upstream value T_∞ .

If the free-stream temperature is constant, the ideal-flow solution therefore yields a uniform fluid temperature $T = T_\infty$ everywhere. This solution does not satisfy in general the boundary condition at the wall, where the temperature T_w may be different from T_∞ . We therefore anticipate the existence of a thin fluid layer adjacent to the wall, termed thermal boundary layer, where the effect of heat conduction is significant, causing the temperature to evolve from the outer value T_∞ found outside the boundary layer to reach the value T_w at $y = 0$.

The boundary-layer form of the energy equation

For a constant-density fluid, the energy equation written in cartesian coordinates $x - y$ is

$$\underbrace{\rho c \frac{\partial T}{\partial t}}_{\rho c \frac{(T_\infty - T_w)}{t_o}} + \underbrace{\rho c v_x \frac{\partial T}{\partial x} + \rho c v_y \frac{\partial T}{\partial y}}_{\rho c \frac{U_\infty (T_\infty - T_w)}{L}} = \underbrace{\phi_v}_{\mu \frac{U_\infty^2}{\delta^2}} + \underbrace{k \frac{\partial^2 T}{\partial x^2}}_{k \frac{(T_\infty - T_w)}{L^2}} + \underbrace{k \frac{\partial^2 T}{\partial y^2}}_{k \frac{(T_\infty - T_w)}{\delta_T^2}} \quad (10.54)$$

where the viscous dissipation function is given by

$$\phi_v = \mu \left[\underbrace{2 \left(\frac{\partial v_x}{\partial x} \right)^2}_{\mu \frac{U_\infty^2}{L^2}} + \underbrace{2 \left(\frac{\partial v_y}{\partial y} \right)^2}_{\mu \frac{V^2}{\delta^2}} + \underbrace{\left(\frac{\partial v_x}{\partial y} \right)^2}_{\mu \frac{U_\infty^2}{\delta^2}} + \underbrace{\left(\frac{\partial v_y}{\partial x} \right)^2}_{\mu \frac{V^2}{L^2}} \right] \simeq \mu \left(\frac{\partial v_x}{\partial y} \right)^2. \quad (10.55)$$

Order-of-magnitude estimates with $\delta \ll L$ and $V \ll U_\infty$ can be used to show the, in the first approximation, the energy equation simplifies to give

$$\rho c \frac{\partial T}{\partial t} + \rho c v_x \frac{\partial T}{\partial x} + \rho c v_y \frac{\partial T}{\partial y} = k \frac{\partial^2 T}{\partial y^2} + \mu \left(\frac{\partial v_x}{\partial y} \right)^2. \quad (10.56)$$

If the problem is unsteady (i.e., $T_\infty(t)$), for the integration one needs to provide the initial temperature distribution in the boundary layer, i.e., $T = T_i(x, y)$ at $t = 0$. Boundary conditions have to be specified at $x = 0$ for $0 \leq y < \infty$ and also for $x > 0$ at $y = 0$ and $y = \infty$. For the steady intake problem addressed above, these boundary conditions reduce to

$$x = 0 : T = T_\infty \quad \text{for} \quad 0 \leq y < \infty \quad (10.57)$$

and

$$x > 0 : \begin{cases} T = T_w & \text{at} \quad y = 0 \\ T = T_\infty & \text{as} \quad y \rightarrow \infty. \end{cases} \quad (10.58)$$

Reynolds analogy

In many applications, including that of flow in pipes, it is of interest to compute the heat flux from the fluid to the wall

$$q_w = k \left(\frac{\partial T}{\partial y} \right)_{y=0} \quad (10.59)$$

(amount of energy transferred per unit surface per unit time). We use as an example the Blasius problem treated above (flow of a uniform stream over a semi-infinite flat plate), for which the velocity and temperature evolution are determined by integration of

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \quad (10.60)$$

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \frac{\partial^2 v_x}{\partial y^2} \quad (10.61)$$

$$v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} = \frac{k}{\rho c} \frac{\partial^2 T}{\partial y^2} + \frac{\nu}{c} \left(\frac{\partial v_x}{\partial y} \right)^2 \quad (10.62)$$

with boundary conditions

$$\begin{aligned} x = 0, \quad 0 < y < \infty : & v_x - U_\infty = T - T_\infty = 0 \\ x > 0 : & \begin{cases} y = 0 : v_x = v_y = T - T_w = 0 \\ y \rightarrow \infty : v_x - U_\infty = T - T_\infty = 0. \end{cases} \end{aligned} \quad (10.63)$$

The energy equation can be alternatively written in terms of the so-called recovery temperature

$$T_r = T + \frac{v_x^2}{2c} \quad (10.64)$$

by adding (10.61) times v_x/c and (10.62) to give

$$v_x \frac{\partial T_r}{\partial x} + v_y \frac{\partial T_r}{\partial y} = \nu \frac{\partial^2 T_r}{\partial y^2} + \left(\frac{k}{\rho c} - \nu \right) \frac{\partial^2 T}{\partial y^2}, \quad (10.65)$$

which further reduces to

$$v_x \frac{\partial T_r}{\partial x} + v_y \frac{\partial T_r}{\partial y} = \nu \frac{\partial^2 T_r}{\partial y^2} \quad (10.66)$$

when the Prandtl number $Pr = \nu/[k/(\rho c)] = \mu c/k$ equals unity. Boundary conditions for (10.66) are $T_r = T_\infty + U_\infty^2/2c$ at $x = 0$ for $0 \leq y < \infty$ and $T_r = T_w$ at $y = 0$ and $T_r = T_\infty + U_\infty^2/2c$ as $y \rightarrow \infty$ for $x > 0$. Note that equations (10.61) and (10.66) are identical, so that their normalized solutions satisfy

$$\frac{v_x}{U_\infty} = \frac{T - T_w + v_x^2/(2c)}{T_\infty - T_w + U_\infty^2/(2c)}. \quad (10.67)$$

Taking the derivative of this last equation with respect to y and evaluating the result at $y = 0$ yields the so-called Reynolds analogy

$$\frac{q_w}{c[T_\infty - T_w + U_\infty^2/(2c)]} = \frac{\tau_w}{U_\infty}, \quad (10.68)$$

relating the heat transfer rate to the wall and the wall shear stress. This equation, strictly valid for $Pr = 1$, applies also to gas flow by replacing c with c_p .

Chapter 11

Shock and Expansion Waves

As mentioned in the previous chapter, for the solution of the Euler equations to represent adequately a given large-Reynolds-number flow, we need to consider in general the existence of discontinuity surfaces, across which the fluid properties or their derivatives exhibit finite jumps. Mach cones, for instance, represent an example of weak discontinuities with jumps in derivatives. Boundary layers (and also vortex sheets and mixing layers), treated in the previous chapter, are examples of tangential discontinuities. We shall focus below on the analysis of normal discontinuities, showing that they are always compression waves, also called shock waves, with specific results given below for the jump conditions across normal and oblique shocks. Conversely, expansion waves cannot be discontinuities; rather, they are continuous isentropic waves, to be analyzed below in the specific case of steady flow around a sharp corner (the so-called Prandtl-Meyer expansion flow).

Jump conditions across discontinuity surfaces

To determine the conditions that relate the different flow properties that exist on the sides of a given discontinuity surface we shall apply the conservation equations to the infinitesimally small control volume shown in figure 11.1. As can be seen, the control volume includes faces of area $d\sigma$, lying on the two sides of the surface, separated by a small distance ε . In a reference frame moving with the discontinuity, mass conservation can be expressed in the form

$$\frac{d}{dt} \int_{V_c} \rho dV + \int_{\Sigma_c} \rho \bar{v} \cdot \bar{n} d\sigma = 0. \quad (11.1)$$

The first integral is proportional to the volume of the control volume, of order $\varepsilon d\sigma$, and therefore becomes negligibly small when an infinitesimally thin control volume satisfying $\varepsilon \ll (d\sigma)^{1/2}$ is considered. In that limit, only fluxes across the two faces of area $d\sigma$ need to be computed when evaluating the second integral, which therefore provides the simplified continuity equation

$$\rho_1 v_{1n} = \rho_2 v_{2n} = m, \quad (11.2)$$

where m denotes the local value of the mass flux across the discontinuity and the subscript n is used to denote normal velocity components, with the subscript t employed below for the tangential components. Volume integrals can also be neglected when evaluating the momentum

and energy conservation equations, which can be seen to provide

$$\rho_1 v_{1n}^2 + p_1 = \rho_2 v_{2n}^2 + p_2 \quad (11.3)$$

$$\rho_1 v_{1n} v_{1t} = \rho_2 v_{2n} v_{2t} \quad (11.4)$$

and

$$m \left(e_1 + \frac{p_1}{\rho_1} + \frac{v_1^2}{2} \right) = m \left(e_2 + \frac{p_2}{\rho_2} + \frac{v_2^2}{2} \right). \quad (11.5)$$

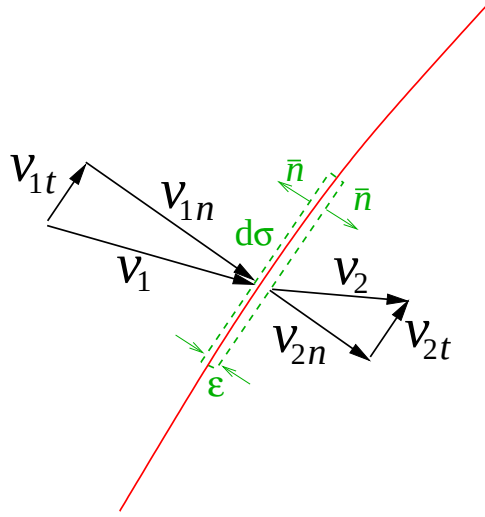


Figure 11.1: Jumps across a discontinuity surface.

Equations (11.2)–(11.5) apply to all discontinuities. Tangential discontinuities, such as boundary layers and vortex sheets, are defined by the condition that the mass flux is identically zero, that is, $m = 0$. From (11.2) it then follows that $v_{1n} = v_{2n} = 0$ and, from (11.3), it is seen that $p_1 = p_2$, indicating that the jump in pressure across tangential discontinuities is zero, a condition used before when analyzing jets and boundary layers. With $m = 0$, the conservation of tangential momentum and energy, expressed in (11.4) and (11.5), are identically satisfied, regardless of the values of $v_{1t} \neq v_{2t}$, $\rho_1 \neq \rho_2$, and $T_1 \neq T_2$.

Normal discontinuities

When the mass flux across the discontinuity is nonzero, then it follows from (11.4) that

$$v_{1t} = v_{2t}, \quad (11.6)$$

whereas (11.5) reduces to

$$h_1 + \frac{v_1^2}{2} = h_2 + \frac{v_2^2}{2}, \quad (11.7)$$

indicating that, in a reference frame moving with the discontinuity, the stagnation enthalpy (and, therefore, the stagnation temperature) is conserved.

For given values of v_{1n} , ρ_1 , and p_1 , equations (11.2), (11.3), and

$$\frac{\gamma}{\gamma-1} \frac{p_1}{\rho_1} + \frac{v_{1n}^2}{2} = \frac{\gamma}{\gamma-1} \frac{p_2}{\rho_2} + \frac{v_{2n}^2}{2} \quad (11.8)$$

determine v_{2n} , ρ_2 , and p_2 , with (11.8) corresponding to the energy conservation balance (11.7) written with use made of (11.6) and of the thermodynamic identity $h = [\gamma/(\gamma-1)]p/\rho$.

Straightforward manipulation of (11.2) and (11.3) leads to the equation of the so-called Rayleigh line

$$\frac{p_2}{p_1} - 1 = \frac{m^2}{\rho_1 p_1} \left(1 - \frac{\rho_1}{\rho_2} \right). \quad (11.9)$$

On the other hand, writing (11.8) in the form

$$\frac{\gamma}{\gamma-1} \left(\frac{p_2}{\rho_2} - \frac{p_1}{\rho_1} \right) = \frac{1}{2} (v_{1n}^2 - v_{2n}^2) = \frac{1}{2} m^2 \left(\frac{1}{\rho_1} - \frac{1}{\rho_2} \right) \left(\frac{1}{\rho_1} + \frac{1}{\rho_2} \right) \quad (11.10)$$

and using (11.9) to eliminate m^2 yields

$$\frac{p_2}{p_1} = \frac{\frac{\gamma+1}{\gamma-1} - \frac{\rho_1}{\rho_2}}{\frac{\gamma+1}{\gamma-1} \frac{\rho_1}{\rho_2} - 1}. \quad (11.11)$$

This last equation, relating the jumps in pressure and density across the discontinuity, is known as Hugoniot curve. Observation of the numerator and denominator in (11.11) indicates that the solution for the density ratio across the discontinuity must lie in the range

$$\frac{\gamma-1}{\gamma+1} < \frac{\rho_1}{\rho_2} < \frac{\gamma+1}{\gamma-1} \quad (11.12)$$

while the range of possible pressure ratios extends in principle for $0 < p_2/p_1 < \infty$, giving the solution shown in Fig. 11.2 for $\gamma = 1.4$.

For a given mass flux m , the values of p_2/p_1 and ρ_1/ρ_2 are determined by the point where the Rayleigh line (11.9) and the Hugoniot curve (11.11) cross in Fig. 11.2 (of course, a second solution $p_2/p_1 = \rho_1/\rho_2 = 1$ always exists, but that is just the trivial solution corresponding to a continuous flow with no jumps).

To investigate the different possible solutions, we begin by noting that the slope of the Rayleigh line can be written in terms of the Mach number associated with the normal component of the incident velocity $M_{1n} = v_{1n}/a_1$ according to

$$-\frac{m^2}{\rho_1 p_1} = -\frac{v_{1n}^2}{p_1/\rho_1} = -\gamma M_{1n}^2. \quad (11.13)$$

This value is to be compared with the slope of the Hugoniot curve at $\rho_1/\rho_2 = 1$, which can be computed by straightforward differentiation of (11.11) to give

$$\left(\frac{d(p_2/p_1)}{d(\rho_1/\rho_2)} \right)_{\rho_1/\rho_2=1} = -\gamma. \quad (11.14)$$

For $M_{1n} = 1$ the Rayleigh line is tangent to the Hugoniot curve at $p_2/p_1 = \rho_1/\rho_2 = 1$, which is therefore the only solution of (11.9) and (11.11), whereas a nontrivial crossing point appears for $M_{1n} \neq 1$ according to

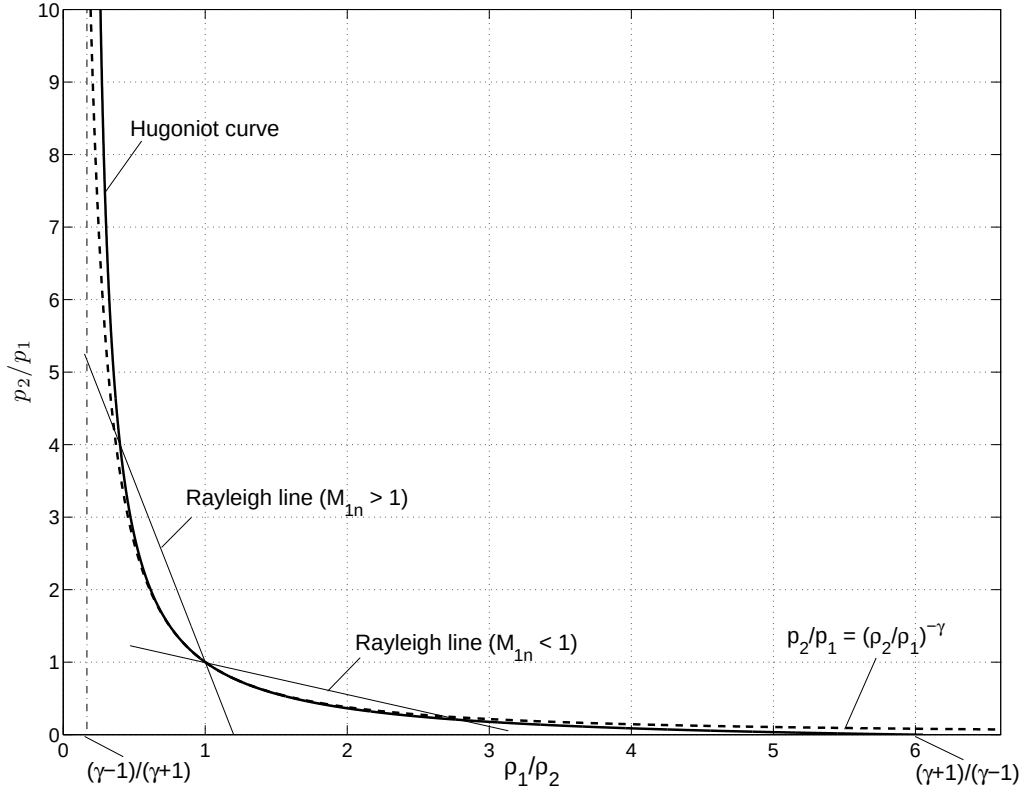


Figure 11.2: Hugoniot curve and Rayleigh line.

- If the normal component of the incident velocity is supersonic ($M_{1n} > 1$), then the Rayleigh line crosses the Hugoniot curve along the upper stretch, so that the solution corresponds to a compression with

$$\infty > \frac{p_2}{p_1} > 1 \quad \text{and} \quad \frac{\gamma - 1}{\gamma + 1} < \frac{\rho_1}{\rho_2} < 1$$

- If the normal component of the incident velocity is subsonic ($M_{1n} < 1$), then the Rayleigh line crosses the Hugoniot curve along the lower stretch, so that the solution corresponds to an expansion with

$$1 > \frac{p_2}{p_1} > 0 \quad \text{and} \quad 1 < \frac{\rho_1}{\rho_2} < \frac{\gamma + 1}{\gamma - 1}$$

Although both compressions and expansions are apparently valid solutions of (11.9) and (11.11), only compression waves, termed *shock waves*, may exist in reality. To prove this point, one may use thermodynamic arguments based on the second principle, which indicates that, in an adiabatic process, the entropy change $s_2 - s_1$ can never be negative, i.e., the entropy can never be destroyed. For a perfect gas,

$$\frac{s_2 - s_1}{c_v} = \ln \left(\frac{p_2/p_1}{(\rho_2/\rho_1)^\gamma} \right), \quad (11.15)$$

and one may use (11.11) to compute the entropy variation. As seen in Fig. 11.2, for $\rho_1/\rho_2 < 1$ the Hugoniot curve (11.11) lies above the isentrope $p_2/p_1 = (\rho_2/\rho_1)^\gamma$. Compression waves, therefore, are associated with positive entropy increments

$$\frac{s_2 - s_1}{c_v} = \ln \left[\frac{\left(\frac{\gamma+1}{\gamma-1} - \frac{\rho_1}{\rho_2} \right) / \left(\frac{\gamma+1}{\gamma-1} \frac{\rho_1}{\rho_2} - 1 \right)}{(\rho_2/\rho_1)^\gamma} \right] > 0, \quad (11.16)$$

as can be calculated from (11.15). Conversely, in expansion waves with $\rho_1/\rho_2 < 1$ the Hugoniot curve (11.11) lies below the isentrope $p_2/p_1 = (\rho_2/\rho_1)^\gamma$, so that the entropy change computed from (11.15) would be negative ($s_2 - s_1 < 0$), thereby violating the second principle of thermodynamics.

Hence, normal discontinuity surfaces are always shock waves, with jumps of thermodynamic properties in the ranges

$$\infty > \frac{p_2}{p_1} > 1, \quad \frac{\gamma-1}{\gamma+1} < \frac{\rho_1}{\rho_2} < 1, \quad \text{and} \quad \infty > \frac{T_2}{T_1} > 1, \quad (11.17)$$

the latter to be computed from the equation of state $T_2/T_1 = (p_2/p_1)/(\rho_2/\rho_1)$. The normal component of the incident velocity v_{1n} must be supersonic, so that the inequalities

$$M_1 > M_{1n} > 1 \quad (11.18)$$

are always satisfied. Use of (11.3) and (11.6) leads to the additional conditions

$$\frac{\gamma-1}{\gamma+1} < \frac{v_{2n}}{v_{1n}} < 1 \quad \text{and} \quad \frac{v_2}{v_1} < 1. \quad (11.19)$$

It can also be seen that

$$M_{2n} < 1. \quad (11.20)$$

It should be noted that, despite this last result, in oblique shock waves the downstream velocity can be however supersonic, i.e., $M_2 > 1$.

Jump conditions across shock waves

To resolve for the jumps across a generic shock wave, it is convenient to consider first the change in the normal component of the velocity along with the jumps in thermodynamic properties, and address subsequently the deflection of the flow δ as a function of the incident angle β (see Fig. 11.3). The development begins by using (11.2), (11.3), and (11.8), which are rewritten below for convenience

$$\rho_1 v_{1n} = \rho_2 v_{2n}, \quad (11.21)$$

$$\rho_1 v_{1n}^2 + p_1 = \rho_2 v_{2n}^2 + p_2, \quad (11.22)$$

$$\frac{\gamma}{\gamma-1} \frac{p_1}{\rho_1} + \frac{v_{1n}^2}{2} = \frac{\gamma}{\gamma-1} \frac{p_2}{\rho_2} + \frac{v_{2n}^2}{2}. \quad (11.23)$$

For a gas of known specific-heat ratio γ , these three equations determine the downstream values v_{2n} , p_2 , and ρ_2 in terms of v_{1n} , p_1 , and ρ_1 . The dependence can be simplified by using the II

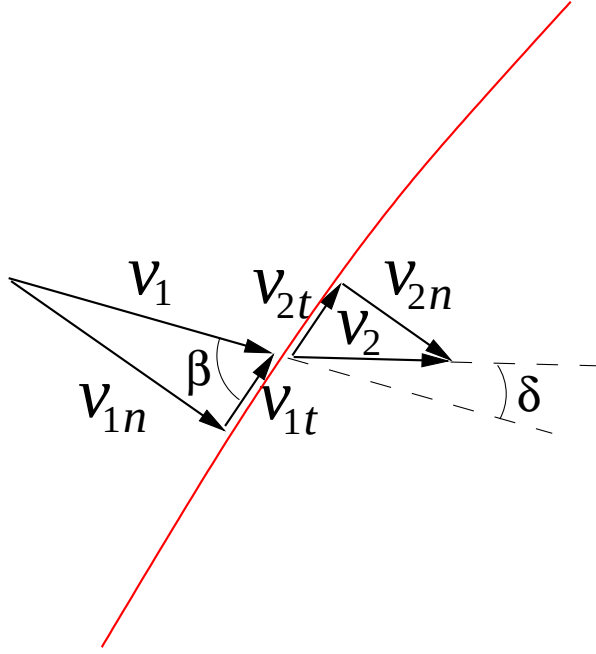


Figure 11.3: Velocity jumps across a normal discontinuity.

theorem of dimensional analysis to give

$$\left. \begin{aligned} v_{2n} &= f_1(v_{1n}, p_1, \rho_1, \gamma) \\ p_2 &= f_2(v_{1n}, p_1, \rho_1, \gamma) \\ \rho_2 &= f_3(v_{1n}, p_1, \rho_1, \gamma) \end{aligned} \right\} \Rightarrow \left\{ \begin{aligned} v_{2n}/v_{1n} &= \varphi_1(M_{1n}, \gamma) \\ p_2/p_1 &= \varphi_2(M_{1n}, \gamma) \\ \rho_2/\rho_1 &= \varphi_3(M_{1n}, \gamma) \end{aligned} \right. \quad (11.24)$$

The functions φ_1 , φ_2 , and φ_3 can be determined by manipulation of (11.21)–(11.23) to give (we leave the needed development as an exercise for the reader)

$$\frac{v_{2n}}{v_{1n}} = \left(\frac{\rho_2}{\rho_1} \right)^{-1} = \frac{2 + (\gamma - 1)M_{1n}^2}{(\gamma + 1)M_{1n}^2} \quad \text{and} \quad \frac{p_2}{p_1} = \frac{2\gamma M_{1n}^2 + 1 - \gamma}{\gamma + 1}, \quad (11.25)$$

which can be combined with $T_2/T_1 = (p_2/p_1)/(\rho_2/\rho_1)$ and $M_{2n}^2 = M_{1n}^2(\rho_2/\rho_1)^{-1}(p_2/p_1)^{-1}$ to give

$$\frac{T_2}{T_1} = \frac{(2\gamma M_{1n}^2 + 1 - \gamma)[2 + (\gamma - 1)M_{1n}^2]}{(\gamma + 1)^2 M_{1n}^2} \quad \text{and} \quad M_{2n}^2 = \frac{2 + (\gamma - 1)M_{1n}^2}{2\gamma M_{1n}^2 + 1 - \gamma}. \quad (11.26)$$

The above formulae take simplified forms in limiting cases of interest, including very strong shocks with $M_{1n} \gg 1$, for which

$$\frac{v_{2n}}{v_{1n}} = \left(\frac{\rho_2}{\rho_1} \right)^{-1} = \frac{\gamma - 1}{\gamma + 1}, \quad \frac{p_2}{p_1} = \frac{2\gamma M_{1n}^2}{\gamma + 1}, \quad \frac{T_2}{T_1} = \frac{2\gamma(\gamma - 1)M_{1n}^2}{(\gamma + 1)^2} \quad \text{and} \quad M_{2n}^2 = \frac{\gamma - 1}{2\gamma}, \quad (11.27)$$

and weak shocks with $\varepsilon = M_{1n} - 1 \ll 1$, for which

$$\frac{v_{2n}}{v_{1n}} = \left(\frac{\rho_2}{\rho_1} \right)^{-1} = 1 - \frac{2\varepsilon}{\gamma + 1}, \quad \frac{p_2}{p_1} = 1 + \frac{2\gamma\varepsilon}{\gamma + 1}, \quad \frac{T_2}{T_1} = 1 + \frac{2(\gamma - 1)\varepsilon}{\gamma + 1} \quad (11.28)$$

and $1 - M_{2n} = \varepsilon$. The associated entropy increase can be computed from (11.15) to give $(s_2 - s_1)/c_v \sim \varepsilon^3$, indicating that these weak shocks are nearly isentropic.

For normal shock waves, such that the incident flow forms an angle $\beta = \pi/2$ with the shock, no flow deflection occurs, so that $M_1 = M_{1n}$ and $M_2 = M_{2n}$, and the above equations (11.25) and (11.26) are all we need to determine the jumps across the shock. As an illustrative example, we can use them to determine the flow induced by a piston moving with constant velocity in a long cylindrical tube containing a gas. The piston, initially at rest, is suddenly accelerated to reach a constant velocity V_P , generating a shock wave that moves with velocity V_{SW} along the tube. The gas ahead of the shock is at rest, with initial thermodynamic properties characterized for instance by the sound velocity a_1 and pressure p_1 . The shock wave sets the fluid in motion, so that the gas between the shock and the piston moves with uniform constant velocity, equal to that of the piston.

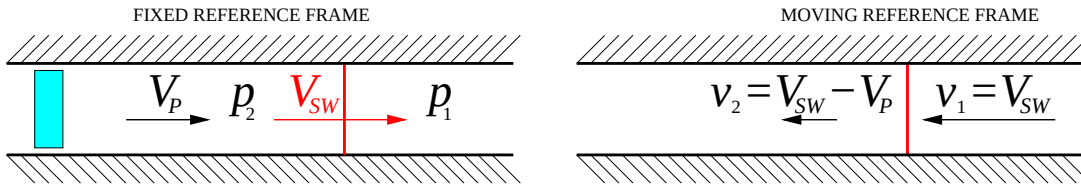


Figure 11.4: Piston-supported shock wave.

The velocities ahead and behind the shock $v_1 = v_{1n}$ and $v_2 = v_{2n}$ and the corresponding Mach numbers M_{1n} and M_{2n} appearing in (11.25) and (11.26) are measured relative to the shock, so in analyzing the problem we need to use a reference frame moving to the right with velocity V_{SW} . In that reference frame, the incident and downstream velocities are $v_1 = V_{SW}$ and $v_2 = V_{SW} - V_P$, whereas the incident Mach number is $M_{1n} = M_1 = V_{SW}/a_1$. In terms of these quantities, the first equation in (11.25) yields

$$\frac{v_2}{v_1} = 1 - \frac{V_P}{V_{SW}} = \frac{2 + (\gamma - 1)(V_{SW}/a_1)^2}{(\gamma + 1)(V_{SW}/a_1)^2}, \quad (11.29)$$

which can be rewritten in the form

$$\left(\frac{V_{SW}}{a_1}\right)^2 - \frac{\gamma + 1}{2} \frac{V_P}{a_1} \left(\frac{V_{SW}}{a_1}\right) - 1 = 0. \quad (11.30)$$

Solving now for V_{SW}/a_1 gives

$$M_{1n} = \frac{V_{SW}}{a_1} = \frac{\gamma + 1}{4} \frac{V_P}{a_1} + \left[1 + \left(\frac{\gamma + 1}{4}\right)^2 \left(\frac{V_P}{a_1}\right)^2 \right]^{1/2}, \quad (11.31)$$

which can be used in (11.25) and (11.26) to determine the pressure, density and temperature in front of the piston as a function of V_P/a_1 .

The above piston problem can be used as a simplified representation of the flow induced by a high-speed train entering a tunnel, the main simplification being that our piston occupies the whole transverse section of the tube, whereas in reality the train occupies only a fraction of the

tunnel section. The results indicate that, as the train enters the tunnel, a sudden overpressure develops immediately ahead. The shock is weak for $V_P \ll a_1$, when

$$M_{1n} - 1 = \frac{\gamma + 1}{4} \frac{V_P}{a_1} \ll 1 \quad \text{and} \quad \frac{p_2}{p_1} - 1 = \frac{\gamma}{2} \frac{V_P}{a_1} \ll 1,$$

which explain why, in conventional trains, the problems encountered at the entrance of a tunnel are minor. Relative pressure differences of order unity appear however as V_P approaches values of the order of the sound velocity a_1 . That is the case of high-speed trains, for which special measures need to be taken to remedy the overpressure problem at the tunnel entrance.

Oblique shocks

For known values of M_1 and β one may compute

$$M_{1n} = M_1 \sin \beta, \quad (11.32)$$

which in turn determines from (11.25) and (11.26) the jumps of thermodynamic properties and the values of

$$M_{2n}^2 = \frac{2 + (\gamma - 1)M_1^2 \sin^2 \beta}{2\gamma M_1^2 \sin^2 \beta + 1 - \gamma} \quad (11.33)$$

and

$$\frac{v_{2n}}{v_{1n}} = \frac{2 + (\gamma - 1)M_1^2 \sin^2 \beta}{(\gamma + 1)M_1^2 \sin^2 \beta}. \quad (11.34)$$

To determine the flow deflection, we use (11.6) written in the form

$$v_t = v_1 \cos(\beta) = v_2 \cos(\beta - \delta). \quad (11.35)$$

Using the equations $v_{1n} = v_1 \sin(\beta)$ and $v_{2n} = v_2 \sin(\beta - \delta)$ to give

$$\frac{\tan(\beta - \delta)}{\tan(\beta)} = \frac{v_{2n}}{v_{1n}} \quad (11.36)$$

and solving for δ with use made of (11.34) yields

$$\tan(\delta) = 2 \cot(\beta) \frac{M_1^2 \sin^2(\beta) - 1}{2 + M_1^2 [\gamma + \cos(2\beta)]}. \quad (11.37)$$

Once δ is known one may easily compute

$$M_2 = \frac{M_{2n}}{\sin(\beta - \delta)}, \quad (11.38)$$

with M_{2n} determined from (11.33).

The results are summarized in Figs. 11.5 and 11.6, where the angles are given in degrees. In particular, Fig. 11.5 indicates that for given values of M_1 and δ two different solutions are found if $\delta < \delta_{\max}$ and no solution exists for $\delta > \delta_{\max}$, where the value of $\delta_{\max}$ increases for increasing M_1 . For $\delta < \delta_{\max}$, the solution with larger β corresponds to a stronger shock wave (i.e., larger value of $M_{1n} = M_1 \sin \beta$). The corresponding value of M_{2n} is sufficiently small that the flow downstream

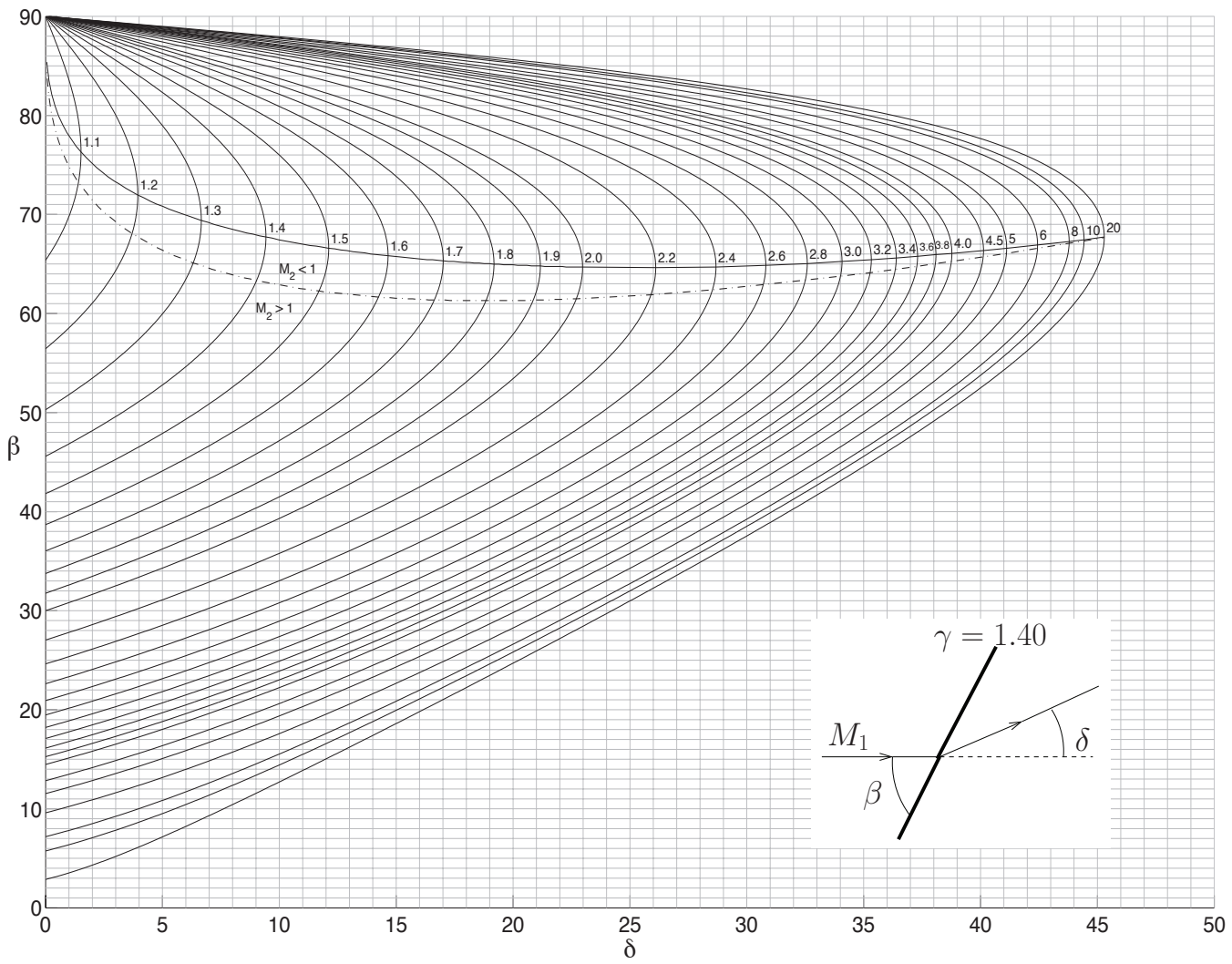


Figure 11.5: The variation of β with δ for constant values of M_1 .

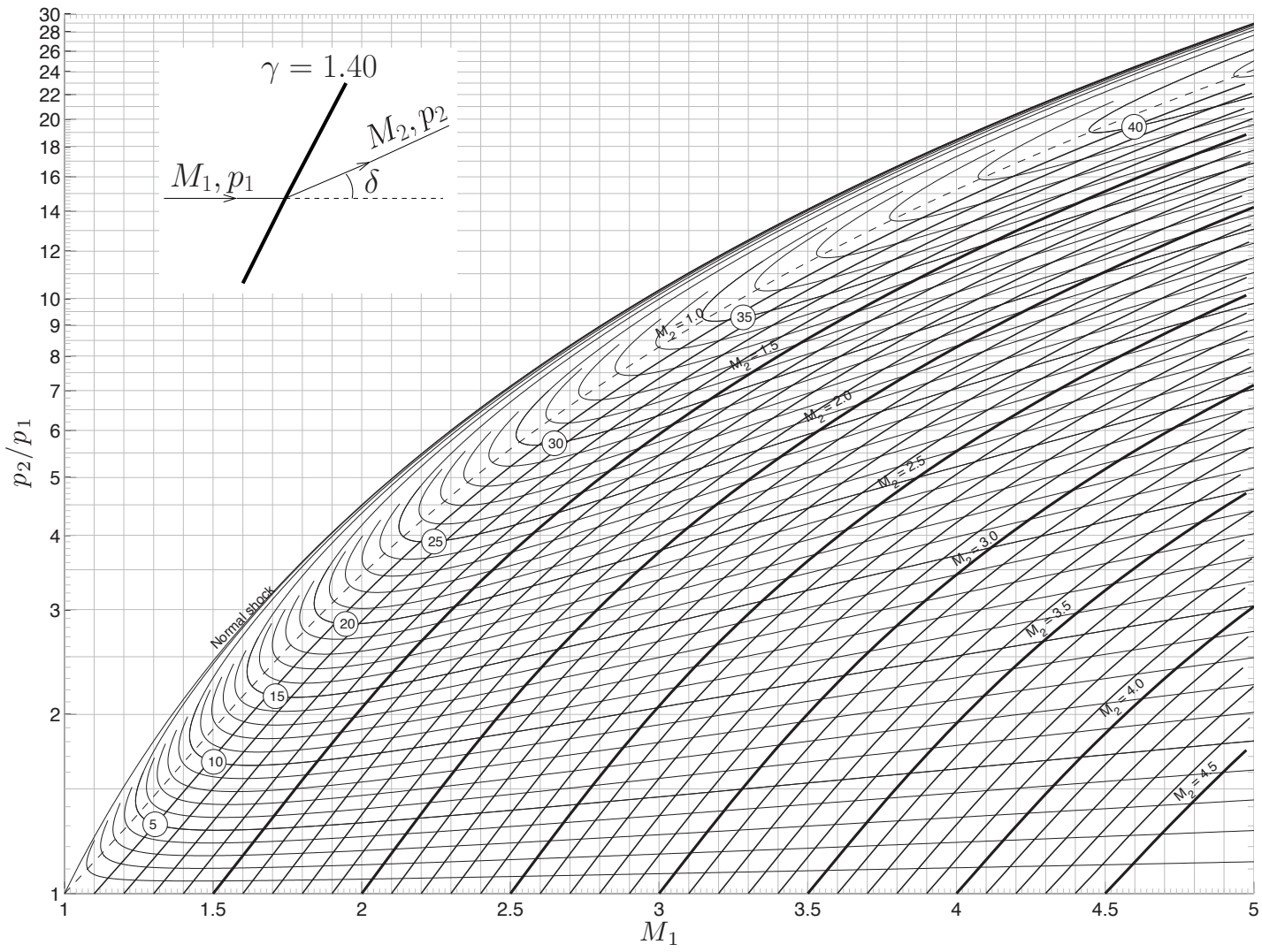


Figure 11.6: The variation of p_2/p_1 with M_1 for constant values of M_2 and δ .

is subsonic, that is, $M_2 < 1$. On the other hand, the solution lying along the lower part of the curve of constant M_1 in Fig. 11.5 corresponds to a lower value of β , and therefore a weaker solution including a values of M_2 that are generally supersonic (except for a small region near $\delta = \delta_{\max}$, as indicated in the figure). Because the associated downstream flow is supersonic, the weak-shock solution is more stable, in that perturbations can never reach it from downstream. Because of their stable character, when both strong or weak solutions may exist, weak solutions tend to prevail in realistic configurations.

For each value of M_1 , there exist two solutions with zero deflection angle $\delta = 0$. One is the normal shock wave, corresponding to $\beta = \pi/2$ and $M_1 = M_{1n}$, which is the strongest possible wave for that value of M_1 . The other, corresponding to $\beta = \mu$ with

$$\mu = \sin^{-1} \left(\frac{1}{M_1} \right), \quad (11.39)$$

is the weakest shock wave for that value of M_1 ; the incident angle is such that $M_{1n} = M_1 \sin \beta = 1$. These limiting solutions are termed *Mach waves*. According to what we mentioned below (11.28), these weak shocks are effectively isentropic compressions. A Mach wave is seen to bound, for instance, the Mach cone sketched in Fig. 8.

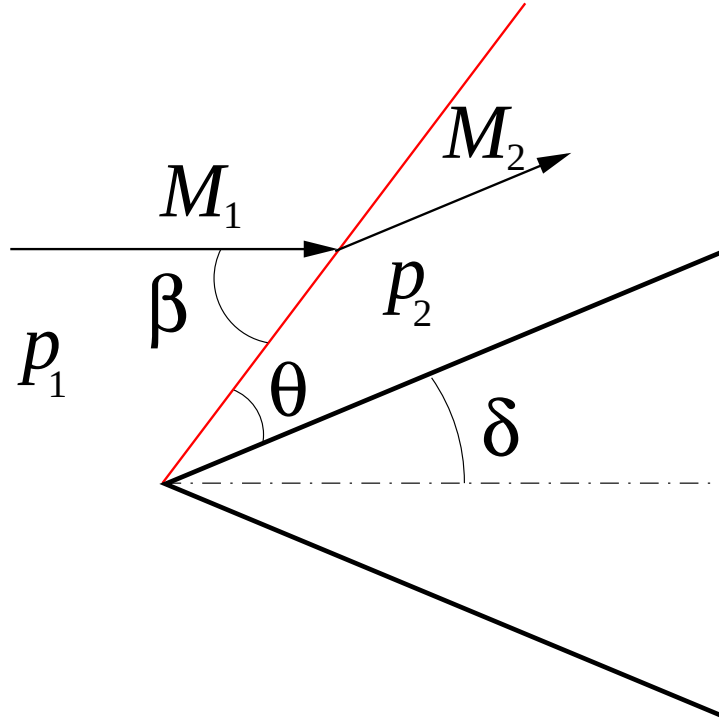


Figure 11.7: Supersonic flow over a wedge.

Figure 11.5 can be used, for instance, to solve the supersonic flow over a wedge, characterized by the appearance of an oblique shock, that forms an angle θ with the wall, deflecting the incoming stream to make it parallel to the wedge, as sketched in Fig. 11.7. If the incident Mach number is $M_1 = 2$ and the wedge semi angle is $\delta = 15^\circ$, then two solutions are possible, as can be seen in Fig. 11.5. Following the arguments given above, we select the weaker shock, corresponding

to $\beta \simeq 45^\circ$, thereby giving $\theta = \beta - \delta = 30^\circ$ and $M_{1n} = M_1 \sin \beta = \sqrt{2}$. Using now (11.25) and (11.26) yields $p_2/p_1 \simeq 2.17$ and $M_{2n} \simeq 0.734$. This last value can be substituted into (11.38) to give $M_2 = 1.47$. The problem could also be solved by using Fig. 11.6 instead, an exercise left for the reader to attempt. Also of interest is that, for the flow over a wedge, no solution involving an oblique shock exists for $\delta > \delta_{\max} \simeq 23^\circ$. In that case, the shock detaches to form a bow shock wave, which is locally planar at the center line and becomes increasingly inclined farther away. These bow shock waves are found in general in front of blunt bodies, as seen in the experimental visualization shown below in Fig. 11.8.

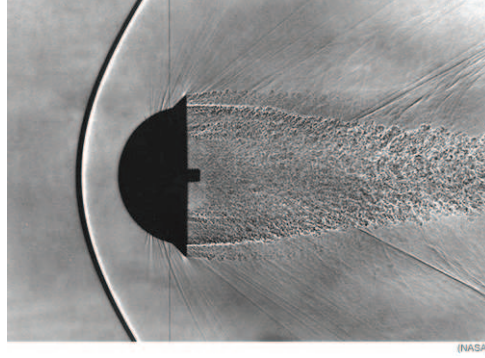


Figure 11.8: Supersonic flow over a blunt body.

Regarding the flow over a wedge (or rather the equivalent case of the flow around a concave corner), it is of interest to consider what happens when, instead of a sharp corner, one encounters a smoothly curved wall, as shown in Fig. 11.9. The perturbations associated with the continuous increase of slope are necessarily weak near the wall, and therefore give rise to a system of Mach waves that propagate into the stream with a local angle of incidence $\beta = \mu$ such that, at each position, $M_n = M \sin \beta = 1$. For the concave wall, the value of M decreases and the pressure increases as the flow crosses the system of Mach waves, giving an increasing value of μ and causing the Mach waves to approach one another. The compression reinforces as a result of the interaction between the different Mach waves, so that a shock wave of finite strength eventually emerges. This complex interaction process is at the origin of the oblique shock that is seen to emerge from the corner when the observation distance is sufficiently large for the curved wall to appear as a sharp corner (see Fig. 11.9).

Prandtl-Meyer expansions

Let us now consider the case of supersonic flow over a convex wall, represented in Fig. 11.10. A system of Mach waves originates from the wall as it curves downwards. In this case, however, the value of M increases and the pressure decreases as the flow crosses the system of Mach waves, giving an decreasing value of μ and causing the Mach waves to open up. As a result, the expansion wave generated as the supersonic stream turns around the convex corner is formed by a fan of Mach waves that cause the stream lines to evolve smoothly. Since the resulting flow is steady and isentropic, all stagnation magnitudes are preserved across the expansion wave.

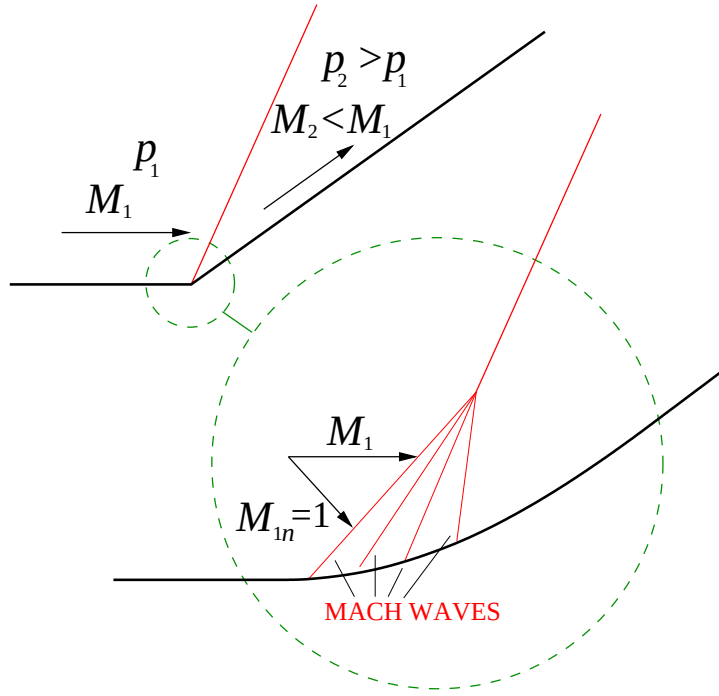


Figure 11.9: Supersonic flow over a concave wall.

Besides the condition of conservation of all stagnation properties,

$$\frac{T_0}{T} = \left(\frac{p_0}{p}\right)^{(\gamma-1)/\gamma} = \left(\frac{\rho_0}{\rho}\right)^{\gamma-1} = \left(\frac{a_0}{a}\right)^2 = 1 + \frac{\gamma-1}{2}M^2, \quad (11.40)$$

the solution for the convex-corner expansion, commonly known as Prandtl-Meyer expansion, requires knowledge of the relationship between the flow deflection and the flow acceleration, which comes from consideration of the infinitesimal evolution due to a given Mach wave. As shown schematically in Fig. 11.11, the condition that the tangential component of the velocity must be conserved as the flow is deflected can be used to relate the velocity increment and the deflection according to

$$d\theta = \frac{dv/\tan \mu}{v}. \quad (11.41)$$

From (11.39) it follows that

$$\tan \mu = \frac{1}{\sqrt{M^2 - 1}}. \quad (11.42)$$

To obtain dv/v we begin by differentiating $v = Ma$ to give

$$\frac{dv}{v} = \frac{dM}{M} + \frac{da}{a}. \quad (11.43)$$

On the other hand, differentiating the last equation in (11.40) for a constant value of a_0 yields

$$\frac{da}{a} = -\frac{1}{2} \frac{(\gamma-1)M dM}{1 + (\gamma-1)M^2/2}, \quad (11.44)$$

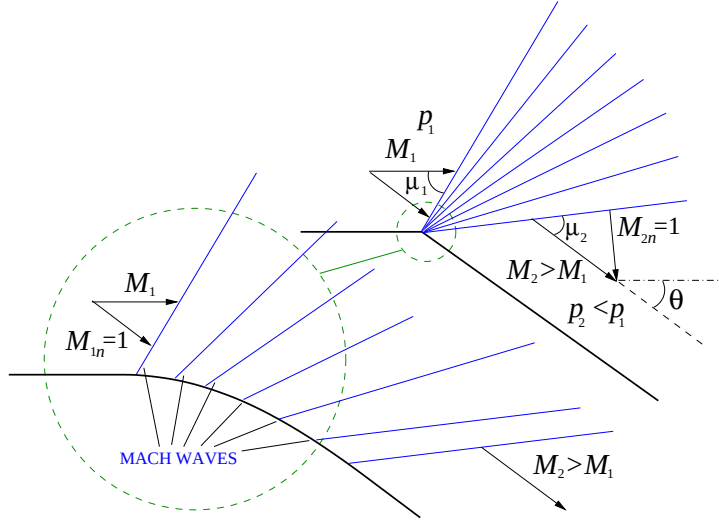


Figure 11.10: Supersonic flow over a convex wall.

which can be combined with (11.43) to provide

$$\frac{dv}{v} = \frac{1}{1 + (\gamma - 1)M^2/2} \frac{dM}{M}. \quad (11.45)$$

Substituting (11.42) and (11.45) into (11.41) finally gives

$$d\theta = \frac{\sqrt{M^2 - 1}}{1 + (\gamma - 1)M^2/2} \frac{dM}{M} \quad (11.46)$$

as a relationship between the deflection and the Mach number increment.

This last equation can be integrated to relate the total deflection θ with the change in Mach number as the supersonic stream in Fig. 11.10 turns around the corner according to

$$\theta = \int_{M_1}^{M_2} \frac{\sqrt{M^2 - 1}}{1 + (\gamma - 1)M^2/2} \frac{dM}{M}, \quad (11.47)$$

which can be alternatively written as

$$\theta = \nu(M_2) - \nu(M_1) \quad (11.48)$$

in terms of the so-called Prandtl-Meyer function

$$\nu(M) = \int_1^M \frac{\sqrt{M^2 - 1}}{1 + (\gamma - 1)M^2/2} \frac{dM}{M} \quad (11.49)$$

given in explicit form by

$$\nu(M) = \sqrt{\frac{\gamma + 1}{\gamma - 1}} \tan^{-1} \left[\sqrt{\frac{\gamma - 1}{\gamma + 1}} (M^2 - 1) \right] - \tan^{-1}(\sqrt{M^2 - 1}) \quad (11.50)$$

with $\tan^{-1}$ representing the arctangent (the inverse of the tangent function).

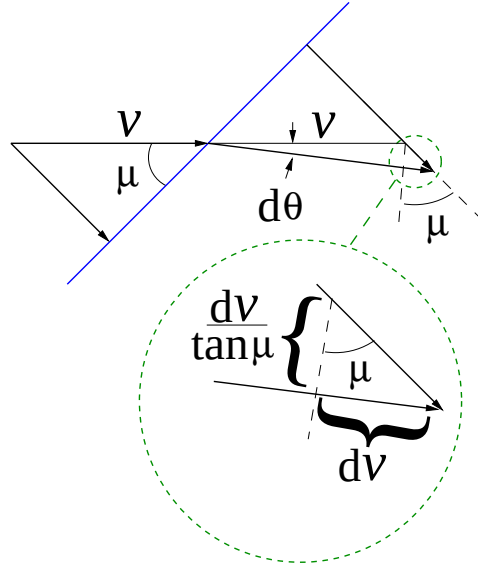


Figure 11.11: Differential evolution across a Mach wave.

For the problem represented in Fig. 11.10, if the corner angle θ and the Mach number of the incoming stream M_1 are known, then one may use (11.48) to determine the value of M_2 and then use (11.40) written in the form

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{(\gamma-1)/\gamma} = \left(\frac{\rho_2}{\rho_1} \right)^{\gamma-1} = \left(\frac{a_2}{a_1} \right)^2 = \frac{1 + (\gamma-1)M_1^2/2}{1 + (\gamma-1)M_2^2/2} \quad (11.51)$$

to obtain the ratios of thermodynamic properties across the expansion. For instance, if $\theta = 15^\circ$ and $M_1 = 2$, then $\nu(M_1) = 26.38^\circ$. Using this value in (11.48) gives $\nu(M_2) = 41.38^\circ$, which can be employed to obtain from (11.50) the value $M_2 \simeq 2.6$.

In other occasions, what determines the flow deflection is the pressure that exists downstream. An example of interest occurs at the exit of nozzles of thrust rockets used for satellite attitude control. Near the nozzle edge the flow is locally planar and corresponds in the first approximation to a Prandtl-Meyer expansion, as indicated in Fig. 11.12. The surrounding pressure is very small, so that the expanding gas reaches very high Mach numbers M_2 . If we assume for simplicity complete vacuum by using $p_2 = 0$, then for any finite values of M_1 and p_1 one obtains $M_2 = \infty$, as can be seen from (11.51). The associated flow deflection must be calculated from (11.48) with $\nu(M_2) = \nu(\infty) = 130.45^\circ$. If the value of M_1 is sufficiently small, then the deflection may exceed 90° , giving rise to a reverse jet flow that may impinge on the satellite wall, possibly damaging it (for instance, for $M_1 = 2$ one obtains $\nu(M_1) = 26.38^\circ$ and $\theta = 104.07^\circ$). This could be avoided by ensuring that the value of M_1 remains above $M_1 \simeq 2.56$, so that $\theta < 90^\circ$.

Flow in convergent-divergent nozzles

After studying shock and expansion waves we have now all the necessary elements to address the flow through a convergent-divergent nozzle. The nozzle geometry is defined by the values of the exit area A_e and throat area A_t (the section of minimum cross-section area). The nozzle

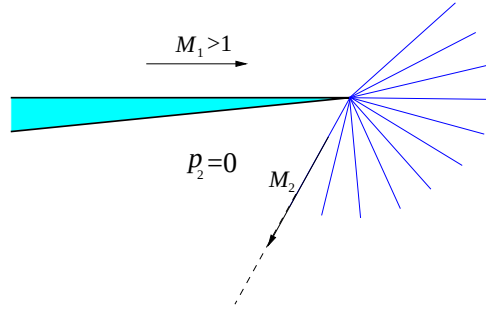


Figure 11.12: Supersonic stream discharging to the vacuum.

is assumed to be connected to a reservoir, where the gas pressure is p_0 , and discharges to the atmosphere, where the pressure is p_a . Different solutions emerge depending on the value of p_a/p_0 , as indicated in Fig. 11.13.

Let us begin by considering the case where the ambient pressure is only slightly smaller than p_0 (case *a* in Fig. 11.13). As a result of the pressure difference, there appears a slow isentropic steady outflow. For the resulting subsonic flow, the pressure decreases as the Mach number increases in the convergent part of the nozzle, and opposite behaviors are observed in the divergent stretch. The minimum pressure and the maximum Mach number are therefore reached at the throat. The flow discharges to the ambient as a subsonic jet. The condition that the pressure at the exit equals the ambient value can be used to compute the Mach at the exit section from

$$\left(\frac{p_0}{p_a}\right)^{(\gamma-1)/\gamma} = 1 + \frac{\gamma-1}{2} M_e^2, \quad (11.52)$$

which in turn determines the mass flow rate

$$\begin{aligned} G &= \rho_0 a_0 A_e M_e \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \\ &= \rho_0 a_0 A_e \left(\frac{2}{\gamma-1}\right)^{1/2} \left(\frac{p_0}{p_a}\right)^{-\frac{\gamma+1}{2\gamma}} \left[\left(\frac{p_0}{p_a}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]^{1/2}, \end{aligned} \quad (11.53)$$

with the stagnation properties for the flow being those found in the reservoir. The Mach number at the throat $M_t > M_e$ can be obtained from

$$A_e M_e \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}} = A_t M_t \left(1 + \frac{\gamma-1}{2} M_t^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \quad (11.54)$$

obtained by equating the values of G evaluated at the throat

$$G = \rho_0 a_0 A_t M_t \left(1 + \frac{\gamma-1}{2} M_t^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \quad (11.55)$$

and at the exit.

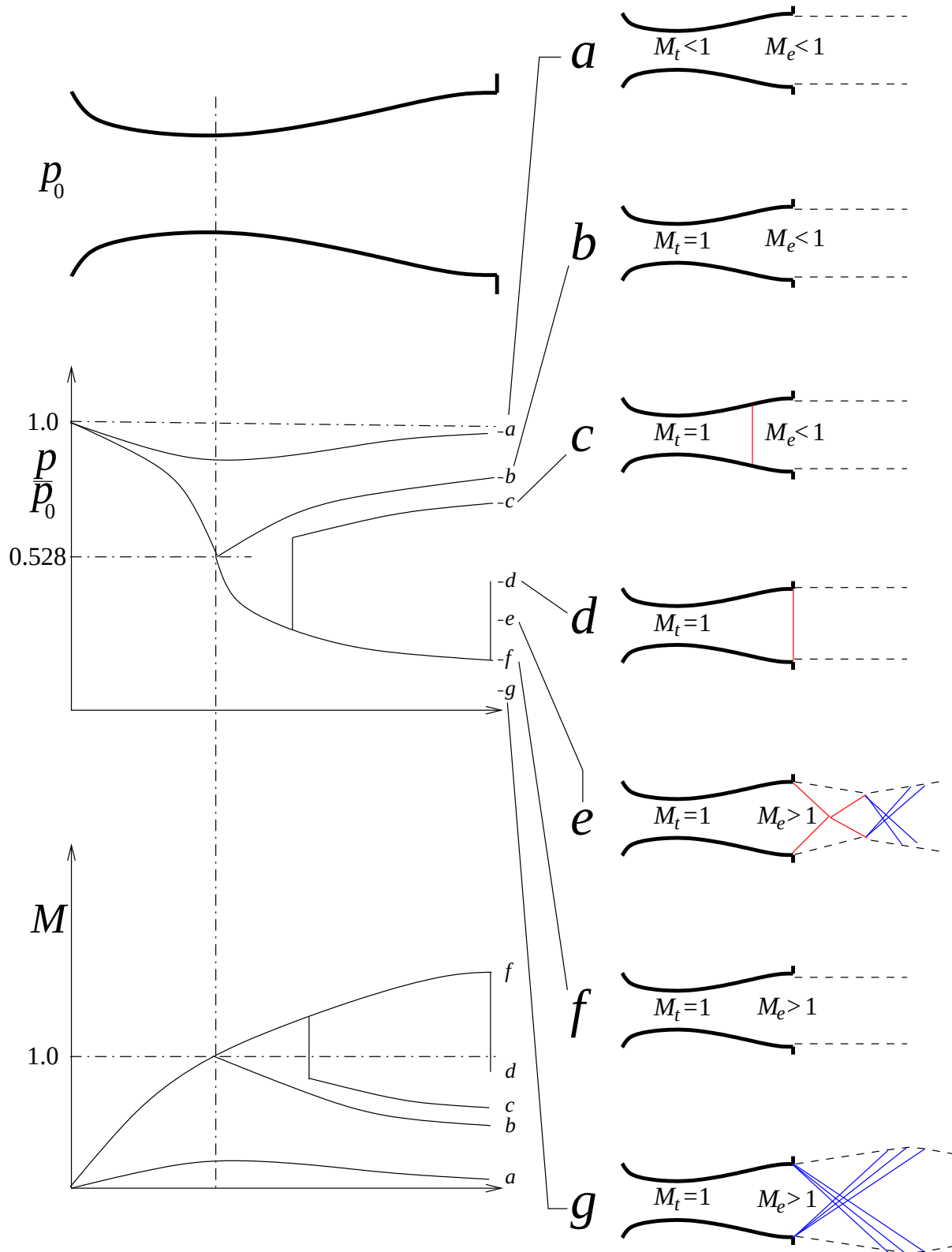


Figure 11.13: Flow in a convergent-divergent nozzle.

The value of M_t continues to increase for decreasing values of p_a/p_0 , reaching sonic conditions when $p_a = p_{CH}$ (case *b* in Fig. 11.13). The value of p_{CH}/p_0 can be determined as a function of A_t/A_e by equating the mass flow rate at the throat,

$$G = \rho_0 a_0 A_t \left(\frac{\gamma + 1}{2} \right)^{-\frac{\gamma+1}{2(\gamma-1)}} \quad (11.56)$$

evaluated from (11.55) with $M_t = 1$, to that crossing the exit section, evaluated from (11.53), yielding

$$\frac{A_t}{A_e} = \left(\frac{\gamma + 1}{2} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \left(\frac{2}{\gamma - 1} \right)^{1/2} \left(\frac{p_0}{p_a} \right)^{-\frac{\gamma+1}{2\gamma}} \left[\left(\frac{p_0}{p_a} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]^{1/2}. \quad (11.57)$$

It can be seen that, for a given value of A_t/A_e , this last equation is satisfied by two different values of p_a/p_0 . One of the two solutions, corresponding to the choking value $p_a = p_{CH}$, is associated with an exit Mach number $M_e < 1$, to be evaluated from (11.52) with $p_a = p_{CH}$. There exists, however, a second isentropic solution with $p_a < p_{CH}$ in which the flow discharges as a supersonic jet with $p_e = p_a = p_{SJ}$ and $M_e > 1$. For instance, for $A_t/A_e = 0.477$ the two solutions of (11.57) and (11.52) are $p_a/p_0 = p_{CH}/p_0 = 0.943$ with $M_e = 0.29$ and $p_a/p_0 = p_{SJ}/p_0 = 0.0865$ with $M_e = 2.25$.

The results indicate that the convergent-divergent nozzle can be choked more easily than the convergent nozzle (note that for $A_t/A_e = 0.477$ a value $p_a/p_0 = 0.943$ is sufficient to choke the flow, to be compared with the value $p_a/p_0 = 0.528$ required in convergent nozzles). The flow remains choked for $p_a < p_{CH}$, which implies that, regardless of the value of p_a , the flow in the convergent part of the nozzle is identical, with the mass flow rate taking always the constant value (11.56). In the isentropic solution discharging as a supersonic jet (case *f* in Fig. 11.13) the flow is therefore subsonic in the convergent stretch, sonic at the throat, and supersonic in the divergent stretch, leaving the nozzle with $p_e = p_a$, so that neither expansions nor compressions are needed to adapt the pressure to the ambient value.

It is of interest to consider what happens when the ambient pressure is below p_{SJ} and also when it takes intermediate values in the range $p_{SJ} < p_a < p_{CH}$. For $p_a < p_{SJ}$ (case *g* in Fig. 11.13), the flow conditions at the exit are exactly those found for $p_a = p_{SJ}$, so that $p_e = p_{SJ} > p_a$ (underexpanded flow). An expansion is therefore needed outside the nozzle to decrease the pressure to the ambient value. The expansion wave that appears is locally planar near the nozzle rim and therefore corresponds there to a Prandtl-Meyer expansion, which deflects the stream outwards as it decreases its pressure to p_a while increasing its Mach number beyond that found at the exit section.

Similarly, when the ambient pressure is slightly above p_{SJ} (case *e* in Fig. 11.13), the flow conditions at the exit are also exactly those found for $p_a = p_{SJ}$. In this case, however, the flow is overexpanded, and a shock wave is needed to increase the pressure to the ambient value. When p_a is only slightly above p_{SJ} , an oblique weak shock is sufficient. As the value of p_a further increases, the needed shock wave becomes stronger, and the associated incidence angle increases towards $\beta = \pi/2$. There is a critical value of p_a for which the compression occurs through a normal shock wave standing at the exit section (case *d* in Fig. 11.13). For even larger values of p_a the shock wave migrates into the divergent stretch of the nozzle, so that downstream from the throat there is a region of supersonic flow that ends at the inner shock, which is followed by a region of subsonic flow eventually discharging to the ambient as a jet, as indicated in Fig. 11.13. These steady solutions including planar shock waves within the divergent part of the nozzle are

hard to observe in reality, because they are often affected by boundary-layer separation immediately downstream from the shock, which may lead to significant changes in the downstream flow structure, including at times unsteady flow variations and asymmetric flow patterns.

Chapter 12

Turbulent Flow in Pipes

Introduction

Most of the presentation so far has been devoted to the study of flows that are *laminar*, in that the fluid moves in an orderly fashion forming a flow pattern with parallel fluid layers (or laminae). This was assumed to occur, for instance, in flows in circular pipes of diameter D , where the flow was seen to evolve from the uniform profile found for $x \ll ReD$ to a steady parabolic profile

$$v = 2v_m \left[1 - \left(\frac{r}{D/2} \right)^2 \right] \quad (12.1)$$

with parallel stream lines for $x \gg ReD$, as indicated in Fig. 12.1, with

$$Re = \frac{\rho v_m D}{\mu} \quad (12.2)$$

representing the flow Reynolds number based on the fluid density and viscosity ρ and μ and on the mean velocity

$$v_m = \frac{Q}{A} \quad (12.3)$$

associated with the pipe volume flux Q and cross section area $A = \pi D^2/4$.

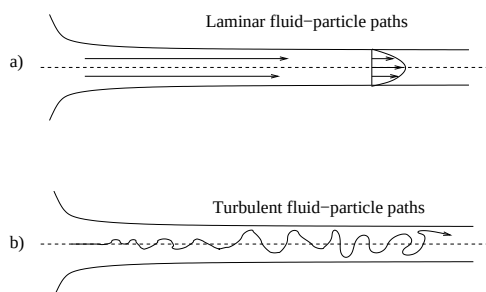


Figure 12.1: Pathlines in laminar and turbulent pipe flow.

For their existence, the laminar solutions of the Navier-Stokes equations have to be stable, in that, when they are perturbed, the perturbation must decay as the flow evolves, so that the

original laminar solution is restored. Viscosity tends to dissipate the perturbations, while the convective terms present in the equations of motion provide a nonlinear mechanism that tends to promote perturbation growth. Therefore, when the Reynolds number is sufficiently small, laminar flows are stable, whereas at large Reynolds numbers, the perturbations grow to destroy the orderly motion, which is replaced by a complicated unsteady flow pattern termed *turbulent* flow. For each flow configuration, there exists a critical value of the Reynolds number Re_c that characterizes its stability, in that for $Re < Re_c$ the laminar solution is stable and it is therefore the solution found in reality, while for $Re > Re_c$ the laminar solution loses stability, and a turbulent flow replaces the regular flow pattern. In turbulent pipe flow, the resulting particle paths are no longer parallel to the pipe walls, as indicated in Fig. 12.1.

For pipe flow, the value of Re_c depends on the flow conditions, so that, for instance, it is possible to find laminar flow for values of the Reynolds number as large as $Re = 10,000$ in laboratory experiments designed to avoid perturbations. In engineering applications, where perturbations are inevitable, the flow is found to be laminar for $Re < 2300$ and turbulent for $Re > 4000$, whereas for intermediate values in the range $2300 < Re < 4000$ the flow switches intermittently from laminar to turbulent.

Fundamental flow differences are revealed when the instantaneous velocity is measured at a given transverse location within the pipe. For $Re < 2300$ the velocity is steady, with a value that depends solely on the radial distance to the pipe center as dictated by (12.1), so that the local measurement would provide a constant value, as indicated in Fig. 12.2. In the intermediate range $2300 < Re < 4000$ the flow is said to be transitional and is characterized by the spontaneous appearance of perturbation velocities that are eventually attenuated, giving rise to an intermittent signal. For $Re > 4000$ velocity fluctuations are permanently observed, giving a local instantaneous value $v(t)$ that oscillates around an average value $\langle v \rangle = \frac{1}{T} \int_t^{t+T} v dt$, where T denotes here an arbitrary integration period, selected to be sufficiently large for $\langle v \rangle$ to be constant. For pipe flow, typical turbulent fluctuations are on the order of $v(t) - \langle v \rangle \sim 0.03\langle v \rangle$.

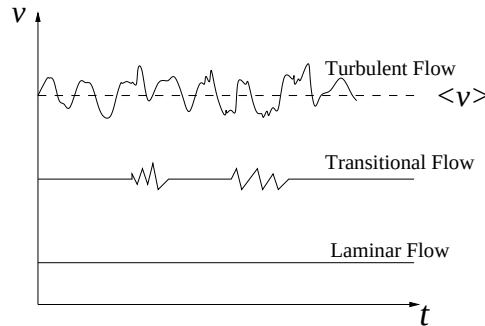


Figure 12.2: Instantaneous velocity measurements.

The unsteady turbulent velocity fluctuations are associated with the existence of vortices that are convected downstream by the flow stream. These vortices, involving a wide range of characteristic sizes and turnover times, increase significantly the rate of transverse transport of mass, momentum and energy. Because of this increased transport rate, the resulting profile of average velocity $\langle v \rangle(r)$ is nearly uniform across the pipe, except in a thin boundary-layer region near the wall, across which the velocity decreases to reach a zero value at the wall. Similarly, the profile of average temperature $\langle T \rangle(r)$ is also nearly uniform, except across the boundary layer, where it

evolves rapidly to reach the wall value $\langle T \rangle = T_w$.

Wall shear stress in turbulent pipe flow

The structure of the turbulent boundary layer is very important for pipe flow, because it determines the value of the viscous stresses on the wall

$$\tau_f = -\mu \left(\frac{\partial \langle v \rangle}{\partial r} \right)_{r=D/2}, \quad (12.4)$$

which in turn determines the streamwise pressure gradient required to maintain the flow rate. In this introductory study, we approach the problem by using dimensional analysis. We begin by assuming the functional dependence

$$\tau_f = \tau_f(\rho, \mu, D, v_m, \epsilon) \quad (12.5)$$

for the viscous stresses. As can be seen, besides the fluid properties, represented by its density ρ and viscosity μ , the pipe diameter D , and the flow rate, characterized by the mean velocity v_m defined in (12.3), we anticipate in (12.5) an additional dependence on the wall roughness ϵ (the average size of the roughness bumps characteristic of the pipe-wall material). Straightforward application of the Π theorem leads to

$$f = \frac{\tau_f}{\frac{1}{8}\rho v_m^2} = f\left(\frac{\rho v_m D}{\mu}, \frac{\epsilon}{D}\right). \quad (12.6)$$

As can be seen, the dimensionless wall shear stress f , commonly referred to as *friction coefficient* or *friction factor*, is a function of the Reynolds number Re , defined above in (12.2), and of the relative roughness ϵ/D , with an arbitrary scaled factor $\frac{1}{8}$ introduced customarily in its definition. The results, obtained here for a pipe of circular cross section, can be generalized to a pipe of arbitrary cross section by averaging the wall viscous stresses around the perimeter according to

$$\tau_o = \frac{1}{l_p} \int_{\text{PERIMETER}} \tau_f dl, \quad (12.7)$$

finally yielding the general expression

$$f = \frac{\tau_o}{\frac{1}{8}\rho v_m^2} = f\left(\frac{\rho v_m D_e}{\mu}, \frac{\epsilon}{D_e}\right). \quad (12.8)$$

in terms of the mean velocity v_m defined in (12.3) and the effective hydraulic diameter

$$D_e = \frac{4A}{l_p}, \quad (12.9)$$

with A and l_p representing the cross-section area and the perimeter length for the pipe section at hand. Note, in particular, that for a pipe of circular cross section (12.9) provides $D_e = D$, so that (12.8) reduces to (12.6). The variation of f with Re and ϵ/D_e is displayed in the so-called Moody diagram, given in Fig. 12.3. Among the different simplified expressions that approximate (12.8), the most widely used is the so-called Colebrook's formula

$$\frac{1}{\sqrt{f}} = \log_{10} \left(\frac{\epsilon/D_e}{3.7} + \frac{2.51}{Re\sqrt{f}} \right). \quad (12.10)$$

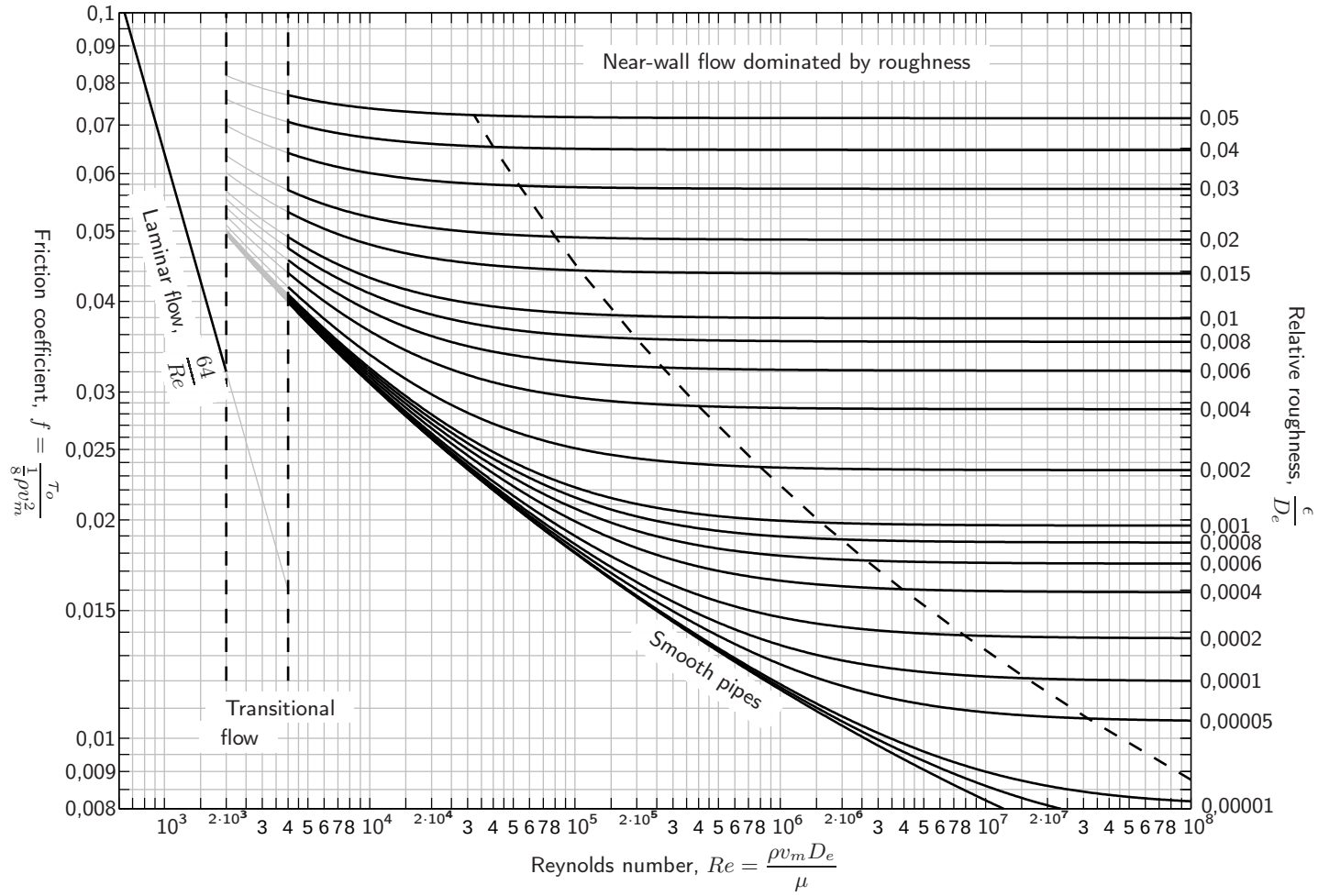


Figure 12.3: Moody Diagram. The expression $f = 64/Re$ used for laminar flow corresponds to the friction coefficient in a pipe of circular cross section, for which $\tau_f = 8\mu v_m/D$, as can be obtained by straightforward derivation of (12.1).

Conservation equations for turbulent flow in pipes

We are now in a position to write the conservation equations of mass, momentum and energy for fully developed turbulent flow in pipes. All fluid properties will be assumed to be uniform when computing the transverse fluxes of the different quantities, thereby neglecting the presence of the boundary layer, so that, in particular, the velocity across the pipe will be taken to be $\langle v \rangle = v_m$. The differential equations for a pipe of variable area $A(x)$ are obtained by applying the integral conservation principles in the control volume of small differential length dx depicted in Fig. 12.4

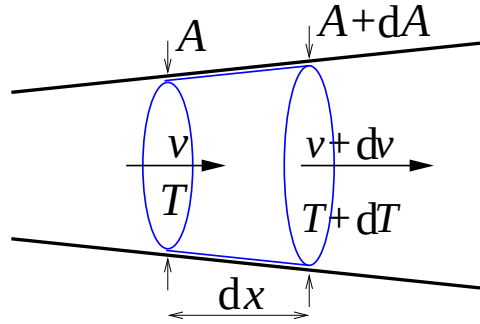


Figure 12.4: Differential control volume.

If the motion is steady, in the sense that the different average flow properties do not vary with time, then the terms involving time derivatives can be neglected when writing the different conservation equations. In particular, the continuity equation

$$\frac{d}{dt} \int_{V_c} \rho dV + \int_{\Sigma_c} \rho(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = 0 \quad (12.11)$$

yields

$$(\rho v A)_{x+dx} - (\rho v A)_x = (\rho v A)_x + dx \frac{d}{dx}(\rho v A) - (\rho v A)_x = 0 \quad (12.12)$$

when a Taylor expansion is used to express the mass flux at $x + dx$, thereby leading to

$$\frac{d}{dx}(\rho v A) = 0 \quad \Rightarrow \quad \rho v A = G, \quad (12.13)$$

where G is the constant mass flux in the pipe.

Similarly, the momentum equation

$$\frac{d}{dt} \int_{V_c} \rho \bar{v} dV + \int_{\Sigma_c} \rho \bar{v}(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = - \int_{\Sigma_c} p \bar{n} \cdot \bar{n} d\sigma + \int_{\Sigma_c} \bar{\tau}' \cdot \bar{n} d\sigma + \int_{V_c} \rho \bar{f}_m dV \quad (12.14)$$

projected in the streamwise direction provides

$$(\rho v^2 A)_{x+dx} - (\rho v^2 A)_x = -(pA)_{x+dx} + (pA)_x + p dA - \tau_o l_P dx - \rho \frac{dgZ}{dx} A dx. \quad (12.15)$$

In writing the last term it is assumed that gravity is the only mass force acting on the fluid, so that $\bar{f}_m = -\nabla(gZ)$, with a streamwise component given by $f_{mx} = -d(gZ)/(dx)$ and Z representing the vertical coordinate. Using in (12.15) Taylor expansions to express the quantities evaluated

at $x + dx$ in terms of those at x , along with the continuity equation (12.13), and dividing the result by $\rho A dx$ yields finally

$$\frac{d}{dx} \left(\frac{v^2}{2} \right) + \frac{1}{\rho} \frac{dp}{dx} + \frac{d}{dx} (gZ) = -\frac{fv^2}{2D_e} \quad (12.16)$$

when the mean wall shear stress τ_o is written in terms of the friction coefficient f given in (12.8). To write the energy balance for the temperature T , assumed to be uniform across the pipe, we use the conservation equation written in the form

$$\begin{aligned} \frac{d}{dt} \left[\int_{V_c} \rho (e + v^2/2 + gZ) dV \right] + \int_{\Sigma_c} \rho (h + v^2/2 + gZ) \bar{v} \cdot \bar{n} d\sigma = \\ \int_{\Sigma_c} \bar{v} \cdot \bar{\tau}' \cdot \bar{n} d\sigma - \int_{\Sigma_c} \bar{q} \cdot \bar{n} d\sigma \end{aligned} \quad (12.17)$$

obtained from (8.112) by considering the case of fixed control volume with gravity as the only mass force. For steady fluid motion, application of (12.17) to the differential control volume of Fig. 12.4 yields after appropriate manipulation

$$\rho v A \frac{d}{dx} \left(h + \frac{v^2}{2} + gZ \right) = -q_o l_p \quad (12.18)$$

where

$$q_o = \frac{1}{l_p} \int_{\text{PERIMETER}} q_w dl \quad (12.19)$$

is the mean value of the heat flux to the wall $q_w = (\bar{q} \cdot \bar{n})_w$ (amount of energy transferred from the fluid to the wall per unit surface per unit time). To evaluate q_w , we shall make use of the so-called Reynolds analogy, which enables the heat flux to be written in terms of the wall shear stress, as discussed earlier in connection with the heat transfer problem in laminar boundary layers. The corresponding expression, given earlier in (10.68), can be written for pipe flow as

$$\frac{q_w}{c[T - T_w + v^2/(2c)]} = \frac{\tau_f}{v}, \quad (12.20)$$

thereby yielding

$$q_o = \frac{c}{v} \left(T - T_w + \frac{v^2}{2c} \right) \tau_o \quad (12.21)$$

and

$$q_o = \frac{f}{8} \rho v \left[c(T - T_w) + \frac{v^2}{2} \right], \quad (12.22)$$

expressions that are also valid for gas flow, provided that we replace c with c_p to give

$$q_o = \frac{f}{8} \rho v \left[c_p(T - T_w) + \frac{v^2}{2} \right]. \quad (12.23)$$

Note that, although the Reynolds analogy is strictly valid only when the Prandtl number $Pr = \mu c/k$ equals unity ($Pr = \mu c_p/k = 1$ for gas flow), for turbulent pipe flow the above equations provide sufficient accuracy under most conditions.

Substituting (12.22) into (12.18) gives

$$\frac{d}{dx} \left(h + \frac{v^2}{2} + g\mathcal{Z} \right) = -\frac{f}{2D_e} \left[c(T - T_w) + \frac{v^2}{2} \right], \quad (12.24)$$

which admits simplified forms for liquid flow

$$\frac{dT}{dx} = -\frac{f}{2D_e} (T - T_w) + \frac{f}{4D_e} \frac{v^2}{c}, \quad (12.25)$$

obtained by subtracting (12.16), and also for gas flow

$$\frac{d}{dx} \left(c_p T + \frac{v^2}{2} \right) = -\frac{f}{2D_e} \left[c_p (T - T_w) + \frac{v^2}{2} \right], \quad (12.26)$$

obtained with use made of $h = c_p T$ by assuming gravity to have a negligible effect on the temperature field¹. Equation (12.25) can be further simplified when, as occurs in most cases, the temperature increment associated with viscous dissipation $\Delta T \sim (v^2/c)$ is small. For gas flow, viscous dissipation is seen to be important whenever the flow Mach number $M = v/a$ is of order unity, but can be neglected for $M \ll 1$.

Turbulent liquid flow in pipes

The equations that determine the evolution of v , p , and T for a liquid moving in turbulent motion along a pipe of variable cross section $A(x)$ are

$$vA = \frac{G}{\rho} = Q, \quad (12.27)$$

$$\frac{d}{dx} \left(\frac{v^2}{2} + \frac{p}{\rho} + g\mathcal{Z} \right) = -\frac{fv^2}{2D_e}, \quad (12.28)$$

$$\frac{dT}{dx} = -\frac{f}{2D_e} (T - T_w) + \frac{f}{4D_e} \frac{v^2}{c}, \quad (12.29)$$

the first two being derived from the general continuity and momentum equations (12.13) and (12.16) by taking the density to be constant.

Pipes of constant cross section

The velocity and pressure can be obtained, independently of the temperature field, by integrating (12.27) and (12.28). The problem is simplified when the pipe has constant cross section, so that (12.27) gives $v = \text{constant}$ and (12.28) reduces to

$$\frac{d}{dx} (p + \rho g\mathcal{Z}) = -\frac{f\rho v^2}{2D_e}. \quad (12.30)$$

¹The effect of gravity leads to temperature differences ΔT on the order of $g\Delta\mathcal{Z}/c_p$, as can be anticipated from (12.24), so that a variation of one kelvin is associated with differences in altitude of the order of $\Delta\mathcal{Z} \sim 100$ m.

Since in pipes of constant section D_e is independent of x , the Reynolds number $\rho D_e v / \mu$ and the relative roughness ϵ / D_e do not change along the pipe, thereby giving a constant value of f . Straightforward integration of (12.30) from the entrance section then yields

$$(p + \rho g \mathcal{Z})_e - (p + \rho g \mathcal{Z}) = \frac{fx}{D} \frac{\rho v^2}{2}, \quad (12.31)$$

which can be evaluated at the outlet to give

$$(p + \rho g \mathcal{Z})_e - (p + \rho g \mathcal{Z})_o = \frac{fL}{D} \frac{\rho v^2}{2} \quad (12.32)$$

for the drop of reduced pressure along the pipe. On the other hand, integration of (12.29) subject to a boundary temperature $T = T_e$ at the entrance provides

$$T = T_w + \frac{v^2}{2c} + \left(T_e - T_w - \frac{v^2}{2c} \right) \exp \left(-\frac{fx}{2D_e} \right), \quad (12.33)$$

which predicts an outlet temperature

$$T_o = T_w + \frac{v^2}{2c} + \left(T_e - T_w - \frac{v^2}{2c} \right) \exp \left(-\frac{fL}{2D_e} \right). \quad (12.34)$$

Note that, for long pipes such that $fL/D_e \gg 1$, the temperature in the liquid becomes equal to the recovery temperature $T_w + v^2/(2c)$ for $x \gg D_e/f$. For water, the temperature increment due to viscous dissipation $v^2/(2c)$ is always very small, giving for instance $\Delta T \sim 0.012$ K for $v = 10$ m/s.

Minor pressure losses

Typical hydraulic systems are composed of several straight (or almost straight) pipes connected through joints of different shape, including bends, tees, contractions, expansions, along with valves designed to control the flow. The flow across each individual element provides an additional loss of stagnation pressure, which needs to be accounted for in computing the system performance. The pressure loss is associated with the effect of viscous dissipation acting in regions of unsteady separated flow inside the given element. To see it, consider for instance the flow across the straight bend shown in Fig. 12.5, which joins two pipes of different cross section. While the flow pattern inside can be quite complicated, when we move sufficiently upstream and downstream the average velocity recovers the almost uniform profile corresponding to turbulent flow in long pipes.

To evaluate the pressure loss, we write the integral conservation equation for the kinetic energy (4.24) in the form

$$\begin{aligned} \frac{d}{dt} \left[\int_{V_c} \rho \left(\frac{v^2}{2} + g\mathcal{Z} \right) dV \right] + \int_{\Sigma_c} \rho \left(\frac{p}{\rho} + \frac{v^2}{2} + g\mathcal{Z} \right) \bar{v} \cdot \bar{n} d\sigma = \\ \int_{\Sigma_c} \bar{v} \cdot \bar{\tau}' \cdot \bar{n} d\sigma - \int_{V_c} \phi_v dV, \end{aligned} \quad (12.35)$$

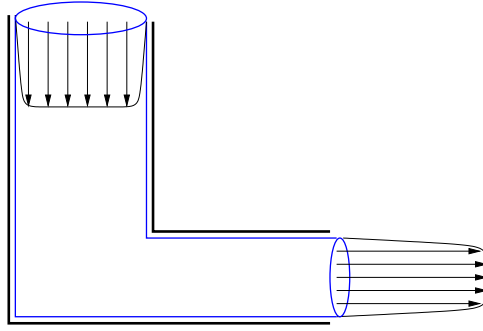


Figure 12.5: Bend joining two pipes of different section.

which applies to constant-density flow in a fixed control volume. Assuming uniform profiles at the entrance and outlet of the control volume of Fig. 12.5 gives

$$(p + \rho v^2/2 + \rho g\mathcal{Z})_e - (p + \rho v^2/2 + \rho g\mathcal{Z})_o = \frac{\int_{V_c} \phi_v dV}{Q}, \quad (12.36)$$

where $Q = -\int_{\Sigma_e} \bar{v} \cdot \bar{n} d\sigma = \int_{\Sigma_o} \bar{v} \cdot \bar{n} d\sigma$ is the volume flux.

Just as we did before when analyzing pressure losses in a pipe, one may use now dimensional analysis to reduce the parametric dependence of the pressure loss between the entrance and the outlet of a given element. The analysis provides the general expression

$$(p + \rho v^2/2 + \rho g\mathcal{Z})_e - (p + \rho v^2/2 + \rho g\mathcal{Z})_o = K \frac{1}{2} \rho v^2, \quad (12.37)$$

where the loss coefficient K is in general a function of the Reynolds number and of the geometry of the element. If the Reynolds number is sufficiently large, then the value of K is constant for a given geometry. It is worth mentioning that, when there is a jump in velocity across the element, as occurs in the bend of Fig. 12.5, it is customary to define the loss coefficient such that the accompanying factor $\frac{1}{2} \rho v^2$ is evaluated where the velocity is maximum, i.e., at the outlet for a contraction and at the inlet for an expansion.

Turbulent gas flow in pipes

The evolution of v , p , and T in steady turbulent gas motion in a pipe of variable cross section $A(x)$ is determined by integration of

$$\rho v A = G, \quad (12.38)$$

$$\frac{d}{dx} \left(\frac{v^2}{2} \right) + \frac{1}{\rho} \frac{dp}{dx} = -\frac{f v^2}{2 D_e}, \quad (12.39)$$

$$\frac{d}{dx} \left(c_p T + \frac{v^2}{2} \right) = -\frac{f}{2 D_e} \left[c_p (T - T_w) + \frac{v^2}{2} \right], \quad (12.40)$$

supplemented with the equation of state

$$\frac{p}{\rho} = R_g T \quad (12.41)$$

and subject to the boundary conditions

$$x = 0 : \quad p - p(0) = T - T(0) = v - v(0) = 0. \quad (12.42)$$

As can be seen, the effect of gravity has been neglected in (12.39) and (12.40), an excellent approximation under most conditions of interest. For pipes connected to a container, the condition of isentropic steady motion in the entrance region allows the thermodynamic properties in the container to be related to those at the entrance according to

$$\left(\frac{T_c}{T(0)} \right) = \left(\frac{p_c}{p(0)} \right)^{(\gamma-1)/\gamma} = \left(\frac{\rho_c}{\rho(0)} \right)^{\gamma-1} = \left(\frac{a_c}{a(0)} \right)^2 = 1 + \frac{\gamma-1}{2} M^2(0). \quad (12.43)$$

Simplifications for long pipes $fL/D_e \gg 1$

In general, numerical integration is needed to solve the problem (12.38)–(12.43). An approximate solution emerges for long pipes such that

$$\frac{fL}{D_e} \gg 1. \quad (12.44)$$

Order-of-magnitude estimates of (12.39)

$$\underbrace{\frac{d}{dx} \left(\frac{v^2}{2} \right)}_{\frac{v^2}{L}} + \underbrace{\frac{1}{\rho} \frac{dp}{dx}}_{\frac{\Delta p}{\rho L}} = - \underbrace{\frac{fv^2}{2D_e}}_{\frac{fv^2}{D_e}} \quad (12.45)$$

reveal that the acceleration along the pipe is negligibly small, so that in the resulting motion the pressure force balances the wall friction according to

$$\frac{1}{\rho} \frac{dp}{dx} = - \frac{fv^2}{2D_e}, \quad (12.46)$$

giving characteristic velocities of order

$$v^2 \sim \left(\frac{fL}{D_e} \right)^{-1} \frac{\Delta p}{\rho}. \quad (12.47)$$

The gas moves therefore at low Mach numbers

$$M^2 = \frac{v^2}{\gamma p / \rho} \sim \left(\frac{fL}{D_e} \right)^{-1} \frac{\Delta p}{p} \ll 1. \quad (12.48)$$

As a result, the conditions at the entrance (12.43) reduce to

$$T(0) - T_c = p(0) - p_c = \rho(0) - \rho_c = a(0) - a_c = 0, \quad (12.49)$$

and the kinetic energy can be neglected compared with the enthalpy in (12.40) to give

$$\underbrace{\frac{dT}{dx}}_{\frac{\Delta T}{L}} = - \underbrace{\frac{f}{2D_e} (T - T_w)}_{\left(\frac{fL}{D_e} \right) \frac{\Delta T}{L}}. \quad (12.50)$$

As can be seen, if $L \gg D_e/f$ then heat transfer from the wall is dominant in the pipe, thereby reducing the solution for the temperature in the pipe simply to

$$T = T_w. \quad (12.51)$$

In the first approximation, the pressure becomes linearly proportional to the density

$$p = R_g T_w \rho, \quad (12.52)$$

as dictated by (12.41). Multiplying (12.46) by $R_g T_w \rho^2$ and using (12.38) to write $\rho v = G/A$ provides

$$p \frac{dp}{dx} = -\frac{f R_g T_w}{2 D_e} \left(\frac{G}{A} \right)^2 \quad (12.53)$$

which can be integrated with initial condition $p(0) = p_c$ to give

$$p_c^2 - p^2(x) = R_g T_w G^2 \int_0^x \frac{f}{D_e A^2} dx \quad (12.54)$$

for the pressure variation along the pipe. If the pipe discharges to the atmosphere, where the pressure is p_a , one can evaluate this last expression at $x = L$ to give

$$G = \left(\int_0^L \frac{f}{D_e A^2} dx \right)^{-1/2} \left(\frac{p_c^2 - p_a^2}{R_g T_w} \right)^{1/2}, \quad (12.55)$$

which further reduces to

$$G = A \left(\frac{f L}{D_e} \right)^{-1/2} \left(\frac{p_c^2 - p_a^2}{R_g T_w} \right)^{1/2} \quad (12.56)$$

when the pipe has constant cross section.

Note that these friction-dominated flows are a special case of low-Mach-number flows. In previous examples we have seen that gas flow with $M \ll 1$ is associated with small pressure differences $\Delta p/p \ll 1$, with the balance of acceleration and pressure forces giving $M^2 \sim \Delta p/p$. In turbulent flow in long pipes the dominant effect of wall friction yields the balance (12.48) instead, so that a low-Mach-number flow is established in long pipes with relative pressure differences of order unity, i.e., $p_c - p_a \sim p_a$. The accompanying relative density difference $[\rho_c - \rho(L)]/\rho_c \sim (p_c - p_a)/p_a$ is also of order unity, according to (12.52) so that, although $M \ll 1$, the gas does not behave as a constant-density fluid.

Also of interest is that the simplified solution (12.51) for the temperature does not satisfy the boundary condition $T(0) = T_c$, revealing that there exist an entrance region where the temperature adapts from the entrance value $T(0) = T_c$ to the wall value $T = T_w$. The evolution of $T(x)$ in this region would be determined by integration of (12.50) with initial condition $T(0) = T_c$ to give

$$T = T_w + (T_c - T_w) \exp \left(-\frac{f x}{2 D_e} \right). \quad (12.57)$$

As can be seen, transition occurs over distances of order $x \sim D/f$, much smaller than L , so that the approximation (12.51) is valid for most of the pipe (i.e., at distances $D/f \ll x \leq L$).

Simplifications for small driving pressures $p_c/p_a - 1 \ll 1$

Another case of interest for which a simple solution is available appears when the container pressure is not far from the discharge pressure, $(p_c - p_a) \ll p_c$, while $(T_w - T_c)/T_c$ and fL/D_e take arbitrary values (e.g., of order unity). The analysis is to be carried out for pipes of constant cross section, for which the mass flux per unit surface

$$\mathcal{G} = \rho v = G/A \quad (12.58)$$

is constant along the pipe. According to the order-of-magnitude estimates given in (12.45), with $fL/D \sim O(1)$, all three terms must be retained in the momentum balance equation. The resulting characteristic velocity, of order $v \sim (p_c - p_a)/\rho$, corresponds, again, to small values of the Mach number

$$M^2 = \frac{v^2}{\gamma p / \rho} \sim \frac{p_c - p_a}{p_c} \ll 1, \quad (12.59)$$

so that the energy equation reduces to

$$\frac{dT}{dx} = -\frac{f}{2D_e}(T - T_w). \quad (12.60)$$

Since the Mach number everywhere is small, the boundary value $T(0)$ needed in integrating (12.60) reduces to $T(0) = T_c$, as can be inferred from (12.43), so that the integration yields (12.57) for the temperature variation along the pipe. Since the pressure changes by a small amount, one may then use (12.41) and (12.58) to obtain the accompanying distributions of velocity and density along the pipe

$$v = \frac{\mathcal{G}}{\rho} = \frac{\mathcal{G}R_g}{p_c}T = \frac{\mathcal{G}R_g}{p_c} \left[T_w + (T_c - T_w) \exp\left(-\frac{fx}{2D_e}\right) \right]. \quad (12.61)$$

As can be seen, while the pressure differences are very small, for $(T_w - T_c)/T_c \sim O(1)$ and $fL/D_e \sim O(1)$, the temperature, density and velocity vary by amounts of order unity as one moves along the pipe.

To close the problem, we need to determine the mass flow rate $G = A\mathcal{G}$, which depends on the small pressure difference $p_c - p_a$. The solution requires integration of the momentum equation (12.45), which can be written in the alternative form

$$\frac{dp}{dx} = -\frac{\mathcal{G}^2 R_g}{p_c} \left(\frac{dT}{dx} + \frac{f}{2D_e} T \right) \quad (12.62)$$

when use is made of the first two equations in (12.61). Combining linearly with (12.60) provides then the simple expression

$$\frac{dp}{dx} = -\frac{\mathcal{G}^2 R_g}{2p_c} \frac{fT_w}{D_e}, \quad (12.63)$$

indicating that the pressure varies linearly along the pipe, so that

$$p(0) - p(x) = \frac{\mathcal{G}^2 R_g T_w}{2p_c} \frac{fx}{D_e}. \quad (12.64)$$

The boundary value $p(0)$ is determined from the analysis of the entrance region, where the pressure varies by an amount that, albeit small, is comparable to the pressure drop along the

pipe, so that the approximation $p(0) = p_c$ cannot be used. Instead, we linearize (12.43) for $M(0) \ll 1$ to give

$$\frac{p_c}{p(0)} = \left(1 + \frac{\gamma-1}{2} M^2(0)\right)^{\gamma/(\gamma-1)} \simeq 1 + \frac{\rho(0)v^2(0)}{2p(0)}, \quad (12.65)$$

leading to

$$p(0) = p_c - \frac{\mathcal{G}^2 R_g T_c}{2p_c}. \quad (12.66)$$

Using this last equation in (12.64) and evaluating the resulting expression at $x = L$ where $p(L) = p_a$ provides

$$p_c - p_a = \frac{\mathcal{G}^2 R_g T_w}{2p_c} \left(\frac{fL}{D_e} + \frac{T_c}{T_w} \right) \quad (12.67)$$

finally yielding

$$G = A\mathcal{G} = A \left(\frac{fL}{D_e} + \frac{T_c}{T_w} \right)^{-1/2} \left(\frac{2p_c(p_c - p_a)}{R_g T_w} \right)^{1/2} \quad (12.68)$$

for the mass flow rate. Naturally, the result obtained from this last expression in the limit $fL/D_e \gg 1$ coincides with that obtained from (12.56) when $p_c - p_a \ll p_a$.

Thermally insulated pipes of constant cross section

An implicit solution to (12.38)–(12.43) exists for flows in pipes of constant cross section when the heat transferred to the gas from the wall is negligible (for instance, because the pipe wall is insulated), so that the equations of motion become

$$\rho v = \mathcal{G} = G/A, \quad (12.69)$$

$$\frac{d}{dx} \left(\frac{v^2}{2} \right) + \frac{1}{\rho} \frac{dp}{dx} = -\frac{fv^2}{2D_e}, \quad (12.70)$$

$$\frac{d}{dx} \left(c_p T + \frac{v^2}{2} \right) = 0. \quad (12.71)$$

The last equation indicates that the stagnation enthalpy $h_0 = h + v^2/2$ is conserved along the pipe, that is,

$$h_0 = c_p T_0 = c_p T + \frac{v^2}{2}. \quad (12.72)$$

Combining this last result written in the form

$$R_g T = \frac{\gamma-1}{\gamma} \left(h_0 - \frac{v^2}{2} \right) \quad (12.73)$$

with the equation of state (12.41) yields

$$p = \frac{\gamma-1}{\gamma} \mathcal{G} \left(\frac{h_0}{v} - \frac{v}{2} \right). \quad (12.74)$$

Differentiating to obtain

$$\frac{1}{\rho} \frac{dp}{dx} = -\frac{\gamma-1}{\gamma} \left(\frac{h_0}{v^2} + \frac{1}{2} \right) \frac{d}{dx} \left(\frac{v^2}{2} \right) \quad (12.75)$$

and substituting the result into (12.70) yields

$$\frac{1}{2\gamma} \left(\frac{\gamma+1}{2} - (\gamma-1) \frac{h_0}{v^2} \right) \frac{dv^2}{dx} = -\frac{fv^2}{2D_e}, \quad (12.76)$$

which can be integrated to determine the evolution of $v^2(x)$ from its entrance value $v^2(0)$. Before proceeding with the integration, it is convenient to discuss some aspects of the solution. To that end, we use (12.73) to write the expression

$$M^2 = \frac{v^2}{a^2} = \frac{1}{\gamma-1} \frac{v^2}{h_0 - v^2/2} \quad (12.77)$$

linking the velocity and the Mach number, which can be used to express (12.76) in the alternative form

$$\frac{1}{2\gamma} \left(\frac{1}{M^2} - 1 \right) \frac{dv^2}{dx} = \frac{fv^2}{2D_e}. \quad (12.78)$$

As can be seen, if $M < 1$, the effect of friction increases v as one moves along the pipe, thereby decreasing ρ , T , and p , as can be inferred from (12.69), (12.72), and (12.75), and increasing the associated Mach number $M = v/(\gamma R_g T)^{1/2}$. The Mach number in the pipe remains however subsonic, with a singularity emerging in (12.78) as $M \rightarrow 1$ (conversely, for supersonic flow in a pipe the effect of friction decreases the Mach number towards sonic conditions). For subsonic flow, the peak Mach number $M(L) \leq 1$ is therefore reached at the pipe exit. For a given value of $M(0)$ there exists a length $L_{M=1}$ for which the pipe would choke ($M(L_{M=1}) = 1$), and if the pipe length is $L > L_{M=1}$ then there does not exist a solution with that value of $M(0)$.

Integrating (12.76) with boundary condition $v(0)$ at $x = 0$ gives

$$\ln \left(\frac{v^2(0)}{v^2(x)} \right) - \frac{2(\gamma-1)}{(\gamma+1)} \left(\frac{1}{v^2(x)} - \frac{1}{v^2(0)} \right) h_0 = \frac{\gamma}{\gamma+1} \frac{2fx}{D_e}, \quad (12.79)$$

which can be written by virtue of (12.77) in terms of the Mach number to give

$$\ln \left[\frac{M^2(0)}{M^2(x)} \frac{1 + (\gamma-1)M^2(x)/2}{1 + (\gamma-1)M^2(0)/2} \right] + \frac{2}{\gamma+1} \left(\frac{1}{M^2(0)} - \frac{1}{M^2(x)} \right) = \frac{\gamma}{\gamma+1} \frac{2fx}{D_e}. \quad (12.80)$$

This last expression provides the value of $L_{M=1}$ for which $M(L) = 1$ according to

$$\frac{\gamma}{\gamma+1} \frac{2fL_{M=1}}{D_e} = \ln \left[\frac{(\gamma+1)M^2(0)}{2 + (\gamma-1)M^2(0)} \right] + \frac{2}{\gamma+1} \left(\frac{1}{M^2(0)} - 1 \right). \quad (12.81)$$

On the other hand, an expression relating the Mach number and the pressure follows from writing (12.74) with use made of (12.77) to give

$$p^2 = \frac{\gamma-1}{\gamma^2} \frac{\mathcal{G}^2 h_0}{M^2[1 + (\gamma-1)M^2/2]}, \quad (12.82)$$

which can be used to compute the pressure at a given section in terms of that found at the entrance according to

$$\left(\frac{p(0)}{p(x)} \right)^2 = \frac{M(x)^2[1 + (\gamma-1)M(x)^2/2]}{M^2(0)[1 + (\gamma-1)M^2(0)/2]}. \quad (12.83)$$

In solving the discharge of a container through a pipe, it is convenient to rewrite this last equation in terms of the stagnation pressure at the inlet

$$p_0(0) = p(0) \left(1 + \frac{\gamma-1}{2} M^2(0) \right)^{\gamma/(\gamma-1)} \quad (12.84)$$

to give

$$\left(\frac{p_0(0)}{p(x)} \right)^2 \frac{M^2(0)}{\left(1 + \frac{\gamma-1}{2} M^2(0) \right)^{(\gamma+1)/(\gamma-1)}} = M(x)^2 \left(1 + \frac{\gamma-1}{2} M(x)^2 \right) \quad (12.85)$$

where $p_0(0) = p_c$, as dictated by (12.43).

Evaluating (12.80) and (12.85) at the pipe exit provides

$$\ln \left[\frac{M^2(0)}{M^2(L)} \frac{1 + (\gamma-1)M^2(L)/2}{1 + (\gamma-1)M^2(0)/2} \right] + \frac{2}{\gamma+1} \left(\frac{1}{M^2(0)} - \frac{1}{M^2(L)} \right) = \frac{\gamma}{\gamma+1} \frac{2fL}{D_e} \quad (12.86)$$

and

$$\left(\frac{p_0(0)}{p(L)} \right)^2 \frac{M^2(0)}{\left(1 + \frac{\gamma-1}{2} M^2(0) \right)^{(\gamma+1)/(\gamma-1)}} = M(L)^2 \left(1 + \frac{\gamma-1}{2} M(L)^2 \right), \quad (12.87)$$

which can be used to compute the solution for a given value of $2fL/D_e$ according to

- If the pipe is not choked, then $p(L) = p_a$ and the simultaneous solution of (12.86) and (12.87) determines $M(0)$ and $M(L)$.
- If the pipe is choked, then $M(L) = 1$ and (12.86) determines $M(0)$, while (12.87) gives $p(L)/p_0(0)$.

Once the value of $M(0)$ is known, the mass flow rate can be calculated from

$$G = \rho(0)v(0)A = M(0) \left(1 + \frac{\gamma-1}{2} M^2(0) \right)^{-\frac{(\gamma+1)}{2(\gamma-1)}} \rho_0(0)a_0(0)A, \quad (12.88)$$

where the stagnation properties are those of the container, i.e., $\rho_0(0) = \rho_c$ and $a_0(0) = a_c$.

The computation is facilitated by representing the solution of (12.86) and (12.87) in graphical form by plotting $p_0(0)/p(L)$ as a function of $2fL/D_e$ for constant values of $M(0)$ and also for constant values of $M(L)$, giving the results shown in Fig. 12.6. When the pipe is not choked, the flow conditions are identified in the diagram by the values of $M(0)$ and $M(L) \leq 1$ corresponding to $p_0(0)/p(L) = p_c/p_a$ and $2fL/D_e$. The procedure fails when the point falls above the curve $M(L) = 1$, an indication that the starting assumption $p(L) = p_a$ was wrong and that the flow is choked instead. In that case, the values of $M(0)$ and $p_0(0)/p(L)$ are given by the point along the curve $M(L) = 1$ corresponding to the value of $2fL/D_e$.

Frictionless flow in pipes of constant cross section

An implicit solution exists also for flows in pipes of constant cross section when the wall friction is negligible, thereby enabling (12.38) and (12.39) to be written in terms of the fluid properties at the pipe entrance as

$$\rho(x)v(x) = \mathcal{G} = \rho(0)v(0), \quad (12.89)$$

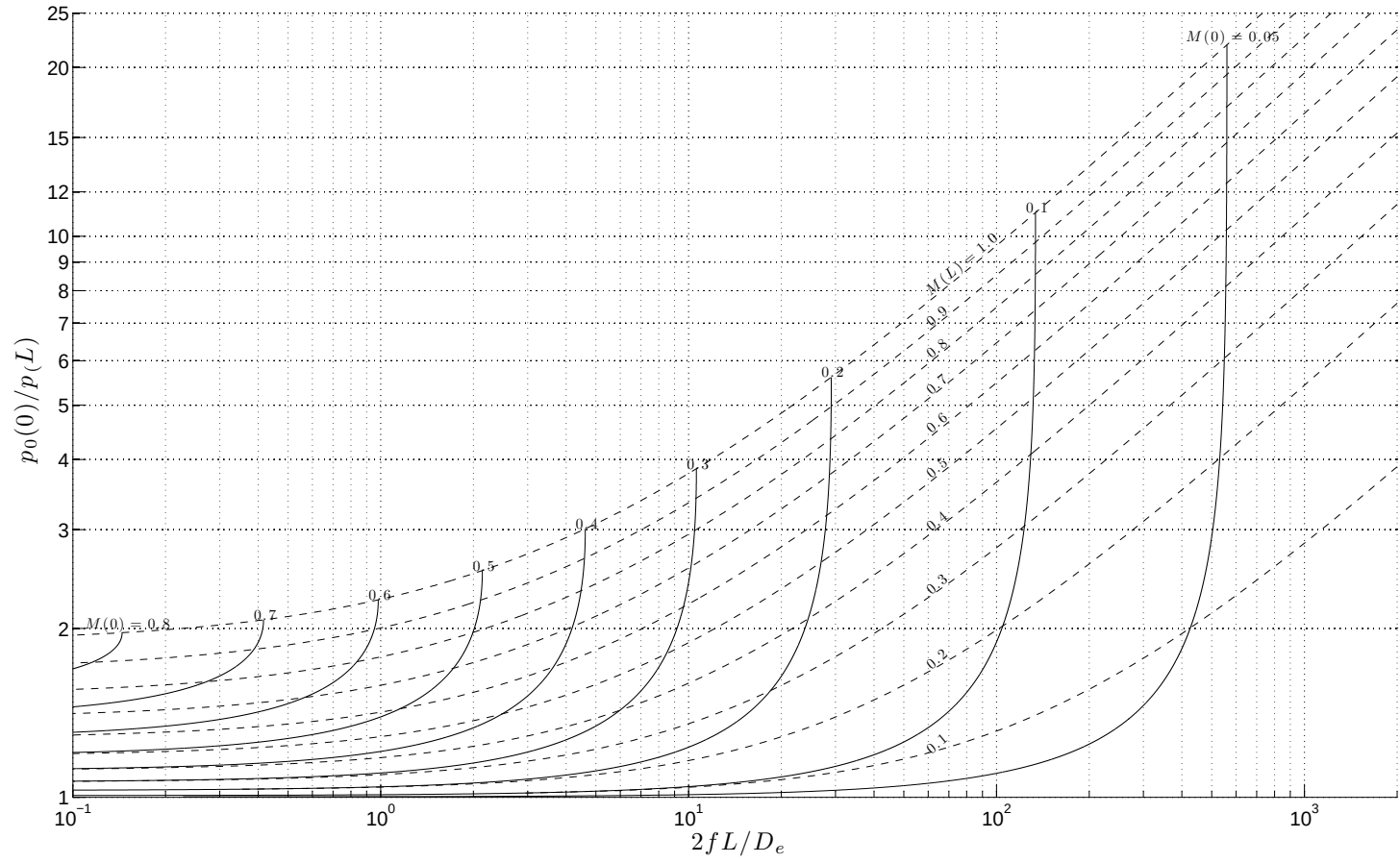


Figure 12.6: The variation of $p_0(0)/p(L)$ with $2fL/D_e$ for constant values of $M(0)$ (solid curves) and for constant values of $M(L)$ (dashed curves) as obtained from (12.86) and (12.87).

and

$$\rho v \frac{dv}{dx} + \frac{dp}{dx} = 0 \quad \rightarrow \quad p(x) + \rho(x)v(x)^2 = p(0) + \rho(0)v^2(0). \quad (12.90)$$

All terms need to be retained in the energy equation (12.40)

$$\rho v \frac{d}{dx} \left(h + \frac{v^2}{2} \right) = -\frac{4q_o}{D_e}, \quad (12.91)$$

where $-q_o$ is the heat added to the fluid per unit surface per unit time, with

$$q_o = \frac{f}{8} \rho v \left[c_p(T - T_w) + \frac{v^2}{2} \right]. \quad (12.92)$$

correspondingly representing, as in (12.23), the heat flux from the fluid to the wall. Integrating (12.91) along the pipe gives

$$h(x) + \frac{v(x)^2}{2} = h(0) + \frac{v(0)^2}{2} - \frac{4}{\mathcal{G}} \int_0^x \frac{q_o}{D_e} dx = h_0(0) + Q(x), \quad (12.93)$$

where $h_0(0)$ is the stagnation enthalpy at the pipe entrance and

$$Q(x) = -\frac{4}{\mathcal{G}} \int_0^x \frac{q_o}{D_e} dx \quad (12.94)$$

is the amount of heat added in the pipe stretch $0 - x$ per unit mass of gas. For a given value of $Q(x)$, equations (12.89), (12.90), and (12.93) supplemented with the equation of state written in the form

$$\frac{h(x)}{h(0)} = \frac{T(x)}{T(0)} = \frac{p(x)/\rho(x)}{p(0)/\rho(0)} \quad (12.95)$$

determine the values of $\rho(x)$, $T(x)$, $p(x)$, $v(x)$, and $h(x)$ in terms of their entrance value $\rho(0)$, $T(0)$, $p(0)$, $v(0)$, and $h(0)$.

In solving the problem, it is convenient to introduce the Mach number by rewriting (12.90) in the alternative form

$$\frac{p(x)}{p(0)} = \frac{1 + \gamma M^2(0)}{1 + \gamma M^2(x)} \quad (12.96)$$

Combining the energy equation (12.93) written in the form

$$\frac{T(x)}{T(0)} = \frac{1 + (\gamma - 1)M^2(0)/2}{1 + (\gamma - 1)M^2(x)/2} \left(1 + \frac{Q(x)}{h_0(0)} \right). \quad (12.97)$$

with the equation of state (12.95) and the continuity equation (12.89) provides a second equation linking the pressure and the Mach number according to

$$\begin{aligned} \left(\frac{p(x)}{p(0)} \right)^2 &= \left(\frac{\rho(x)T(x)}{\rho(0)T(0)} \right)^2 = \left(\frac{v(0)T(x)}{v(x)T(0)} \right)^2 \\ &= \left(\frac{M(0)}{M(x)} \right)^2 \frac{1 + (\gamma - 1)M^2(0)/2}{1 + (\gamma - 1)M^2(x)/2} \left(1 + \frac{Q(x)}{h_0(0)} \right). \end{aligned} \quad (12.98)$$

Eliminating the pressure between (12.96) and (12.98) yields

$$\frac{h_0(x)}{h_0(0)} = \frac{M^2(x)[1 + (\gamma - 1)M^2(x)/2]/[1 + \gamma M^2(x)]^2}{M^2(0)[1 + (\gamma - 1)M^2(0)/2]/[1 + \gamma M^2(0)]^2} = 1 + \frac{Q(x)}{h_0(0)} \quad (12.99)$$

for the variation of the Mach number along the pipe.

This last equation can be expressed in the compact form

$$\frac{h_0(x)}{h_0(0)} = \frac{T_0(x)}{T_0(0)} = \left(\frac{F[M(x)]}{F[M(0)]} \right)^2 = 1 + \frac{Q(x)}{h_0(0)} \quad (12.100)$$

by introducing the Mach-number function

$$F(M) = \frac{M}{1 + \gamma M^2} \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{1/2}, \quad (12.101)$$

which is plotted in Fig. 12.7 for $\gamma = 1.4$. As can be seen, for subsonic flow F increases with increasing M to reach a peak value $F \simeq 0.456$ at $M = 1$ and then decreases in the supersonic regime. If the flow conditions at the pipe entrance are subsonic, i.e., $M(0) < 1$, then according to (12.100) adding heat would increase the value of $F(M)$, producing a larger Mach number, as indicated in the figure. As a result, in subsonic flow with heat addition the Mach number increases continuously towards sonic conditions. Just as it happens in convergent nozzles and also in pipes with friction, the flow may choke in pipe flow with heat addition. For each value of $M(0)$ there exists a critical value of $Q/h_0(0)$ given by

$$\left(\frac{Q}{h_0(0)} \right)_{M=1} = \left(\frac{0.456}{F[M(0)]} \right)^2 - 1 \quad (12.102)$$

for which $M(L) = 1$. For larger values of $Q/h_0(0)$ there does not exist a solution with that entrance Mach number $M(0)$. Note that cooling the supersonic stream would conversely tend to reduce the flow Mach number along the pipe. Also of interest is that for supersonic flow the effects of heat addition and heat removal are just the opposite as those discussed above for subsonic flow.

In solving the discharge of a container through a pipe, it is convenient to rewrite (12.96) in terms of the stagnation pressure at the inlet

$$p_0(0) = p(0) \left(1 + \frac{\gamma - 1}{2} M^2(0) \right)^{\gamma/(\gamma-1)}. \quad (12.103)$$

and evaluate the resulting expressions at the outlet to give

$$\frac{p(L)}{p_0(0)} = \frac{[1 + \gamma M^2(0)]/[1 + \gamma M^2(L)]}{[1 + \frac{\gamma-1}{2} M^2(0)]^{\gamma/(\gamma-1)}}. \quad (12.104)$$

On the other hand, evaluating also (12.99) at $x = L$ gives

$$\frac{M^2(L)[1 + (\gamma - 1)M^2(L)/2]/[1 + \gamma M^2(L)]^2}{M^2(0)[1 + (\gamma - 1)M^2(0)/2]/[1 + \gamma M^2(0)]^2} = 1 + \frac{Q(L)}{h_0(0)}. \quad (12.105)$$

If the pipe is not choked, then the above two equations determine $M(0)$ and $M(L)$ for given values of $p(L)/p_0(0) = p_a/p_c$ and $Q(L)/h_0(0)$, while for choked flow the solution with $M(L) = 1$ provides $p(L)/p_c$ and $M(0)$ for a given value of $Q(L)/h_0(0)$, with the mass flow rate given in (12.88) in terms of $M(0)$.

The computation is facilitated by representing the solution of (12.104) and (12.105) in graphical form by plotting $p_0(0)/p(L)$ as a function of $Q(L)/h_0(0)$ for constant values of $M(0)$ and also

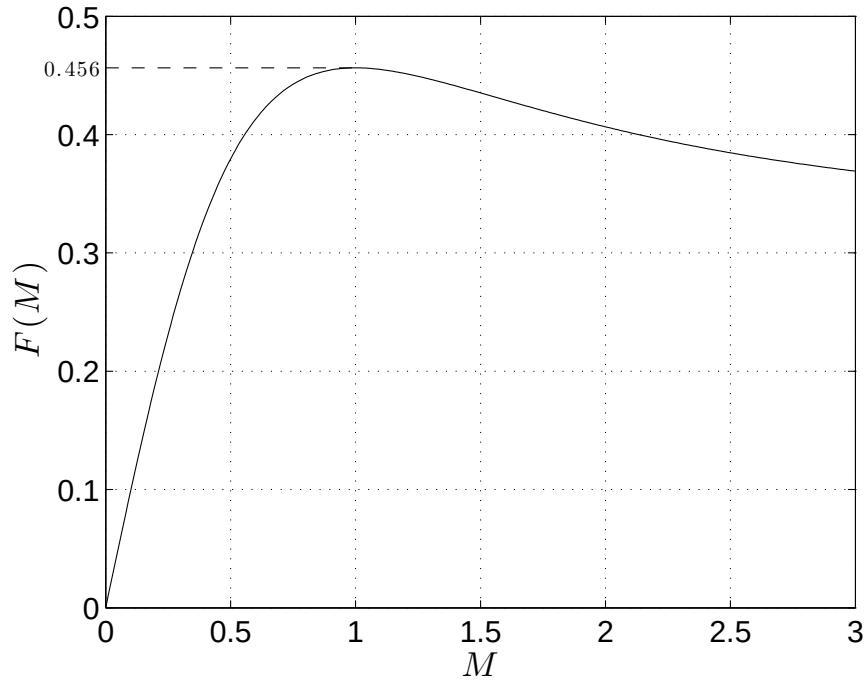


Figure 12.7: The variation of the function F as obtained from evaluating (12.101) for $\gamma = 1.4$.

for constant values of $M(L)$, giving the results shown in Fig. 12.8. The lookup procedure is identical to that explained before in connection with the results of insulated pipes. Thus, when the pipe is not choked, the flow conditions are identified in the diagram by the values of $M(0)$ and $M(L) \leq 1$ corresponding to $p_0(0)/p(L) = p_c/p_a$ and $Q(L)/h_0(0)$. The procedure fails when the point falls above the curve $M(L) = 1$, an indication that the starting assumption $p(L) = p_a$ was wrong and that the flow is choked instead. In that case, the values of $M(0)$ and $p_0(0)/p(L)$ are given by the point along the curve $M(L) = 1$ corresponding to the value of $Q(L)/h_0(0)$.

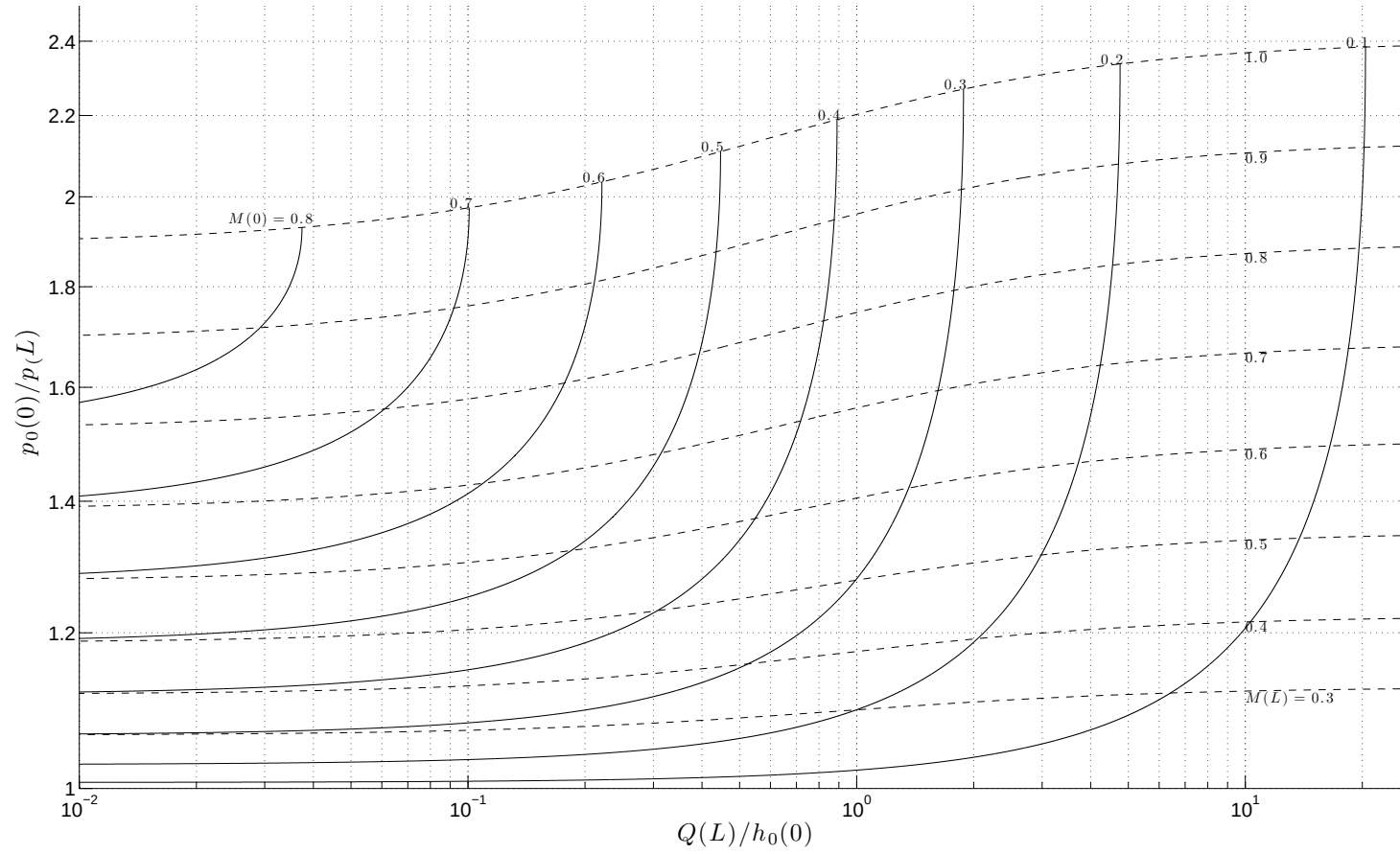


Figure 12.8: The variation of $p_0(0)/p(L)$ with $Q(L)/h_0(0)$ for constant values of $M(0)$ (solid curves) and for constant values of $M(L)$ (dashed curves) as obtained from (12.104) and (12.105).